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博士学位论文

气候变化背景下突尼斯干旱风险的综合评估：基于遥感与建模
方法的研究及其对可持续土地-水纽带管理的影响

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摘要

制定有效的可持续水与土地管理政策对解决区域水资源短缺问题至关重要，也是干旱地区研究的核心议题。气候变化、频发干旱以及土地退化和生物多样性丧失，对全球旱地可持续发展构成重大威胁。这些问题在水资源匮乏的干旱国家尤为严峻，其农业用地日益干旱且更易受长期干旱影响。以突尼斯为代表的干旱国家正面临严峻环境挑战，包括干旱加剧和大范围土地退化，而气候变化对农业、地下水、生态系统和社会经济稳定的影响或将使这些问题进一步恶化。

尽管形势紧迫，突尼斯在干旱特征、演变机制及未来趋势方面的系统性评估仍存在空白。本研究采用跨学科方法，融合遥感技术、气候模型和地理空间分析，深入探究干旱、气候变化、水资源短缺与土地退化间的复杂关系。具体而言，研究通过多模型水文气象模拟，系统考察干旱的多维特征、演变机制、农业影响、与气候遥相关及土地退化的关联，以及气候变化情景下的未来预测，旨在深化认知并为突尼斯及其他脆弱旱地的可持续发展提供策略支持。

主要研究发现如下：

(1) 创新性地引入 GRACE 卫星衍生的地下水干旱指数 (GGDI)，结合标准化降水指数 (SPI)，和标准化径流指数 (SRI)，标准化降水蒸散指数 (SPEI)，全面解析了 2002-2017 年突尼斯水文气象干旱格局。基于一阶马尔可夫链模型对月尺度 SPEI-GGDI 的随机分析显示，突尼斯干旱具有持续性特征，极端干旱稳定性高且恢复概率低。2014 年 11 月至 2015 年 10 月为最严重地下水干旱期 (GGDI 均值-1.36)，2002-2003 年则是 SPEI、SPI 和 SRI 表征的最干旱期 (均值-1.9)。相关性分析揭示气象与水文干旱的复杂相互作用，GGDI 识别的干旱在频次、强度与持续时间上均高于气象干旱，暗示显著人为影响。小波相干分析指出厄尔尼诺-南方涛动 (ENSO) 是突尼斯干旱最强驱动因子，其强烈负相位会加剧干旱。研究支持将农业干旱分析纳入水文气象干旱评估体系以揭示多类型干旱互作机制。

(2) 针对气象干旱 (MD) 向农业干旱 (AD) 转变机制的研究空白，本研究构建新框架，整合 Spearman 秩相关分析 (SRC)、动态时间规整 (DTW) 和转移概率分析，通过 SPEI 与 MODIS 植被健康指数 (VHI) 量化不同时段、不同

强度等级的 MD-AD 传导时间与概率关联。DTW 分析显示植被健康对干旱存在滞后响应，SPEI-12 与 VHI-1 间最优对齐（DTW 距离=136.36），表明长期降水亏缺对 AD 影响最大。植被健康对中长期气候条件更为敏感，VHI 与 SPEI-6（ $\rho=0.65$ ）、SPEI-12（ $\rho=0.50$ ）相关性最高，而短期波动（SPEI-1: $\rho=0.16$ ）影响微弱。传导时间呈现季节异质性：夏季秋季因高蒸散发导致响应时间短（1–3 个月），冬季春季因土壤水分作用显著使滞后延长至 6–9 个月。严重 AD 发生概率随 MD 强度递增，28.6% 的严重 MD 会演变为严重 AD，19.2% 的中度 MD 亦会升级，凸显持续干旱的放大效应。时空分析显示 AD 存在显著区域与季节差异，南部突尼斯为干旱热点区，频繁遭遇极端干旱导致土地生产力丧失并加剧土地退化。

(3) 聚焦脆弱性突出的突尼斯东南部杰法拉流域，首次集成 MODIS EVI 净初级生产力（NPP）数据与 SPI，解析干旱特征并评估 NPP 响应。时空分析证实该区域易发中度（10%）至严重（7%）干旱，极端事件罕见（2%）。干旱是干旱生态系统 NPP 波动的关键诱因，其响应主要取决于干旱强度与持续时间。SPI 在 12 个月尺度上与 NPP 异常呈显著正相关（0.5–0.9， $p<0.05$ ），显示干旱对土地生产力的滞后效应。研究强调东南部对长期干旱的高度脆弱性，此类干旱不仅降低土地生产力，更加速脆弱生态系统的土地退化。

(4) 针对干旱与土地退化协同研究不足，本研究构建 GIS 综合评估框架，分析突尼斯南部 Negueb 流域的干旱侵蚀耦合关系。基于 MODIS EVI（MOD13A2）和地表温度（MOD11A2）构建植被状态指数（VCI）与温度状态指数（TCI），合成植被健康指数（VHI）表征干旱；采用修正通用土壤流失方程（RUSLE）量化土壤侵蚀，并通过 Pearson 相关分析二者关联。结果显示 2000–2016 年间频发轻度至极端干旱，平均土壤侵蚀速率 $1.8 \text{ t}/(\text{hm}^2 \cdot \text{a})$ 。西部山地子流域对干旱侵蚀最敏感，坡度坡长因子（LS）是主要侵蚀驱动力。干旱与侵蚀呈显著正相关（ $r=0.3$ ），空间相关性-0.5 至 0.8，在干旱高发与强侵蚀区最为显著，印证二者在东南部的紧密关联，强调需制定水土保持综合策略应对未来风险。

(5) 基于 CMIP6 气候模型与共享社会经济路径（SSP1-2.6 和 SSP5-8.5），首次系统预测突尼斯未来干旱特征（强度、范围、频率与持续时间）。SPEI 应用于三组 GCM 集合输出，其中 SSP5-8.5 情景下 MIROC6 模型可靠性最高（ $r=0.7$ ，RMSE 与 MAE 最低）。近期（2016–2040）、中期（2041–2070）和远期（2071–

2100) 预测均显示干旱将更趋频繁、强烈和持久, SSP5-8.5 情景尤甚。干旱影响区预计扩大 47.2% (SSP1-2.6) 和 46.1% (SSP5-8.5), 轻中度干旱占主导。严重干旱可能增加近三倍 (10.8%), 极端干旱超四倍 (4.9%)。至 2100 年, 81.7% 国土或受干旱影响, 南部将遭遇最强干旱, 北部则面临持久但强度较低的干旱。这些发现对突尼斯制定抗旱减灾战略具有重要参考价值。

关键词: 干旱、气候变化、土地退化、遥感、干旱地区.

Abstract

Developing effective sustainable water and land management policies is crucial for addressing regional water scarcity and remains a key research focus in arid regions. Climate change, recurrent droughts, along with land degradation and biodiversity loss pose major threats to sustainable development in drylands worldwide. These challenges are particularly severe in arid water-scarce countries, where agricultural lands are becoming drier and more vulnerable to prolonged droughts. Tunisia, with regards to other arid countries, faces critical environmental challenges, including increasing droughts and widespread land degradation, which are expected to intensify with climate change affecting agriculture, groundwater, ecosystems, and socio-economic stability.

Despite these challenges, comprehensive assessments of drought characteristics, propagation, and future trajectories in Tunisia remain lacking. This study adopts an interdisciplinary approach, integrating advanced remote sensing, climate modeling, and geospatial techniques to explore the complex interactions between drought, climate change, water scarcity, and land degradation. Specifically, it investigates the multifaceted nature of droughts, their characteristics, propagation mechanisms, agricultural impacts, linkages to climate teleconnections and land degradation, and future projections under climate change scenarios using multi-model hydrometeorological simulations. The findings aim to enhance understanding and inform strategies for sustainable development in Tunisia and other vulnerable drylands. The main findings are:

- (1) The novel introduction of the Gravity Recovery and Climate Experiment (GRACE) satellite-derived Groundwater Drought Index (GGDI), along with traditional hydrometeorological drought indices, namely Standardized Precipitation Index (SPI), Standardized Runoff Index (SRI), and Standardized Precipitation-Evapotranspiration Index (SPEI), comprehensively examined hydro-meteorological drought patterns during 2002-2017. The stochastic analysis of monthly SPEI-GGDI using a first-order Markov chain model, revealed persistent drought conditions in Tunisia, with extreme droughts demonstrating high stability and low recovery probabilities. The most severe groundwater drought occurred from November 2014 to October 2015, averaging a GGDI value of -1.36. The period of 2002-2003 was the driest based on SPEI, SPI, and SRI, averaging -1.9. Correlation analysis highlighted complex interactions between meteorological and hydrological droughts, with GDDI-identified droughts exhibited

greater severity pertaining to frequency, intensity, and duration compared to meteorological droughts, suggesting significant anthropogenic influence. Wavelet coherence analysis revealed El Niño-Southern Oscillation ENSO as the strongest drought contributor in Tunisia, with intense negative phases amplifying severity. This study supports complementing hydrometeorological drought assessment with agricultural drought analysis to explore the complex interactions between different drought types.

(2) To address the gap in understanding the transition from meteorological drought (MD) to agricultural drought (AD), this study introduces a novel framework that integrates statistical methods, including Spearman rank correlation (SRC) analysis, Dynamic Time Warping (DTW), and transition probability analysis, to quantify MD-AD propagation time and probabilistic linkages at different severity levels over time, through SPEI and MODIS-derived Vegetation Health Index (VHI). DTW analysis identified a delayed propagation response, with the strongest alignment observed between SPEI-12 and VHI-1 (DTW distance = 136.36), indicating long-term precipitation deficits have the greatest impacts on AD. Vegetation health is more sensitive to medium- and long-term climate conditions, with VHI correlating highest at SPEI-6 ($\rho = 0.65$) and SPEI-12 ($\rho = 0.50$), while short-term fluctuations (SPEI-1: $\rho = 0.16$) have minimal influence. Propagation time varies seasonally, with shorter drought response times (1–3 months) during summer and autumn due to high evapotranspiration and longer lags (6–9 months) in winter and spring, where soil moisture plays a more significant role. The occurrence probability of severe AD increases with MD intensity, where 28.6% of severe MDs evolving into severe ADs and 19.2% of moderate MDs evolve into severe ADs, highlighting the amplification effect of prolonged droughts. Spatiotemporal analysis revealed significant regional and seasonal AD variations, identifying southern Tunisia as a drought hotspot, frequently experiencing prolonged extreme droughts that induce land productivity loss and exacerbate land degradation.

(3) Given the region's heightened vulnerability, this study refines its focus to southeastern Tunisia at a finer basin scale (Jeffara) to assess drought impacts on Net Primary Productivity (NPP), marking the first study in Tunisia to integrate MODIS EVI-based NPP datasets with SPI, to analyze drought characteristics and assess NPP responses to droughts. Spatiotemporal analysis confirmed the region's high susceptibility to recurrent moderate (10%) to severe (7%) droughts, with rare extreme events (2%). Drought emerged as a key trigger of NPP variations in Tunisian arid

ecosystems, with NPP responses depending mainly on drought intensity and duration. A significant lagged effect on land productivity was evident, with SPI showing a strong positive correlation (0.5–0.9, p -value < 0.05) with NPP anomalies at a 12-month scale. Findings further highlight southeastern Tunisia's critical vulnerability to prolonged droughts, which not only reduce land productivity but also accelerate land degradation in these fragile ecosystems.

(4) In response to the current lack of studies combining drought conditions and land degradation processes, this study introduces a comprehensive Geographic Information System (GIS)-based approach to assess both drought soil erosion at Negueb basin scale in southern Tunisia. Drought conditions were analyzed using VHI, by combining the Temperature Condition Index (TCI), derived from MODIS Land Surface Temperature (LST) MOD11A2, and the Vegetation Condition Index (VCI), derived from MODIS Enhanced Vegetation Index (EVI) MOD13A2. The Revised Universal Soil Loss Equation (RUSLE) model quantified soil erosion, and its relationship with drought was assessed using Pearson correlation. Results revealed recurrent mild to extreme droughts between 2000 and 2016, with an average soil erosion rate of 1.8 t/(hm²•a). The mountainous western watershed was the most vulnerable to both soil erosion and droughts, with slope length and steepness factor (LS) emerged as key erosion driving factor. A significant positive correlation ($r=0.3$) between drought and soil erosion was revealed, with spatial correlations ranging from -0.5 to 0.8, strongest in drought-prone and high-erosion regions. This underscores the interconnected nature of drought and land degradation in southeastern Tunisia, emphasizing the need for integrated soil and water conservation strategies to mitigate future risks.

(5) This study provides the first comprehensive projection of future drought characteristics (intensity, area, frequency, and duration) across Tunisia using CMIP6 climate models and Shared Socioeconomic Pathways (SSP1-2.6 and SSP5-8.5). SPEI was applied to ensemble outputs from three GCMs, among which MIROC6 under SSP5-8.5 showed the highest reliability ($r = 0.7$, lowest RMSE and MAE). Projections for the near-term (2016–2040), mid-term (2041–2070), and late-term (2071–2100), revealed an unequivocal trend toward more frequent, intense, and prolonged droughts, particularly under SSP5-8.5. Drought-affected areas are projected to expand by 47.2% (SSP1-2.6) and 46.1% (SSP5-8.5), with mild and moderate droughts becoming the most widespread. Severe droughts could nearly triple (10.8%), while extreme droughts could

more than quadruple (4.9%). By 2100, droughts could impact up to 81.7% of Tunisia's land area, with southern regions experiencing the most intense droughts, while northern areas will face prolonged but less severe conditions. These findings are crucial for strengthening drought-related risk reduction strategies in Tunisia.

Key Words: Droughts, Climate change, Land degradation, Remote sensing, Drylands.

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Abbreviations/Acronyms

AD	Agricultural Drought
C	Cover Management
CHIRPS	Climate Hazards Group Infrared Precipitation with Stations
CMIP6	Coupled Model Intercomparison Project 6
CORDEX	Coordinated Regional Climate Downscaling Experiment
CRU	Climatic Research Unit
CV	Coefficient of Variation
DEM	Digital Elevation Model
DRR	Disaster Risk Reduction
DTW	Dynamic Time Warping
EEMD	Ensemble Empirical Mode Decomposition
ENSO	El Niño Southern Oscillation
ERA5	European Re-Analysis
ESRL	Earth System Research Laboratory
ET	Evapotranspiration
EVI	Enhanced Vegetation Index
FAO	Food And Agriculture Organization
GCM	General Circulation Models
GD	Groundwater Drought
GGDI	GRACE-Derived Groundwater Index
GIS	Geographic Information System
GLDAS	Global Land Assimilation System
GRACE	Gravity Recovery and Climate Experiment
GWS	Groundwater Storage
GWSA	Groundwater Storage Anomalies
HD	Hydrological Drought
K	Soil Erosivity
LS	Slope Length Steepness
LULC	Land Use Land Cover
ISIMIP2b	Inter-Sectoral Impact Model Inter-Comparison Project Group2
IPCC	Intergovernmental Panel on Climate Change
JPL	Jet Propulsion Laboratory
LST	Land Surface Temperature
MAE	Lowest Mean Absolute Error
Mascon	Mass Concentration
MBE	Mean Bias Error
MD	Meteorological Drought
ME	Mean Error
MENA	Middle East and North Africa
MK	Mann-Kendall
MMK	Modified Mann-Kendall
MODIS	Moderate Resolution Imaging Spectroradiometer
NAO	Northern Atlantic Oscillation
NASA	National Aeronautics and Space Administration
NDVI	Normalized Difference Vegetation Index
NOAA	National Oceanic and Atmospheric Administration
NPP	Net Primary Productivity
P	Precipitation

PDSI	Palmer Drought Severity Index
PET	Potential Evapotranspiration
PHDI	Palmer Hydrological Drought Index
R	Rainfall Erosivity
R ²	Coefficient Of Determination
RCP	Representative Concentration Pathways
RDI	Reconnaissance Drought Index
RF	Random Forest
RMSE	Root Mean Square Error
RUSLE	Revised Universal Soil Loss Equation
SDG	Sustainable Development Goal
SH	Spherical Harmonics
SM	Soil Moisture
SPEI	Standardized Precipitation and Evapotranspiration Index
SPI	Standardized Precipitation Index
SRC	Spearman Rank Coefficient Correlation
SRI	Surface Runoff Index
SRI	Standardized Runoff Index
SRTM	Shuttle Radar Topography Mission
SSP	Shared Socioeconomic Pathway
TCI	Temperature Condition Index
TRMM	Tropical Rainfall Measuring Mission
TWS	Terrestrial/Total Water Storage
TWSA	Total Water Storage Anomalies
TWSC	Total Water Storage Change
P	Conservation Support Practice
UCAS	University Of Chinese Academy of Sciences
UNCCD	United Nations Convention on Combating Desertification
USDA	United States Department of Agriculture
VCI	Vegetation Condition Index
VHI	Vegetation Health Index
WC	Wavelet Coherence

“Drought is not just the absence of rain; it is often fueled by land degradation and climate change. Together, we can overcome its devastating effects on people and nature around the world and start preparing now to drought-proof our future.”

—Teresa Ribera Rodríguez,

Vice-President of the Government of Spain
and Minister for Ecological Transition
and Demographic Challenge

Chapter 1 General Introduction

1.1. Background and significance of the research

Since antiquity, land has served as a vital foundation of natural resources, holding numerous ecosystem services that mankind had ultimately gone through where food, water, and energy were the basic actors and/or providers for ecological functioning and stable economic. However, the rapid exploitation of natural resources in the past century has intensified environmental degradation, a trend exacerbated by climate change. Once considered a regional concern, climate change has now evolved into a global crisis with far-reaching implications (Stevanović, 2017).

Climate change is increasingly impacting agriculture and water supply by altering precipitation patterns, increasing the frequency of extreme events, and intensifying drought conditions (FAO, 2016). The complex interplay between climate change, water scarcity, and land degradation represents a global challenge. Droughts, in particular, disrupt water availability, impact soil quality, and reduce agroecosystem resilience, with cascading effects on food security and livelihoods. Influences of climate change could lead to a median crop yield decline of about 2% per decade until the end of the century. Food insecurity projections by 2050, point out towards an increase by 10-20%, compared to a climate change-free scenario (UNCCD, 2014).

Recurrent extreme events, particularly droughts, under climate change effects, combined with land degradation and biodiversity loss, have been recognized as major challenges to sustainable development in global drylands. These factors are contributing to increased vulnerability to drought in water-scarce countries and their agricultural areas (Davies et al., 2015). With regard to other arid water-scarce countries, droughts and land degradation are significant environmental challenges in Tunisia, as up to 75% of its land faces degradation risk (MEHAT, 2012). Degradations encompassed water supply scarcity, decrease and disappearance of vegetation cover, soil degradation, decline in biological diversity, and other limiting factors (Aronson et al., 1993; Jeddi and Chaieb, 2010; Dhaouadi et al., 2020; Besser and Hamed, 2021). Droughts are becoming more frequent and intense due to global climate change, severely impacting the agricultural sector and the national economy (Pachauri et al., 2014).

Historically, Tunisia has experienced frequent and severe droughts, which have had devastating effects on agriculture, water resources, and the livelihoods of its population (Touchan et al., 2008). The increasing variability in rainfall patterns and rising temperatures owing to climate change have intensified drought frequency and severity in Tunisia. Due to its diverse climate zones, spanning from northern humid coastal areas to southern dry desert regions, drought severity and impacts vary significantly (Verner et al., 2018a). Understanding historical drought patterns is crucial for developing effective mitigation and adaptation strategies. Assessing long-term climatic trends, particularly precipitation and temperature changes, helps identify key drought drivers and predict future trends. Tunisia exhibits significant heterogeneity not only in climate, but also in terrain, and land use/land cover (LULC) features. The findings from Chapters 1 and 2 further underpin this variability, identifying southern Tunisia as the most drought-prone region. Given this, two basins, Jeffara and Negueb, were selected within the southern region as case studies to monitor drought effects on land productivity (Chapter 3) and its linkages with land degradation (Chapter 4). These basins represent key socio-agroecological zones where water scarcity, drought, and land degradation are particularly pronounced. Moreover, they are critical for water and land resource management, making them representative of the broader challenges faced by Tunisian drylands. This structured approach ensures a comprehensive understanding, transitioning from a national-scale drought assessment to localized impact evaluations, thereby enhancing insights into sustainable water and land resource management in Tunisia's drylands. In arid and semi-arid regions like Tunisia, where extreme climate events exacerbate water scarcity and land degradation, it is essential to integrate climate variability into resource management planning. Additionally, the combined pressures of climate change and land degradation significantly impact groundwater and subsurface water resources, further stressing the urgency for establishing sustainable land-water use strategies.

To date, growing concerns about Total Water Storage (TWS) variability and its associated impacts have been significantly influenced by extreme events, anthropogenic activities, and climatic pressures on water resources across various regions. As highlighted by several studies, TWS changes, encompassing variations in water mass from surface waters, soil moisture, and groundwater, are crucial indicators of potential long-term modifications in the water cycle due to natural or anthropogenic factors (Pokhrel et al., 2016; Scanlon et al., 2018). This issue is particularly critical in

arid and semi-arid regions, where conditions are deteriorating (Hasan et al., 2019). In Tunisia, a country facing significant economic and agricultural challenges, water resources, including surface water, subsurface water, and soil moisture, are under severe stress (Boutheina et al., 2022). Rapid economic and population growth, along with increasing urbanization, has led to increased water demand, causing fluctuations in water availability and quality (Gaaloul, 2011; Soula et al., 2021). Over the past decades, Tunisia has experienced the negative impacts of climate change, such as frequent droughts and occasional floods, which have resulted in reduced agricultural productivity and strain on water resources (Khordagui, 2014). Despite these challenges, the full extent of the adverse impacts of drought on TWS and groundwater drought conditions remain unclear so far.

Given that these environmental challenges are not specific to Tunisia, the alarming nature of this issue has prompted the international communities to implement rapid and robust policies and actions to tackle this human-induced and natural hazard. These policies include collaborative efforts with global communities like the Drought Initiative of 2018–2019, Sendai Framework for Disaster Risk Reduction (DRR) (2015–2030), and United Nations Convention on Combating Desertification (UNCCD) (UNISDR, 2015). These policies promote a varied set of actions aimed at the sustainable management of natural resources. This is created on the basis that the preservation, protection, and restoration of freshwater resources, along with land, are crucial elements of every policy aimed at natural resources preservation. Additionally, underpinning resilience of communities to drought and risks linked to water scarcity and land degradation is another key element. In this context, the global community established several SDGs in 2015. Most importantly, SDG 15.3 stating “By 2030, combat desertification, restore degraded land and soil, including land affected by desertification, drought and floods, and strive to achieve a land degradation-neutral world” (Minelli et al., 2017).

Droughts, land use, and water use, are closely interconnected. Droughts, for instance, aggravate land degradation by increasing the frequency of dust and sand storms (Belal et al., 2014; Middleton and Kang, 2017; Reichhuber et al., 2019a). On the other hand, land degradation may also amplify water scarcity and increase vulnerability to droughts. Land degradation is strongly interconnected with water scarcity. Degraded land loses its ability to reserve and filter water, which healthy land naturally provides (Keesstra et al., 2018). Moreover, changes in land-use categories,

such as the conversion of wetlands and forests to other land uses, disrupt the water cycle and hydrological functions (Behrend, 2016). Conversely, water scarcity and droughts can accelerate the processes of land degradation, for instance due to poor irrigation management and drainage and altered hydrology, leading to poorer quality soils (Gerber et al., 2019; Masroor et al., 2022). To effectively tackle drought and secure water resources for the long-term, managing and conserving both water and land is crucial (Guerquin, 2020).

Recognizing the nexus of water and land management, along the several advantages of mitigating drought impacts and restoring degraded lands to achieve water security and sustainable development, a holistic assessment of droughts is essential in Tunisia. This study addresses the existing gaps in the comprehensive understanding of droughts in Tunisia by targeting various facets of drought, and its interconnections with climate teleconnections and land degradation (Figure.1-1). This approach provides a holistic assessment of droughts in Tunisia, emphasizing the synergies between water security, drought mitigation, and land degradation neutrality. Such an assessment aligns with global initiatives aimed at achieving Sustainable Development Goals (particularly SDG 13, SDG 6, and SDG 15), to combat desertification and risks induced by climate within the DRR-Sendai framework and UNCCD.

This study adopts an interdisciplinary approach, using remote sensing, climate modeling, and geospatial techniques to assess drought characteristics, propagation mechanisms, and long-term risks. By incorporating multi open-source of earth observation datasets, and robust quantitative and statistical methodologies, this research contributed to an enhanced drought risk assessment and developing effective mitigation strategies that underpin resilience to drought, particularly in the Tunisian drylands. The study approach fully integrates the links among drought, land degradation, and climate change to efficiently tackle drought future risks. Produced data serve as a basis for drought risk prevention, preparedness, and reduction as well as climate change adaptation in regional and global drylands. Therefore, the findings of this study can help in establishing more targeted and robust conservation strategies for arid environments in Tunisia. Additionally, they will provide valuable insights for policymakers and stockholders to enhance long-term management of land and water resources through informed land-use planning.

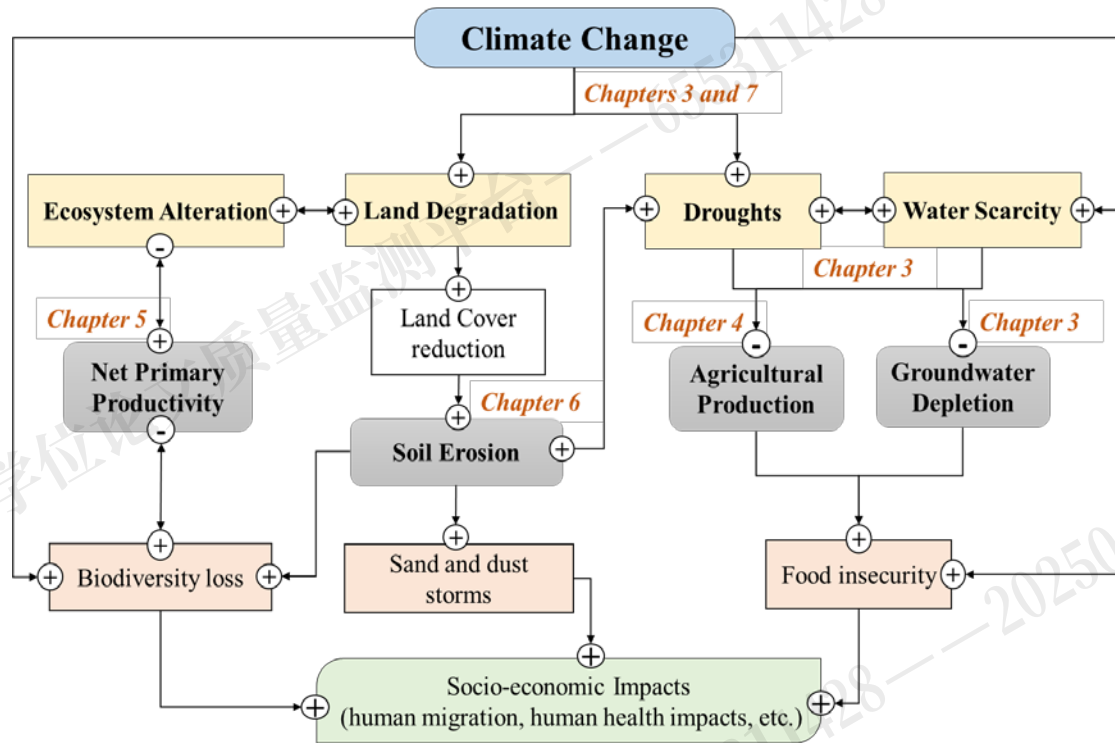


Figure 1-1. The conceptual framework and theoretical foundation of this study: Cascading effects and interactions within droughts, land degradation, water scarcity, and socioeconomic systems affected by climate change. Arrows show the direction of influence, while symbols of plus and minus indicate whether the impact on these processes is intensifying or diminishing.

1.2. Literature Review (Research Status)

1.2.1. Drought types and characteristics

Drought is a pervasive natural hazard that is expected to occur more frequently in the context of a changing climate (Pachauri et al., 2014). Described as a slow-onset creeping natural hazard, drought has crosscutting impacts on a broad range of environment and society sectors (Dai et al., 2004; Tian et al., 2020a). Droughts represent a water deficit situation relative to normal conditions (Lloyd-Hughes 2014), and they are divided into four categories namely hydrological (HD), meteorological (MD), agricultural (AD), and socioeconomic (Nalbantis and Tsakiris, 2009; Wu et al., 2016).

- **MD** is described as a prolonged period of precipitation shortage over a region, where analyses are mainly focused on precipitation.

- **HD** refers to a period of insufficient surface and groundwater to meet water needs. Streamflow and runoff data are the typical indicators of to identify hydrological droughts. A more specific type of hydrological drought was defined as groundwater droughts (GD) referring to a decrease of groundwater levels, recharge, or discharge (Van Lanen and Peters, 2000)
- **AD** establishes when soil moisture rates decline over a prolonged period leading to crops failure. This type of drought usually follows a meteorological drought but precedes a hydrological drought. Drought indices combining usually precipitation, evapotranspiration and soil moisture have been developed to analyze agriculture droughts.
- **Socioeconomic drought** occurs if the imbalance between economic goods and supply resulting from shortage in water supply induced by climate variability, or if human activities are influenced by decreased precipitation and water supply (Wilhite and Glantz, 1985; Spinoni et al., 2015; Parente et al., 2019).

While each drought category has its distinct characteristics, they are all interconnected by the common impact of precipitation shortage, and progressing to more severe socio-economic consequences. Generally, though not invariably, meteorological drought serves as the precursor to other types of drought (Wang et al., 2021; Jiang et al., 2023). Given this interdependency, our study targeted these different types of droughts, and the complex interactions among their effects.

The increasing frequency of droughts coupled with climate change, pose significant challenges to sustainable development in global arid regions, leading to higher degradation vulnerability and drier agriculture (Cervigni and Morris, 2016; Banerjee and Pandey, 2021). Population threatened by hunger risk is expected to increase by 10-20% by 2050, due to the impacts of climate extremes (UNCCD, 2014). Therefore, droughts disrupt ecosystems, hinder food production, and compromise water supply, resulting in profound effects on human health and survival (Pachauri et al., 2014; Mohammad et al., 2018; Ullah et al., 2019). Furthermore, there are significant connections between droughts, water usage, and land degradation.

Droughts and water scarcity further lead to the expansion of degraded lands by exacerbating dust and sand storms, promoting inefficient irrigation practices, and disrupting hydrological cycles, resulting in deteriorated soil quality (Belal et al., 2014; Middleton and Kang, 2017; Reichhuber et al., 2019a). Therefore, manage effectively

land and water resources is crucial to mitigating drought impacts, neutralizing degraded lands and ensure water security at long-term.

This alarming situation prompted global communities to take decisive actions through policies like the UNCCD Drought Initiative (2018–2019) and the DRR within Sendai Framework (2015–2030) (UNISDR, 2015). These frameworks advocate for integrated measures aiming at reducing drought risks and enhancing preparedness. In this context, different drought indices were developed to address the characteristics of each drought type (Zargar et al., 2011). For instance, SPEI and SPI are widely employed to evaluate meteorological droughts, relying basically on precipitation and evapotranspiration (McKee et al., 1993; Vicente-Serrano et al., 2010).

Hydrological drought is characterized by the negative anomalies in groundwater and streamflow (Mishra and Singh, 2010). Nevertheless, it can be assessed using indices involving precipitation, streamflow, runoff, groundwater levels, and water balance. For example, SRI based on SPI approach, is a widely employed index for HD monitoring (Shukla and Wood, 2008).

HD and MD are interlinked via different processes in the context of the water cycle (Van Loon and Laaha, 2015). In a broader context, increased evaporation from oceans could alter precipitation patterns on land, leading to increased water storage on continents (Gimeno et al., 2012). Conversely, scarce precipitation over land, coupled with increased evaporation resulting from meteorological drought, can result in a decrease of water storage, inducing hence hydrological drought (Van Lanen et al., 2013; Van Loon, 2015). Therefore, comprehensive monitoring is necessary to understand the interplay between AD, HD, and MD. A new hybrid drought index, namely Precipitation Evapotranspiration Difference Condition Index (PEDCI), incorporates components of several recognized indices. This index is useful for tracking agricultural and meteorological droughts because it makes use of an intuitive water balance model and a unique normalization technique (Tian et al., 2020b). To provide a more precise assessment of drought conditions, it demonstrates strong correlations with data on crop yield and soil moisture. Several studies suggested an association between meteorological drought and hydrological droughts where they claimed that prolonged meteorological droughts lead to alterations in hydrological characteristics (Van Loon, 2015; Liu et al., 2019c; Xu et al., 2019b; Pan et al., 2020). The transition from MD to HD involves complex interactions between climatic and terrestrial factors. Seasonal groundwater recharge cycles and timing of precipitation play significant roles in

drought propagation (Apurv et al., 2017; Han et al., 2019; Ding et al., 2021; Liu et al., 2023d).

Similarly, prolonged periods of low precipitation can lead to soil moisture depletion, ultimately resulting in transitioning from MD to AD (Berg and Sheffield, 2018). Therefore, gaining a precise understanding of how MD evolves into AD and linkages among them is crucial for early warnings, helping to minimize the adverse AD effects. The most widely used method for examining the MD-AD relationships is correlation analysis, which compares AD indices at specific timescales (typically monthly) with MD indices at various timescales. The MD index timescale yielding the highest correlation coefficient is regarded as the drought propagation time (Dai et al., 2022; Liu et al., 2023c; Feng et al., 2024). For instance, Xu et al. (2023c) revealed that MD propagate to AD in 48 days across Yangtze basin, by correlating the Improved Soil Moisture Anomaly Percentage Index (ISMAPI), with different time-scale SPI. Possega et al. (2023), identified the response time scale of 1-2 months for drought propagation in the Iberian Peninsula, with the probability of occurrence increasing with the MD severity. Similarly, Xu et al. (2021) reported a propagation time between 1 and 3 months in humid regions of China, while it can range between 1 and 8 months in arid and semi-arid regions.

The fundamental drought characteristics are intensity, duration, severity, and frequency. Climate change has intensified the characteristics of drought, leading to more frequent and severe droughts in many regions across the globe with varying intensity and duration (Kiem et al., 2016; Cook et al., 2018b; Cavus and Aksoy, 2019; Grillakis, 2019; Haile et al., 2020; Zhang et al., 2022). This is evident in the increasing area and frequency of drought-affected regions in Australia under enhanced greenhouse conditions (Kirono et al., 2011). In Africa, drought risk is primarily impacted by climate change, population growth, and socioeconomic vulnerability. Central African countries are particularly at high risk due to the rising population and high vulnerability, while northern Africa experiencing greater drought hazards (Ahmadalipour and Moradkhani, 2018). The frequency and intensity of drought events in the Middle East and North Africa (MENA) region demonstrate the region's susceptibility to severe and protracted droughts (Bazza et al., 2018b).

The Mediterranean region is experiencing severe consequences from water scarcity, and are projected to intensify according to climate change projections (Castro et al., 2023). Compared to other mediterranean regions, Tunisia is a water-scarce country,

where droughts are a recurrent phenomenon and expected to be more frequent at an alarming pace and intensity (Nasr et al., 2008; Radhouane, 2013; Pachauri et al., 2014; Bahri et al., 2019). Tunisia is periodically affected by droughts as an inherent part of its semi-arid to arid climate and the irregular nature of its rainfall regime. Throughout the history, these dry events have adversely impacted the environmental and socioeconomical development of the country. For example, droughts of 2015-2016, has caused a drop in economic growth resulted from the loss of agricultural productions and the high charge of mitigation measures (Verner et al., 2018a).

Multiyear drought and alterations in rainfall are therefore the most challenging issues for agriculture in Tunisia. The IPCC projections exhibited a strong possibility of continuing temperature increases in Tunisia, with a significant rise of 0.50°C every 10-year starting from 1970s, surpassing the global increase of 0.15°C (Radhouane, 2013). Several studies highlighted that variabilities in climate over the mediterranean region will have detrimental hydrological and environmental impacts through the decreased water availability and increased drought events (Candela et al., 2012; Copetti et al., 2013; Van Loon, 2015; Trambly et al., 2020a). As a result, groundwater is extensively extracted to supplement irrigation needs, adapting to climate change. Such measures require enhanced management of water resource along with better regulations. However, the Tunisian drought management system remains underdeveloped. Tunisia still lacks a comprehensive national drought management strategy (Verner et al., 2018a). Initial step towards the implementation of such system is providing a comprehensive of drought monitoring and a nuanced understanding of its characteristics, propagation mechanisms, and future projections in Tunisia in order to cope with drought impacts and underpin resilience to droughts.

Despite this increasing drought risk in Tunisia, existing drought studies remain very limited, and focused mainly on temperature and rainfall trends analysis, without considering various drought indices and the impact of other parameters involving in droughts like PET, surface runoff, streamflow, soil moisture, and groundwater (Ben Boubaker et al., 2003; Bergaoui and Louati, 2010; Chebil et al., 2019). So far, there is no country-level scale assessment has been carried out to evaluate how meteorological droughts evolve and propagate into different other types of droughts. This study addresses these gaps by introducing a novel framework integrating remote sensing, multi-source hydro-climatic data and advanced geospatial and statistical techniques, to provide a holistic assessment of different types of droughts and delve into their

specific characteristics, spatiotemporal dynamics, and propagation mechanisms across Tunisia.

Additionally, there is a significant lack of studies devoted to projected droughts and their characteristics variations. Previous climate change studies in Tunisia have primarily relied on CMIP3 and CMIP5 models, which have outdated emission pathways. CMIP6, with its Shared Socioeconomic Pathways (SSPs), offers improved future projections, allowing for better drought risk assessment. To date, there have been no studies using these newer models, which creates uncertainty in predicting Tunisia's long-term water security challenges. This study is the first to employ CMIP6 multi-model hydrological simulations across different SSP scenarios, offering improved projections of future drought conditions and aiding long-term drought adaptation strategies.

Moreover, almost all investigations on droughts in Tunisia were applied on small catchments or areas, where an overall assessment of drought conditions and characteristics across the whole country is still lacking. To address this gap, our study expands the assessment of drought conditions across the whole country, including several drought-hotspots basins were selected as study cases to thoroughly investigate drought impacts and their linkages to land degradation issues.

By addressing these various facets, our study is conceived as the first country-level study that can serve as national database platform providing a holistic assessments of drought dynamics and can be linked to local management actions. It provides critical insights for policymakers and stakeholders to implement a straightforward and reliable early-warning system for drought risk reduction and preparedness. This could involve creating a mechanism for broad coordination prior to major drought-related impacts.

1.2.2. Remote sensing of droughts: Progress, Challenges, and Opportunities

Comprehensive regional drought assessments, based on field observations, are crucial to endorse environmental restoration and hydrological research. However, such assessments are often unsystematic as they are limited in both space and time and require considerable financial resources (Creutzfeldt et al., 2012). Remote sensing techniques have emerged as an effective substitute for traditional field-based observation methods. They provide continuous, cost-effective, and spatially distributed information on parameters associated with various types of droughts, particularly for groundwater droughts where monitoring through traditional observations is challenging

(Thenkabail and Gamage, 2004; Chao et al., 2016). Remote sensing has been evidenced to be an effective for evaluating drought factors from hydro-meteorological perspectives and also evaluating drought impacts from an ecosystem perspective. The estimation of drought-related factors, like rainfall, temperature, evapotranspiration, or soil moisture, have been largely generated from different satellite data sources, including thermal infrared, multispectral, or microwave (AghaKouchak and Nakhjiri, 2012; Zhang and Jia, 2013; Dorigo et al., 2017). These parameters are used as inputs to derive different drought indicators used to evaluate drought conditions. This review examines the advancements of remote sensing applications to estimate drought-related parameters and establishment of satellite-derived indices based on these parameters to assess the different types of droughts.

1.2.2.1. Precipitation-based Monitoring

The advent of precipitation products generated from remote sensing has greatly improved the effectiveness and spatial-temporal extent of the assessments of rainfall variability and droughts (Huffman et al., 2007; Tian and Peters - Lidard, 2010; Wu et al., 2023). The initial endeavor in this realm was the Tropical Rainfall Measuring Mission (TRMM), and it has been operational since 1997 until 2015. TRMM was dedicated to measuring rainfall in tropical and subtropical regions and marked the pioneering use of a dedicated microwave precipitation radar on a satellite (Kummerow et al., 1998). TRMM dataset is a key reference in global rainfall measurement and continues to be used for analyzing rainfall patterns and drought drivers (Zhang and Jia, 2013; Zhao et al., 2018). Its successor, the Global Precipitation Measurement (GPM) mission, with improved spatial resolution (0.1-0.25 degrees) and broader coverage, presents high accuracy data when compared to in-situ and TRMM data (Tang et al., 2016; Sun et al., 2018). ERA5, created by the European Centre for Medium-Range Weather Forecasts (ECMWF), has recently emerged as potential tool for drought monitoring with greater resolution (Gu and Wang, 2024). It succeeds the ERA-Interim dataset and aims to provide a comprehensive and a thorough archive of the Earth's land surface, atmosphere, land surfaces, and oceans (Hersbach et al., 2020).

Several studies have leveraged ERA5's capabilities for drought analysis. For instance, Albergel et al. (2019) proved the reliability of ERA5 in assessing the soil moisture dynamics, a crucial parameter in monitoring drought conditions. Similarly, Muñoz-Sabater et al. (2021) underscored the reliability of ERA5's precipitation and

temperature in assessing meteorological droughts through the computation of SPI and SPEI. These indices are the most widely used I drought monitoring, as by leveraging remote sensing to access crucial parameters such as precipitation, temperature, and evapotranspiration (McKee et al., 1993; Vicente-Serrano et al., 2010). In the same context, a variety of indices have been developed that utilize remote sensing datasets to effectively monitor and assess drought conditions. A negative SPI value highlights a drought condition, while a positive value reflects a drought-free condition. In Tunisia, several studies adopted SPI to examine drought conditions and characteristics across different regions. For example, Ben Abdelmalek, and Nouri (2020) demonstrated the reliability of SPI in drought characterization in Tunisia, where they indicated Tunisia has undergone multiple prolonged dry periods with varying intensities and durations. They revealed that the year 1981 was marked by a generalized drought, while 70% of the country revealed extreme humidity during 1996. Jemai et al. (2018) found significant variability in SPI values, indicating frequent short-term fluctuations and pronounced long-term droughts in southeastern Tunisia, especially from 2001 to 2005. Significant variations in precipitation and the occurrence of droughts over different periods was detected in Medjerda basin northern Tunisia using SPI (Gader et al., 2022). Mohamed et al. (2023a) identified SPI as a useful tool for long-term MD assessing, characterizing drought extent, duration, and frequency over Tunisia. Overall, drought conditions in Tunisia are effectively evaluated using SPI, which is convenient for characterizing droughts in regions with low precipitation and high vulnerability to droughts. However, SPI relies solely on precipitation, while other parameters could be influencing drought conditions such as the temperature and evapotranspiration. To address this, SPEI, integrating precipitation and potential evapotranspiration (PET), provides more robust drought evaluation. SPEI has facilitated the assessment and evaluation of drought conditions worldwide, while also aiding drought forecasting at long-term. Recent evaluations of long-term drought conditions over Tunisia from 1952 to 2017, using SPEI, revealed increased drought frequency and intensity were observed, which correlated with changes in vegetation and land-cover in the northwest Tunisia (Touhami et al., 2022a). Notably, the SPEI has been instrumental in estimating drought conditions across different regions in Tunisia, highlighting a significant negative trend over the study period from 1955-2021 (Touhami et al., 2023). In an effort to achieve efficient drought assessment, the Palmer Drought Severity Index (PDSI) was introduced by Palmer (1965). The PDSI integrates precipitation and temperature data

along with water balance information. Based on soil moisture levels, this index quantifies long-term drought by considering changes in PET to account for climate change. The index relies on the available water capacity to calculate water balance. Several studies confirmed the reliability of PDSI in drought detection across different regions. For instance, Yuan, and Quiring (2014) compared different methods for estimating PET in PDSI over the U.S. Great Plains. They found that the original PDSI showed a stronger drying trend than equations using more physically based PET methods. In Tunisia, PDSI was found to be more appropriate for central and southern parts of the country, where it detected long-lasting and intense drought events (Ben Mhenni et al., 2021). Critical factors in PDSI's assessment of drought events involve the duration of prolonged dry periods and its ability to reflect global warming impacts (Alley, 1984). An adaptation of PDSI, the Palmer Hydrological Drought Index (PHDI), has demonstrated strong potential in detecting extended drought phases and evaluating effects on streamflow, groundwater levels, and variations in water storage (McKee et al., 1993). Similarly, the self-calibrating PDSI (scPDSI) has greatly enhanced drought assessment, allowing more effective in identifying extreme dry and wet events (Wells et al., 2004). Short-term drought assessment is particularly useful for identifying vegetation responses to dry spells. Sousa et al. (2011) analyzed the spatiotemporal trends of droughts across Mediterranean basin throughout 20th century by using both PDSI and scPDSI. Results showed across the majority of the west and central of the basin, drier conditions have been observed, while northwestern Iberia and Turkey have experienced increased moisture levels. The scPDSI provided a more realistic depiction of drought variability, aligning better with historical records. Therefore, this review improves our understanding of the integration of drought indices in evaluating meteorological drought conditions and capture the spatiotemporal regional characteristics of droughts, by means of remote sensing data.

1.2.2.2. Vegetation based drought index monitoring

Drought conditions can be monitored by analyzing vegetation cover changes using remotely-sensed data (Mariano et al., 2018; Karakani et al., 2021). Severe dry events have significant impacts on land productivity, particularly Net Primary Productivity (NPP), which measures the rate at which plants convert atmospheric carbon into biomass (Clark et al., 2001). NPP is commonly expressed in kg/ha/year and plays a significant role in the global carbon balance (Sun et al., 2016). The increasing

trend of droughts coupled with climate change effects, has led to a global reduction in NPP (Zhao and Running, 2010; Chen et al., 2013; Xu et al., 2020). As a result, investigating NPP responses to droughts at both global and regional levels has become a prominent area of research (Vicca et al. 2016; Vicente-Serrano et al. 2013; Zhang et al. 2016). For instance, Peng et al. (2017) demonstrated that droughts were responsible for 37% of the global NPP decline in recent decades. Similar trends have been observed at regional scales across the globe (Khalifa et al., 2018; Fu et al., 2020; Yin et al., 2024).

AVHRR, introduced in 1997, has transformed drought monitoring through remote sensing by supplying higher resolution products apt at vegetation assessments (Babikir, 2019). In this context, NDVI was the pioneering remote sensing-based metric for assessing drought characteristics and their impacts (Javed et al., 2021). NDVI is the most commonly used due to its effectiveness in identifying plant stress caused by droughts and in evaluating vegetation dynamics at different spatiotemporal scales, as it can effectively distinguish vegetation from the bare soils (Huete et al., 2002). There is a significant relationship between precipitation, soil moisture and NDVI. Accordingly, NDVI and its derivatives are commonly used for assessing drought and monitoring vegetation health (Gu et al., 2008; Liu et al., 2018; Bento et al., 2020). Analyzing vegetation health over time helps to observe vegetation changes across various landscapes globally (Fensholt et al., 2015). This is particularly important for monitoring drought in arid or semi-arid regions with heterogeneous landscapes (Andela et al., 2013; Zewdie et al., 2017). Building on NDVI, other indicators like the Transformed Vegetation Index (TVI), EVI, Soil-adjusted Vegetation Index (SAVI), VCI have been developed (Huete, 1988; Kogan, 1997; Huete et al., 1999). These indices reflect vegetation conditions by integrating spectral information sensitive to biophysical characteristics the amount of chlorophyll, internal leaf structure, and water content.

VCI, which normalizes NDVI values within its minimum and maximum range, serves to differentiate short-term climate fluctuations from long-term ecological patterns in drought assessment (Zhong et al., 2021). Kogan (1995) demonstrated VCI's effectiveness in detecting and monitoring drought onset, duration, and intensity in various ecological environments, proving useful for both prolonged and localized droughts. Building on the insights provided by VCI, TCI adds another dimension by focusing on thermal infrared data to reflect LST anomalies. TCI helps identify thermal stress in vegetation, which can be indicative of drought. Seiler et al. (1998) applied TCI and VCI to monitor drought in Argentina, finding that TCI was particularly useful

for assessing temperature-related vegetation stress and drought severity. Various studies have shown that combining TCI and VCI serves as a reliable approach for assessing vegetation stress and drought conditions, with these indices being extensively applied in diverse regions (Babikir, 2019; Terwayet Bayouli et al., 2023b). For example, VHI integrates both VCI and TCI to capture both moisture and thermal stress in vegetation, ensuring a more comprehensive assessment of vegetation health. VHI has proven to correlate well with the standardized precipitation indices, indicating its reliability in reflecting drought conditions across different climate regions (Bento et al., 2018b; Lima et al., 2023). Zou et al. (2020) assessed the performance of VCI, TCI, and VHI, finding that TCI was particularly effective in detecting drought during the wet-to-dry transition period. In Tunisia, Ghoneim et al. (2017) conducted a comprehensive geospatial investigation using the VHI to monitor drought conditions in Tunisia from 2000 to 2013, where they proved its effective in identifying drought-prone areas, particularly in the southern regions where grasslands and croplands were most affected. The VHI values showed strong correlations with crop yields. Ben Mhenni et al. (2021) employed SPI, PDSI, and Soil Moisture Index (SMI), to assess drought characteristics across Tunisia, integrating meteorological data, remote sensing-based vegetation indices (e.g., NDVI), and agricultural production records to evaluate drought impacts on vegetation and crop yields over different climatic zones. While, Ben Abdelmalek, and Nouiri (2020) analyzed drought patterns based on rainfall data (1973–2016), and examined the drought impacts on agricultural productivity by correlating drought indices with crop production data and Agricultural Gross Domestic Product variations.

Despite these contributions, none have specifically examined the propagation time between meteorological and agricultural drought in Tunisia. This gap in the literature highlights that drought propagation mechanism and how MD translates to AD at different intensities in Tunisia remain unexplored and are a crucial aspect that warrants further investigation. Recent studies in Tunisia have concentrated on a narrow range of indicators, underscoring the necessity for more precise and responsive vegetation metrics to enhance the understanding of vegetation dynamics and its response to droughts. Furthermore, existing research predominantly focuses on identifying specific drought events and analyzing rainfall patterns, without considering the link between droughts and NPP variations. The impact of the increasing drought intensity and severity on NPP remains undiscussed.

Therefore, this study bridges these gaps by providing the first in-depth analysis of spatiotemporal agricultural drought dynamics across Tunisia, and investigating the propagation time and probabilistic linkages between MD and AD, at different drought levels. Subsequently, further examining the correlation between long-term drought indicator, through SPI-12 and vegetation indicators through NPP, to assess drought impacts on land productivity in the drylands of Tunisia, the most vulnerable regions.

1.2.2.3. Groundwater and Terrestrial Water Storage

Severe droughts have adverse direct and indirect impacts on water storage, such as depletion of groundwater resources pertaining quantity and quality, reduced land productivity, and increased soil erosion and degradation (Mosley, 2015; Nasir and Selvakumar, 2018; Nair and Indu, 2020). Drought impact mainly five sources of usable water, namely soil moisture, groundwater, snow, streamflow, and reservoir storage (Masson-Delmotte et al., 2021).

In arid and semi-arid regions like Tunisia, groundwater resources are pivotal for sustaining agriculture and providing water supply. However, these resources are under significant stress due to recurrent droughts, climate change, and overexploitation. The study by Gaaloul (2011) highlights that repeated drought events combined with aquifer depletion cause acute water shortages, affecting the social, economic, and environmental aspects of the country. Consequently, groundwater depletion has significant socioeconomic impacts, particularly on agriculture. For instance, farmers in central Tunisia adopted coping strategies such as crop selection and altered agricultural practices to mitigate the impacts of water shortages (El Amami et al., 2019). However, these strategies often result in income losses and are not sustainable solutions for long-term groundwater management. In the Gafsa basin in Tunisia, Mokadem et al. (2018) observed that intensive groundwater abstraction during drought periods has led to the near extinction of traditional water systems like "Foggara" and springs, further reducing groundwater availability. The increased use of modern wells and pumps has accelerated the drawdown of piezometric levels and degraded water quality. Droughts also negatively impact surface water, leading to reduced river flows, lower reservoir levels, and diminished water availability for various uses in Tunisia. Ben Mhenni et al. (2021) found that droughts in Tunisia trigger prolonged soil moisture deficits, reducing surface water availability. Droughts exacerbate the interdependence between surface and groundwater resources. Lachaal et al. (2018) noted that the Grombalia aquifer system

experienced significant hydrodynamic and hydro-chemical changes due to combined effects of drought and intense agricultural activities.

Subsurface water and groundwater are the primary source for domestic and irrigation purposes during droughts, making its management crucial. Examining the direct effects of droughts on both surface and groundwater storage separately provides a robust basis for policy-makers to sustainably manage these resources. Hydrological drought can be assessed using indices involving precipitation, streamflow, runoff, groundwater levels, and water balance. For example, SRI, building on SPI method, is a widely employed index that integrates hydrological process causative of the seasonal decline in streamflow influenced by climate conditions. SRI at one-month to seasonal time scale serves as a valuable supplement to SPI for portraying hydrological aspects of droughts (Shukla and Wood, 2008). However, drought analysis through proxies of precipitation, temperature, and surface water observations does not reflect direct evidence of water availability deficits. Water is stored in various forms, including surface water, soil moisture, groundwater, snow, vegetation, river and lake water (Giroto and Rodell, 2019). Therefore, integrating measurements of all water storage types is vital to accurately evaluate responses of water resources to drought, a task become easier through remote sensing techniques applications. GRACE satellite gathers precise measurements of gravity field changes of the earth and translates them into variations in total water storage (TWS) across continental regions (Xiao et al., 2015). This serves as a valuable information source to address gaps in data availability and understand hydrological characteristics (Tapley et al., 2004). The TWS data derived from GRACE encompass various water contents, such as ice, snow, surface water, soil moisture, and groundwater. These data are particularly important for monitoring groundwater droughts. For instance, Thomas et al. (2017) introduced the GGDI to estimate the deficit in groundwater storage in California Central Valley. Satish Kumar et al. (2021) analyzed the correlation between the drought severity index derived from different precipitation-based indices with GRACE TWS anomalies over different basins in India. At global scale, Liu et al. (2023d) employed GDI along with the SPI, SRI, and Standardized Streamflow Index (SSI) to examine the propagation mechanisms among the various drought types during 2003-2020. Several studies have investigated the relationship between long-term trends or variations in the amplitude of seasonal precipitation and TWS. For example, Zhao et al. (2022); Liu et al. (2023b) revealed that in China, both climate change and human activities contribute to the reduction of

groundwater storage. Few studies have utilized GRACE observations to explore the spatial and temporal variations of groundwater resources in Tunisia. However, these investigations have solely focused on changes in groundwater storage. For instance, Gonçalves et al. (2021) employed GRACE data to evaluate the groundwater storage variation (GWS) for the period 2002–2016. This study was spatially limited over the Djefara Plain (Tunisia-Lybia), where a severe unsustainable groundwater exploitation was revealed. Gaaloul (2008) underscored the pivotal role of GRACE data in assessing groundwater depletion during droughts. The study identified a good correlation between the GRACE-derived TWS anomalies and in-situ groundwater level measurements, providing a reliable means of tracking groundwater changes in Tunisia. In another study, Lachaal et al. (2018) employed GRACE data to investigate the hydrodynamic and hydrochemical changes in the Grombalia aquifer system. Their findings indicated significant piezometric declines and water quality deterioration due to overexploitation and drought, underscoring the value of GRACE data in water resource management in Tunisia. On a larger scale, Frappart (2020) highlighted considerable water depletion in the Northwestern Sahara Aquifer System during the 2003-2016 period, primarily utilizing GRACE data.

Despite these contributions, groundwater drought in Tunisia, particularly their spatial and temporal dynamics, remains unexplored. Another major shortcoming of the abovementioned studies, is their limited spatial coverage, particularly with GRACE's coarse spatial resolution, which is not suitable for small catchments. In addition, none of the abovementioned studies employed GRACE-based drought indices to evaluate the groundwater drought conditions in Tunisia. Therefore, additional studies are required to fully understand the groundwater drought conditions in Tunisia to effectively target the sustainable management of these resources. To fill the gap, this study develops a comprehensive assessment of spatiotemporal variations of groundwater drought at an annual time-scale using GRACE, GLDAS, and satellite data estimates, considering the correlation coefficients with other hydro-meteorological drought indices.

1.2.3. Droughts and climate teleconnections

Climate change, coupled with periodic climate events named teleconnections, was identified as main contributors for more severe and frequent dry events, particularly in

water-scarce countries like Tunisia. These teleconnections encompass natural climatic phenomena, such as El Niño Southern oscillation (ENSO), the North Atlantic oscillation (NAO), and the Mediterranean oscillation (MO), the Western Mediterranean oscillation (WeMO) trigger droughts by increasing temperatures rise and shaping a more irregular rainfall regime. In particular, Forootan et al. (2019) showed that ENSO has a crucial effect hydrological drought extent and evolution in Australia, north of Asia, during 2006-2011 and in South Asia, and North Africa in the period of 2010-2012. They underscored also a significant regional effect of the Indian ocean Dipole (IOD) and NAO on the evolution of hydrological droughts. Also, Wang et al. (2020) found that ENSO had a strong impact on droughts in north China plain. Another study conducted over Africa showed that ENSO alongside with the Atlantic Multidecadal Oscillation AMO could induce significant changes in recharge to the aquifers and interannual groundwater storage (Carvalho Resende et al., 2019). In arid regions like Ethiopia, the NAO was found to have a significant impact on winter weather patterns. Positive phases of NAO are associated with dry conditions, while negative phases can lead to wetter periods. This oscillation affects the region's rainfall distribution and drought occurrence (Lanckriet et al., 2015).

In Tunisia, extended periods of drought and overall rainfall variability pose significant challenges for agriculture. For example, groundwater resources in Tunisia are considered as the primary buffer for drought impacts, where more than 40% of irrigation water used is supplied by groundwater aquifers. Thus, exerting an additional pressure on groundwater resources which are already declining, which has serious implications for their long-term sustainability. However, limited research has delved into the climatology of droughts, particularly groundwater drought, and their potential associations with the various teleconnections in Tunisia. Previous studies focused particularly on linking teleconnections to weather patterns such as precipitation variability and streamflow. For example, Ouachani et al. (2013) highlighted the complex interplay of multiple teleconnection patterns in determining precipitation and streamflow in the upper Medjerda Basin. They linked the reduced precipitation to positive phases of NAO and the negative phases were associated with increased precipitation. Findings also highlighted the effect of MO on precipitation patterns particularly during the transitional seasons. ENSO influence was more pronounced on the interannual variability of streamflow. Barhoumi et al. (2024) and Rezaei (2023) revealed the significant influence of ENSO and NAO on rainfall variability, with ENSO

showing a stronger impact during the wet season and NAO impacting both dry and wet seasons. Other investigations reported NAO and MO having the strongest influence on the amount and concentration of daily concentration events in Mediterranean region including Tunisia (Taïbi et al., 2022; Mathbout et al., 2023).

Despite these reviewed studies collectively demonstrate that climate teleconnections such significantly influence weather patterns in Tunisia, very few studies have analyzed their direct impact on drought occurrence and severity. For example, Verner et al. (2018a) reported that the 2015–2016 drought in Tunisia was more impacted by NAO and MO, which had significant impacts on regional climate patterns. However, this study poses several research gaps. Long-term studies analyses are needed to understand how teleconnection impacts vary over time in Tunisia instead of focusing on singular drought event. Also, further research should include other significant teleconnections believed to impact the region like Western Mediterranean Oscillations (WeMO). Another potential research gap, is the use of simple correlation analysis to investigate the interlinkages between climate teleconnections and drought, which may overlook the crucial aspects like non-stationary behavior and non-linear interactions. Our study addresses these gaps by conducting long-term analyses to understand how teleconnection impacts vary over time in Tunisia, including other multiple significant teleconnections, like WeMO, ENSO, NAO, and MO. It employs advanced mechanisms like wavelet coherence to provide a comprehensive understanding of the interlinkages between teleconnections and droughts in the region.

1.2.4. Drought and Land degradation: the inherent connection

The interconnectedness of drought and land degradation has garnered significant attention in the context of global environmental challenges. Climate change significantly impacts water resources, agriculture, and ecosystem stability. This literature review delves into the inherent connection between drought and land degradation, highlighting their impacts on agricultural productivity, ecological functions, and human well-being. Climate change is impacting water supply and agriculture by altering the seasonal timing of rainfall, increasing the frequency and intensity of extreme events (i.e. droughts and floods), reducing the resilience of agroecosystems, and diminishing soil quality and resource use efficiency (FAO, 2016; Wang et al., 2022). The interplay between climate change, water scarcity, and land

degradation presents a global challenge. The combined effects of climate change and land degradation are expected to account for a median decline in crop yields of approximately 2% every decade for the remainder of the century (Lobell and Gourdji, 2012). By 2050, the number of individuals at risk of hunger due to climate change is projected to rise by 10-20% more than would occur in the absence of climate change (UNCCD, 2014). This highlights significant negative effects on agricultural productivity and ecological functions, ultimately impacting human well-being and quality of life.

Climate change, coupled with recurring extreme climate events, land degradation, and biodiversity loss, has been recognized as one of the most significant challenges to sustainable development in drylands globally, making water-scarce countries and their agricultural lands increasingly arid and vulnerable to drought (Davies et al., 2015; Lian et al., 2021). In Tunisia, droughts and land degradation are major environmental issues, with up to 75% of the country's territory at risk of land degradation (MEHAT, 2012). These degradations are characterized by water scarcity, the reduction and loss of vegetation cover, soil degradation, biodiversity decline, and other limiting factors. Moreover, droughts are a recurring phenomenon in Tunisia and are projected to become more frequent and intense in the future due to global climate change, posing particularly severe threats to the agricultural sector and having broader consequences for the national economy (Pachauri et al., 2014; Verner et al., 2018a).

Given that these environmental challenges are not specific to Tunisia, the urgency of this matter mobilized the international communities to adopt immediate and assertive actions and policies to combat this anthropogenic and natural phenomenon. These policies involve collaborative efforts with international organizations such as the UNCCD, the Drought Initiative during the 2018–2019 biennium, and the Sendai Framework for DRR 2015–2030 (UNISDR, 2015). They endorse a wide array of actions aimed at improving sustainable natural resource management. This approach is grounded in the understanding that the conservation, protection, and rehabilitation of land and freshwater resources are crucial elements of every policy aimed at safeguarding the environment and natural resources. Another key aspect is strengthening communities' resilience to drought and other risks related to water scarcity and land degradation. In this regard, various Sustainable Development Goals (SDGs) were adopted by the global community in 2015, with a specific goal addressing land degradation. This goal, along with its associated target, aims to achieve a land

degradation-neutral world by 2030. Specifically, goal 15.3 of the UN 2030 Agenda for SDGs states, “By 2030, combat desertification, restore degraded land and soil, including land affected by desertification, drought, and floods, and strive to achieve a land degradation-neutral world” (Minelli et al., 2017).

In arid regions and semi-arid regions, the degradation of natural resources extends beyond droughts to include soil erosion, which is widely identified as a critical environmental challenge (Du et al., 2016; Machiwal et al., 2022; Terwayet Bayouli et al., 2023b; Quinton and Fiener, 2024). Among the various forms of land degradation, soil erosion is the most prevalent, particularly in arid environments such as Tunisia (Jebari et al., 2010). This phenomenon results from intricate interactions between natural processes and human activities and is projected to intensify over time (Eekhout and de Vente, 2022). The consequences of erosion pose significant risks to agriculture, water availability, and ecosystem functions, further amplifying the impacts of drought (Patil, 2018; Qiu et al., 2021). Alongside its adverse effects on soil fertility and agricultural yields, erosion accelerates desertification, creating formidable obstacles for sustainable land management and economic resilience in affected regions (Kumar and Pani, 2013).

Over the past two decades, various methods for estimating soil erosion have been developed, including simple empirical models and complex physically-based models. Developed by Wischmeier, and Smith (1965), the Universal Soil Loss Equation (USLE), is most widely used due to its simplicity and reliability. USLE is a standard approach for annual water erosion estimation. In 1975, (Williams) modified USLE to create the MUSLE (Modified Universal Soil Loss Equation), replacing the rainfall aggressiveness factor with a runoff energy factor to better predict sediment yield. This model was further improved by Renard (1991), resulting in the RUSLE. RUSLE integrates additional research and data to enhance erosion prediction and management strategies (Prasannakumar et al., 2012). RUSLE model involves five input parameters, namely soil erodibility, land cover, slope length and steepness, erosivity, and anti-erosion techniques (Terwayet Bayouli et al., 2023b). These parameters exhibit spatial and temporal variability and are influenced by a range of environmental and anthropogenic variables (Prasannakumar et al., 2012). Physically-based models, including vector-based and raster models, were developed to address RUSLE's limitations. These models use mathematical equations to represent the environment and sediment transport processes more accurately (Alencar et al., 2020). While these models provide a more

dynamic and process-based approach, they require more detailed input data and extensive calibration and validation over several years compared to RUSLE, making them less applicable in data-scarce regions such as Tunisia. Potential improvements in these methodologies could involve integrating remote sensing data to enhance accuracy and predictive capabilities (Zhou et al., 2008; Gumiere et al., 2009; Borrelli et al., 2017).

As a part of Mediterranean basin, Tunisia faces significant environmental challenges related to drought and soil erosion, particularly in its arid southern regions. With approximately 10 million hectares designated for agriculture, nearly 3 million hectares of Tunisia's agricultural lands are at risk of water-induced soil erosion (Kefi et al., 2012). Despite its arid climate, southeastern Tunisia experiences widespread water erosion, primarily due to extreme climatic conditions, including intense torrential rainfall that triggers severe runoff, shallow soils with limited water retention, sparse vegetation cover, and unsustainable land-use practices such as overgrazing (Taamallah, 2003; Gamoun, 2016a).

Droughts and soil erosion are interlined through different processes. As a key driver of land degradation, soil erosion often stems from prolonged dry conditions, which in turn contribute to ecosystem alterations (Peri et al., 2021). Drought conditions amplify erosion by increasing both wind-driven sand movement and surface runoff (Trnka et al., 2016; Middleton and Kang, 2017). Furthermore, prolonged droughts hinder vegetation growth, particularly in arid environments, leading to the depletion of soil organic carbon. This loss of organic material, coupled with disruptions in soil structure, creates a feedback loop that further accelerates erosion processes (Kraamwinkel et al., 2021). The loss of topsoil, which contains the highest concentrations of nutrients and organic matter, is a direct consequence of erosion. Changes of LULC, driven by droughts, like the withdrawal of wetlands or cultivated fields into barren land disrupt hydrological cycles, increasing soil susceptibility to erosion, especially during prolonged drought periods (Baumhardt and Blanco-Canqui, 2014). At the same time, erosion itself can heighten drought vulnerability by degrading soil structure and reducing its water retention capacity, ultimately worsening the impacts of water scarcity (Reichhuber et al., 2019b).

Despite the significance of the issue, research on the connection between droughts and soil erosion remains scarce. Globally, limited research has directly explored this connection. For instance, Santra, and Santra Mitra (2020) discovered that regions in West Bengal, India, with frequent droughts experienced nearly double the soil loss rate

compared to areas with less frequent drought events. Similarly, studies in southwest China and western India have reported a positive correlation between droughts and soil erosion (Yu et al., 2021; Masroor et al., 2022). However, these few studies are geographically limited to specific regions, leaving a notable research gap, particularly in Mediterranean areas, which are highly vulnerable to both soil erosion and drought. There is a notable lack of research in other regions, particularly in the Mediterranean, which is highly prone to soil erosion and droughts. Current research often relies on limited datasets. Integrating more extensive and diverse data sources, such as remote sensing data, could enhance the accuracy and applicability of the findings. Our research is pioneering the examination of the link between droughts and soil erosion in the Mediterranean region, within a study case in southeast Tunisia, a region highly susceptible to droughts and soil erosion. Expanding the geographical scope and addressing the outlined gaps will contribute to a more holistic understanding of this critical environmental issue to tackle the adverse impacts of these phenomena.

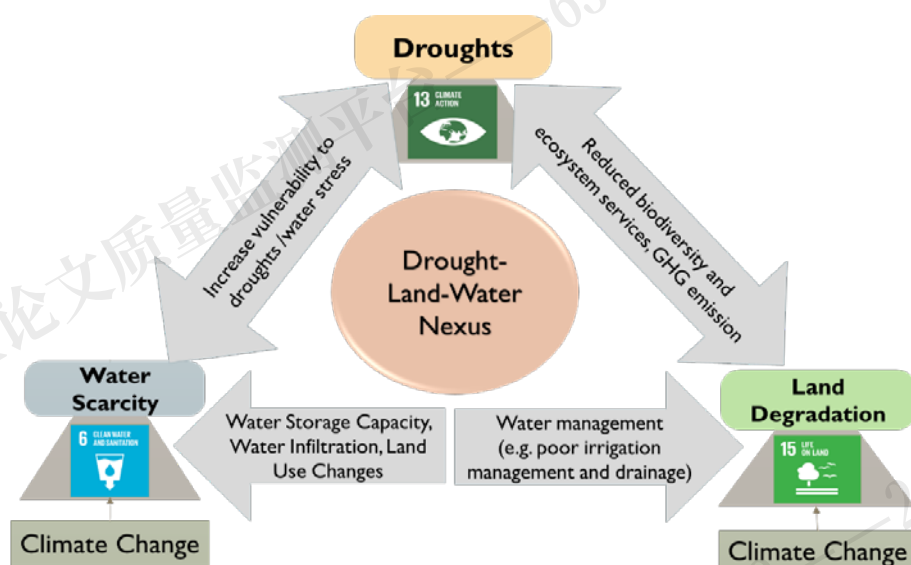


Figure 1-2. Schematic representation of the interlinkages among climate change, land degradation, and droughts to induce water scarcity.

Chapter 2 General Research Approach and Data

2.1. Research content and overall scheme

2.1.1. Scientific Research Questions

As part of north Africa, which is appropriately described as an arid and water-scarce region, Tunisia faces the above-mentioned circumstances and existing research problems. In response, this research was devoted to provide a holistic assessment of drought dynamics in Tunisia by targeting its multifaceted nature through the investigation of its characteristics, trends, propagation mechanisms, impacts and interlinkages with climate teleconnections and soil erosion processes, one of the most widespread forms of land degradation in arid regions of Tunisia. Besides, the future projections of droughts under climate change scenarios were also investigated. This research addresses the following key research questions:

- i. What are the spatiotemporal patterns, characteristics, and propagation mechanisms of different drought types (meteorological, hydrological, and agricultural) in Tunisia, and how do they vary across its diverse climatic regions?
- ii. How can GGDI be employed to assess groundwater drought at a national scale in Tunisia, and what are its relationships with other hydro-meteorological indices?
- iii. How do major climate teleconnections (such as ENSO, NAO, MO, and WeMO) influence the frequency, severity, and spatial distribution of droughts in Tunisia?
- iv. How do future climate projections under different climate change scenarios influence the frequency and intensity of drought events in Tunisia?
- i. What are the impacts of drought on land productivity in Tunisia's most drought-prone regions, and what are its relationships with land degradation, particularly soil erosion, as the most two prominent environmental challenges in Tunisia?

2.1.2. Objectives of the study

The main objective of this thesis is to deliver a holistic, integrated assessment of drought risk in Tunisia within the context of climate change, while unfolding the

intricate interactions among drought, climate change, and land degradation, and how these factors collectively impact environmental sustainability.

The specific objectives of the study are:

1. Analyze the spatiotemporal patterns of different types of droughts (meteorological, hydrological, and agricultural), their characteristics, hazards, and their propagation mechanisms and probabilistic linkages across Tunisia.
2. Evaluate the applicability of GRACE-based groundwater drought index for assessing groundwater drought in Tunisia at a national scale and its relationship with other hydro-meteorological indicators.
3. Examine the relationship between droughts and land degradation, focusing on soil erosion processes, specifically in Tunisia's arid, drought-prone regions.
4. Assess drought impacts on land productivity through Net primary Productivity (NPP), specifically in the arid drought-prone regions of Tunisia.
5. Explore the interlinkages between major climate teleconnections and drought variability in Tunisia using wavelet-based quantitative approach.
6. Project future drought characteristics (intensity, frequency, area, duration), under different climate change scenarios using multi-model hydrological simulations, analyzing their implications for Tunisia's drought risk and sustainability.

Overall, this study aimed to address the key research questions to support effective drought management and ensure the sustainable use of water resources. Additionally, through a critical review of the literature, this dissertation highlights both the advancements and limitations of current research, providing a foundation for a deeper understanding of past trends and identifying opportunities for future studies.

2.2. Methodological approaches

To address the overall research questions, build upon previous studies in the field, and achieve the stated research objectives, the conceptual framework and theoretical foundation of this study are summarized in Figures 1-3. This is followed by an overview of the methodological approaches employed in each chapter and the overall structure of the dissertation in Figures 2-1.

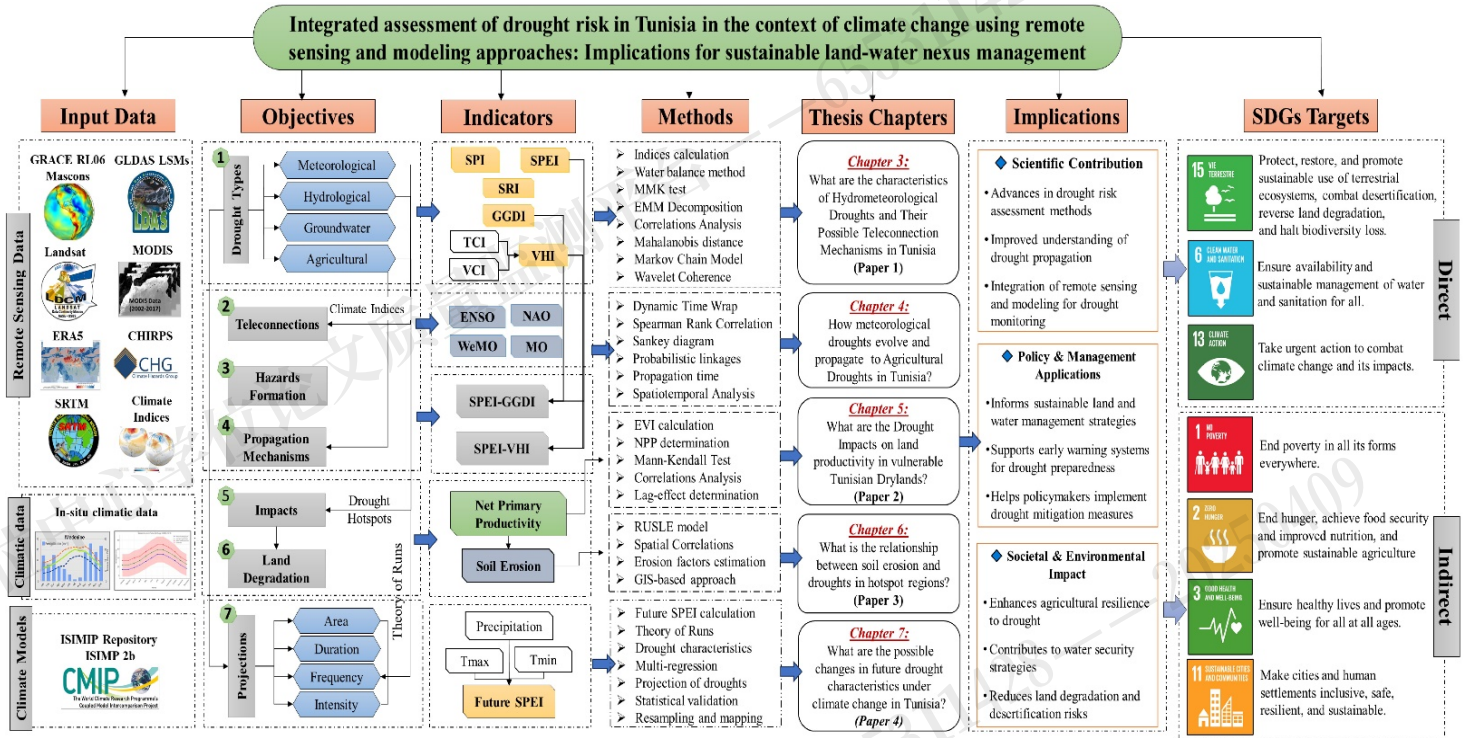


Figure 2-1. Graphical overview of the detailed research approach integrating remote sensing and modeling approaches for achieving land-water resources implications and SDGs.

2.2.1. Research overall detailed methodology

Figure 2-2 presents the overall overview of the research methodological flowchart adopted for this study. This develops a novel approach not only by considering different facets of droughts, including mainly evaluation of its different types, characteristics, impacts, hazard evolution, propagation mechanisms, linkages with teleconnection factors, and future projections, but also by interrelating droughts to land degradation, two major environmental challenges in Tunisia. By integrating a wide array of earth observation datasets, modelling tools, GIS and remote sensing techniques, statistical methods, this work establishes a robust scientific foundation at a country-level scale to inform national adaptation policies and measures, supporting efforts in early warning systems and drought risk reduction.

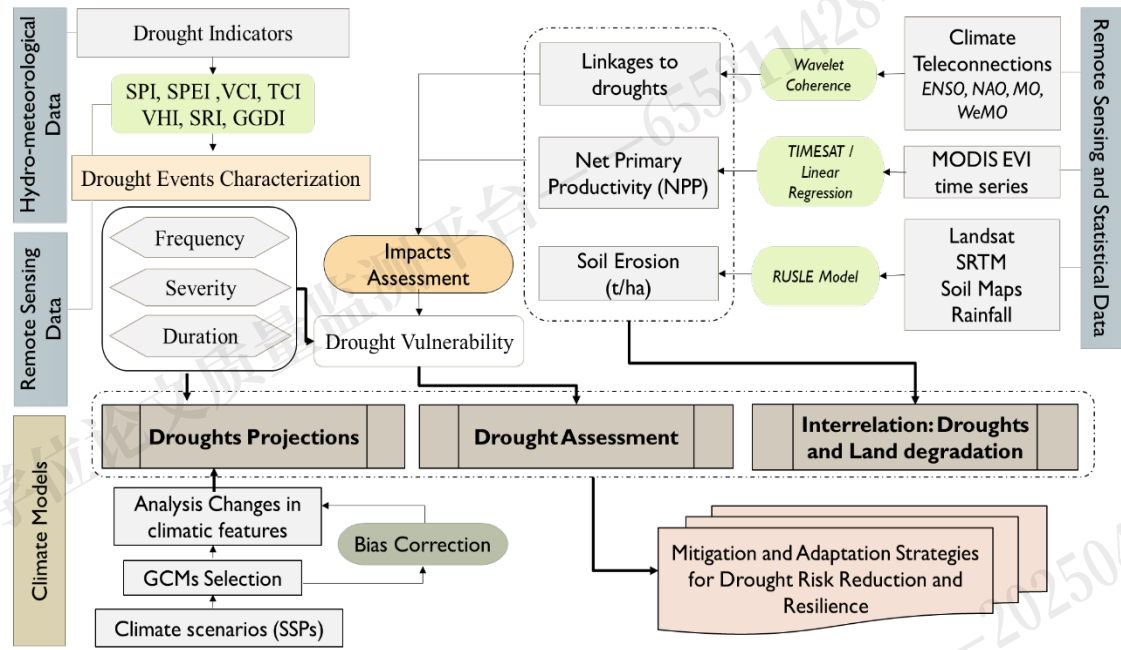


Figure 2-2. Overview of the methodological flowcharts of the research.

2.2.2. Study area

Tunisia (32°–38°N; 7°–12°E), nestled in North Africa along the Mediterranean Sea, boasts a wide array of natural landscapes (Fig. 2-2). It is located in a transitional area that connects the temperate European climate zone with the desert. Tunisia covers an area of 164,000 km², with a 1,300 km coastline extending along two coastal fronts (northern and eastern). The geographical position and the topographic orientation of Tunisia crosscuts two opposite climate zones: the Mediterranean Sea in the north and the Sahara in the south. It exhibits a large variety of climate conditions, wherein the majority of its area (67%) falls under semiarid and arid climates, while the remainder is marked by subhumid and humid conditions. This climate variety has resulted in irregular rainfall regime across the country, where the annual rainfall varies from 100 mm in the southernmost regions to 1200 mm in the northernmost regions (WBG, 2021). Similarly, temperature patterns vary across space and time, with winter temperatures dropping below 10°C (December to February) and summer temperatures reaching up to 32°C (June to August). As a result of this climate variability, Tunisia is experiencing noteworthy climate events such as droughts, heatwaves, occasional heavy rainfall, and flooding.

Tunisia is among the most water-scarce countries in the Mediterranean region, facing increasing pressure from both climate change and human activities (Chouchane et al., 2015). The agricultural sector plays a crucial social and economic role. The majority of the Tunisian national territory is devoted for agriculture where the agricultural land is estimated at 10 million ha or being 61% of the total country area. The main agricultural products are olives, cereals, citrus fruits, and palm dates. Besides to arable lands (arboriculture, field crops, and fallows), other land types such as forests and rangelands are among the land use system in Tunisia (Verner et al., 2018a). The agriculture is largely rain-fed and has a major influence on the regional economy, with agriculture providing a livelihood to the collective population in Tunisia. However, the country's agricultural production is increasingly threatened by climate change-induced droughts and declining water availability. Water scarcity in Tunisia is driven not only by natural aridity but also by unsustainable water management practices.

The groundwater resources are counted for 2140 Mm³/year (44%), while surface water resources are estimated at 2700 Mm³/year (56%). More than 40% of irrigation water currently used in Tunisia is provided by groundwater aquifers, most of which are in a critical state of depletion due to its overexploitation during the last three decades (Frija et al., 2015). Groundwater resources have been intensively mobilized in Tunisia since the 1980s, causing overexploitation. Among 273 aquifers in Tunisia 71 are detected as overexploited, with an average extraction rate reaching 146% of their natural recharge capacity (Frija et al., 2015; Mekki et al., 2017). The persistent depletion of aquifers, combined with increasing salinization, poses a severe challenge to sustainable water management.

Compared to other Mediterranean countries, Tunisia is particularly susceptible to the adverse effects of climate change due to its high exposure to increasing temperatures, prolonged droughts, and erratic rainfall patterns (Pachauri et al., 2014). These climatic shifts pose serious risks to water resources, food security, and economic stability, particularly in arid and semi-arid regions, where prolonged droughts have become more frequent and severe (Verner et al., 2018a). The combination of rising temperatures, increased evapotranspiration, and declining precipitation levels is expected to further exacerbate drought conditions, placing additional strain on the already fragile water-land nexus.

Southern Tunisia is particularly exposed to these challenges due to its arid climate, limited and irregular rainfall regime, fragile soils, and increasing land degradation. This

region faces acute environmental challenges, including desertification, soil erosion, and declining groundwater levels, which exacerbate the risks to agricultural productivity and local livelihoods.

Given the southern region's vulnerability, we selected two basins in southeastern Tunisia, Jeffara and Negueb (Figure 2-2), as ideal case studies to examine the impacts of droughts on land productivity and to investigate the relationship between droughts and soil erosion, the most widespread form of land degradation in the region. These basins represent critical socio-agroecological zones of the country where water scarcity, droughts and land degradation are highly pronounced. Their strategic importance for water and land resource management, as well as their representativeness of broader challenges in Tunisian drylands, makes them ideal study sites for assessing climate-induced drought risks and land degradation, aimed at supporting land and water resource management.

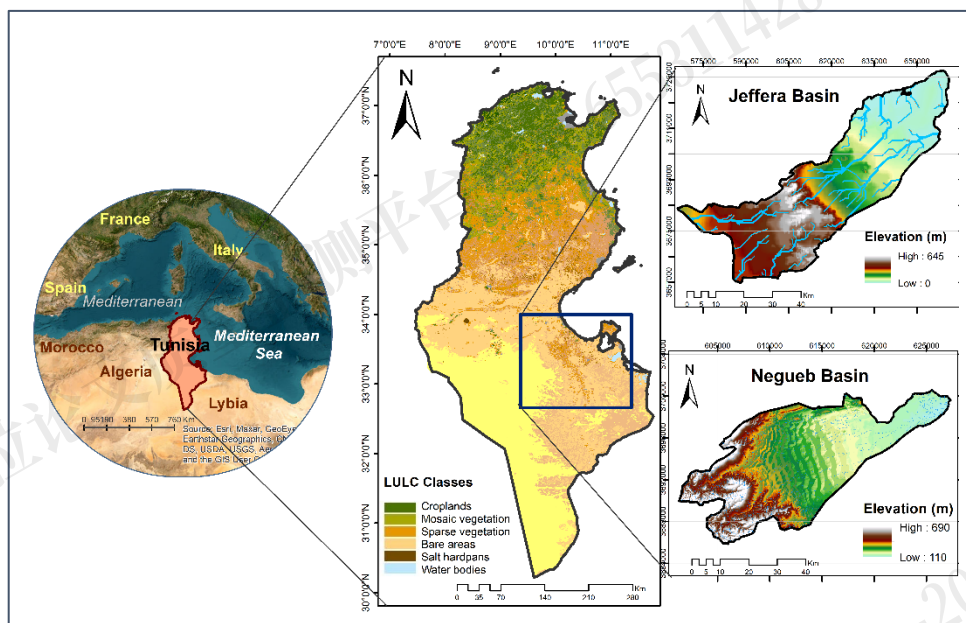


Figure 2-3. Study area map and selected basins for the case studies in Tunisia. (a) Study area describes the location Tunisia, (b) The study cases selected in the Southeastern as representative for the Tunisian drylands.

2.3. Thesis Structure

This dissertation is structured on eight chapters following the rules and regulations for the PhD Dissertation writing guide provided by the University of Chinese Academy of Sciences (UCAS), the studies are independent and related research components:

Chapter 1 presents the general research background and a comprehensive literature review, outlining the state of the art in drought assessment and its impacts. It examines previous scientific contributions relevant to this thesis and explores various aspects of drought, including its characteristics, interlinkages with climate teleconnection factors, and delves into the inherent connection between droughts and land degradation.

Chapter 2 highlights the general research approach and data, including the significance of the research, current challenges in the field, research questions, the defined objectives of the present research, and an overview about the adopted methodological approaches in this study. It encloses also a detailed description of the study area and the selected watersheds as study cases, due to their importance as they are regions highly prone to recurrent drought events and land degradation, particularly to soil erosion.

Chapter 3 is the first research part, which delves into a holistic assessment of the characteristics, trends, and hazards of hydro-meteorological droughts in Tunisia, along with an assessment of their teleconnections with large-scale climate indices. This study pioneers the GGDI application over Tunisia, linking surface and subsurface conditions via various hydro-meteorological drought indices (SRI, SPI, and SPEI) during 2002-2017. A stochastic analysis of monthly SPEI-GGDI values was performed using a first-order Markov chain model, to investigate regional drought hazards formation, persistence, and evolution. Pearson's correlation coefficient and wavelet coherence were applied to evaluate interactions among indices and their teleconnections with large-scale climate patterns. This study introduces new insights of drought monitoring using GRACE gravity satellite in Tunisia, and could act as a basis for implementing drought mitigation and adaptation strategies to devise sustainable development.

Chapter 4 is the third research chapter that focuses on agricultural droughts to complement the analysis of hydrological and meteorological droughts in Tunisia, providing further insights into the complex interaction between different types of droughts by employing MODIS-derived VHI, generated by combining TCI and VCI. This study introduces a novel framework that integrates statistical methods, including Spearman rank correlation (SRC) analysis, Dynamic Time Warping (DTW), and transition probability analysis, to quantify MD-AD propagation time and probabilistic linkages at different severity levels over time, through SPEI and MODIS-derived VHI

drought indices. This study provides critical insights strengthening early warning systems and for drought risk mitigation.

Chapter 5 is the fourth research chapter, developed following a key identification from Chapter 4 as the southern Tunisia is highly vulnerable to droughts, that comes as a case study, to investigate the drought impacts on land productivity, applied on a strategically important basin in southern where problems of droughts and land degradation are highly pronounced. This study involves an investigation of their impacts on net primary productivity (NPP) as a response of the lack of the studies related to this issue. It takes a leading role in unraveling NPP responses to droughts in southeastern Tunisia. The assessment was performed based on SPI extracted from long-term observation data and satellite-based annual NPP datasets derived from MOD13Q1 EVI products. In this study, drought events were first identified and their characteristics were analyzed in terms of severity and spatiotemporal variability. Accordingly, we assessed NPP variation and explored the relationship between NPP and droughts at annual timescale through SPI and NPP anomalies. This study could constitute a scientific basis for regional planners and policy-makers to effectively develop adaptation and restoration strategies to cope with drought impacts and enhance ecosystem services in arid environments.

Chapter 6 explores the relationship between drought variability and land degradation through soil erosion processes in southern Tunisia, a region highly vulnerable to droughts and land degradation. This study develops a comprehensive Geographic Information System (GIS)-based approach to assess soil erosion and droughts, and reveal their relationship, under an arid climate. To monitor droughts, VCI and TCI derived respectively from MOD13A2 EVI and MOD11A2 LST products were combined to generate VHI. The VHI was used as an efficient tool to monitor droughts in Tunisia. RUSLE model was applied to quantitatively estimate soil erosion, and its relationship with drought was analyzed through Pearson correlation. The current study uniquely interrelates drought to soil erosion, two major environmental challenges in southeastern Tunisia. The findings offer guidance to land managers and stakeholders in implementing targeted management strategies that align with sustainable development goals, ensuring better resilience in these fragile environments.

Chapter 7 focuses on the projection of future droughts under climate change scenarios using datasets from three GCMs, namely MRI-ESM2-0, IPSL-CM6A-LR, and MIROC6, available from CMIP6. CMIP6 was selected for its reliability,

bias-correction, and inclusion of high-resolution GCMs, enabling a robust assessment of future drought conditions, using SSP1-2.6 and SSP5-8.5 scenarios. The analysis covered the period from 1981 to 2100, with baseline data from 1981-2015 sourced from ERA5. This study explores the future trends of different drought characteristics, including magnitude, frequency, duration and intensity across Tunisia. Notably, it represents the first comprehensive assessment of future drought trends in the region using CMIP6 outputs under different SSP scenarios. Providing such insights into the projected impacts of climate change in Tunisia is crucial for policymakers and stakeholders to effectively manage local disaster risks, mitigate their major impacts and build adaptation and resilience strategies.

Finally, Chapter 8 is dedicated to summarizing the key findings of the thesis, drawing the main conclusions, and highlighting the major innovations of this research. It also outlines future perspectives, offering directions for further studies and potential applications.

Chapter 3 A Comprehensive Analysis of Hydro-meteorological Drought in Tunisia: Insights from GRACE Gravity Satellite and Teleconnections

3.1. Introduction

In the context of global warming along with anthropogenic influences, the global hydrological cycle is undergoing significant modulations (Xie et al., 2010). Yoon et al. (2015) suggest that global warming is impacting the regional climate by inducing hotter and drier conditions that lead to considerable declines in soil water. These factors collectively exacerbate drought conditions, especially in drylands where water resources are limited and unevenly distributed. Drought is categorized into four main types: meteorological (MD), hydrological (HD), agricultural (HD) and socioeconomic. MD stemming from prolonged precipitation shortages, and hydrological droughts, characterized by sustained deficits in water supply affecting surface water and groundwater resources, stand as the two most widespread types of droughts (Palmer, 1965; Dracup et al., 1980). Generally, though not invariably, meteorological drought serves as the precursor to other types of drought (Wang et al., 2021; Jiang et al., 2023). A more specific type of hydrological drought was defined as groundwater droughts (GD) referring to groundwater levels reduction, decreasing recharge, or discharge (Van Lanen and Peters, 2000). Droughts are projected to be more frequent in the future under changing climate (Pachauri et al., 2014). Thus, assessment of drought characteristics is crucial to target a sustainable management of water resources. In this context, different drought indices were developed to address the characteristics of each drought type (Zargar et al., 2011). For instance, SPI and SPEI are commonly used to evaluate meteorological droughts, relying basically on precipitation and evapotranspiration (McKee et al., 1993; Vicente-Serrano et al., 2010). HD is characterized by negative anomalies in streamflow and groundwater (Mishra and Singh, 2010). Nevertheless, it can be assessed using indices involving precipitation, streamflow, runoff, groundwater levels, and water balance. SRI, based on SPI approach, is a widely employed index that integrates hydrological process causative of the seasonal decline in streamflow influenced by climate conditions. SRI, (from 1-month to seasonal scale), serves as a valuable supplement to SPI for portraying hydrological characteristics of droughts (Shukla and Wood, 2008).

Practically, evaluating hydro-meteorological droughts involves linking SPI or SPEI with a streamflow or groundwater-based index, or integrating the variables utilized in formulating these indices (Carrão et al., 2016; Zhao et al., 2022). HD and MD are interlinked via different processes involved in the hydrological cycle (Van Loon and Laaha, 2015). In a broader context, elevated evaporation rates over oceans can influence precipitation patterns on land, leading to an increase in water storage on continents (Gimeno et al., 2012). Conversely, scarce precipitation over land, coupled with MD-driven increased evaporation can trigger HD (Van Lanen et al., 2013; Van Loon, 2015). Several studies suggested an association between meteorological drought and hydrological droughts where they claimed that prolonged meteorological droughts lead to alterations in hydrological characteristics (Van Loon, 2015; Liu et al., 2019c; Xu et al., 2019b; Pan et al., 2020).

Comprehensive regional drought assessments, based on field observations, are crucial to endorse environmental restoration and hydrological research. However, such assessments are often unsystematic as they are spatiotemporally limited and require considerable financial resources (Creutzfeldt et al., 2012). Earth observation tools have emerged as an effective substitute for traditional field-based observation methods. They provide continuous, cost-effective, and spatially distributed information on parameters associated with various types of droughts, particularly for groundwater droughts where monitoring through traditional observations is challenging (Thenkabail and Gamage, 2004; Chao et al., 2016). In addition, drought analysis through proxies of precipitation, temperature, and surface water observations does not reflect direct evidence of water availability deficits. Water is stored in various forms, including soil moisture, surface water, snow, vegetation, groundwater, river and lake water (Giroto and Rodell, 2019). Therefore, integrating measurements of entire water storage types is vital to accurately evaluate responses of water resources to drought, a task facilitated by the use of remote sensing technologies.

GRACE gathers precise measurements of Earth's gravity field changes and translates them into variations in total water storage (TWS) across continental regions. This serves as a valuable information source to address gaps in data availability and understand hydrological characteristics (Tapley et al., 2004). Based on GRACE data, Thomas et al. (2017) developed the GGDI to measure deficit in groundwater storage and applied it over California Central Valley. At global scale, Liu et al. (2023d) employed GDI along with the SPI, SRI, and Standardized Streamflow Index (SSI) to

examine the propagation relationships between drought types at global scale. Several studies revealed that climate change along with anthropogenic activities lead to groundwater storage depletion (Zhao et al., 2022; Liu et al., 2023b).

Climate change, coupled with recurring climate events, namely teleconnections, was identified as main contributors for more severe and frequent dry events, particularly in water-scarce countries like Tunisia. These teleconnections encompass natural climatic phenomena, such as El Niño Southern oscillation (ENSO), the North Atlantic oscillation (NAO), and the Mediterranean oscillation (MO), the Western Mediterranean oscillation (WeMO) trigger droughts by increasing temperatures and making rainfall more erratic. In particular, Forootan et al. (2019) showed that ENSO has a crucial impact on HD evolution in North Africa from 2010 to 2012. Also, Wang et al. (2020) found that ENSO had a strong impact on droughts in north China plain. Another study conducted over Africa showed that ENSO alongside with the Atlantic Multidecadal Oscillation AMO could induce significant changes in recharge to the aquifers and interannual groundwater storage (Carvalho Resende et al., 2019).

Drought, one of the most widespread natural disasters in the world, is projected to be more frequent in the future as a result of the increasing climate variability (Zhang and He, 2016). Drought exhibits particularly distinctive features of high frequency, prolonged duration, and widespread influence in arid and semi-arid regions endowed with scarce water resources, such as Tunisia. The frequent and prolonged incidence of drought not only engenders to substantial losses in industrial and agricultural production in such fragilized regions, but also induces adverse environmental consequences, including water resource shortages, land desertification, and ecological environment degradation (Kahil et al., 2015). For example, extended periods of drought and overall rainfall variability pose significant challenges for agriculture in Tunisia. For example, groundwater resources in Tunisia are considered as the primary buffer for drought impacts, where more than 40% of irrigation water used is supplied by groundwater aquifers. Thus, exerting an additional pressure on groundwater resources which are already declining, which has serious implications for their long-term sustainability. However, limited research has delved into the climatology of droughts, particularly groundwater drought, and their potential associations with the various teleconnections in Tunisia.

To address these gaps, this study analyzes hydrometeorological droughts in Tunisia (2002-2017), integrating advanced remote sensing, traditional drought indices, and

climate teleconnection analyses. Key objectives include: (a) introduce GGDI application in Tunisia to provide new insights into GD variations, (b) evaluate hydro-meteorological indices (SPI, SPEI, SRI, GGDI) for assessing drought characteristics and hazards, and (c) investigate the interlinkages between climate teleconnections factors (MO, ENSO, NAO, and WeMO) and hydro-meteorological drought indices. This study contributes to understanding regional drought dynamics, highlighting GGDI's role in GD monitoring, and exploring the links between drought indices and teleconnections. Such insights are vital for tackling water scarcity issues in Tunisia.

3.2. Materials and methods

3.2.1. Datasets

3.2.1.1. GRACE data

GRACE data are processed and available at different levels, with Level 2 providing spherical harmonic coefficients and Level 3 providing gridded TWS estimates Mass Concentration blocks (Mascons) based on Level 2 or Level 1b. Mascons are a priori processed products that could be directly employed, catering to diverse scientific interests (Nasa/Jpl, 2023). Monthly total water storage anomalies (TWSA) deduced from GRACE satellite processed at Jet propulsion Laboratory (JPL) using the monthly mass concentration Mascon approach (RL06.1Mv03) (https://grace.jpl.nasa.gov/data/get-data/jpl_global_mascons/), was employed to retrieve TWS in Tunisia from 2002 to 2017. These datasets, with a spatial resolution of 0.25° , present a reduced leakage errors signal by employing a Coastal Resolution Improvement (CRI) filter (Nasa/Jpl, 2023). Any missing periodic data were addressed using interpolation techniques. GRACE-derived gravity measurements have been extensively used to evaluate TWS changes and identify hydrological drought characteristics and impacts (Ndehedehe et al., 2016; Wang et al., 2020; Foroumandi et al., 2023). Monthly TWSA time series were computed from April 2002 (GRACE start date) to January 2017. GRACE ensure an integrated TWS assessment in a hierarchic system that includes soil moisture, surface water, and groundwater resources.

3.2.1.2. GLDAS Data

Given data scarcity in our study area, outputs from Global Land Data Assimilation System (GLDAS) of NASA Version 2 (GLDAS-2.1) from 2000 to 2020

(<https://disc.gsfc.nasa.gov/datasets?keywords=GLDAS>), were involved to quantify surface-water components: surface runoff, soil moisture, and canopy storage. Given its agreement with GRACE-derived TWS (Tangdamrongsab, 2023), the land surface model NOAH version 2.1 (spatial resolution of 0.25°) was employed for this study, to reflect the fluctuations in water storage components and to compute SRI for HD characterization in Tunisia (Table 3-1).

3.2.1.3. Meteorological Data

In the current study, monthly precipitation and temperature data from ERA5 (<https://app.climateengine.com/climateEngine>), at $0.25^\circ \times 0.25^\circ$ spatial resolution, were employed to compute precipitation-based indices (SPI and SPEI) at a 12-month timescale, for 2002-2017 period over Tunisia.

3.2.1.4. Climate data

Monthly climate oscillations, namely ENSO and NAO were sourced from the Earth System Research Laboratory (ESRL) of the United States National Oceanic and Atmospheric Administration (NOAA) (https://psl.noaa.gov/gcos_wgsp/Timeseries/). In addition, MO and WeMO were obtained from the Climatic Research Unit datasets of University of East Anglia (<https://crudata.uea.ac.uk/cru/data/>). A detailed description of these indices is available from the mentioned website. These datasets were used to characterize the association between drought events and teleconnection factors during 2002-2017.

Table 3-1. Summary of used datasets.

Data	Products	Sources	Spatiotemporal resolution, Data span
TWS	GRACE RL06 Mascon solutions	https://grace.jpl.nasa.gov/	0.25° × 0.25 °, 2002 - 2017 (monthly)
Soil Moisture, ETP, Surface/Sub-Surface Runoff and TWS	GLDAS Version 2 (GLDAS-2.1)	https://disc.gsfc.nasa.gov/datasets?keywords=GLDAS_NOAH025_M_2.1&page=1	0.25 ° × 0.25 °, 2002 – 2017 (monthly)
Precipitation and Evapotranspiration	European Re Analysis (ERA5)	https://app.climateengine.com/climateEngine	0.25 ° × 0.25 °, 2002 to 2017 (monthly)
ENSO and NAO	Earth System Research Laboratory (ESRL) of NOAA	https://psl.noaa.gov/gcos_wgsp/Timeseries/	0.25 ° × 0.25 °, 2002 to 2017 (monthly)
MO and WeMO	Climatic Research Unit time-series (CRU)	https://crudata.uea.ac.uk/cru/data/	0.25 ° × 0.25 °, 2002 to 2017 (monthly)

3.2.2. Methods

3.2.2.1. Groundwater storage change estimation

GRACE gravity satellite was used to detect changes in GWS over Tunisia and subsequently to calculate GGDI. GRACE-derived TWSA reflects variations within surface water, groundwater, soil moisture, snow water, and canopy water. Equation (3-1) was calculated to determine GWS changes through GRACE (Wei et al., 2022):

$$GWSA = TWSA - SMSA - CWSA \quad (3-1)$$

Where, SMSA, GWSA, and CWSA are the anomalies of Soil Moisture Storage, Groundwater Storage, and Canopy Water Storage, respectively, simulated by GLDAS. GRACE and GLDAS data are provided in millimeters, at 1-month timescale.

3.2.2.2. GGDI

The impacts of droughts on GWS, particularly using GRACE satellite data, has not yet been investigated in Tunisia. This study introduces GGDI as a novel and reliable index to evaluate GD characteristics, particularly for water-stressed regions. As first step, monthly climatology was determined to adjust for seasonal GWS variations (Equation 3-2) (Thomas et al., 2017).

$$Ci = \frac{1}{ni} \sum_1^{ni} GWSAi \quad (3-2)$$

With, Ci: climatology for month i (i=1,2, 3, ..., 12).

The GWS Deviation (GSD) was generated by abstracting the seasonal average GWSA, to emphasize GWS variations unconnected to regular seasonal changes. For GSD standardization, the mean value ($\bar{x}GSD$) was deducted from GSD, after that divided by the standard deviation ($sGSD$) (Equation.3-3).

$$GGDI = \frac{GSDi - \bar{x}GSD}{sGSD} \quad (3-3)$$

GGDI is conceived as a robust indicator of HD, and portraying the normalized net changes in GWS volumes.

Categories of drought severity classes are given in Table. 3-2.

Table 3-2. Drought severity levels based on drought indices.

Drought condition	SPI/SPEI/SRI/GGDI
No drought	$-0.5 < I$
Mild drought	$-1.0 < I \leq -0.5$
Moderate drought	$-1.5 < I \leq -1.0$
Severe drought	$-2.0 < I \leq -1.5$
Extreme drought	$I \leq -2.0$

3.2.2.3. Standardized drought indices

Droughts are typically defined by means of standardized indices (Zargar et al., 2011). In this research, SPI, SPEI, and SRI were computed to characterize hydro-metrological droughts in Tunisia, compare with GGDI, and subsequently identify teleconnections with different climate indices (refer to appendix chapter Table.A.1). Figure. 3-1 displays the overall methodological framework.

- **SPI**

SPI was originally introduced by McKee et al. (1993) as a metric assessing MD based on long-term precipitation time-series adjusted to conform to a probability distribution (Equation 3-4). SPI was chosen as it is widely validated capturing precipitation deficits over multiple timescales, ideal for semi-arid climates like Tunisia (Hao et al., 2015; Ashraf et al., 2022). SPI was calculated at a 12-month timescale through an ERA5-derived rainfall record (2002–2017), by means of Gamma method

(Equation 3-5) and converted to a normal distribution. Although it is a temporally flexible index, SPI focus primarily on precipitation, and potentially overlooking the influence of other drought parameters such as temperature.

$$SPI = (P_i - P_m) / S \quad (3-4)$$

With P_i : year i precipitation, P_m : average precipitation, and S : rainfall standard deviation.

$$g(x) = (1/\beta^\alpha \Gamma(\alpha)) x^{\alpha-1} e^{-x/\beta}, \quad (x > 0) \quad (3-5)$$

With, $\Gamma(\alpha)$: gamma function; x (mm): precipitation ($x > 0$); α and β : shape and scale parameters (> 0).

• SPEI

SPEI takes into account potential evapotranspiration (PET) along with to precipitation, leveraging all the SPI benefits, reflecting an improved assessment by accounting for climate change impacts (Equation 3-10) (Vicente-Serrano et al., 2010). SPEI adopts a three-parameter based distribution for adjusting deficit values (Equation 3-9), making it appropriate for arid regions like Tunisia, with common negative moisture deficit. SPEI was computed at an annual timescale from 2002 to 2017, via the Log-logistic probability distribution (Equation 3-8) for the normalized water balance (Equation 3-6), with PET calculated through Thornthwaite method (Thornthwaite, 1948).

$$D_i = P_i - PET_i \quad (3-6)$$

The D_i values are computed and then averaged across annual time-scale as follows:

$$D_n^k = \sum_{i=0}^{k-1} P_{n-1} - (PET)_{n-1} \quad (3-7)$$

With $k= 12$ months and n : calculation month.

The Log-logistic distribution's probability density function is:

$$f(x) = \beta/\alpha (x - \gamma/\alpha)^{\beta-1} \left(1 + (x - \gamma/\alpha)^\beta\right)^{-2} \quad (3-8)$$

α , β , and γ represent the scale, shape, and origin parameters, ($\gamma > D < \infty$).

The D series distribution function:

$$f(x) = \left[1 + (\alpha/x - \gamma)^\beta\right]^{-1} \quad (3-9)$$

SPEI normalizes the water balance with this distribution, which is suited for extreme negative values in drought conditions (Hernandez and Uddameri, 2014). Using $f(x)$, SPEI is derived as the standardized $F(x)$ values (Abramowitz and Stegun, 1964):

$$SPEI = W - (c_0 + c_1 w + c_0 w^2 / 1 + d_1 w + d_1 w^2 + d_1 w^3) \quad (3-10)$$

Where: $W = -2\ln(P)$ for $P \leq 0.5$

P represents the probability of surpassing a specified Di value: $P=1-f(x)$, with $C0 = 2.515517$, $C1 = 0.802853$, $C2 = 0.010328$, $d1 = 1.432788$, $d2 = 0.189269$, and $d3 = 0.001308$.

- **SRI**

Developed by Shukla, and Wood (2008), SRI is a HD index akin to SPI but employing runoff data. SRI can be computed over different time scales, enabling the short-term and long-term analysis (monthly, seasonal, or annual periods) of runoff variations. Similar to SPI, SRI calculation includes: (1) Creating a runoff time series and applying a probability distribution, (2) calculating the cumulative probability of the observed runoff, and (3) transforming it into a standard normal deviate. As Tunisia is highly dependent on surface water, SRI is crucial for capturing HD, providing valuable insights into surface water deficits. In this research, SRI was computed at a 12-month timescale using runoff data from GLDAS at 0.25° resolution over Tunisia from 2000 to 2020.

3.2.2.4. Modified Mann-Kendall (MMK) trend test

MK test is a non-parametric ranked method commonly used to evaluate trends in MD and HD (Mann, 1945; Kendall, 1975). It calculates z scores, which help determine the significance and direction of monotonic trends. Positive z scores indicate upward trends throughout the time series, while negative z scores suggest downward trends (Fensholt et al., 2013). It is important to note that the autocorrelation within hydrometeorological datasets can influence MK test outcomes (Yue et al., 2002). To address this issue, the Modified Mann-Kendall test (MKK) was developed by Hamed, and Ramachandra Rao (1998) to correct for autocorrelation in the time series. In this study, the MKK test was applied to assess spatial drought trends in Tunisia from 2002 to 2017.

3.2.2.5. SPEI-GGDI indicator for assessing hydrometeorological droughts

In this study, we developed a combined SPEI-GGDI indicator to monitor hydro-meteorological drought hazards by combining monthly SPEI and GGDI data. This indicator concurrently evaluates MD through SPEI and HD through GGDI. The drought indices were integrated by pairing the monthly values of SPEI and GGDI. First, the Mahalanobis distance was determined using Eyton (1984), in order to detect whether a SPEI-GGDI pair falls within an equal probability ellipse (Equation 3-11):

$$D_M = \sqrt{(X - \mu)^T \Sigma^{-1} (X - \mu)} \quad (3-11)$$

Where:

- X: vector of paired values (SPEI_t, GGDI_t)
- M: mean vector of SPEI and GGDI values over the reference period
- Σ^{-1} : inverse covariance matrix of SPEI and GGDI.

This enabled the classification of hydrometeorological conditions into five categories, with values within a 0.5 range considered normal (Class 0). Values falling outside this range are categorized into different hydro-meteorological classes: wet conditions (Class 1), wet hydrological and dry meteorological conditions (Class 2), dry hydro-meteorological conditions (Class 3), and dry hydrological and wet meteorological conditions (Class 4). The time series of SPEI-GGDI classes were examined as a stochastic process, utilizing first-order Markov chain properties to calculate transition probabilities, return periods (Equation 3-13), and the expected duration of each class (Equation 3-14) (Hermanns and Hermanns, 2002). This analysis offers a probabilistic framework for evaluating hydrometeorological drought hazards.

The Markov chain, a stochastic process (X_t), defines states (S) and transitioning probabilities (p_{ij}) between them. The probability p_{ij} (Equation 3-12) signifies the likelihood transitioning to state j from state i at the next time step. In a first-order Markov chain, the transition probability at time $t+1$ relies only on time t state.

$$P = [p_{ij}] = P(X_{t+1} = j | X_t = i)$$
$$p_{ij} = n_{ij} / \sum_{i=0}^S n_{ij} \quad (3-12)$$

n_{ij} : transitions frequency from state i at time $t-1$ to state j at time t .

The return period for a state i is:

$$R_i = T / n_i \quad (3-13)$$

Where, T: total time series length, and n_i : state i frequency.

The expected residence times in state i is the expected steps number a Markov chain spends in state i before transitioning:

$$E_i = 1/(1 - P_{ii}) \quad (3-14)$$

where P_{ii} is the probability of staying in state i.

3.2.2.6. Decomposition of drought indices using Ensemble Empirical Mode Decomposition (EEMD) method

The intricate nature of nonlinear and non-stationary drought systems requires sophisticated analytical approaches. EEMD, an enhanced form of Empirical Mode Decomposition, mitigates mode mixing by incorporating white noise, generating diverse scenarios, and averaging outcomes across multiple ensemble iterations (Wu and Huang, 2009). EEMD has been extensively used for decomposing hydrological data, especially in multi-scale drought analysis (Liu et al., 2019a). In this study, EEMD was employed to decompose the monthly time series of drought indices (GGDI, SPEI, SPI, and SRI; 2002–2017) into intrinsic mode functions (IMFi) and residual trend components (Resi). This decomposition facilitated the examination of short-term fluctuations and long-term trends, providing insights into dominant drought timescales—an essential factor in understanding the MD-HD relationship. The decomposition process is provided in the following equations:

$$IMF_j(t) = \frac{1}{M} \sum_{i=1}^M IMF_{ij}(t) \quad (3-15)$$

$$Res(t) = \frac{1}{M} \sum_{i=1}^M Res_i(t) \quad (3-16)$$

$$x(t) = \sum_{j=1}^J IMF_j(t) + Res(t) \quad (3-17)$$

Here, $IMF_j(t)$: the jth IMF component, $Res(t)$: the term-related trend. M: the frequency of white noise additions, and J: IMFs total number.

3.2.2.7. Wavelet coherence (WC)

WC is a potential method for examining the interdependency in the time-frequency domain by assessing their correlation. Following the methodology of Torrence, and Webster (1999) and Grinand et al. (2017), WC coefficient is expressed as:

$$R^2(s, \tau) = \frac{|S(s^{-1}W_{xy}(s, \tau))|^2}{S(s^{-1}|W_x(s, \tau)|^2) \cdot S(s^{-1}|W_y(s, \tau)|^2)} \quad (3-18)$$

where $R^2(s, \tau)$: coherence coefficient with minimum coherence of 0 and maximum of 1, and $W_{xy}(s, \tau)$: cross wavelet transforms between the two series. This equation, similar to determination coefficient, indicates that WC ranges from 0 to 1 (Liu, 1994). The smoothing operator is formulated as follows (Seifi et al., 2021):

$$S(W) = S_{scale}(S_{time}(W(s, \tau))) \quad (3-19)$$

This study used WC to analyze the relationship between drought indices (GGDI, SPEI, SPI, and SRI) and large-scale climate teleconnections (ENSO, MO, NAO, and WeMO) in the time-frequency domain. The analysis reflected the teleconnection patterns: lead, lag, or coincide with drought events. A 5% significance level (95% confidence interval) was applied, with smoothing across the wavelet axis denoted as S_{scale} and S_{time} .

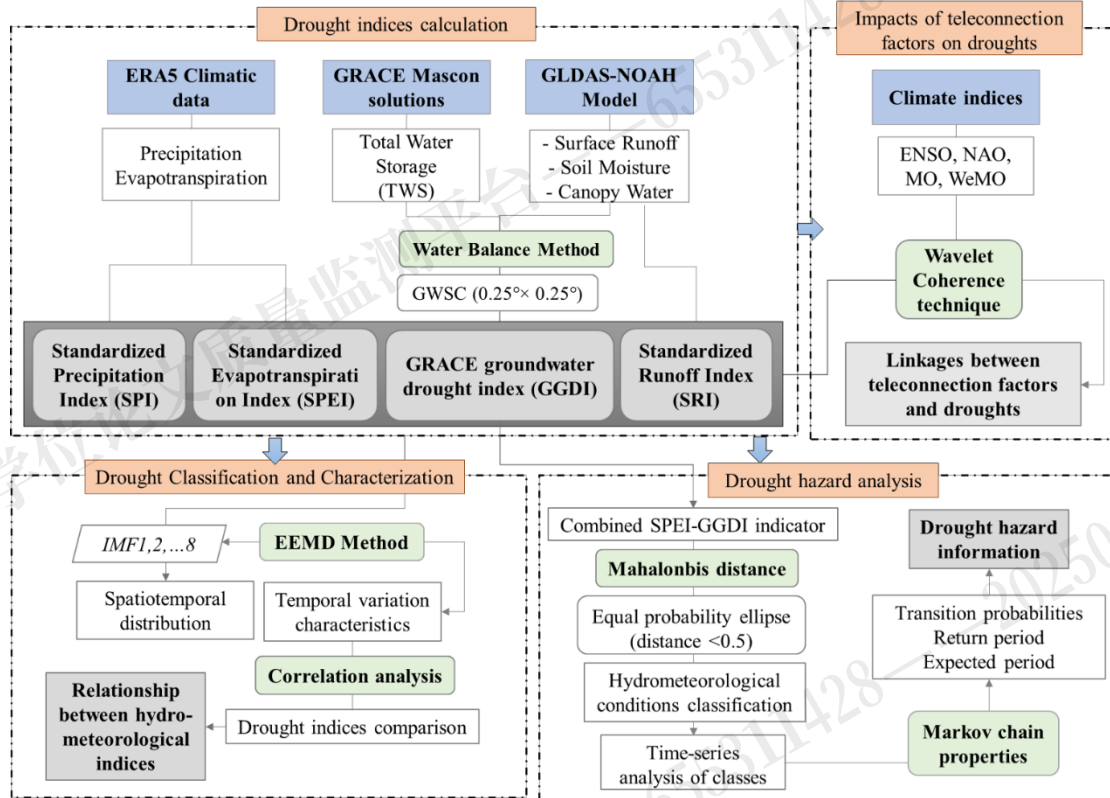


Figure 3-1. Methodological framework used in this study.

3.3. Results

3.3.1. TWS and GWS variations

The monthly anomalies of TWS and GWS varied between -102.54 mm and 56.32 mm, reflecting an overall decline in water storage levels (Fig. 3-2). Both datasets exhibited

similar temporal patterns, with the most pronounced decrease occurring between 2016 and 2017, while peak values were observed in 2006. A persistent negative trend emerged from 2007 onward, with average annual declines of -15.78 mm for TWS and -16.68 mm for GWS (Fig. 3-2b, Fig. 3-2c). Precipitation data (Fig. 3-2d) demonstrated clear seasonal variability, accompanied by lagged responses in water storage. An increasing trend in both GWS and TWS was evident from 2002 to 2006, followed by a prolonged decline. These results underscore significant temporal variations and highlight the intricate relationship between water storage dynamics and precipitation patterns.

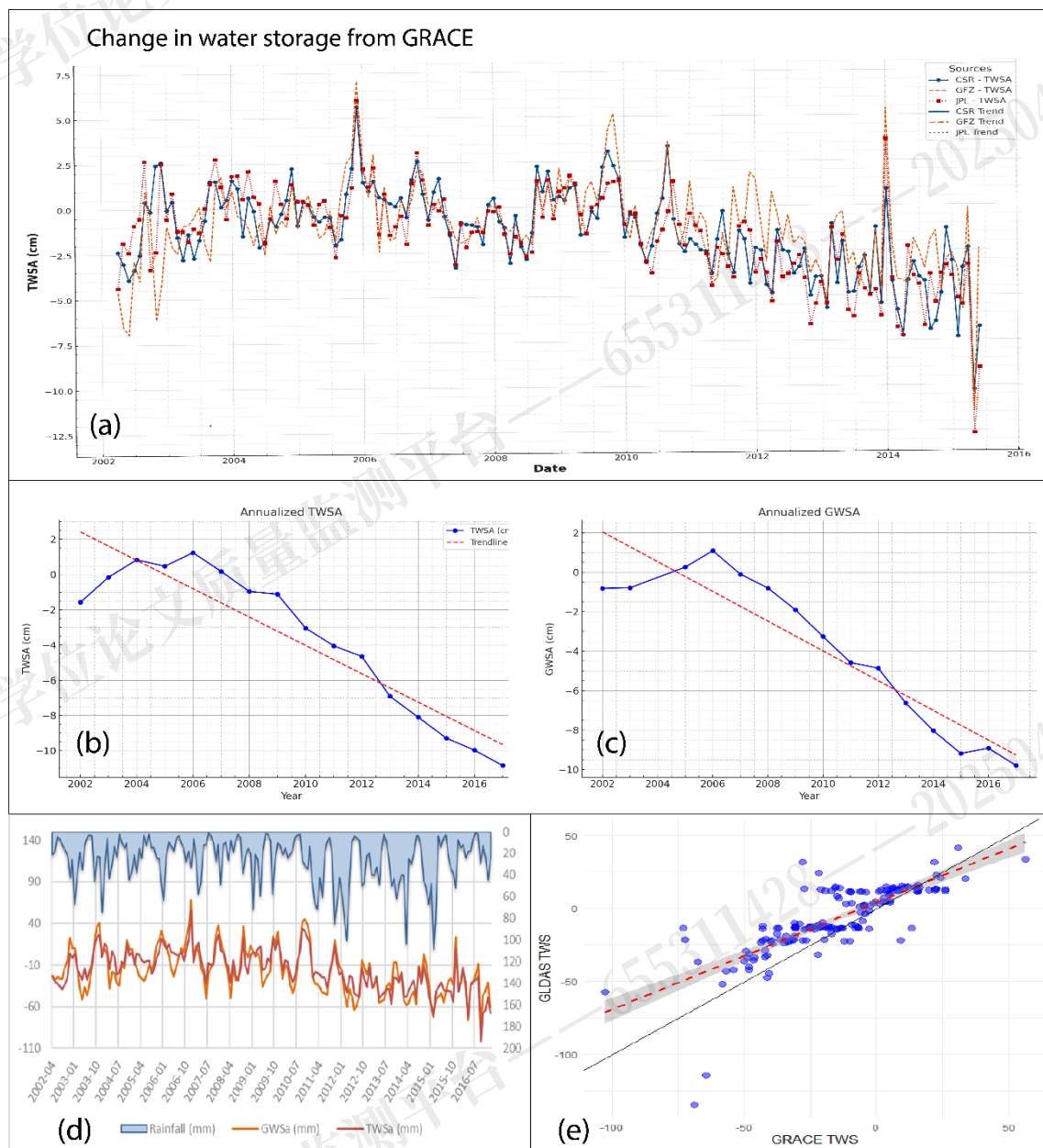


Figure 3-2. Time series representation of (a) GRACE-derived TWSA, (b) annualized TWSA trend, and (c) annualized GWSA trend, (d) ERA5-derived precipitation and GRACE-

derived GWSA and TWSA, (e) monthly GLDAS-TWSA vs GRACE-TWSA, for Tunisia (2002-2017).

MKK trend test identified a significant declining trend in both TWS and GWS, with Sen's slope estimator indicating monthly reductions of -3.38 mm for TWS and -4.10 mm for GWS ($p < 0.005$). To assess the reliability of GRACE-derived TWSA, a comparison was performed with GLDAS-derived TWSA (Cao et al., 2015). GRACE-TWSA varied between -102.54 mm and 56.32 mm, whereas GLDAS-TWSA ranged from -134.37 mm to 41.33 mm. The observed disparities likely stem from GRACE capturing both surface and groundwater variations, while GLDAS primarily focuses surface water storage (Wang et al., 2020). Nevertheless, a strong correlation was revealed between GRACE-derived TWSA and GLDAS-derived TWSA ($r = 0.78$, $p < 0.05$) (Fig. 3-2e), indicating that GRACE data effectively mirrors GLDAS trends, reflecting similar fluctuation patterns.

3.3.2. GGDI-identified drought

Groundwater plays a crucial role in water supply of Tunisia, supplying 48% of agricultural needs and acting as a primary buffer during droughts when surface water is scarce (Hamdane, 2014). The prolonged depletion of groundwater stemming from climate variability, overextraction, and limited recharge has turned GD depletion into an challenging matter, underscoring the importance of understanding these dynamics for ensuring water security (El Amami and Hermassi, 2018). Despite its importance, the response of GWS to droughts, particularly through GRACE data, has not been thoroughly studied in Tunisia. To address this, GGDI was applied to evaluate GD events in Tunisia from 2002 to 2017 (Fig. 3-3). Temporal variations in GGDI showed a general downward trend, accompanied by an increasing frequency of drought events in recent years. Dry events lasting more than three months were specifically analyzed to characterize groundwater depletion.

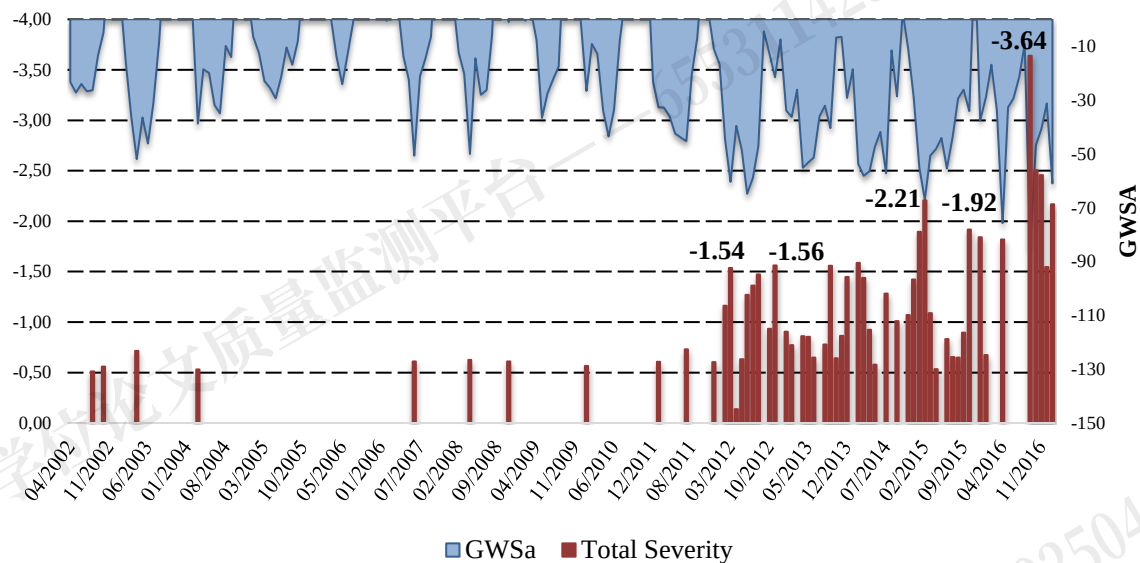


Figure 3-3. Instantaneous severity and associated total drought severity values with groundwater storage anomalies determined for GRACE-observed data.

The total groundwater drought severity detected by GGDI over Tunisia was plotted in Fig.3-3 following Thomas et al. (2017) method for better quantification and analysis of drought severity within the 178-month study period. The values of severity are represented alongside drought events (Fig.3-3). Higher GGDI values with longer durations reflect the more severe drought event. Six major groundwater drought events were recorded in Tunisia (Table.2), i.e. (i) 09-2012/08-2012; (ii) 04-2013/06-2013; (iii) 08-2013/12-2013; (iv) 02-2014/05-2014; (v) 11-2014/10-2015; and (vi) 09-2016/01-2017 (Table 3-3). The longest event, lasting from November 2014 to October 2015, spanned 12 months and had the greatest total severity (-13.60). The GGDI dropped as low as -3.68 in 2016 to cause a significant water storage decline that lasted for 5 months, claiming thus the most severe drought event in Tunisia. Other GD events lasting less than two months with different intensity and severity (from mild to severe) have occurred in the region during the studied period (2002, 2003, 2004, 2007, 2008, 2010, 2011, 2012, 2013, 2014, 2015, 2016).

Table 3-3. Characteristics of GRACE-identified drought events.

No.	Time period	Duration (no, of months)	Average deficit (mm)	Peak deficit (mm)	Total severity (mm)
1	05-2012 / 08-2012	4	-1,18	-1,47	-4,73
2	04-2013/06-2013	3	-0,79	-0,86	-2,36
3	08-2013/12-2013	5	-1,06	-1,56	-5,29
4	02-2014/05-2014	4	-1,13	-1,59	-4,53
5	11-2014/10-2015	12	-1,14	-2,21	-13,60
6	09-2016/01-2017	5	-2,46	-3,64	-12,31

GGDI was computed for each pixel across Tunisia to evaluate the spatial distribution of drought. Figure 3-4 presents the spatial patterns of severe GD, showing considerable variability across different areas. The northern regions, especially the northeastern part, experienced more frequent and intense GD events. The MKK test on the GGDI time series yielded a Z value of -3.19 ($p = 0.00141$), with Sen's slope indicating a decrease of around -0.01. This suggests a statistically significant reduction in GGDI from 2002 to 2017.

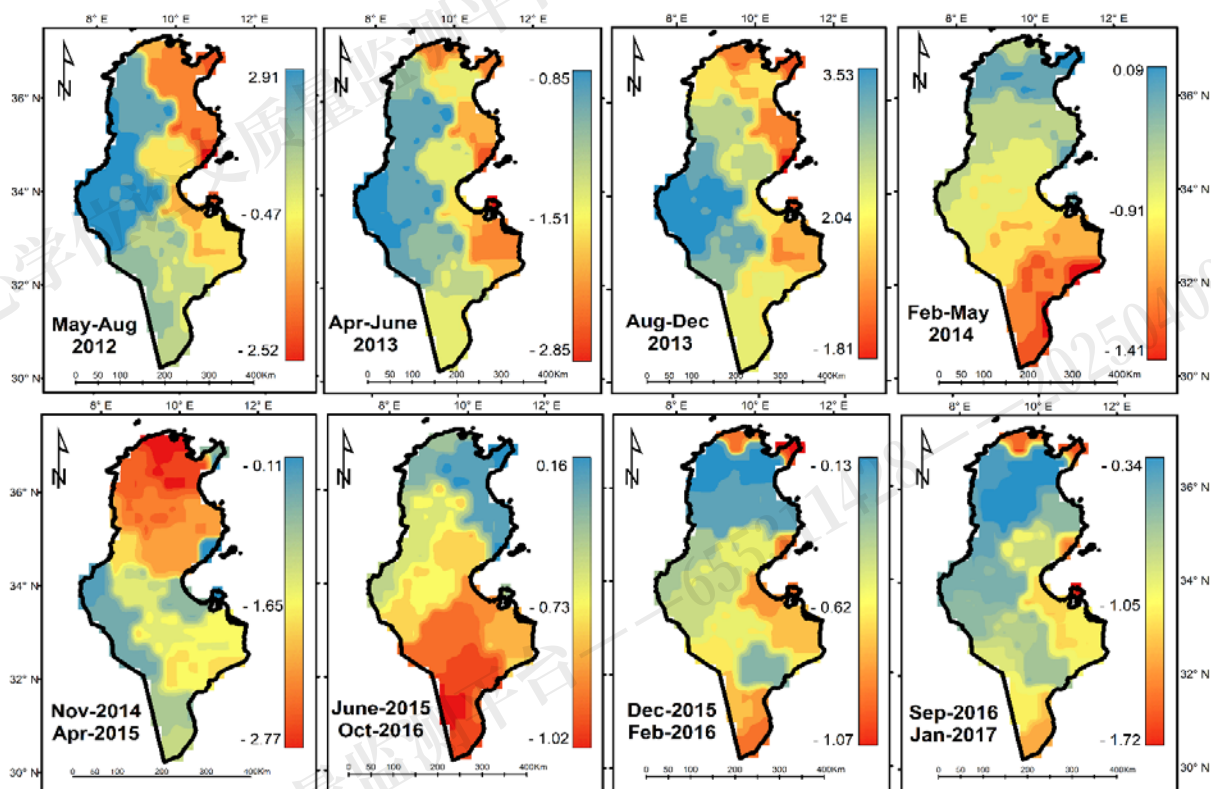


Figure 3-4. Spatial distribution of the most severe groundwater drought events (in terms of duration and intensity) identified by GGDI in Tunisia during 2002-2017.

3.3.3. Comparison of GGDI with other hydrometeorological drought indices

Fig.3-5 illustrates a comparison between GGDI and three widely used standardized drought indices, namely SPI, SPEI, and SRI. These indices are commonly used to reflect hydro-meteorological drought. GGDI had a different variation pattern from the other calculated drought indices. GGDI fluctuations aligned with other standardized indices, particularly the SRI in some months for the years of 2002, 2005, 2007, 2008, 2010, 2011, and 2016. In these years, the strong correlation between GGDI and the other standardized indices, occurred mostly during the rainy season in Tunisia in autumn and winter (September to February). However, for most of instances, GGDI displayed an antagonist variation pattern, particularly in the years of 2002, 2006, 2008, and significantly from 2010 to the beginning of 2016.

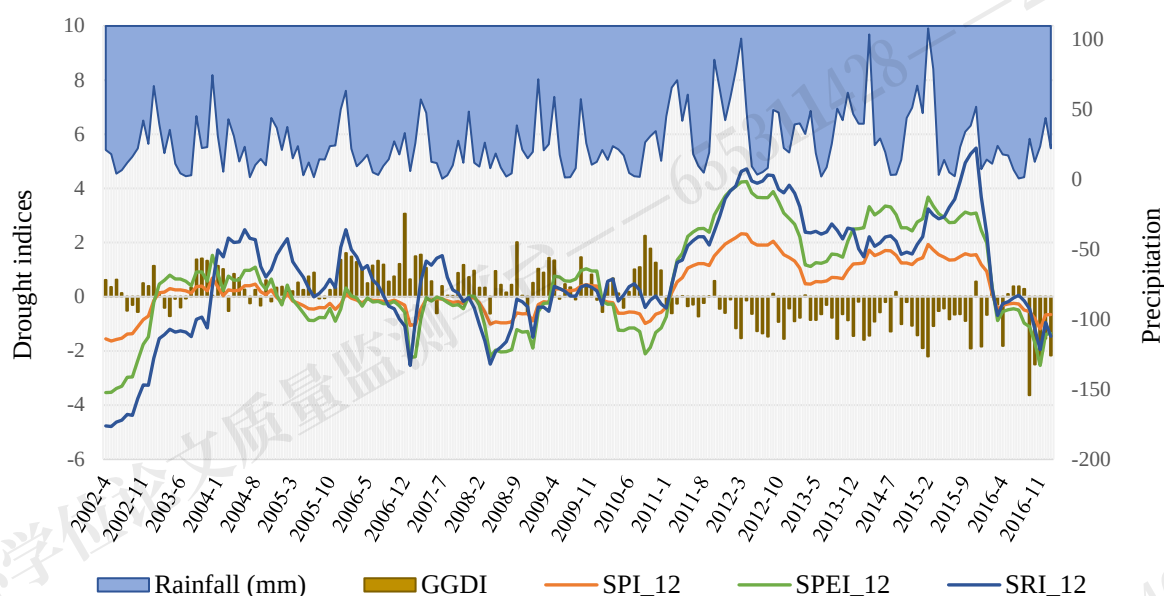


Figure 3-5. Time series depicting hydro-meteorological drought indices (SPI-12, SPEI-12, and SRI-12) alongside GGDI over Tunisia from 2002 to 2017.

3.3.4. Drought indices decomposition through EEMD

The time series of drought indices were analyzed using EEMD technique to explore their interrelationships more effectively. Figure 3-6 displays the annual trends of GGDI, SPI, SPEI, and SRI in Tunisia from 2002 to 2017. The Seasonal-Trend decomposition using LOESS (STL) was also applied to GGDI for comparison with EEMD results (refer to appendix chapter Fig. A1). EEMD analysis revealed distinct fluctuation patterns for GGDI, showing partial alignment with SRI. Prior to 2010/2011, GGDI

values were consistently higher than those of the other indices, followed by a decline lasting until late 2016, with a significant turning point around 2010/2011. The trend component (Res) initially increased for all indices, with SPI and SPEI peaking in 2012 at 3.06 and 2.32, respectively. In contrast, SRI reached its peak in 2010 and declined thereafter. GGDI trend decreased from a peak of 3.06 in 2005 to a low of -3.64 in 2017. While meteorological indices showed improvement until 2012, Res's trend of GGDI indicated worsening drought conditions, particularly after 2006, signaling increased GD severity in Tunisia. The decomposition results revealed distinct temporal dynamics, reflecting the unique physical processes and sensitivities captured by each index. SPI and SPEI showed high-frequency components, representing short-term meteorological drought variability, while GGDI displayed dominant low-frequency components,

emphasizing the delayed and prolonged groundwater response to droughts.

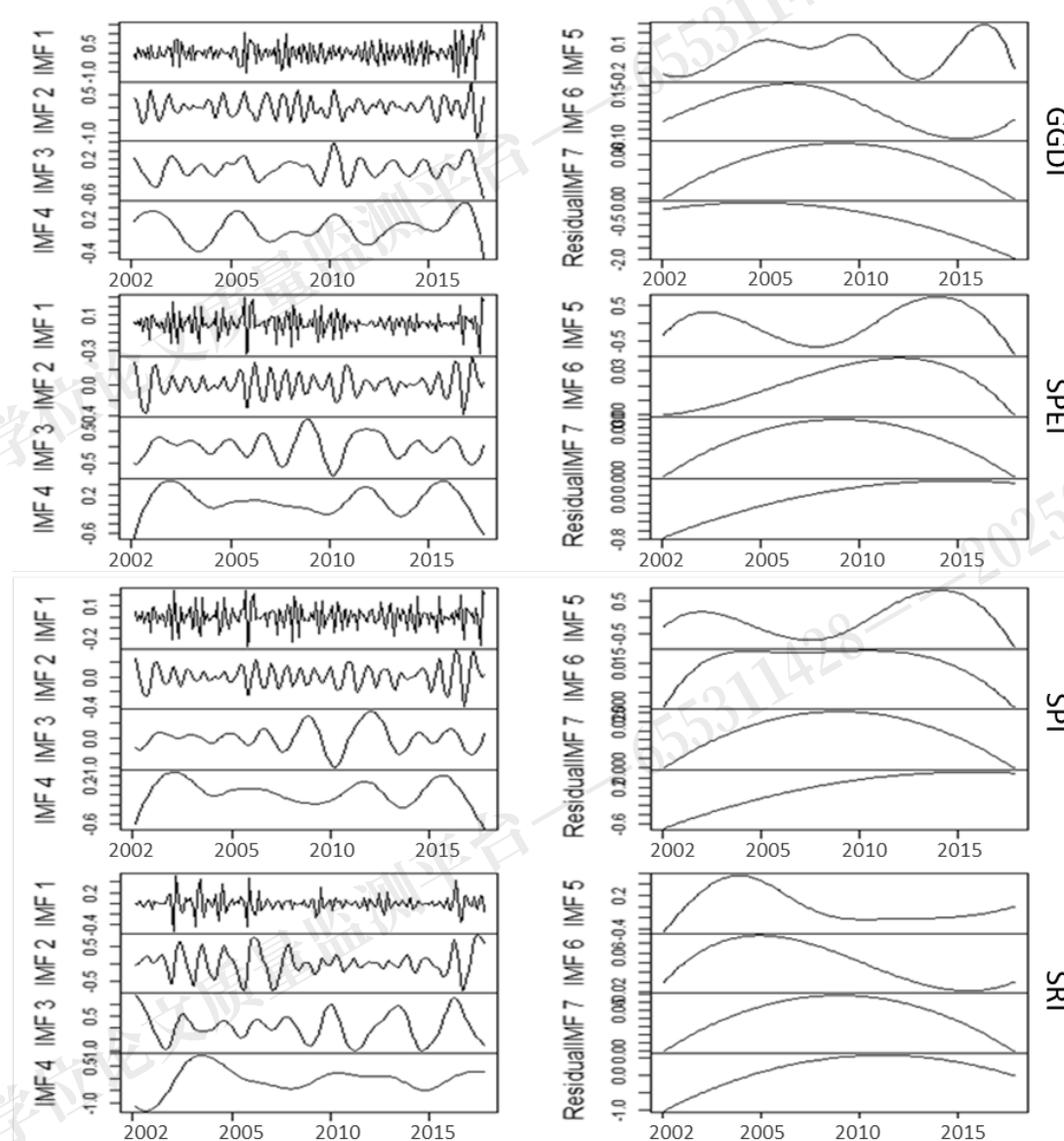


Figure 3-6. Temporal changes of annual drought indices (GGDI, SPEI, SPI, and SRI) based on EEMD method over Tunisia during 2002–2017.

3.3.5. Relationship between GGDI and hydro-meteorological drought indices

The correlation matrix of hydro-meteorological drought indices provided valuable insights (Fig. 3-7). A moderate negative correlation was observed between GGDI and SPI ($r = -0.43$), and GGDI and SPEI ($r = -0.40$), indicating that HD is not only influenced by climatic conditions. The weak positive correlation ($r=0.1$) between GGDI and SRI showed a subtle tendency for increased GD alongside higher SRI values. A strong positive correlation between SPI and SPEI ($r = 0.99$) highlighted their shared reliance on precipitation, while the weak correlations between SRI and SPI ($r = -0.06$) and SRI and SPEI ($r = -0.05$) reflected limited linear connections between the

precipitation-related indices and SRI, potentially due to terrain effects. This analysis underscores that MD and HD do not straightforwardly lead to GD, reflecting the complex interplay of factors driving drought dynamics.

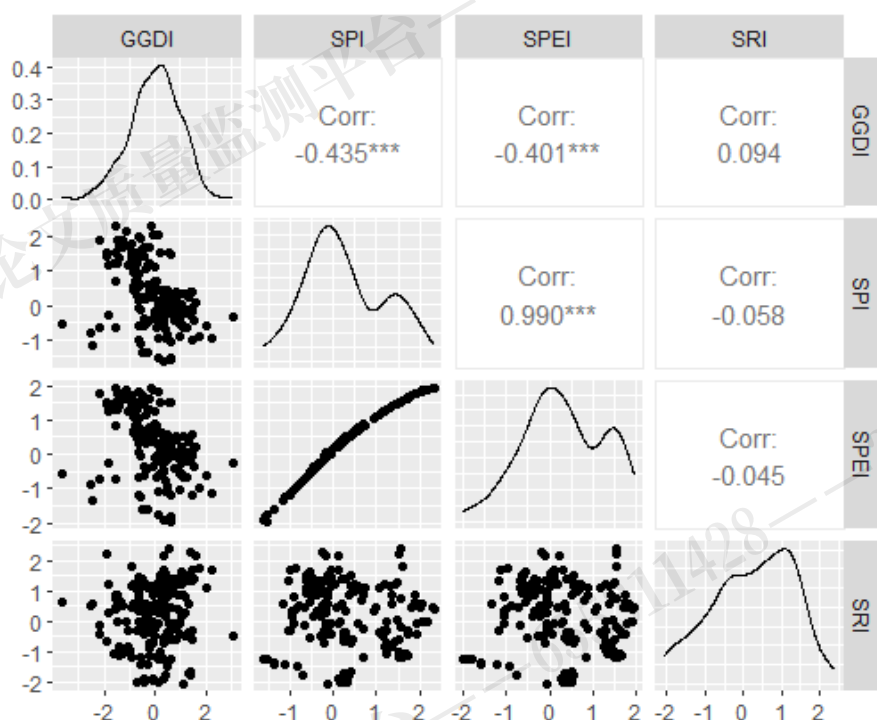


Figure 3-7. Pearson correlation matrix (R) between MD indices (SPI and SPEI) and HD indices (SRI and GGDI).

To further explore the relationships between the different hydrometeorological drought indices (SPI, SPEI, SRI, and GGDI) and identify potential groupings, Principal Component Analysis (PCA) was applied. This dimensionality reduction technique facilitated the visualization of clusters among the indices, revealing underlying patterns in their variability. The results of this analysis are presented in Appendix A (Figure A.4).

3.3.6. Hydro-meteorological drought assessment

3.3.6.1. Drought characteristics

Analysis of hydro-meteorological droughts through SPI, SPEI, SRI, and GGDI highlighted that GGDI captured the highest frequency of drought events (54). Mild droughts were the most frequent compared to all indices. SRI drought events (37) outnumbered SPI (26) and SPEI (31), because of the smaller magnitude of MD that lacked the severity to induce HD, leading to higher values of SRI. Both GGDI and SRI

detected more hydrological droughts than meteorological droughts. Table 3-4 presents the drought severity, with GGDI indicating the highest severity (-64.36), reflecting frequent droughts, including a prolonged event from January 2011 to January 2017, and the lowest severity (-3.64) in September 2016. SRI identified fewer droughts but with greater intensity, with the most severe event (-2.06) in January 2003, linked to reduced water retention capacity and steep terrain. SPI and SPEI both recorded 2002 as the driest year, with values of -1.64 and -1.94, respectively. HD showed greater severity than meteorological droughts. The frequency of droughts across all selected indices exceeded 20%, with GGDI exhibiting the highest frequency (32%), emphasizing the influence of external factors on drought conditions in Tunisia. While the frequency of droughts in SPI, SPEI, and SRI was relatively similar, frequency of GGDI was approximately 10% higher.

Table 3-4. Summary of the identified drought characteristics.

	Events	Frequency	Severity
SPI	26	21.35%	-33.30
SPEI	31	21.35%	-35.94
SRI	37	20.22%	-44.77
GGDI	54	32.02%	-64.36

3.3.6.2. Analysis of drought hazard formation

Figure 3-8 illustrates the normalized return periods for adopted drought hazard classes, highlighting their frequency and severity. Class 2, with the shortest return period (4.09), characterizes the most frequent drought condition, demonstrating that moderate droughts, characterized by a mix of dryness and wetness across both hydrological and meteorological domains, are common in Tunisia. Class 3, the most severe drought condition, has the longest return period (10.8), implying that while it occurs less often, it tends to be the highest in terms of intensity. Class 1, with a return period of 7.30, corresponds to recovery phases, which are less frequent than moderate droughts. Class 4, with a moderate return period (3.12), indicated intermittent yet impactful events, likely due to localized or short-term anomalies in drought patterns. Class 0, with a return period of 3.50, reflects conditions within the expected range, indicating a tendency toward frequent hydrometeorological droughts in Tunisia interspersed with short periods of normal or wet conditions.

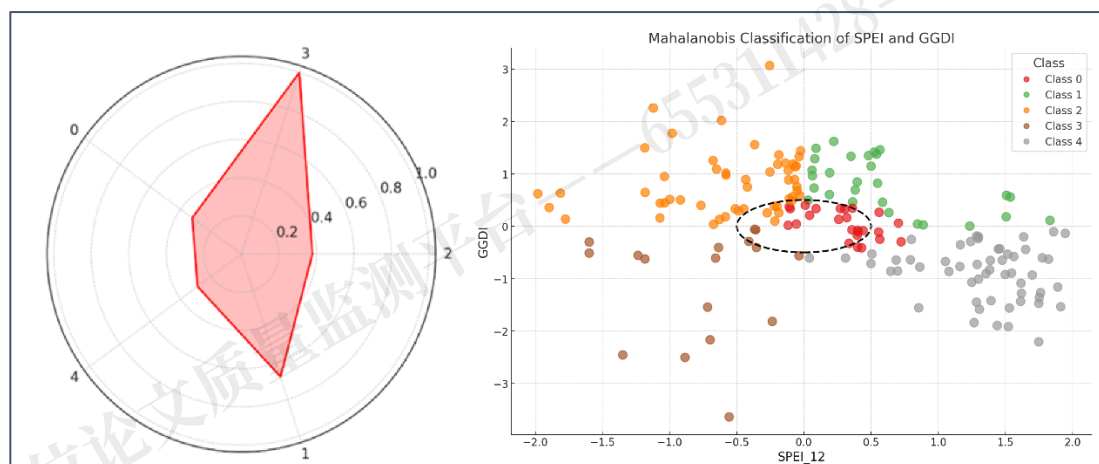


Figure 3-8. Radar plot representing classified values of SPEI-GDDI indicator (right) and hydro-meteorological drought hazard established for Tunisia (left). With, Class 0: normal, Class 1: wet hydro-meteorological, Class 2: dry meteorological and wet hydrological, Class 3: dry hydro-meteorological, and Class 4: dry hydrological and wet meteorological, conditions.

Table 3-5 illustrates the expected residence time for each drought class, indicating their persistence. Class 4 outlines the most prolonged residence time, with an average of about five months, implying extended impacts on agriculture and water resources. Class 3 follows with a 2-month average duration, indicating a challenging recovery. Class 2 lasts for approximately 3.07 months, while Classes 1 and 0 have shorter average durations of 1.62 and 2.24 months, respectively.

The transition probability matrix portrays key aspects of shifts in monthly drought condition. Class 4 displays the greatest stability, with a persistence probability of 80%, signifying that once extreme drought conditions occur, they are likely to last for extended periods. The dry hydrometeorological conditions presented by Class 3, exhibited a 50% persistence probability, reflecting the enduring nature of droughts owing to sustained water deficits. Remarkably, Class 3 has a 25% probability of shifting to Class 2, suggesting partial recovery where MD may improve, but HD could still persist. This underscores the intricate interplay of hydro-meteorological processes in shaping drought recovery dynamics. Class 0 returns a stability probability value of 55%. Particularly, the matrix displays a 19% likelihood of a rapid transition from Class 1 (wet conditions) to Class 4, indicating the region's susceptibility to abrupt shifts towards extreme drought. Once extreme conditions in Class 4 are emerged, there is no opportunity to transit to less severe conditions, underlining the severity and persistence

of drought hazards in Tunisia. These results highlight the significant difficulty in addressing long-term drought consequences and stress the importance of continuous mitigation measures.

Table 3-5. Transition Probabilities, Return Periods, and Expected Residence Times for Hydro-Meteorological Developed Classes

Probability of future classes					
Class	0	1	2	3	4
0	0.55	0.11	0.17	0.04	0.13
1	0.38	0.38	0.05	0.00	0.19
2	0.07	0.09	0.67	0.16	0.00
3	0.25	0.00	0.25	0.50	0.00
4	0.12	0.08	0.00	0.00	0.80
Return period (months)					
Class	0	1	2	3	4
	3.50	7.30	4.10	10.81	3.12
Residence time (months)					
Class	0	1	2	3	4
	2.24	1.62	3.07	2.00	5.00

3.3.8. Interlinkages between droughts and teleconnections factors

Indeed, there are significant links between droughts and climate variables (Sun et al., 2021; Zhang et al., 2024). In this research, ENSO, NAO, MO, and WeMO, were chosen to evaluate the potential links between hydro-meteorological droughts and climate patterns in Tunisia, through WC analysis (2002-2017). Fig 3-9 portrays the WC results between drought indices and climate factors at annual time-scale, with high correlation values provided by black-colored contours at 95% level of confidence. The influence cone, which diminishes boundary impacts and adjusts the inaccuracy of high-frequency data, is presented by thin solid line. The vector direction designates the phase variance between drought indices and climate factors.

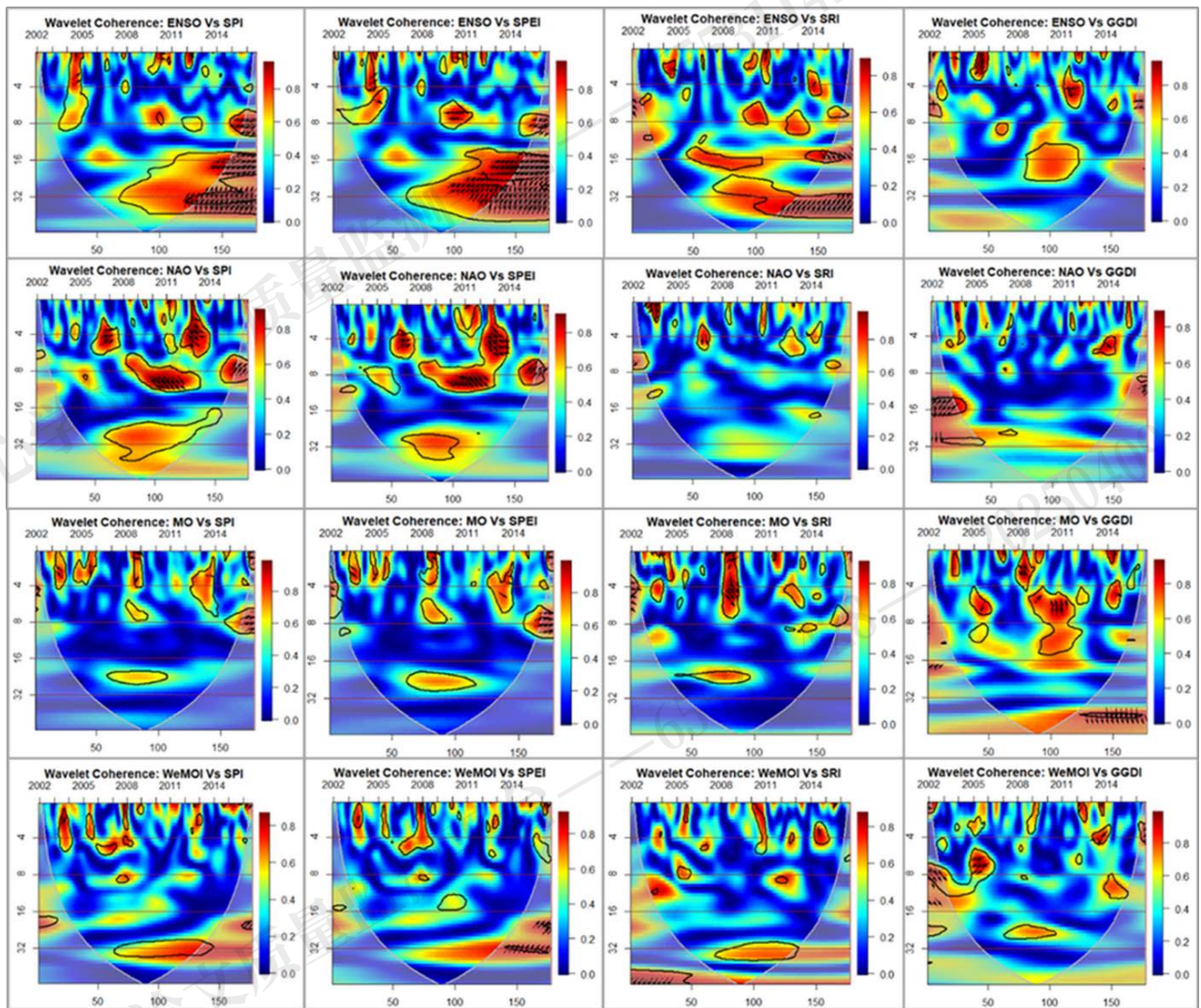


Figure 3-9. The wavelet coherence between yearly drought indices (SPI, SPEI, SRI) and climate indices (ENSO, MO, WeMO, NAO) over Tunisia during 2002-2017.

3.3.8.1. ENSO-related droughts

ENSO demonstrated the most significant coherence in the 14–34-month time scale between 2007 and 2017, showing an in-phase relationship with SPI, SPEI, and SRI. Notable ENSO-driven drought connections were also identified during 2004-2005, where in-phase signals were observed. Interannual variability was particularly evident in the 6–12-month period from 2009 to 2011 and 2015 to 2017. These results highlight that during peak phases, ENSO-induced droughts exhibit strong interactions, with intense negative ENSO phases (El Niño) contributing to severe to extreme droughts in Tunisia.

3.3.8.2. NAO-related droughts

NAO exhibited the most significant common power in the 2-12 months band. The annual variability was detected in 2003, 2003-2005, 2006-2007, 2008-2014, and 2015-2017 in the 2-12 months band, with an anti-phase relationship. NAO exhibited a strong relationship with the meteorological drought indices (SPI and SPEI) during the 2006-2011 and 2005-2013 on period of 16-34 month. For GGDI, the NAO exhibited a significant coherence during 2002-2005 and 2015-2017 in the 8–32-month band with an in-phase relationship, as well as during 2013-2016 at a 4-6-month timescale. NAO effect on SRI was weak, relative to other teleconnection across different scales during various years.

3.3.8.3. MO-related droughts

WC analysis between MO and the multiple drought indices often showed common low power on periods less than 12-month. MO demonstrated the most significant influence on GGDI in the 4–18-month timescale from 2008 to 2012, exhibiting an anti-phase relationship. Additionally, a strong correlation was observed between 2007 and 2017 for timescales exceeding 34 months, showing an in-phase relationship. These findings suggest that during periods of a strong MO-hydrological drought relationship, the interaction between the two phenomena is stronger than that between meteorological droughts.

3.3.8.4. WeMO-related droughts

WC analysis of WeMO compared to SPI, SPEI, SRI displayed joint strong power on periods more than 24-month and low power on periods less than 12-month during 2008-2017, with an in-phase relationship. Generally, WeMO impact on GGDI was weak. A strong yearly variability between GGDI and WeMO was observed from 2002 to 2006, with an anti-phase relationship occurring within the 6–30-month timescale.

3.4. Discussion

This study achieved a holistic investigation of hydro-meteorological droughts over Tunisia spanning a 178-month period (April 2002 to January 2017), using GGDI, SPEI, SPI, and SRI. GGDI showed a notable declining trend and weak correlation with meteorological indices (Fig. 3-7), suggesting limited impact from runoff and soil

moisture on GD, likely linked to high rates of temperatures and evapotranspiration, particularly in the arid regions of southern Tunisia. The 2002-2003 drought event was identified as the most severe drought event as revealed by SPI, SPEI, and SRI. Whereas, GGDI characterized 2016 drought event as the most severe. Notably, during the extreme drought events, all the computed indices exhibited similar declining trend, implying thus the response of HD to MD is intensity-dependent. This aligns with Liu et al. (2023d), who found a weak relationship between GDI and meteorological and surface drought indices in arid regions, particularly in North Africa ($r=0.12$), ascribing it to a disconnected water cycle in such areas (Ding et al., 2021). Similarly, Liu et al. (2023b) revealed weak correlations between MD and GD, with a propagation time of 16.3-month in Southwest China, highlighting that MD do not immediately propagate into GD. The lagged response of groundwater to meteorological conditions, due to slow aquifer recharge, results in prolonged droughts even after meteorological recovery (Wang et al., 2020; Semlali and Vanclooster, 2021). EEMD analysis (Fig. 3-6) supports this, showing the temporal lag in drought propagation. Spatiotemporal distribution of GGDI (Fig.3-4) highlighted significant spatial variability, with northern regions experiencing episodic severe droughts, while central and southern regions, especially the southeast, showed continuously high GGDI values. This variability is influenced by several factors such as the vegetation type, topography, and the deep groundwater depth. For instance, vegetation could greatly affect the drought propagation time (Li et al., 2023). Forests which are mainly located in the northern part of Tunisia, with their developed roots could absorb groundwater to preserve soil moisture. GGDI proved effective in capturing spatial drought patterns, supporting its reliability in detecting drought variability in Tunisia's central and southern regions. Recent studies by Kumar et al. (2021); Huang et al. (2023) have proven the effectiveness of GGDI in capturing spatial drought patterns, supporting its reliability in revealing drought variability in southern and central regions of Tunisia.

EEMD analysis revealed increased low-frequency GGDI, prolonging groundwater drought even during recovery periods from MD, likely due the overextraction of groundwater resources. This is consistent with findings of significant water reduction in the NWSAS, where extraction increased from 0.03 to 1.0 km³/year between 1950 and 2011 (Frappart, 2020; Gonçalves et al., 2021). Groundwater depletion is primarily driven by human activities more than precipitation, as supported by studies in the Mediterranean region of Henao Casas et al. (2022) and global studies (Wang et al.,

2020; Jasechko et al., 2024; Mohasseb et al., 2024; Wang and Liu, 2025). The rate of groundwater depletion notably increased after 2010 (Fig. 3-5), coinciding with Tunisia's 2011 revolution, when insufficient regulation led to unauthorized over-extraction, exacerbating the depletion. This finding is consistent with Soula et al. (2021), who reported significant groundwater overextraction following post-2011. Therefore, human activities play a major role in drought occurrence in Tunisia, more so than climate change.

Drought analysis in Tunisia indicates a pattern of more frequent, but mild droughts (Fig. 3-8), marked by extended periods of decreased rainfall and depleted groundwater levels, although the rarity of extreme droughts. Numerous studies have confirmed this trend, noting a deterioration in drought conditions in the country (Ben Abdelmalek and Nouiri, 2020; Terwayet Bayouli et al., 2022, 2023b). This presents substantial threats to agriculture, water resources, and the environment, as evidenced by significant agricultural losses in 2000-2002 drought period (Verner et al., 2018a). The return periods and frequencies of drought classes are shaped by climate variability and phenomena like ENSO, which influence rainfall and hydrological patterns. Prolonged, severe droughts represented by Class-3 and Class-4, with long residence times and low recovery probabilities, pose serious threats to water management and food security in Tunisia. Persistence of these extreme droughts induce land degradation, decreased crop yields, and increased stress on water resources, potentially causing socio-economic instability. Supporting our findings, Yao et al. (2023) claimed that drylands, including Tunisia, experience extended recovery periods after severe droughts. The difficulty in bouncing back from such droughts highlights the importance of comprehensive mitigation plans, alongside improved predicting and early warning systems.

WC analysis demonstrated that teleconnection factors play a role in the evolution of droughts in Tunisia, with varying levels of intensity, and ENSO having the most significant impact (Fig. 3-9). Negative ENSO phases (El Niño) were associated with severe or extreme drought events, supporting the conclusions of Hassan, and Nayak (2021). Taibi et al. (2019) further supports the strong connection between ENSO and prolonged drought in Tunisia. Abdelkader, and Yerdelen (2022) highlighted substantial correlations between ENSO phases and drought occurrences, further emphasizing ENSO's key role in shaping drought variability in the Mediterranean region. Global studies suggest that ENSO's influence is growing due to climate change (Chen et al., 2023). Additionally, MO was found to interact more strongly with hydrological

droughts than with meteorological droughts, influencing factors such as regional precipitation, evaporation, soil moisture, surface runoff, and groundwater levels, consistent with the results of Ouachani et al. (2013); Jelassi et al. (2015).

3.5. Conclusions

This study provides a comprehensive analysis of hydrometeorological droughts in Tunisia from 2002 to 2017, utilizing multiple datasets from GRACE-derived TWS monthly data, GLDAS water balance factors, and ERA5 climate. It represents the first instance in which GGDI has been evaluated over Tunisia linking it to both surface and subsurface conditions through different hydro-meteorological drought indices (SRI, SPI, and SPEI). Wavelet coherence analysis was applied to these drought indices to elucidate their teleconnection with large-scale climate indices. The results highlight significant disparities among indices, with GGDI exhibiting distinctive patterns. The driest period, as determined by SPEI, SPI, and SRI, was 2002-2003, averaging a value of -1.9. GGDI-detected droughts were more frequent, severe, and prolonged compared to meteorological indices, identifying six main groundwater drought events lasting longer than three months, with the most prolonged lasting from November 2014 to October 2015, having a total severity of -13.60 and an average GGDI of -1.36. The study highlighted a declining trend in GGDI, that drop as low as -3.64 in 2017, with significant spatial variations, underscoring critical groundwater depletion primarily induced by human activities and climate change. The correlation analysis illustrated the complex relationships between MD and HD, indicating that MD do not directly translate into groundwater droughts. Furthermore, the low probabilities of recovery from extreme droughts and the long residence times emphasize that droughts in Tunisia are persistent and have potential shifts in severity. This underscores the need for incorporating these dynamics into drought management and mitigation strategies. The wavelet coherence technique proves effective in unfolding connections between drought and teleconnection factors. Climate indices, i.e. ENSO, NAO, MO, and WeMO contributed to the evolution of hydrometeorological droughts in Tunisia at a variant intensity. ENSO teleconnection had the strongest influence on meteorological droughts in Tunisia where the intensive negative ENSO phases (El Nino) generated severe and/or extreme drought events.

By considering these various facets, the current study embraces a holistic lens to capture a nuanced and comprehensive understanding of droughts in Tunisia. It provides thus a basis for drought monitoring and for conducting in-depth analyses of their causes using basically remote sensing data, particularly GRACE-satellite data which evidenced as important tool to unfold drought conditions particularly in regions where hydro-meteorological databases are lacking like Tunisia. However, GRACE outputs will be more reliable if applied on longer time series. Groundwater droughts were found to be significantly influenced by human interventions thus the effect of human activities on water storages should be considered in future studies. This study holds practical significance to formulating effective plans and strategies addressing the sustainable management of land and water resources in regions where these resources are under a persistent pressure.

Chapter 4 From Meteorological to Agricultural Drought: Propagation and Probabilistic Linkages with Spatiotemporal Analysis Based on MODIS-Derived Vegetation Time Series

4.1. Introduction

Drought is among the most damaging natural disasters, causing significant losses in agricultural production, ecosystems, and the economy (Kheyruri et al., 2023; Al-Kilani et al., 2024; Tanarhte et al., 2024). It affects various climatic regions, with agriculture being the most impacted sector, yet it remains one of the least understood natural hazards (Bharambe et al., 2023). While definitions of drought vary across disciplines and sectors, it is generally described as an extended period of reduced rainfall, leading to water shortages that affect the environment, economy, and society (Wilhite and Glantz, 1985). Dracup et al. (1980) classified drought into four main categories: meteorological (MD), agricultural (AD), socioeconomic, and hydrological (HD). MD, characterized by precipitation levels less than the long-term average (>30 years), is often considered the primary trigger for other drought types (Guo et al., 2020). While, agricultural drought, is associated with insufficient soil moisture, leading to vegetation stress and potential crop failure (Zhang et al., 2023). Despite these distinctions, all drought events share common features, such as their frequency and duration, which influence their impact on agriculture and ecosystems. For instance, short but intense droughts can severely affect crops during early growth stages, while prolonged, low-intensity droughts can have devastating effects on water resources (Fahad et al., 2017; Li et al., 2022b).

Historically, drought assessments have largely relied on in-situ climatic indices, such as SPI (McKee et al., 1993) and SPEI (Vicente-Serrano et al., 2010). While they are among the most reliable approaches for drought monitoring (Wilhite and Glantz, 1985), these station-based indices are often lacking the capability to provide large-scale and near-real-time information (Ha et al., 2023), particularly in regions with limited observational databases. Moreover, traditional climate-based indices do not directly capture the effects of drought on vegetation health or productivity (AghaKouchak et al., 2015). In contrast, earth observation data have emerged as an effective alternative, offering timely and spatially detailed drought information. For example, remote sensing-based vegetation indices provide direct insights into how drought impacts

vegetation and ecosystem conditions (Kumar et al., 2024). As open-access remote sensing datasets became widely available, long-term global and regional drought monitoring has become more feasible, particularly with advancements such as the Moderate Resolution Imaging Spectroradiometer (MODIS) and the Advanced Very High-Resolution Radiometer (AVHRR). Numerous vegetation-based remote sensing indices have been developed for monitoring drought, including VHI (Liu and Kogan, 1996), the Normalized Difference Vegetation Index (NDVI) (Tarpley et al., 1984), and the Soil-Adjusted Vegetation Index (SAVI) (Huete, 1988). Among these, NDVI has become among the widely used indices for drought-related monitoring due to its ability to reflect plant health and density based on the greenness of the land surface (Zhu et al., 2024). However, NDVI may be limited by its sensitivity to atmospheric effects and soil reflectance. To address these limitations, several alternative vegetation-based indices were developed. For instance, the SAVI was introduced to minimize soil influences, while the VHI was designed to enhance both atmospheric and soil components (Liu and Kogan, 1996). It is effective in assessing drought severity by integrating analysis of vegetation health through VCI and temperature influences through TCI (Bento et al., 2018a). The VCI, in particular, has gained widespread adoption as a drought indicator, with recommendation from World Meteorological Organization (WMO) for both global and local drought monitoring (Svoboda and Fuchs, 2016). VHI has been successfully applied to detect agricultural and vegetative drought events globally, including in regions such as Africa (Gidey et al., 2018; Gbaguidi et al., 2024; Makula et al., 2024), Asia (Ali et al., 2019; Jiang et al., 2021), United States (Alzurqani et al., 2024), and Australia (Kogan et al., 2018).

A strong connection exists between MD and other types of droughts, where MD often emerges as a precursor to other drought types (Wang et al., 2021; Jiang et al., 2023). For instance, prolonged periods of low precipitation can lead to soil moisture depletion, ultimately resulting in agricultural drought (Berg and Sheffield, 2018). Since precipitation data is widely available on a global scale, MD is relatively easy to assess. Therefore, gaining a precise understanding of how MD evolves into AD and establishing probabilistic linkages between them is crucial for early warnings, helping to minimize the adverse AD effects. To achieve this, integrating vegetation-based indices (e.g. VHI) along with meteorological-based drought indices (e.g. SPI, SPEI), at different time-scales (monthly, SPI-3, SPI-6, etc.), provides a more comprehensive perspective on drought dynamics. This approach enables the identification of drought

propagation, which characterizes the transition from MD to AD (Guo et al., 2020; Ding et al., 2021). The most widely used method for examining the propagation time between MD and AD is correlation analysis, which compares AD indices at specific timescales (typically monthly) with MD indices at various timescales. The MD index timescale yielding the highest correlation coefficient is regarded as the drought propagation time (Dai et al., 2022; Liu et al., 2023c; Feng et al., 2024). For instance, Li et al. (2020b), using spatial autocorrelation and cross-wavelet analysis, found that the propagation time from meteorological to hydrological drought was within one month, from hydrological to soil moisture drought within two months, and from hydrological to AD between 1-3 months across most areas of the Yangtze basin. Possega et al. (2023), identified the response time scale of 1-2 months for drought propagation in the Iberian Peninsula, with the probability of occurrence increasing with the MD severity. Similarly, Xu et al. (2021) reported a propagation time between 1 and 3 months in humid regions of China, while it can range between 1 and 8 months in arid and semi-arid regions.

Compared to other southern mediterranean countries, which described as drought-prone regions, Tunisia is increasingly vulnerable to drought-related hazards due to the combined effects of climate change and human activities (Verner et al., 2018a). Drought has significantly impacted Tunisia's agricultural sector in recent years, leading to substantial economic losses. In 2023, the country's grain harvest was approximately 750,000 tons, a sharp decline from the average of 1.5 million tons over the past decade, according to the Tunisian Agriculture and Fishing Union (UTAP). Despite the increasing risks of drought, Tunisia has received limited focus in remote sensing-based drought studies in recent decades. For instance, Ben Mhenni et al. (2021) employed SPI, PDSI, and Soil Moisture Index (SMI) to assess drought characteristics across Tunisia, integrating meteorological data, remote sensing-based vegetation indices (e.g., NDVI), and agricultural production records to evaluate drought impacts on vegetation and crop yields over different climatic zones. While, Ben Abdelmalek, and Nouri (2020) analyzed drought patterns based on rainfall data (1973–2016), and examined the drought impacts on agricultural productivity by correlating drought indices with crop production data and Agricultural Gross Domestic Product variations.

Despite these contributions, none have specifically examined the propagation time between meteorological and agricultural drought. This gap in the literature highlights that the temporal dynamics of drought propagation in Tunisia remain unexplored and are a crucial aspect that warrants further investigation. Therefore, this study aims to

bridge this gap by providing the first in-depth analysis of spatiotemporal agricultural drought dynamics across Tunisia from 2001 to 2020, and investigating the propagation time and probabilistic linkages between MD and AD, at different drought levels. The purposes of this study are the following: (1) assess and analyze the spatiotemporal trends and variability of agricultural drought conditions across Tunisia, through the VCI, TCI, and VHI method based on monthly MODIS vegetation time-series data from 2000 to 2021. (2) analyzing trends in drought conditions (3) examine the temporal variability of drought propagation time from meteorological to agricultural drought; and (4) investigate the propagation probabilistic linkages and transition probabilities of different drought levels.

4.2. Materials and methods

4.2.1. Data sources and preprocessing

This study used data from Google Earth Engine (GEE) cloud platform (Fig. 4-1), which facilitates efficient geospatial analysis. GEE leverages Google's cloud computing infrastructure (<https://earthengine.google.org/>) to significantly reduce processing time (Tamiminia et al., 2020). It provides real-time access, analysis, and processing of vast geospatial datasets sourced from the US Geological Survey (USGS) and other institutions. These datasets include historical and contemporary satellite imagery, along with earth observations datasets such as Landsat, MODIS, and meteorological data (Pérez-Cutillas et al., 2023). A key advantage of using GEE is access to both raw and pre-processed images, including cloud-cleared and mosaicked datasets.

In this study we accessed the data through GEE, to analyze drought conditions over Tunisia from 2001 to 2020. The study utilized imagery from MODIS, which features 36 spectral bands ranging from 0.4 μm to 14.4 μm and has been operational since its launch in 1999. Specifically, LST were derived from MOD11A2 products (8-Day L3 Global, 1km) and NDVI were obtained from MOD13A2 products (16-Day Global, 1Km), were employed for AD assessment. Additionally, CHIRPS rainfall dataset and MOD11A2 evapotranspiration datasets were incorporated for MD assessments through SPEI computation. CHIRPS is particularly valuable for drought monitoring in regions with heterogenous terrains like Tunisia (Funk et al., 2015).

4.2.2. Methods

This study employed two distinct approaches to assess drought patterns and conditions in Tunisia over a 21-year period (2001–2020) (Fig. 4-1). The first method involved generating monthly drought maps using VHI, which integrates the VCI and TCI to evaluate drought-induced stress on vegetation. A spatiotemporal analysis of VHI was then performed to examine AD trends on a monthly timescale to effectively identify patterns of drought-induced vegetation stress. The second approach focused on correlating VHI with SPEI to analyze the linkages among MD and ADs at different time scales. The correlation analysis allowed for the determination of propagation time and the probabilistic linkages between the different drought levels.

To process and analyze the data, JavaScript was used within GEE platform, while GIS, Python, and R software were employed for spatial analysis, statistical analysis, correlations, and the determination of propagation time and transition probabilities between drought types. This comprehensive analysis provided valuable insights into the interaction and propagation of MD and AD, helping to better understand their impact on vegetation health in Tunisia.

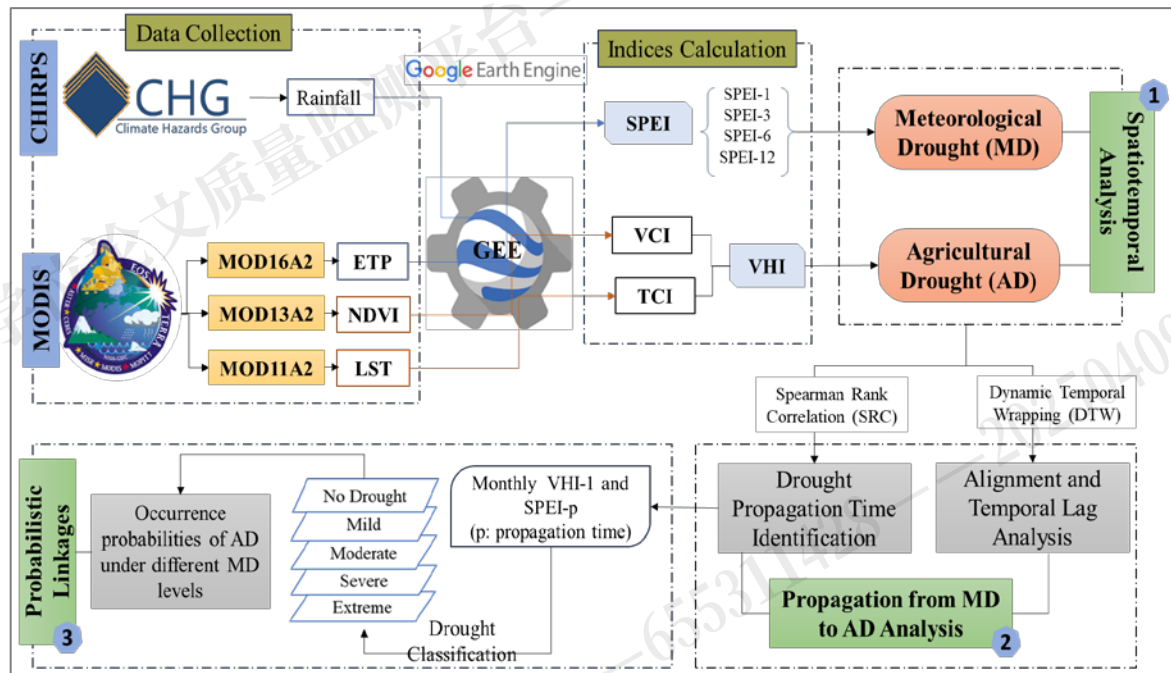


Figure 4-1. Framework of this study.

4.2.2.1. Drought indices development

4.2.2.1.1. Vegetation-based indices

- **VCI**

It assesses the current vegetation state by comparing it to baseline values recorded during the same previous period (Liu and Kogan, 1996). It serves as an indicator of vegetation health over time, with higher values representing favorable conditions and lower values indicating vegetation stress. VCI indirectly reflects drought conditions. In this study, NDVI was used to compute the VCI (Equation 4-1). NDVI data from MOD13A2 products were chosen due to their ability to account for background and atmospheric effects while effectively capturing the spectral response of green vegetation (Terwayet Bayouli et al., 2023b). VCI was determined using the following formula:

$$VCI_i = \frac{NDVI_i - NDVI_{min}}{NDVI_{max} - NDVI_{min}} \times 100\% \quad (4-1)$$

where VCI_i and $NDVI_i$ are VCI and NDVI of a certain month (%), respectively; and $NDVI_{min}$ represent the lowest NDVI values and $NDVI_{max}$ represent the highest NDVI values recorded for the corresponding month throughout the study period.

- **TCI**

TCI evaluates vegetation stress caused by elevated temperatures and soil moisture deficits (Adeyeri et al., 2024). TCI is derived from LST-based thermal brightness temperature measurements, where lower TCI values correspond to more severe drought conditions (Choudhury, 2024). In this study, TCI was calculated using LST data from MOD11A2 products, following the formula outlined in Equation 4-2.

$$TCI_i = \frac{LST_i - LST_{min}}{LST_{max} - LST_{min}} \times 100\% \quad (4-2)$$

where TCI_i is TCI of a certain month (%); LST_i is the LST of a certain month (Kelvin); and LST_{max} represent the lowest LST values and LST_{min} represent the highest LST values recorded for the corresponding month throughout the study period (Kogan, 1995).

- **VHI**

VHI serves as an indicator of overall vegetation health by integrating both VCI and TCI, thereby accounting for moisture availability and thermal stress in vegetation

(Seiler et al., 1998). This indicator has been effectively applied for drought monitoring across various soil and climate conditions, particularly in arid regions (Rhee et al., 2010; Ghoneim et al., 2017; Terwayet Bayouli et al., 2023b). The VHI is calculated using the following formula (Equation 4-3):

$$VHI = \alpha(TCI) + \alpha(VCI) \quad (4-3)$$

Where α is a weight equal to 0.5 for both indices (Bento et al., 2020). In this study, the VHI classification proposed by Kogan (1995) was utilized (Table 4-1).

Table 4-1. VHI Classification based on Kogan (1995)

VHI value	Drought condition
<10	Extreme drought
10–20	Severe drought
20–30	Moderate drought
30–40	Mild drought
>40	No drought

4.2.2.1.2. Meteorological-based index: SPEI

SPEI has been extensively used to assess MD, based on precipitation and evapotranspiration data across multiple timescales (Vicente-Serrano et al., 2010). Each SPEI timescale reflects the cumulative precipitation anomalies over consecutive months. Shorter timescales (i.e. 1-month and 3-months) are useful for tracking short-term fluctuations, while longer timescales (such as 12- and 24-months) highlight long-term trends. For SPEI calculation method refer to Section 3.2.2.3 of Chapter 3. Drought conditions are identified based on threshold of -0.5, over a specific timescale, signaling the onset of MD (refer to Table 3-2, Chapter 3, Section 3.2.2.2). In this study, SPEI was computed over a monthly timeframe from 02-2001 to 12-2020, at different time-scales: SPEI-1, SPEI-3, SPEI-6, and SPEI-12.

4.2.2.2. Propagation from meteorological to agricultural drought: Statistical and temporal linkages

4.2.2.2.1. Identification of temporal shifts in VHI-SPEI relationships

The Dynamic Time Warping (DTW) is a powerful method for comparing two time series that may vary in time alignment or speed (Müller, 2007), making it highly relevant for analyzing the propagation between SPEI (MD) and VHI (AD). It relies on

an algorithm that aligns two time series by minimizing their pointwise distance, even if the series exhibit temporal distortions (e.g., one series is lagged relative to the other). Unlike methods requiring strict temporal alignment (e.g., Pearson correlation), DTW flexibly matches patterns, and is thus ideal for detecting time-lagged relationships (Stübinger and Walter, 2022). Formally, for two time series:

$$X = \{x_1, x_2, \dots, x_N\} \text{ and } Y = \{y_1, y_2, \dots, y_M\}$$

we define a local distance measure $d(x_i, y_j)$, typically the absolute or squared difference. We then construct an $N \times M$ cumulative cost matrix C , where each cell $C(i, j)$ is given by:

$$C(i, j) = d(x_i, y_j) + \min\{C(i-1, j), C(i, j-1), C(i-1, j-1)\} \quad (4-4)$$

Subject to appropriate boundary conditions (e.g., $C(1,1) = d(x_1, y_1)$).

The final warping cost, $C(N, M)$, represents the minimal cumulative distance between X and Y . By tracing back through this cost matrix, DTW identifies the optimal alignment path, which can reveal any lags or stretches between the two series.

In this study, DTW was employed to highlight how much VHI lags behind SPEI, reflecting initial insights into drought propagation time. Because it does not assume linearity or stationarity, DTW is adaptable when dealing with irregular propagation patterns, long-term trends, or seasonal cycles common in agricultural drought indices.

4.2.2.2.2. Response of VHI to SPEI at different timescales

Correlations analysis method is widely used to identify the drought propagation time. In this study, we used the Spearman rank correlation coefficient to assess the relationship between SPEI and VHI across multiple timescales, thereby allowing for the determination of VHI's response time to SPEI. The Spearman's rank correlation is a non-parametric measure designed to quantify the monotonic relationship between two variables (Sedgwick, 2014). Unlike the Pearson correlation, which assumes a linear relationship and normally distributed data, Spearman's correlation operates on ranked values and is therefore robust against outliers and capable of detecting both linear and non-linear associations (Zar, 2005). Spearman's ρ is defined as (Equation 4-5):

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2-1)} \quad (4-5)$$

Where d_i is the difference between the ranks of the i th pair of observations, and n is the total number of paired observations. The correlation coefficient ranges from -1 to $+1$. Values approaching $+1$ indicate a strong positive correlation, values approaching -1

denote a strong negative correlation, and values near 0 suggest little or no monotonic relationship. A higher SRC indicates greater sensitivity of VHI to SPEI, and vice versa.

The SRC method is based on determining the correlative relationship between the time series of the AD index at a single timescale (e.g., VHI-1) and that of the MD index at different timescales (e.g., SPEI- n) and denote the highest correlation coefficients as their propagation time. For instance, if VHI-1 and SPEI-12 have the highest correlation coefficients relative to other timescales of SPEI, the response time of VHI-1 to SPEI- n was almost 12 months. In this study, we initially employed the SRC method to evaluate the overall correlation between the time series of VHI-1 and SPEI- n . We then combined the VHI and SPEI data on a monthly basis and conducted a deeper analysis of the correlation between VHI-1 and SPEI- n at different time scales, aiming to evaluate how VHI-1 responds to seasonal variations in SPEI- n .

4.2.2.3. Propagation relationships and transition probabilities of different drought levels severity

To illustrate the propagation between different levels of MD and AD (i.e., no drought, mild, moderate, severe, and extreme), a Sankey diagram was employed in this study. This type of visualization effectively represents the flow of elements within complex systems, showcasing how various components interact and transition (Schmidt, 2008). In the context of drought analysis, the diagram helps map how MD conditions propagate into corresponding agricultural drought categories. The nodes of the Sankey diagram, signify different drought levels, while the connecting bands indicate the flow and transition relationships between these levels. If no band is present, it suggests no linkage between specific drought stages. The width of each band corresponds to the volume of flow, with thicker bands representing stronger transitions between drought categories. By analyzing these flows, the propagation probabilities between different drought levels can be determined. For instance, if a drought category appears 100 times and shifts to three other categories 50, 30, and 20 times, the propagation probabilities would be 0.50, 0.30, and 0.20, respectively.

4.3. Results and discussions

4.3.1. Spatiotemporal distribution of agricultural droughts

The spatiotemporal VHI analysis from 2001 to 2020 in Tunisia reveals distinct spatial and temporal patterns in AD severity (Figure.4-2). A clear north-south gradient in drought intensity was observed. Northern Tunisia consistently exhibiting milder drought conditions, with no drought or mild drought being the dominant categories, particularly after 2005, due to higher precipitation, leading to shorter and less intense drought episodes, typically limited to short moderate drought in winter. In contrast, central and southern Tunisia experienced more frequent and severe droughts. Central Tunisia, in particular, experienced cyclical droughts, with severe and extreme events in the early 2000s (e.g., 2001, 2005, 2008, 2011, 2017), although some improvement was noted in later years. Southern Tunisia remained the most drought-prone, with prolonged moderate to severe conditions and extreme drought episodes in 2001, 2005, 2007, 2011, and 2016. The persistence of drought stress in the south, particularly in desert regions, continued into the 2010s, notably in 2008, 2014, and 2020. Previous studies on the spatiotemporal precipitation variability in southern Tunisia and similar climates confirm these findings (Ellouze et al., 2009; Feki et al., 2012; Jemai et al., 2018; Jasim and Awchi, 2020).

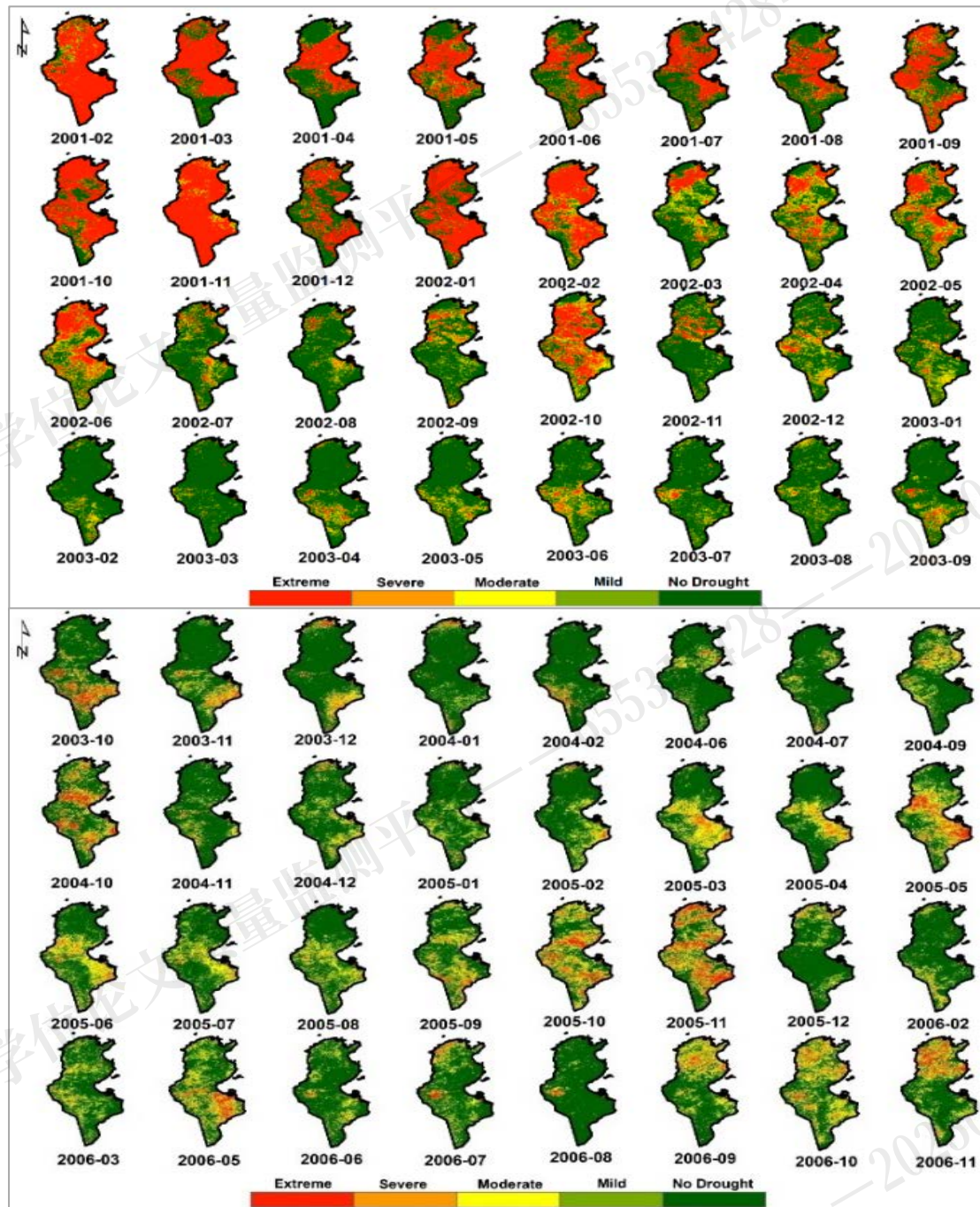


Figure 4-2. Monthly spatial distribution of agricultural drought over Tunisia (2001-2020), using VHI.

The temporal analysis further highlights cyclical drought patterns, with the early 2000s (2001–2005) marked by widespread severe droughts, especially in central and southern Tunisia. The 2005 drought was particularly severe, impacting agricultural productivity across the country (Bazza et al., 2018a). These findings align with previous research conducted in Tunisia and North Africa (Ezzine et al., 2014; Ghoneim et al., 2017; Ballah and Benaabidate, 2021; Ben Mhenni et al., 2021). From the mid-2000s to early 2010s (2006–2015), gradual improvements were observed, particularly in

northern and central Tunisia, with more mild droughts replacing severe ones. However, 2008 was an exception, with severe drought in the south and extreme drought in some areas. Southern Tunisia remained highly vulnerable, experiencing frequent moderate droughts, with 2011 standing out as a particularly dry year. The late 2010s to early 2020s (2016–2020) experienced relative stabilization in the north and central regions, but severe droughts still occurred, particularly in 2016 and 2020. The 2020 drought, in particular, underscored persistent drought stress in southern Tunisia.

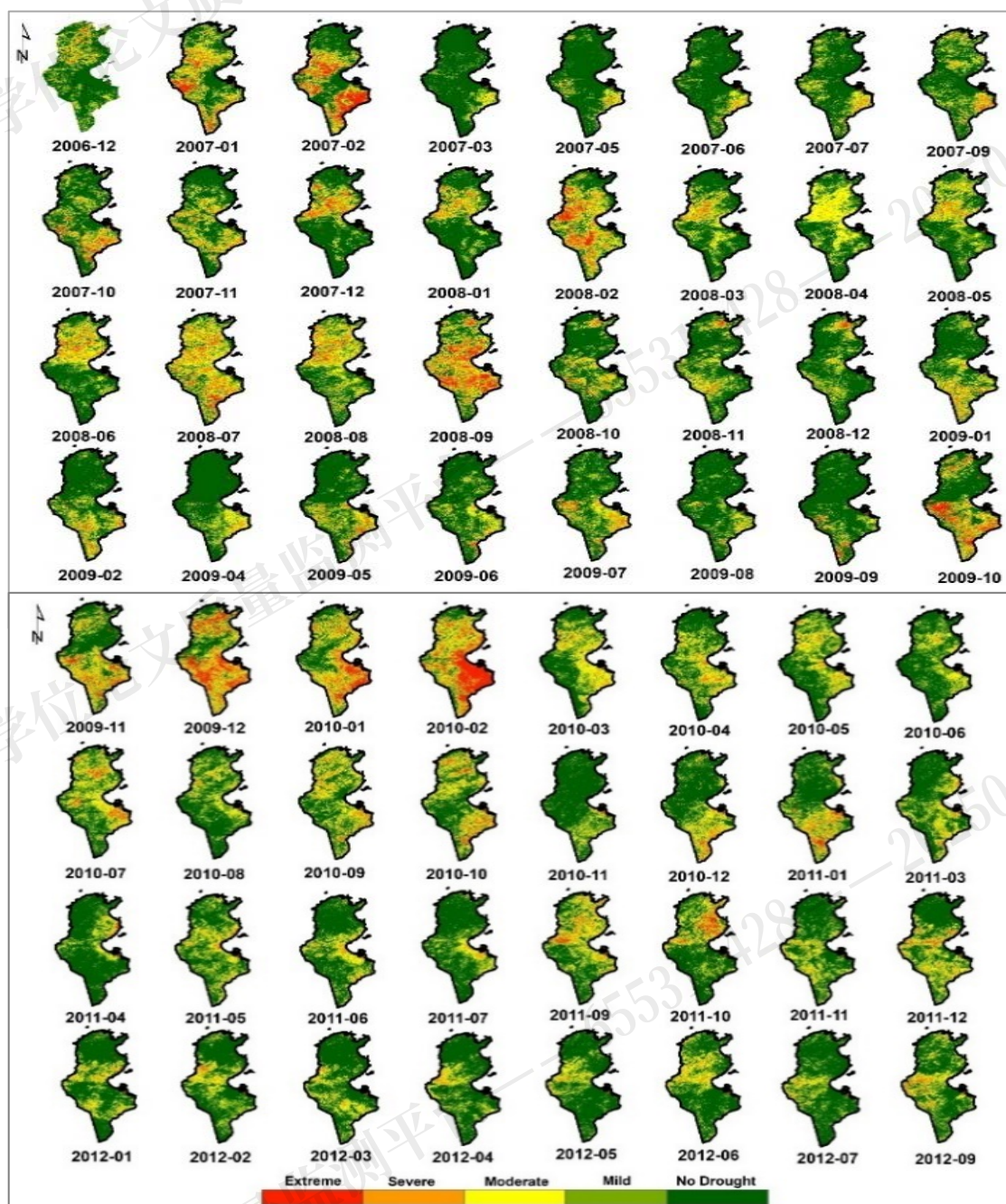


Figure 4-2. (Continued).

Seasonal variations also reveal significant impacts on agriculture, particularly during the crop growth season. Winter months (January–March) typically exhibited mild to moderate droughts in central and southern Tunisia, while the northern regions remained largely unaffected. Droughts during this period, particularly in southern Tunisia, reduced soil moisture availability for rainfed farming, particularly for cereals such as wheat and barley. Severe droughts in 2002, 2005, and 2016 had a major impact on rainfed farming areas, limiting crop survival (FAO, 2016; Verner et al., 2018a). Similarly, droughts during March–April negatively impacted grain-filling, affecting grain quality and overall production. For perennial crops like olives and vineyards, spring droughts (March–May) weakened flowering and fruit-setting, reducing yields, particularly in southern Tunisia. The 2016 and 2020 droughts caused significant agricultural losses, emphasizing the region’s vulnerability to prolonged drought stress (Garrido-Perez et al., 2024). Consistent with our results, Ben Abdelmalek, and Nouri (2020) reported all drought classes significantly impacted the main crop productions, in particular olives, citrus, and cereals.

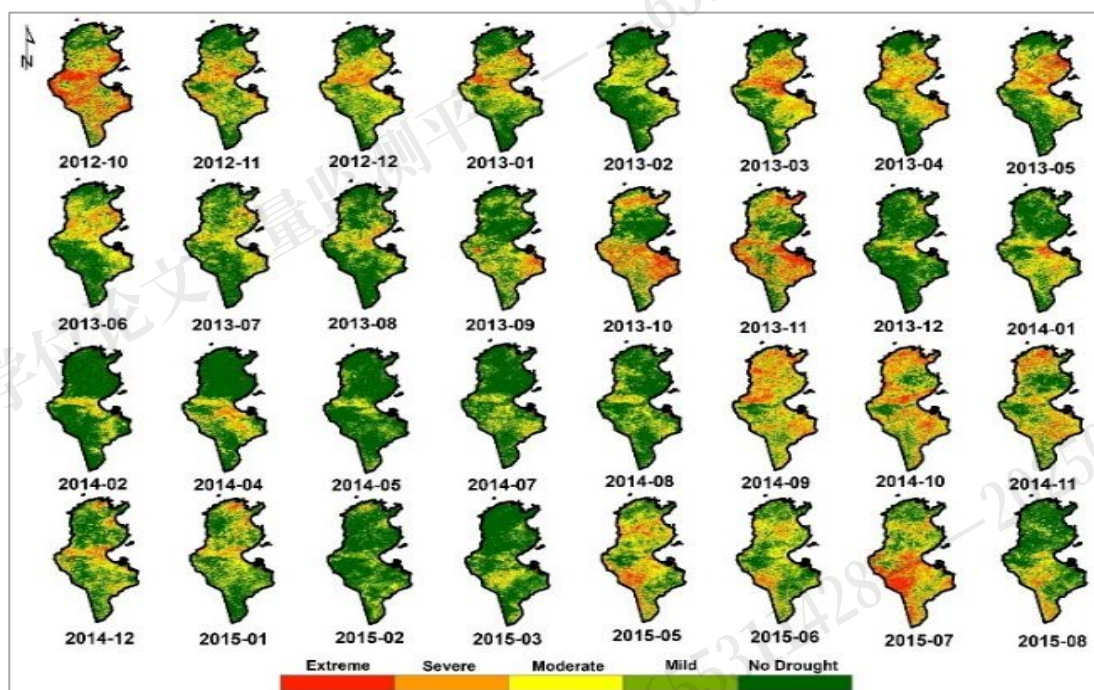


Figure 4-2. (Continued).

Summer months (June–August) exhibited more variability, with moderate droughts affecting northern and central Tunisia, while southern Tunisia faced more severe conditions, including extreme drought episodes. While summer droughts had less impact on cereal crops, which had already been harvested, they severely affected pastoral farming, leading to degraded grazing lands and increased reliance on costly

feed supplements (Gamoun, 2013). The most extreme drought years (2001, 2005, 2011) forced many livestock farmers in the south to reduce herd sizes due to pasture shortages (Verner et al., 2018a). These conditions highlighted the need for long-term water resource management strategies. Despite improvements in vegetation health after 2005, the recurrence of droughts, particularly in the south, underscores the need for sustained monitoring, drought-resilient agriculture, and adaptive water management to mitigate long-term impacts on agriculture and water resources.

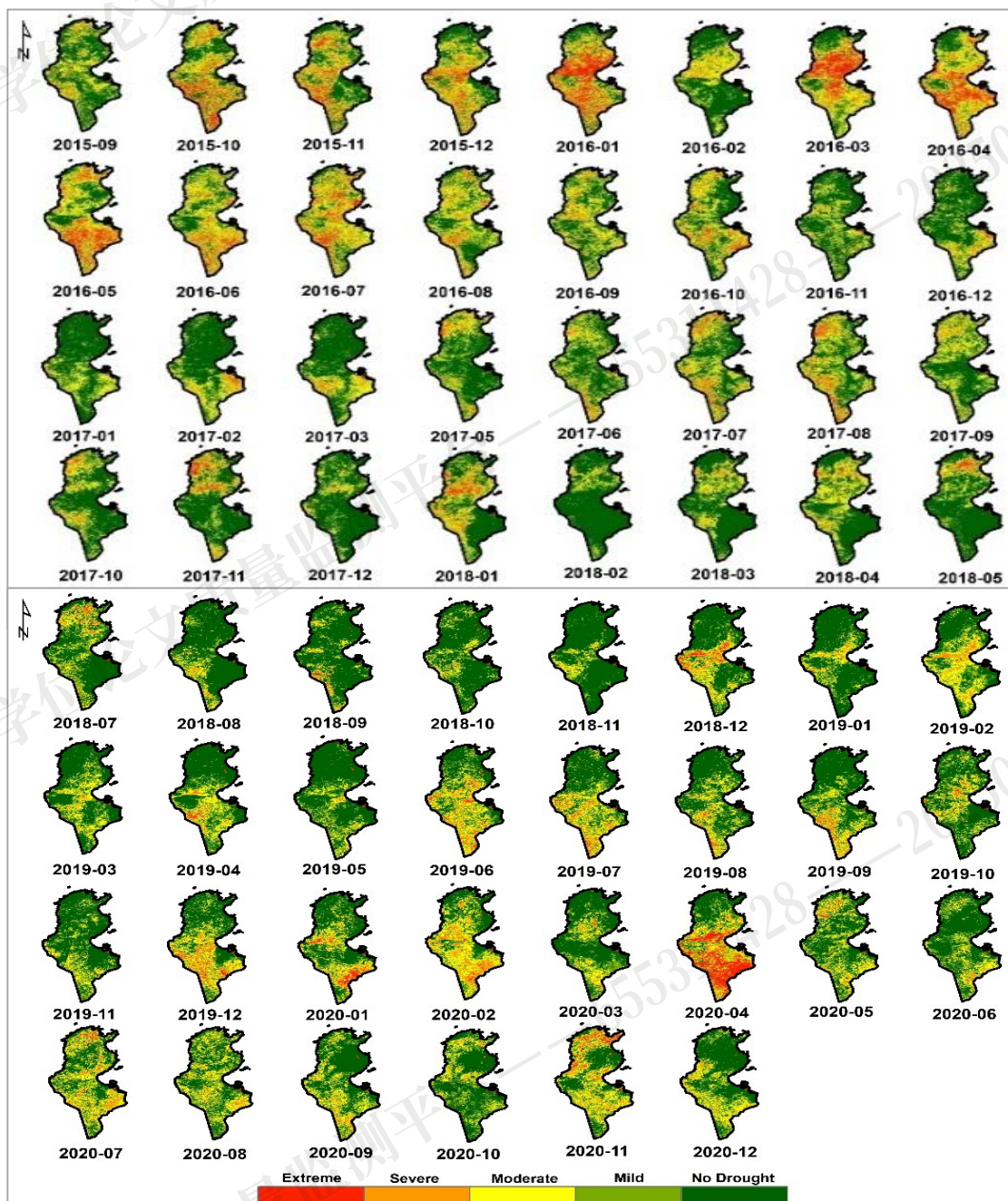


Figure 4-2. (Continued).

The spatial distribution SPEI maps over Tunisia from 2001 to 2020, along with their corresponding analysis, are provided in the Appendix (refer to appendix chapter Fig.A.4). Meanwhile, Chapter 3 provides temporal analysis trends for SPEI.

4.3.2. Analysis of temporal lags in VHI-SPEI relationships using DTW

DTW plot comparing SPEI-1 and VHI-1 (Fig. 4-3-(a)) reveals how the two-time series align over time, with the warping path (shown as a series of connecting lines) illustrating the relationship between MD (SPEI) and AD (VHI). Ideally, if the two series were perfectly aligned, the path would form a diagonal line. However, a complex and winding path suggests that SPEI-1 and VHI-1 do not follow a simple, consistent pattern over time, indicating the potential for nonlinear relationships or temporal distortions in MD propagation into AD. The presence of lags or phase shifts in certain segments of the plot further highlights the varying degrees of influence or timing between the two series, suggesting that AD may lag behind MD at different points in time, possibly due to soil moisture levels or vegetation type. For example, Li et al. (2022c) revealed that vegetation responses to changes in volumetric soil water content in lagged by an average of 9-35 days, emphasizing the critical role of soil moisture in mediating vegetation's delayed response to water deficits. Overall, the DTW plot confirms that the relationship between MD and AD is complex and nonlinear, with varying degrees of synchronization over time.

The DTW analysis of SPEI-3 and VHI-1 reveals several key insights into the relationship between meteorological and Ads (Fig. 4-3- (b)). The warping path also suggests that AD (VHI) tends to follow MD (SPEI-3) with a delay. The DTW distance returned a moderate value of 152.72, implying that while SPEI-3 and VHI-1 generally follow similar trends, there is some misalignment between them. This suggests the presence of other factors influencing the relationship or potential lags between the two series. Consistent with these results, Wang et al. (2023) indicates that approximately 66.41% of vegetated lands show time-lagged responses to drought, with timescales ranging from 1 to 4 months, highlighting the significant cumulative impact of prolonged precipitation deficits on vegetation health.

The DTW distance of 139.72 between SPEI-6 and VHI-1 (Fig. 4-3-(c)), suggests a better level of alignment between the two-time series, indicating that while precipitation anomalies influence vegetation health better than shorter-term precipitation indices (SPEI-1, and -3), the relationship is still not immediate. The black lines represent the

warping path, which, when stretched, indicates delayed effects of precipitation on vegetation, suggesting a time lag between the occurrence of precipitation anomalies and their impact on vegetation health. The warping path also shows peaks and troughs, which may reflect localized climatic events such as heatwaves or sudden fluctuations in rainfall, as well as variations in soil moisture or land management practices (i.e. *Tabias, Jessours, Mgouds*, etc.) that tackle the drought effects on vegetation health (Gasmi et al., 2014; Calianno et al., 2020; Gaaloul et al., 2020).

The DTW analysis of SPI-12 and VHI-1 (Fig. 4-3d) reveals the lowest DTW distance (136.36), indicating the best alignment between the two-time series. The DTW warping path shows a general alignment between SPI-12 and VHI-1, but deviations indicate delayed effects, where vegetation health doesn't immediately respond to precipitation deficits. This lag is influenced by factors such as soil moisture retention, vegetation type, and local climatic conditions like temperature or irregular rainfall. This analysis also suggests a lag time between MD and AD, typically ranging from 3 to 9 months. These findings align with global and local studies reporting that vegetation responses to precipitation deficits are both cumulative and exhibit time lags. For example, Xu et al. (2023b) found that the average response time of vegetation to cumulative and lagged effects of drought is predominantly within 4 to 6 months in the arid regions of central Asia, underscoring the delayed yet substantial impact of sustained precipitation shortages on vegetation dynamics. In Tunisia, Farhani et al. (2022) indicated that vegetation response to drought is a gradual process that alters soil moisture, leading to a time lag effect. Similar findings were also reported in different regions of north Africa and mediterranean basin (Belala et al., 2018; Verner et al., 2018b; Baronetti et al., 2024). Overall, the DTW analysis highlights that while precipitation deficits influence vegetation health, the drought propagation from meteorological to AD in Tunisia typically occurs with a delay due to various environmental factors.

The varying degrees of synchronization observed in the DTW analysis underscore the complexity of how MD translates into agricultural drought. Accurately quantifying the time lag in drought propagation is therefore essential for effective drought monitoring and mitigation strategies.

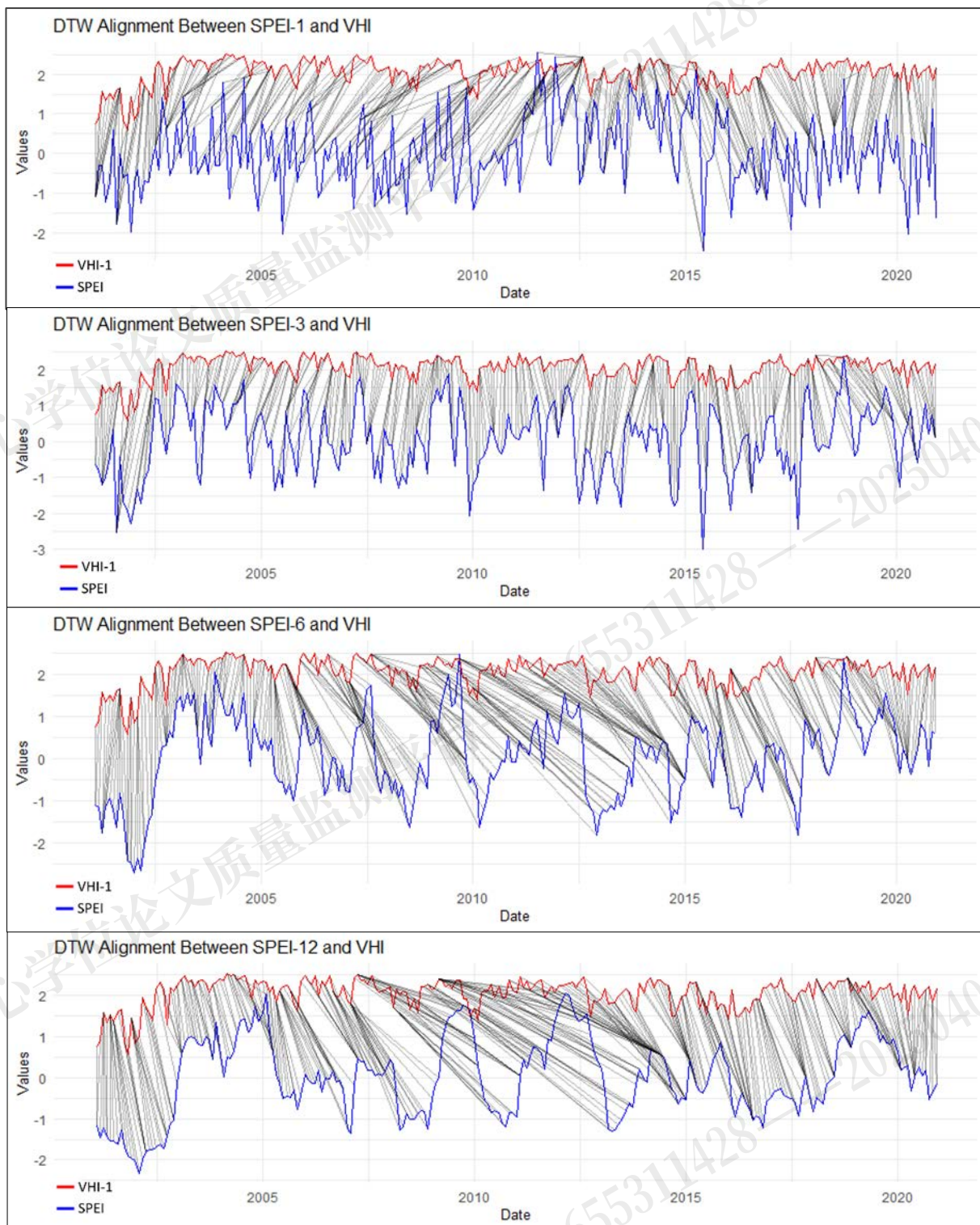


Figure 4-3. Warping path between VHI and SPEI at Different Timescales for Tunisia (2001–2020): (a) SPEI-1, (b) SPEI-3, (c) SPEI-6, and (d) SPEI-12.

4.3.3. VHI Response to multi-time scale of SPEI across different periods

SRC analysis (Fig. 4-4) reveals that the VHI sensitivity to SPEI varies significantly across different time scales. VHI showed the weakest correlation with SPEI-1 (0.16),

indicating that short-term climatic fluctuations have a minimal impact on vegetation health. In contrast, VHI displayed moderate to strong correlations with longer SPEI scales, particularly SPEI-3 (0.55) and SPEI-6 (0.64), suggesting that vegetation health is more sensitive to medium-term and cumulative climate conditions. These time scales better reflect the seasonal and multi-seasonal impact of climate conditions on vegetation.

The maximum correlation was observed between VHI and SPEI-6, indicating the strongest response of vegetation health to climatic variations over a 6-month period. The correlation with SPEI-12 was also substantial (0.47), reflecting the influence of long-term climatic trends, though to a lesser extent. These results indicate that VHI is most responsive to medium- and long-term (6-12 months) environmental changes, with significant seasonal and multi-seasonal impacts on vegetation health. At these scales, lack of rainfall could have a direct impact on soil moisture and crop health, making AD conditions more likely. These findings are consistent with different global studies of vegetation response to MD in arid regions (Waseem et al., 2021; Nxumalo et al., 2022; Achite et al., 2024; C.V. et al., 2024; Cammalleri et al., 2024). For instance, Xu et al. (2023c) pointed out that the propagation time from MD to AD is generally 1 to 8 months in the arid and semi-arid regions of China. This highlights the importance of considering medium- to long-term climate variations, particularly for monitoring drought stress and assessing the health of vegetation in ecological and agricultural contexts. Overall, the varying sensitivity of VHI to different SPEI time scales emphasizes the need for multi-time scale analysis to better understand the dynamics of vegetation response to climate variability.

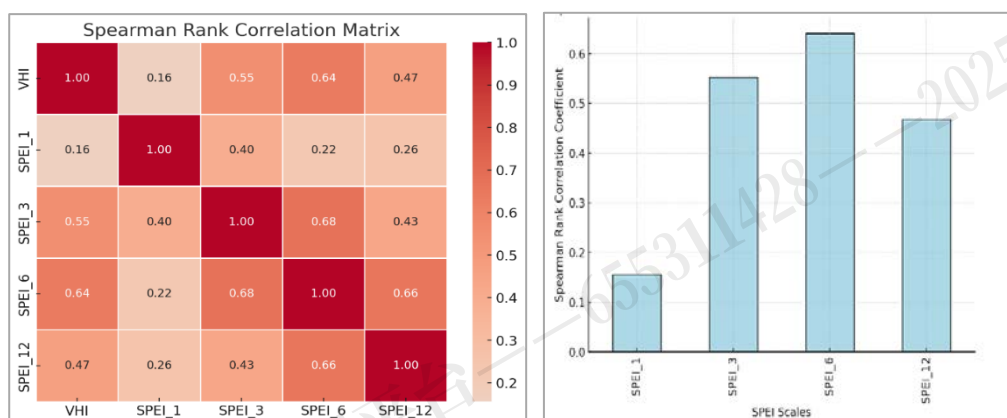


Figure 4-4. Spearman rank coefficient correlation (SRC) between VHI-1 and SPEI-n at different Temporal Scales.

Additional statistical analyses were conducted through cross-correlation and rolling correlation techniques, to further examine the temporal dynamics and lagged relationships between the two variables over different time scales. Cross-correlation helped in identifying shifts in the alignment of VHI with different SPEI time scales, while rolling correlation provided insights into how the strength of the relationship evolved over time. Both methods allowed for a deeper understanding of the response sensitivity of VHI to short-term and long-term climate variations, offering a more nuanced view of the temporal interaction between vegetation health and climatic conditions. The results of these analyses are provided in Appendix chapter Figure A.5 and Figure A.6, where detailed plots and correlations for each time scale are presented.

4.3.4. Drought propagation time and seasonal variability

4.3.4.1. Response sensitivity variations throughout the year

Fig. 4-5 illustrates the seasonal variations in the response sensitivity of VHI to SPEI at different time scales across Tunisia. It highlights the seasonal variations and propagation of droughts in the region. The weak correlation between SPEI-1 and VHI indicates that short-term MD do not immediately translate into agricultural droughts, implying that agricultural droughts in Tunisia require longer drought periods (3-6 months) to develop.

During winter season (December, January, February), long-term precipitation (SPEI-6 and SPEI-12) is most influential, with strong positive correlations between VHI and SPEI. In December, for example, the correlation between VHI and SPEI-12 reaches 0.75, highlighting a strong direct relationship between long-term precipitation and agricultural health. Similarly, January and February show strong correlations (0.66 for SPEI-12 and 0.59 for SPEI-6), emphasizing the significant impact of long-term precipitation on vegetation health during Tunisia's winter months. As the warmer months (March to August) approach, correlations weaken. In March, moderate correlations (0.45 for SPEI-6 and SPEI-12) show that long-term precipitation still has some influence but to a lesser extent. By May and June, correlations drop further, indicating that short-term precipitation (SPEI-1 and SPEI-3) has less impact on VHI. In July and August, correlations become very weak, particularly for short-term precipitation, confirming that rainfall during the summer is less relevant for vegetation health. This reduction in correlation can be attributed to higher evapotranspiration rates

during summer, which leads to a shorter drought propagation period (Wu et al., 2018). Tunisia experiences peak temperatures during this time, with significant heatwaves, particularly in southern regions, which exacerbate the drought impact. For instance, in years like 2007, 2008, and 2013, maximum daily temperatures exceeded 47°C (Verner et al., 2018a).

When precipitation is insufficient, soil moisture becomes the primary water source for evapotranspiration. However, in summer, soil moisture retention is lower than in spring and winter, further accelerating drought propagation. In autumn (September to November), correlations begin to increase again, but they remain moderate compared to the winter months. In September, the correlations for SPEI-6 (0.59) and SPEI-12 (0.62) indicate that the effects of earlier seasonal rains on vegetation health persist into the autumn months, with similar trends observed in November. Overall, during the growing season (warmer months), agricultural systems in Tunisia are more vulnerable due to increased evapotranspiration and crop water needs, leading to faster drought propagation from SPEI to VHI. In contrast, during cooler months, agricultural droughts are less sensitive to short-term precipitation deficits, exhibiting longer lag times of 6-9 months. This reflects the reliance of agricultural systems on accumulated soil moisture, delaying the onset of drought effects. In Tunisia, the availability of irrigation and the management of soil moisture play essential roles in buffering against the immediate effects of precipitation deficits, which may explain the observed lag between SPEI and VHI, particularly during the non-growing season. These variations are primarily driven by climatic factors, as highlighted in previous studies (Han et al., 2019; Fang et al., 2020; Zhou et al., 2021). Among these, temperature, evapotranspiration, and precipitation are the key meteorological variables influencing drought propagation.

The analysis also highlights that prolonged MDs, as captured by SPEI-6 and SPEI-12, result in more significant AD impacts. This supports studies emphasizing that longer MDs tend to propagate more effectively into AD (Amri et al., 2011; Gamoun, 2016a; Farhani et al., 2022). For instance, Touhami et al. (2022b) reported strong correlations between NDVI and SPEI at longer time scales in Northeast Tunisia, reinforcing our findings. This seasonality of drought propagation, (the notably brief propagation period in summer and the prolonged propagation in winter and spring), aligns with global studies (Gevaert et al., 2018; Li et al., 2022a; Long et al., 2024; Wu et al., 2024). Further analysis of time scales reveals that SPEI-1 (1 month) consistently shows weak correlations with VHI throughout the year, particularly during the summer months,

reflecting the limited impact of short-term rainfall on vegetation health in the dry season. SPEI-3 (3 months) shows moderate correlations, particularly in the spring and autumn months, with stronger relationships for longer-term precipitation. However, correlations weaken in the summer. Long-term precipitation indices (SPEI-12) exhibit the strongest correlations with VHI, particularly in winter, indicating that long-term meteorological conditions significantly affect vegetation health. In December, SPEI-12 shows a strong correlation of 0.75, confirming the crucial role of cumulative precipitation in shaping agricultural conditions in Tunisia. Studies have shown that drought lag effects are more pronounced at longer timescales (6-12 months), particularly in arid and semi-arid regions like Tunisia (Sun et al. 2016, Li et al. 2020, Huang and Xia 2019, Huang et al. 2021, Vicente-Serrano et al. 2013, Wang et al. 2021, Xu et al. 2019). For example, Huang et al. (2016) indicated that in arid regions, lag effects can persist as long as 16–19 months.

Other factors contributing to seasonal variability in drought propagation could stem from Tunisia's diverse geomorphology (Taamallah, 2003; Gamoun, 2016a). Vegetation in Tunisia exhibits strong drought resilience through avoidance and tolerance strategies (Laity, 2009; Knapp et al., 2015). Drought-tolerant rangelands, such as chamaephytes, and olive trees with deep root systems and high-water retention capabilities can withstand short-term water shortages, reducing drought propagation time (Breshears et al., 2005; Vicente-Serrano et al., 2013; Thiery et al., 2017) (Sghaier et al., 2013; Laroussi-Mezghani et al., 2016).

From a drought management perspective, understanding these seasonal variations is essential for improving early warning systems and agricultural planning. Since SPI-6 and SPI-12 exhibit the strongest correlations with VHI, particularly during winter, forecasting models should prioritize long-term drought indices to anticipate agricultural drought impacts. Additionally, adaptive strategies, such as drought-resistant crops, soil moisture conservation techniques, and improved irrigation management, can help mitigate the adverse effects of prolonged droughts in Tunisia's semi-arid regions.

The findings underscore the importance of long-term precipitation accumulation for sustaining agricultural health, particularly in winter. The shorter drought propagation time in summer and extended lag in cooler months highlight the complex interplay between meteorological and agricultural droughts. These findings provide valuable insights for drought risk assessment, seasonal forecasting, and agricultural adaptation strategies to enhance resilience against climate variability in Tunisia.

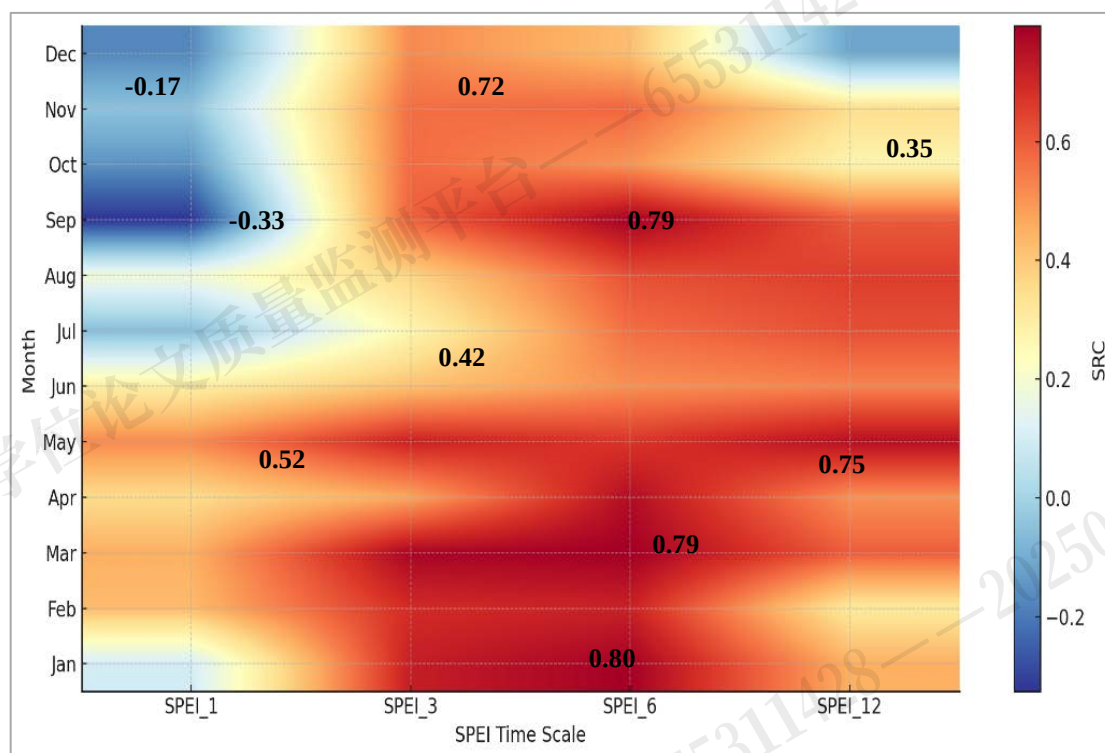


Figure 4-5. Correlations between VHI-1 and SPEI-n across various time scales for each month of the year. Higher SRC coefficient indicates greater VHI sensitivity to SPEI.

4.3.4.2. Propagation linkages of different drought levels severity

The Sankey diagram (Fig. 4-6) illustrates the propagation of drought conditions from MD (SPEI-6) to AD (VHI-1) in Tunisia, revealing complex, non-linear interactions between the two. Most instances of No drought (MD) (163 occurrences) remain as No drought (AD) (85 occurrences), but a significant proportion still transition to Mild (AD) (37 occurrences), Moderate (AD) (40 occurrences), and even Severe (AD) (59 occurrences), highlighting that other factors like soil moisture deficits and irrigation challenges can still create agricultural stress. Mild drought (MD) (37 occurrences) mostly results in Mild (AD) (37 occurrences), but it can also increase into Moderate AD and Severe (AD), indicating that mild MD have the potential to exacerbate into more severe AD, especially in areas with low resilience. Moderate MD (26 occurrences) shows the highest number of transitions, predominantly escalating into Severe (AD) (59 occurrences) and Moderate (AD) (40), suggesting that moderate MD are often amplified into more severe agricultural impacts due to limited water availability and poor agricultural practices. This highlights the critical role of moderate droughts as a turning point, where early intervention becomes essential to prevent escalation into

severe agricultural consequences. Severe drought (MD) (7 occurrences) primarily leads to Severe (AD) (59 occurrences) and a few to Extreme (AD) (18 occurrences), emphasizing the strong link between severe meteorological and agricultural droughts. Lastly, Extreme drought (MD) (6 occurrences) is almost entirely associated with Severe (AD), showing that extreme MD nearly always lead to severe agricultural drought, with minimal recovery.

The overall pattern indicates that more intense MD are strongly correlated with more severe agricultural droughts, with limited transitions to less severe agricultural conditions. This suggests that while MDs of moderate or mild severity may have diverse agricultural outcomes, more extreme MDs consistently lead to severe agricultural consequences, underscoring the amplification effect of severe drought conditions on agriculture. These findings highlight the need for proactive drought management strategies, particularly for moderate droughts, as they are a key tipping point for severe agricultural impacts. Effective early warning systems should be designed to detect moderate droughts early, allowing for timely interventions such as water conservation, drought-resistant crops, and improved irrigation techniques. Furthermore, resilience-building efforts should focus on reducing agricultural sensitivity to droughts of all intensities, ensuring that both meteorological and agricultural indicators are integrated into long-term planning. Given the strong correlation between severe MD and severe AD, policy-makers must implement adaptation strategies that mitigate extreme drought effects, fostering a more drought-resilient agricultural sector in Tunisia.

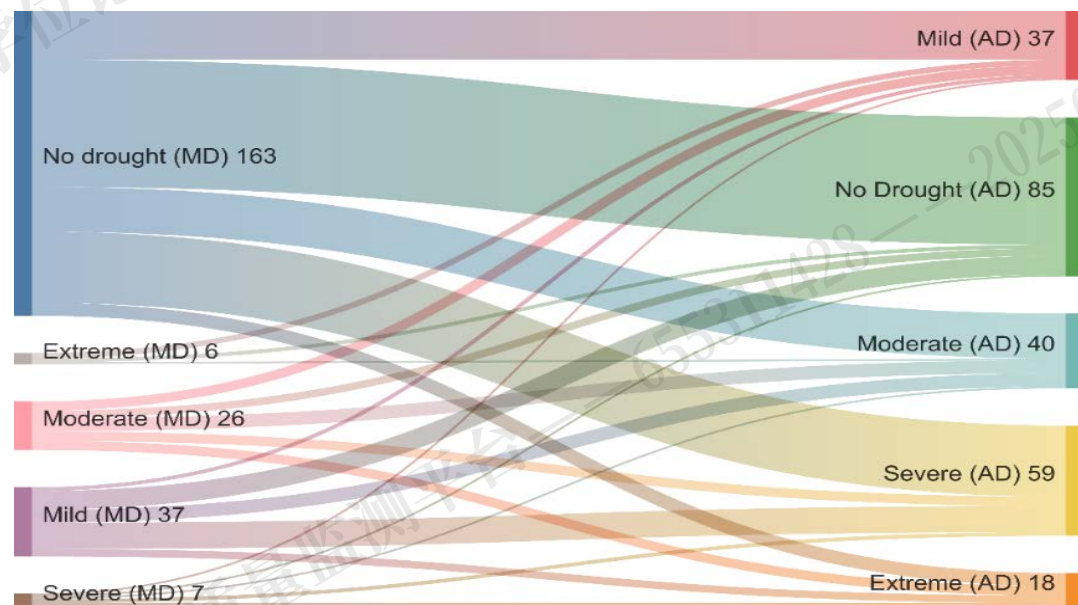


Figure 4-6. Sankey diagram depicting the propagation of drought across various severity levels between MD and AD from 2001 to 2020. The numbers illustrated in the diagram indicate the total occurrences of transitions between drought levels.

4.3.4.3. Occurrence probabilities of AD under different levels of MD

Table. 4-2 displaying the transition probabilities between MD (SPI-6) and agricultural droughts (VHI-1) in Tunisia reveals key insights into how different drought levels propagate from meteorological conditions to agricultural impacts. For extreme MD, the transitions are relatively varied, with 50% of extreme droughts (SPI-6) transitioning to mild agricultural droughts (VHI-1), 16.7% to moderate agricultural droughts, and 33.3% to no drought at all. This suggests that even severe MD do not always lead to severe agricultural droughts, likely due to the buffering effects of soil moisture or irrigation in some areas. Mild MD, which make up a larger portion of the transitions, show significant variability in agricultural outcomes, with the largest proportion (37.8%) transitioning to severe agricultural droughts, indicating that mild droughts still pose a significant risk to agriculture, especially in areas with limited resilience. Moderate MD are particularly concerning, as they have a high probability of worsening into severe agricultural droughts (19.2%) or remaining at moderate levels (26.9%), highlighting the need for early intervention when these conditions arise. In the case of no drought conditions in SPEI-6, 41.7% of these instances result in no agricultural drought, but still, 23.3% transition to severe agricultural drought, showing that even under normal meteorological conditions, agricultural stress can occur due to factors such as soil moisture deficits or poor management practices. Severe MD almost always lead to severe agricultural droughts, with 28.6% of instances transitioning to severe agricultural drought, further emphasizing the strong link between meteorological and agricultural droughts.

These results underscore the importance of long-term precipitation forecasting for early drought detection and management, particularly for moderate and severe MD, which have the potential to cause significant agricultural impacts. Tailored interventions, such as improved soil moisture management and irrigation strategies, are necessary to mitigate the effects of droughts, especially in Tunisia's semi-arid regions, where water scarcity is a persistent challenge.

Table 4-2. Transition probabilities between agricultural drought and meteorological drought under different drought.

MD Level	AD Level	Count	Total Count	Transition Probability
Extreme	Mild	3	6	0.50
Extreme	Moderate	1	6	0.17
Extreme	No Drought	2	6	0.34
Mild	Extreme	4	37	0.11
Mild	Mild	2	37	0.06
Mild	Moderate	7	37	0.19
Mild	No Drought	10	37	0.27
Mild	Severe	14	37	0.38
Moderate	Extreme	5	26	0.20
Moderate	Mild	5	26	0.19
Moderate	Moderate	7	26	0.27
Moderate	No Drought	4	26	0.16
Moderate	Severe	5	26	0.19
No drought	Extreme	7	163	0.04
No drought	Mild	26	163	0.16
No drought	Moderate	24	163	0.15
No drought	No Drought	68	163	0.42
No drought	Severe	38	163	0.24
Severe	Extreme	2	7	0.29
Severe	Mild	1	7	0.14
Severe	Moderate	1	7	0.14
Severe	No Drought	1	7	0.14
Severe	Severe	2	7	0.29

Previous studies have largely concentrated on analyzing the duration of drought propagation across various regions and time periods to uncover spatial patterns and temporal fluctuations in propagation time (Han et al., 2019; Meresa et al., 2023; Zhu et al., 2023). However, there remains a gap in understanding how different drought severity levels influence the duration of drought propagation. By classifying a single type of drought into four distinct severity categories, this study identifies that more intense MD are strongly correlated with more severe AD, with limited transitions to less severe agricultural conditions.

This study offers a novel perspective on drought propagation theory over Tunisia by establishing connections between drought severity, propagation time, and propagation probabilities. This expanded understanding not only enhances predictive capabilities for future drought conditions but also informs practical mitigation strategies to minimize drought impacts.

4.5. Conclusion and future directions

This study provides an in-depth analysis of the spatiotemporal dynamics of AD across Tunisia from 2001 to 2020, exploring the propagation from MD to AD and the probabilities linkages between different drought levels. Using MODIS-derived VHI and SPEI, this research revealed critical insights into the severity, seasonal propagation variability, and propagation time of AD in Tunisia. The investigation concluded the following key findings:

1. Tunisia's agricultural sector faces significant challenges from drought, particularly in the south, which is identified as drought-prone region, frequently experiencing prolonged moderate to severe drought conditions, with extreme drought episodes recorded in 2001, 2005, 2007, 2011, and 2016. December through March tended to be the period with the most widespread mild to moderate droughts.
2. The DTW analysis reveals the intricate and nonlinear relationship between MD and AD, highlighting varying degrees of synchronization over time and a delayed drought propagation in Tunisia due to environmental factors, with the strongest alignment observed between SPEI-12 and VHI-1 (DTW distance: 136.36).
3. Spearman Correlations analysis highlights that vegetation health is more sensitive to medium- to long-term and cumulative climate conditions, with stronger correlations observed at longer SPEI scales (SPEI-6: 0.64 and SPEI-12: 0.5), while short-term climatic fluctuations (SPEI-1: 0.16) have minimal impact.
4. The propagation time was shorter during the warmer season compared to the cooler months (winter and spring), which exhibited longer lag times of 6-9 months. This seasonal variation in drought response is likely due to differences in soil moisture persistence and evapotranspiration, which affect the timing of drought impacts on vegetation health.
5. The occurrence probability of AD increases with MD intensity. On average, approximately 19.2% of moderate MD evolve into severe agricultural droughts, while a significant portion (26.9%) remains at moderate agricultural drought levels. Severe MD (28.6%) almost always leads to severe AD, highlighting the strong and consistent link between MD and AD severity levels in Tunisia.
6. MD severity is crucial in determining AD conditions, with moderate MDs acting as a threshold for more severe agricultural impacts.

This study underscores the importance of long-term drought monitoring systems that integrate both meteorological and agricultural indicators. The results suggest that early warning systems must prioritize medium- to long-term drought indices (such as SPEI-6 and SPEI-12), which are more likely to propagate into severe agricultural droughts. Identifying the southern region of Tunisia as a high drought-prone area further emphasizes the need to assess drought impacts on land productivity and explore its linkages with land degradation processes. These insights are critical for developing targeted adaptive strategies to enhance agricultural resilience. More broadly, this research enhances the understanding of drought propagation dynamics in Tunisia and provides essential guidance for improving drought monitoring, early warning systems, and agricultural planning. It also highlights the necessity of a multi-time scale approach to better predict and manage future drought events.

Future directions

As the southern region of Tunisia has been identified as a high drought-prone area, further investigation is needed to assess the impact of drought on land productivity and its linkages with land degradation processes in these drought-affected regions. Since soil erosion is the dominant form of land degradation in this region, largely influenced by vegetation type and agricultural drought, understanding these interactions is crucial for designing effective land management strategies. Recurrent droughts not only reduce vegetation cover but also accelerate soil degradation, increasing the vulnerability of agroecosystems.

Building on these findings, the next sections of this thesis will focus on assessing the impacts of drought on land productivity in south of Tunisia, and explore its relationships with land degradation. By targeting these aspects, this work will provide deeper insights into the resilience of the region's agroecosystems and inform strategies for sustainable land and water resource management, in these vulnerable regions where drought-driven vegetation loss intensifies land degradation.

Chapter 5 Drought Impacts Assessment in Tunisian Drylands: Assessment of drought characteristics and its impacts on net primary productivity (NPP) in southeastern Tunisia

5.1. Introduction

Drought is a recurrent phenomenon in Tunisia, particularly in its arid and semi-arid regions, where it poses significant challenges to ecosystem stability, agricultural productivity, and land management. Given its geographical location in North Africa, Tunisia is highly vulnerable to increasing drought frequency and severity under global climate change scenarios (Nasr et al., 2008; Stocker, 2014; Verner et al., 2018a). NPP is among the most critical indicators of ecosystem productivity. It refers to the net amount of carbon absorbed by vegetation after accounting for photosynthesis and autotrophic respiration over a given period (Clark et al., 2001; Min and Guang-sheng, 2004). Given its central role in the global carbon balance, understanding how NPP responds to drought events has become a key research priority worldwide (Sun et al., 2016).

Several studies have highlighted the detrimental impact of drought on NPP at global and regional scales (Chen et al. 2013, Peng et al. 2017, Xu et al. 2020, Zhao and Running 2010). The exploration of NPP responses to drought at global and regional scales has recently been the focus of many researchers (Vicca et al. 2016, Vicente-Serrano et al. 2013, Zhang et al. 2016). For example, Peng et al. (2017) reported that droughts accounted for approximately 37% of the global NPP decline in recent decades. Similarly, research conducted in China (Lai et al., 2018), Europe (Vicca et al., 2016) and Africa (Khalifa et al., 2018), confirmed that drought-induced reductions in vegetation productivity significantly impact ecosystem functioning. Actually, understanding NPP responses to droughts contributes in developing drought mitigation and management strategies. The results of these studies confirm that droughts induced remarkably NPP variation (Chen et al. 2013, Zhao and Running 2010).

Despite Tunisia's long history of drought occurrences, research droughts effects on NPP are still lacking. Previous studies focused mainly on specific drought events identification and rainfall regime analysis (Ben Boubaker et al., 2003; Bergaoui and Louati, 2010; Chebil et al., 2019), leaving a critical knowledge gap regarding the direct impact of drought on NPP.

This chapter aims to bridge this gap by investigating the connection between drought and NPP variation in southeastern Tunisia over a 17-year period. Initially, SPI was computed to characterize the spatiotemporal distribution of drought in terms of intensity, duration, severity, and location. Subsequently, the variation of NPP and its response to drought events in dry ecosystems are analyzed. Building on previous findings of chapter 3 and 4, identifying southern region of Tunisia as the most drought-prone regions, this study will focus on assessing the impacts of drought on land productivity mainly in these drought-affected regions of the south. The findings will provide further insights into the nature of drought in southeastern Tunisia and its implications for ecosystem productivity. To the best of our knowledge, this study is the first to comprehensively assess the drought-NPP relationship in southeastern Tunisia, contributing valuable information for developing drought mitigation and management strategies in arid environments, to tackle drought impacts on local ecosystems.

5.2. Data and methods

5.2.1. Study area

As the southern region of Tunisia has been identified as a high drought-prone area (chapter 3 and 4), the study narrows its focus to southeastern Tunisia at a finer spatial scale to the Jeffara region within southeastern Tunisia (Figure.5-1). Situated north of the 30th parallel, between the Mediterranean and the Matmata mountain range, the area experiences an arid climate. The upper part is classified as upper arid, with a temperate winter, whereas the lower part has a mild winter and irregular rainfall (lower arid) (Floret and Pontanier 1982, Ouessar and Yahyaoui 2006). This region is characterized by limited water availability, receiving less than 200 mm of rainfall annually, spread over approximately 30 precipitation days, with a mean annual temperature of 22°C (Genin et al., 2006). The rainfall pattern follows a Mediterranean regime, with the wet season extending from September to April (Ouessar et al. 2009). The soils are formed on a calcareous substratum in the upstream part, while the downstream section consists of gypsum or a mix of gypsum and calcareous formations (Taamallah, 2003). With a sandy to fine sandy texture, these soils are highly susceptible to various forms of disturbance. The sparse and stunted vegetation, predominantly comprising Chamaephyte steppes, exacerbates land degradation processes (Gamoun, 2016b). This region is strategically important to Tunisia, as it is among primary socio-agroecological

zones, experiencing widespread land degradation and recurrent droughts, constituting a representative case for arid southeastern Tunisia.

To mitigate ongoing degradation, Sustainable Land Management (SLM) strategies have been implemented, primarily targeting drought effects, soil erosion, and rangeland degradation (Ouassar et al., 2006). These measures mainly include soil and water conservation (SWC) techniques, such as water harvesting and sand dune stabilization, alongside rangeland management practices. These approaches are fundamental for water scarcity and mitigating the impacts of droughts (König et al., 2012).

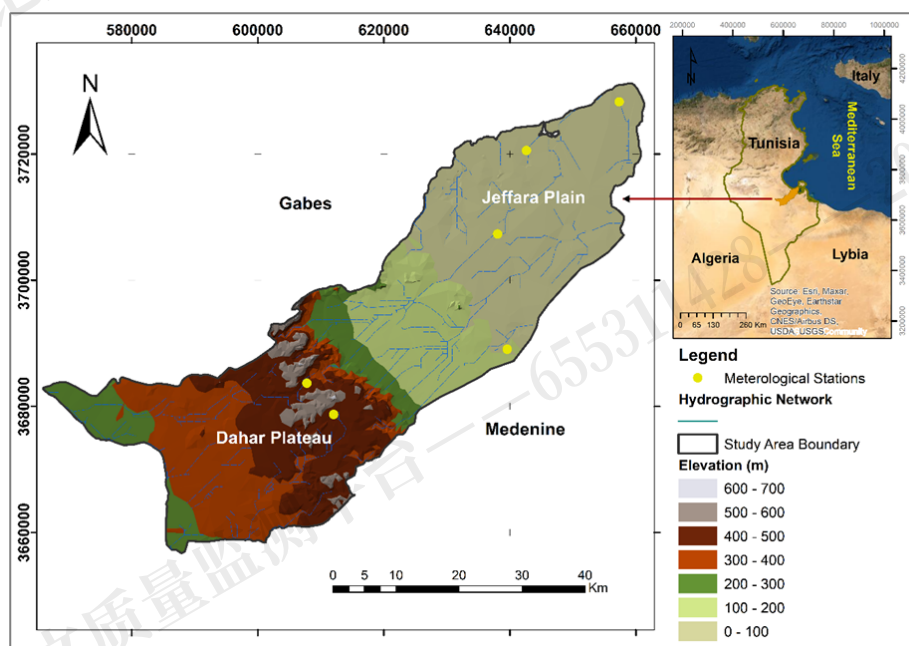


Figure 5-1. Geographical location of the study area with rain-gauge stations Jeffara Region in southeastern Tunisia

5.2.2. Data

5.2.2.1. Rainfall data

Monthly precipitation data from 2000 to 2016 was utilized to calculate SPI values. This long-term dataset was obtained from the hydrological archives of the Regional Commissariat for Agricultural Development of Medenine, based on records from seven rain gauge stations distributed across the study area (Table 5-1).

5.2.2.2. MODIS data

The study employed MOD13Q1 EVI products with a spatial resolution of 250 meters and a temporal resolution of 15 days to analyze the effects of drought on NPP. Due to their effectiveness in capturing the spectral response of green vegetation, a total

of 368 MOD13Q1 images spanning 2000 to 2016 (23 images per year) were retrieved from the USGS database. These images underwent various processing steps using TIMESAT software (Eklundh and Jönsson, 2015) to generate a smoothed EVI time series, which was then used to compute annual “small integral” values on a per-pixel basis as proxies for NPP estimates.

5.2.3. Methods

5.3.2.1. Drought assessment based on SPI

Drought indices serve as critical tools for identifying different types of droughts and detecting climate anomalies (Zargar et al., 2011; Patil et al., 2020; Elhoussaoui et al., 2021). Numerous indices have been developed for identifying drought events (Zargar et al., 2011). This study applies SPI to analyze drought characteristics, including duration, frequency, intensity, and spatial distribution. SPI was computed for various time scales (3 to 24 months) using long-term precipitation data (McKee et al., 1993). Due to its adaptability, SPI is applicable in both short-term agricultural assessments and long-term hydrological studies (Hua et al., 2019). The SPI calculation method (Equation 1) follows the Gamma distribution model (Equation 2) (McKee et al., 1993; Dogan et al., 2012). For this research, SPI was computed on an annual scale using a 21-year precipitation dataset (2000–2016) using the software package of Meteorological Drought Monitoring (MDM) (Salehnia et al., 2017).

$$SPI = \frac{(Pi - Pm)}{S} \quad (5-1)$$

where Pi : Year i precipitation; Pm : mean precipitation; S : the standard deviation of rainfall during the period of study.

$$g(x) = \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-\frac{x}{\beta}}, \quad (x > 0) \quad (5-2)$$

where $\Gamma(\alpha)$: the gamma function; x (mm): precipitation ($x > 0$); α : shape parameter ($\alpha > 0$); and β : scale parameter ($\beta > 0$).

5.3.2.2. Characterization of droughts

Key drought parameters include intensity, duration, frequency, and spatial distribution (Zhang et al. 2015). These aspects were analyzed for each drought event that occurred during the study period.

5.3.2.3. Mann-Kendall Trend Test

MK test is a widely used non-parametric statistical method for detecting trends in meteorological and hydrological datasets (Mann, 1945; Kendall, 1975). This test produces z-scores indicating the trend's direction and significance : $Z \text{ score} < -1.96$: decreasing trend, $Z \text{ score} > 1.96$: increasing trend, $-1.96 < Z \text{ score} < 1.96$: no change (Fensholt et al., 2013; Onyutha et al., 2016). Additionally, the Theil-Sen estimator was applied to assess the trend magnitude in hydro-meteorological time series (Lettenmaier et al., 1994). The combination of the MK test and Sen's slope estimation provided insights into the existence, direction, and statistical significance of trends in the annual SPI series.

5.3.2.4. Drought mapping

The Inverse Distance Weighting (IDW) interpolation method, widely used due to its simplicity and effectiveness (Chen et al., 2017a; Cavus and Aksoy, 2019; Jahangir and Yarahmadi, 2020), was applied to map the spatial extent of droughts. Only negative SPI values were considered for interpolation, with GIS techniques used to delineate drought-prone areas. The resulting maps were classified into different drought severity categories according to Table 3-2 (Chapter 3, Section 3.2.2.2).

5.3.2.5. Impacts of droughts on NPP

In arid regions, increasing drought intensity and frequency due to climate change can lead to biodiversity loss and disrupt ecosystem functioning by reducing NPP (Tubiello et al., 2007; Kath et al., 2019). Tunisia has experienced more frequent droughts in recent decades, exacerbated by growing water demand (Verner et al., 2018a). Analyzing the impact of drought on NPP is crucial for drought risk management, mitigation strategies, and understanding ecosystem resilience to extreme climatic variability.

In this study, NPP was selected as an indicator to evaluate drought impacts on the arid ecosystems of southern Tunisia. Given that NPP is complex and costly to measure directly, satellite-based data was used as a substitute. EVI, developed by the MODIS Land Discipline Group, was used as a proxy for NPP (Huete et al., 1999). Bi-weekly MOD13Q1 EVI products (23 images per year) covering the period 2000–2016 were processed using software package of TIMESAT to compute EVI annual integrals (Figure 5-2) (Eklundh and Jönsson, 2015). A logistic curve smoothing filter was applied

to the EVI time series to minimize noise from cloud contamination. Phenological metrics were then extracted from the smoothed EVI data. An NPP proxy (hereafter referred to as NPP) was estimated for each of the 17 phenological cycles from 2000 to 2016 based on small integral values, which correspond to seasonal productivity (Fensholt et al., 2013; Mao et al., 2015; Olsen et al., 2015). The NPP Standardized Anomaly Index (NPP-SAI), developed by Pei et al. (2013), was employed to detect NPP anomalies, using the following formula:

$$SAI_{npp} = \frac{NPP(i) - NPP}{\sigma_{NPP}} \quad (5-3)$$

where SAI_{npp} : NPP anomalies; $NPP(i)$: the NPP in the year i ; NPP : mean NPP value; and σ_{NPP} : the standard deviation of the NPP.

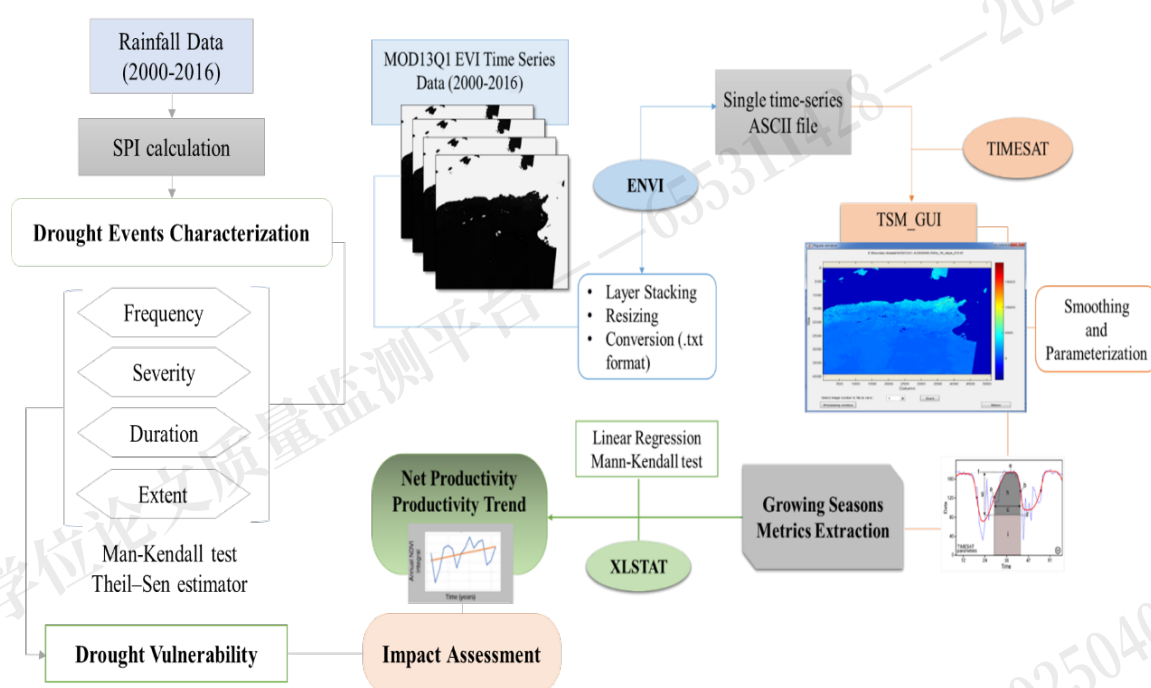


Figure 5-2. Methodological flowchart of this study.

5.3. Results and discussions

5.3.1. Spatiotemporal distribution of droughts

The study area experiences two contrasting climatic influences: Mediterranean and Saharan, resulting in significant spatial and temporal rainfall variability (Terwayet Bayouli et al., 2023b). Southeastern Tunisia has witnessed recurrent drought events of varying intensity and severity (Figure 5-3). Computed SPI values ranged from -2.4 to 2.13, with an average of -0.07, displaying similar trends across most stations. The

driest period, recorded between 2000 and 2002, saw SPI values dropping to -2.4 in the upper part of the study area. During this phase, approximately 47% of the study area (1135 km²) was affected by drought. These findings align with previous studies (Jemai et al., 2018; Verner et al., 2018a), which identified 2000–2002 as one of the most severe drought periods in Tunisia, leading to substantial agricultural losses (Ghoneim et al., 2017). Following this drought episode, the period from 2003 to 2008 experienced predominantly wet conditions, with SPI values between 1.03 and 2.03, except in two downstream stations (Sidi Makhlouf and Grine), where SPI values indicated near-normal conditions. In 2008, SPI dropped to -1.02 at Beni Khedache and Medenine, classifying that year as moderately dry. Further analysis revealed that severe drought conditions affected more than 77% of the study area (1854 km²) in 2009, with SPI ranged between -0.66 and -1.51, in Grine and Beni Khedache stations. Between 2010 and 2012, normal climatic conditions prevailed, except at Sidi Makhlouf, which recorded a moderate drought in 2011, where SPI had a value of -1.01. Drought conditions emerged again in 2013, affecting more than 35% of the study area, marked by the majority of stations ($-1.46 < \text{SPI} < -0.74$). A two-year episode (2014–2015) of normal climate came after the moderate drought event of 2011 with an exception of moderate wet event in Koutine station (SPI = 1.02). Although the predominance of normal climate in 2016, negative SPI were depicted across three stations (43%) where SPI dropped to -1.64 (Tamassent). Hence, a moderate to severe drought condition was specified in coastal area and central part (9%) of the Jeffara region. A climate improvement, highlighted by positive SPI values ($0.58 < \text{SPI} < 2$), was observed in 2017. The final episode (2018–2019) was characterized by normal climate condition except few stations that recorded wet condition in coastal parts (Tamssent and Grine) and moderate dry event in 2019 in the mountainous region (Ksar Halouf).

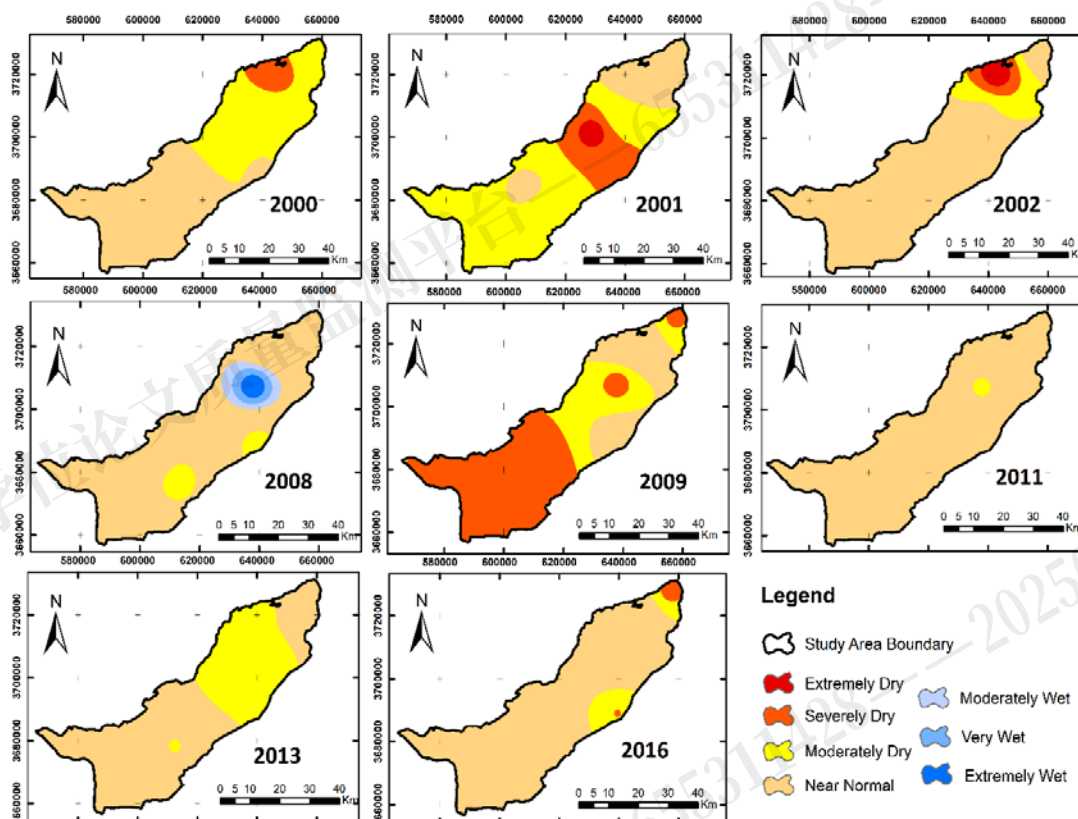


Figure 5-3. Spatial distribution patterns of droughts in Southern Tunisia (Jeffara Region) from 2000 to 2016 period based on SPI.

5.3.1.1. Drought frequency

Although the study area experiences an arid climate, extreme droughts and highly humid events were generally rare across all analyzed stations. Over a 21-year period, SPI calculations identified only two instances of extreme drought, while severe droughts were observed at all stations, accounting for 7% of the recorded events. Additionally, moderate drought conditions were noted in 10% of the cases. The SPI analysis also revealed extreme wet events (3%) at four stations, very wet events (6%) at five stations, and moderately wet events (8%) across all stations. Overall, the climate was predominantly characterized by normal to near-normal years, representing 67% of the total observations.

5.3.1.2. Analysis of drought trends

In this research, the trend curve identification was based on SPI values. The results of Mann-Kendall test and Sen slope estimator are presented in Table 5-1. The test has revealed positive SPI trend values (67%) in most of stations (Medenine, Grine, Tamassent and Sidi Makhlouf) displaying increasing trends. Whereas, negative SPI

trend values were marked in only two stations (Beni khedache and Ksar Hallouf), exhibiting hence a decreasing trend. In fact, the upward trend in SPI data values indicates a tendency towards humid climate, while the downward trend points out a dry climate tendency.

Table 5-1. Mann-Kendall trend Test and Slope Estimators Results of Yearly SPI from 1999 to 2019

Station	Longitude	Latitude	<i>p-value</i> *	Kendall tau	Slope estimator
Beni Khedache	10.2033	33.2458	0.629	-0.081	-0.018
Grine	10.5381	33.6194	0.174	0.219	0.058
Ksar El Hallouf	10.1583	33.2908	0.468	-0.119	-0.036
Mednine	10.5003	33.3358	0.415	0.133	0.041
Sidi Makhoulouf	10.4867	33.5006	0.398	0.138	0.038
Tamassenet	10.6983	33.6869	0.740	0.057	0.016

Note: * non-significant: $p > 0.05$

Positive and negative values indicate increasing and decreasing trends respectively.

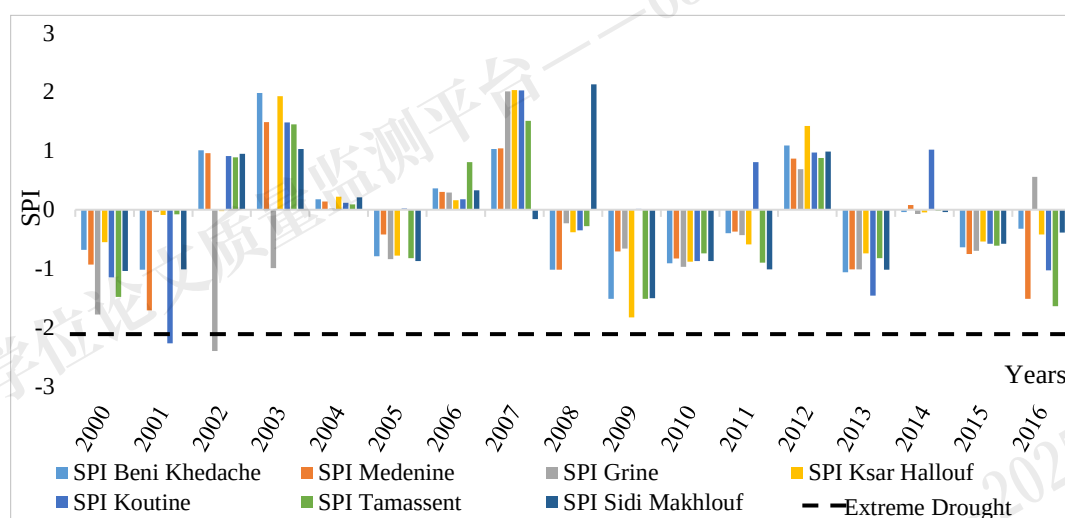


Figure 5-4. Data series of annual SPI from different stations in Jeffara region (Southeastern Tunisia) during 2000-2016.

5.3.2. NPP results

5.3.2.1. Analysis of NPP variations

Trend analysis is used to investigate whether the trend is upward, downward, or no trend regarding its variation. MK test revealed trends detection in the NPP observations that doesn't need normal distribution of data points. The scatter plot showed a non-significant increasing trend in annual NPP over the period 2000-2016, since the

trajectory generated by Mann-Kendall regression, figured out a slope (Figure.5): Z score = $-0.14 > -1.96$ and < 1.96 implying no significant change (Sims et al., 2017). Therefore, when conducting the MK linear regression analysis as based on “EVI small integrals” a marked no significant change ($p = 0.45 > 0.05$) is obtained (Table.2). Although there was a decrease during the extreme droughts period (2000-2002), the overall NPP trend was stable from 2002 to 2016. These results coincided with the unified Arab report (2019) indicating that 62% of the total Arabian lands, including Tunisia, have stable NPP as they haven’t exhibited significant change during the past decade (Darfaoui, 2019). Whereas, at global scale, many studies reported a gradual decline in the global NPP trend triggered essentially by droughts (Guo et al. 2012, Lai et al. 2018, Murray-Tortarolo et al. 2016, Peng et al. 2017). In our study case, moderate and severe droughts are more frequent than extreme droughts where SPI occasionally exceeded -2 (only for 2001 and 2002) during a 21-year period. Thus, the drought severity could explain the overall non-significant NPP trend change from 2002 to 2016. However, during extreme droughts (2000-2001), the annual NPP showed a decreasing trend. Therefore, our findings approve that the annual NPP decrease in dry ecosystems mainly depends on drought intensity (Murray-Tortarolo et al. 2016, Zhao and Running 2010). Murray et al. (2016), suggest that arid ecosystems will continue to experience more severe drought events, leading to an NPP trend decline. They predict a global reduction of 10–13% in total NPP by 2100. Therefore, as a part of dry ecosystem, a declining NPP trend is expected in Tunisian drylands due to recurrent drought events (Murray-Tortarolo et al., 2016; Verner et al., 2018a).

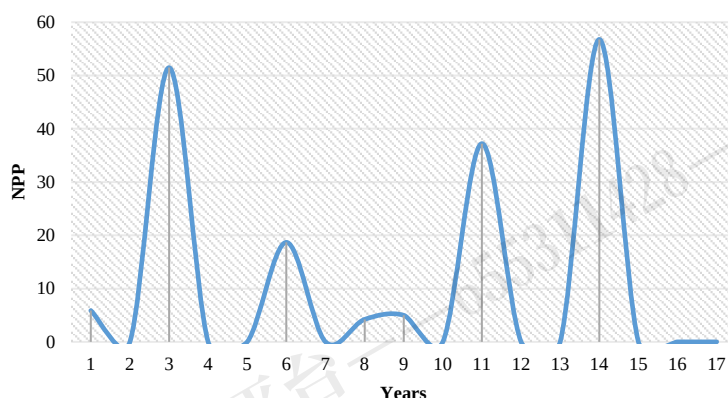


Figure 5-4. Productivity Trend during 2000-2016 using Mann Kendall test

5.3.2.2. Droughts impacts on NPP variations

Decreasing and increasing land productivity detected by vegetation index trends are generally associated to vegetation cover changes, biomass quantity and to the carbon stored quantity in the vegetation. In this study, the EVI datasets were used as proxies for NPP. By means of TIMESAT software, different seasonal information metrics were extracted (Figure 5-5 and Figure A.7 in appendix chapter). Based on our observations, the growth season starts when the vegetation index exceeded 50% of the EVI peak (NPP proxy). The season end was flagged when the vegetation index to below 50% of the peak value, based on their small integral's calculation. It reaches the maximum usually during the spring season (February, March, April). The main season was detected, starting between late January and mid-February (DOY 37) and the season end was on late-April (DOY 118). The middle of season was observed in mid-March (DOY 104). Actually, this season could be associated to spring season when the area becomes more or less covered by annual vegetation. Another season is revealed at the end of the year by December (DOY 336 - DOY 354), this season could be related to the rainfall which occurs usually in the autumn. These analyses indicate a predominant phenological vegetation behavior in Jeffara region, which has a great biomass gain after a rainfall period. During the 17 seasons brought out from 2000 to 2016, the average NPP was 10.5 with a standard deviation of 18.5. The highest NPP peak (56.7) corresponds to the year of 2013. Other high NPP estimates were recorded for the years of 2002 and 2010 with correspondent values of 51.5 and 37.2 kg/ha/y respectively. These NPP peaks could be mainly due to the relatively high rainfall effect since water availability significantly controls the NPP (Vicente-Serrano, 2006; Fernandes et al., 2018). However, in some years (2006/2010/2011/2014) NPP turned out to zero which could be referred to bare soils expansion as a result of droughts conditions. In order to further investigate drought effects on NPP, several statistical analyses were conducted. Temporal correlation between SPI and NPP anomalies at an annual scale were first examined over a 17-year span (2000-2016). As indicated by Ji and Peters (2003) and Pei et al. (2013), the correlation between vegetation index and SPI fluctuates from negative to strong. In this study, the analysis showed a weak positive correlation between these two factors in most of cases (from 0.12 to 0.24). However, for Grine station, the correlation coefficient was negative ($r = -0.1$). As shown in Figure 6, the highest NPP anomalies peaks were surprisingly corresponded to the lowest SPI values

referring to dry years in the majority of stations. Drought events seem to have lag effect on NPP where their impact is visually pronounced in the subsequent year. Many other researches have reported the drought delayed effects specifically depending on ecosystem types (Li et al. 2020b, Shiba and Apan 2011, Zhao et al. 2020).

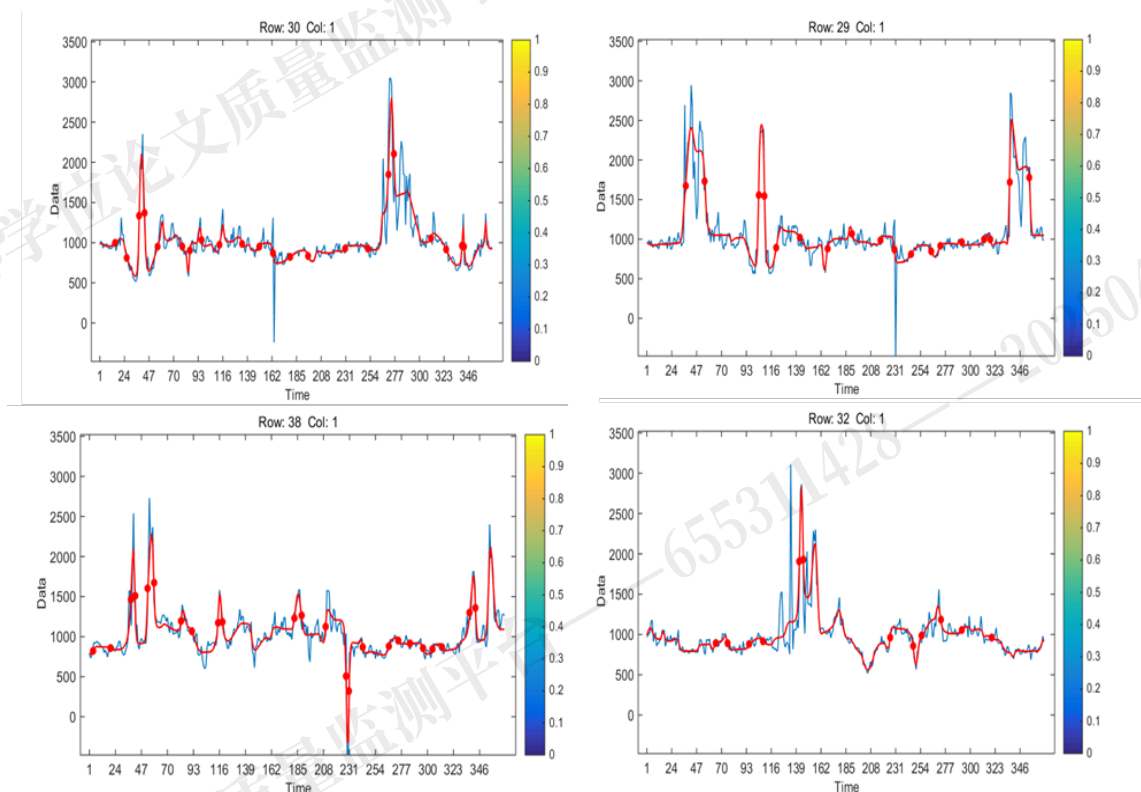


Figure 5-5. Metrics of seasonal dynamics derived from EVI time series through TIMESAT using MOD13Q1 scaled EVI data covering 16-year period.

5.3.2.3. Drought lag effect on NPP

The lagging effect of vegetation responses to droughts was examined through correlation analysis. Firstly, a year-to-year comparison between NPP anomalies and SPI were conducted. Then, a comparison between SPI in the current year with the NPP anomaly in the subsequent year was performed. The correlation between NPP and SPI values without considering the lag effect was generally poor (0.05 to 0.24), and it was even negative in Grine (-0.1). Nevertheless, the correlation coefficients between these two variables significantly increased for all the studied cases when taking the lagging effect into account. For example, in Beni Khedache region the relationship improved as high as 0.9 ($p\text{-value}=0.003 < 0.05$). Overall, the correlation exhibited a significant relationship between NPP anomalies and SPI explained by positive correlation

coefficients fluctuating from 0.5 to 0.9 ($p\text{-value}<0.05$). Visually, an overall strong positive correlation is observed in most of years were (figure.5-6). The SPI and NPP anomalies had a similar trend at an annual scale, where any SPI variations in the current year affect the NPP in the subsequent year. The SPI of 2013 was greatly correlated with NPP of 2014, where the NPP decreased following the drought year of 2013. Similarly, the SPI of 2009 and 2011 (drought years) were significantly correlated with NPP of 2008 and 2010 respectively, inducing thus an NPP drop. NPP peaks were observed clearly in 2004, 2007 and 2013, coinciding with the SPI of 2003, 2006 and 2012 which are normal to wet years elucidating hence NPP increases. Generally, the NPP decline was associated with dry condition of the previous year, proving that droughts are the main trigger for the interannual NPP variability in arid environments of Tunisia. Therefore, our results endorse the insight of droughts lagging effect on land productivity in semi-arid and arid environments, stated by previous studies (Huang et al. 2016, Li et al. 2020a, Pei et al. 2013, Sun et al. 2016, Vicente-Serrano 2006, Wang et al. 2019), where we found that the response to droughts do not normally manifest in the same year when dry condition is proclaimed.

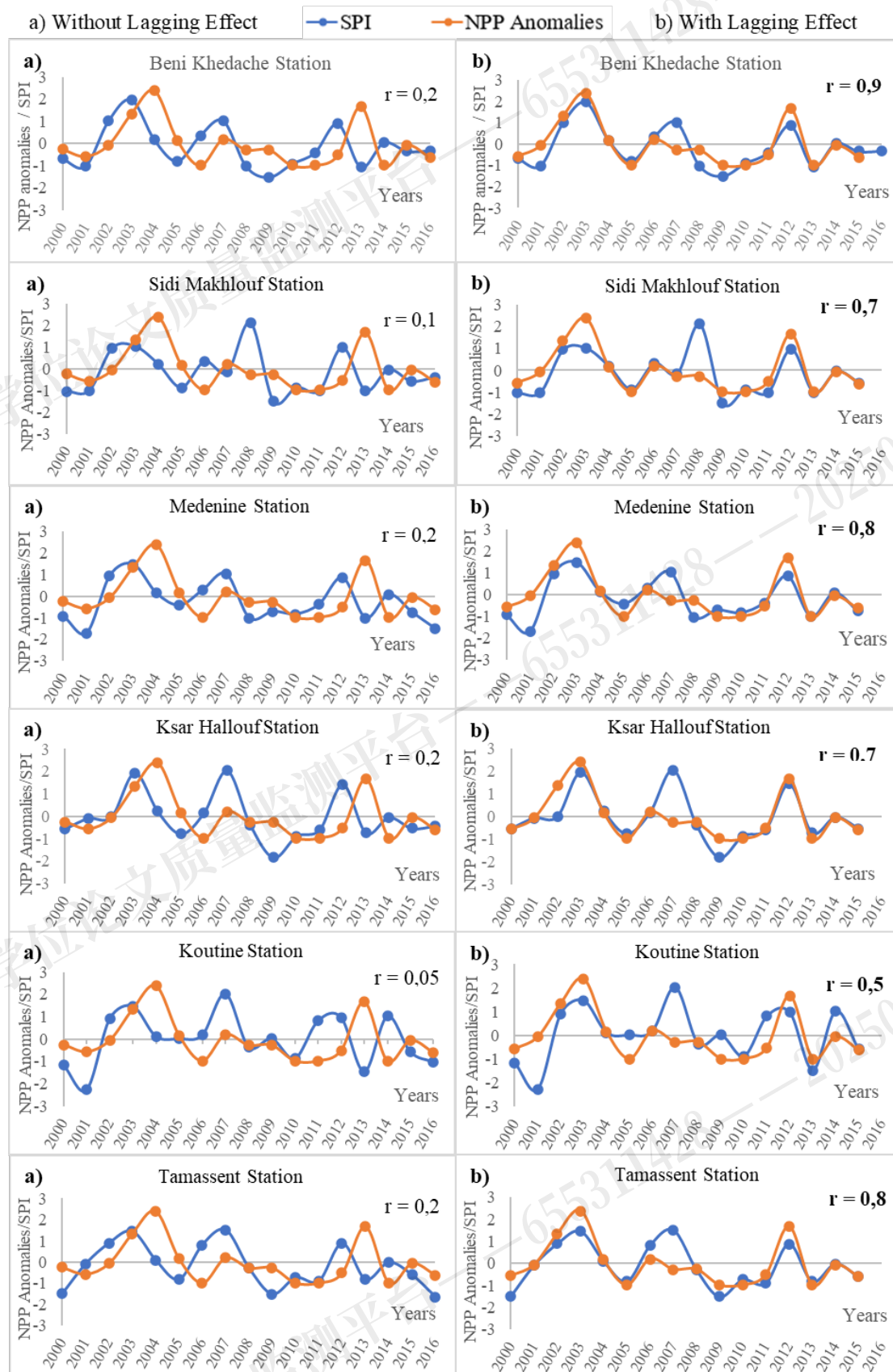


Figure 5-6. Interannual Variation of SPI And NPP Anomalies, along Pearson Correlation Coefficients, without (a) and with (b) drought Lag-effect during 2000-2016 in Southeastern Tunisia (Jeffara Region).

5.4. Discussion

5.4.1. Characteristics of droughts in Southeastern Tunisia

The spatiotemporal analysis of SPI over 21-year period, revealed that the study area was continuously hit by droughts events. Generally, droughts were dispersed most in Jeffara plain region in eastern direction which had lower topographic relief than the Dahar plateau. These results were consistent with previous researches mentioning that air temperature declined along with the elevation increase which led to higher drought occurrence probability in low elevation areas (Benavides et al., 2007; Chen et al., 2017b; Khalil, 2020). Also, it is important to consider the Mediterranean Sea conditions influence on annual rainfall variability. Considering the spatial pattern of droughts occurrence, it was noted that moderate droughts were more recurrent with more than 10% occurrence probability against 5% for severe droughts. While extreme droughts events rarely occurred in the study area with a low probability less than (2%). Previous studies on the spatiotemporal precipitation variability in southern Tunisia and other countries under similar climate condition confirmed the present findings (Ellouze et al., 2009; Feki et al., 2012; Jemai et al., 2018; Jasim and Awchi, 2020). Compared to other years, droughts over the consecutive years from 2000 to 2002 were the most severe in terms of duration, intensity and extent. The seasonal rainfall variability coupled with the low seasonal mean precipitation are the main triggers for this drought period, leading to substantial economic losses (Verner et al., 2018a; Terwayet Bayouli et al., 2023b). Similar findings have been reported in the Mediterranean basin (Bonaccorso et al., 2013; Ezzine et al., 2014; Haied et al., 2017).

5.4.2. NPP response to drought

Drought impacts on NPP were widely studied across the globe where it was been identified that droughts obviously affected the NPP (Zhang et al., 2016; Peng et al., 2017). A drought-induced wide NPP reduction was recorded over Europe ($\sim 16\text{gC/m}^2/\text{month}$ in 2003) (Ciais et al., 2005; Vicca et al., 2016), southeast Asian countries (30% of NPP decline) (Lai et al., 2018), and south American countries (a reduction of 37%) (Peng et al., 2017). At global scale, Chen et al. (2013) confirmed Zhao, and Running (2010) results who admitted that there was a statistic significant correlation between global NPP and droughts at an annual timescale. In fact, the vegetation response to precipitation deficits was cumulative and it had a lagging time (Sala et al. 2012, Pei et al. 2013, Peng et al. 2019). In this study, we investigated the

NPP variation and local droughts events during 2000-2016 period at an annual timescale in arid region based on NPP anomalies. The results of this research, revealed that droughts have a lagging effect on NPP. Overall, droughts often induced the NPP decrease at a 12-month timescale, where their impacts are clearly visible in the subsequent year. Our findings confirm therefore the previous studies reporting that drought impacts are reflected in the following months (Khalifa et al. 2018, Li et al. 2020a, Liu et al. 2015, Sun et al. 2016, Zhao et al. 2018). Researchers identified that drought lagged effects tend to be more significant at longer timescales (6-12 months) in arid and semi-arid ecosystems (Wang et al. 2020, Sun et al. 2016, Li et al. 2020, Huang and Xia 2019, Huang et al. 2021, Vicente-Serrano et al. 2013, Wang et al. 2021, Xu et al. 2019) . Another global study suggested that drought lagging effect could be up to 16-19 months in arid ecosystems (Huang et al., 2016). The current study was conducted in similar climatic conditions in arid context, thereby confirmed the droughts lagging time.

The Jeffara region has a significant diversity of geomorphological (Mountains, heterogeneous foothills and plain) and ecological features (Taamallah, 2003; Gamoun, 2016a). Due to this large variety, the NPP distribution is irregular and depends mainly to regional climate conditions (precipitation and temperature). Despite the detection of negative NPP anomalies, the total NPP trend did not displayed a significant change during 2002-2016. It showed only a decrease during extreme drought period (2000-2001). A slight NPP increase could be witnessed during mild droughts in arid and semi-arid environments (Sun et al., 2016; Khalifa et al., 2018; Xu et al., 2019a). Therefore, we suggest that the annual NPP decrease in dry ecosystems depends mainly on the drought intensity. This result tied strongly also with previous findings of Tortarolo et al. (2016) and Zhao and Runing (2010).

The vegetation in arid environments had a greater potential to tolerate water deficits with an important resilience capacity owing to strategies namely avoidance and tolerance allowing them to tackle droughts and survive water shortage period (Laity (2009) Knapp et al. (2015). Rangelands are the most dominant land cover category in our study area (Gamoun et al., 2018). They host a rich flora known by their high adaptation to drought conditions (e.g. *Chamoephytes* and *Plantago Albicans*) (Jauffret and Visser, 2003; Gamoun et al., 2012). Thus, this vegetation type found evidence for the insignificant change of total NPP trend in response to mild recurrent drought events. Furthermore, olive trees occupy an important place among the local agriculture (about

4 million olive trees on an area of 188,250 ha) and they are mostly rainfed (Sghaier et al., 2013; Laroussi-Mezghani et al., 2016). They are apt to grow in arid environments and resist dry conditions through their deep root system and their water-retention capacity (Dhaou et al., 2009). Thus, they could tolerate short term water shortages and minimize drought impacts on land productivity (Breshears et al., 2005; Vicente-Serrano et al., 2013; Thiery et al., 2017). Consequently, this could be added to explain the delayed response of local ecosystems to droughts. Furthermore, the local communities have developed for a long time several traditional water harvesting infrastructures namely *Jessours*, *Tabias*, *Cisterns*, etc (Ouassar et al., 2009). These techniques increased soil water retention underpinning hence their resilience to droughts. Therefore, the NPP response to droughts in Tunisian arid environments is chiefly depending not only to the drought intensity and duration but also to the ecosystem's types and soil management techniques. The current study highlighted also the importance of arid ecosystems to resist dry periods contributing thus to the terrestrial carbon cycle enhancement and ecosystem vulnerability to future droughts.

5.4.3. Factors possibly inducing NPP variation

The droughts effects on NPP are obvious where they have been identified as chief trigger of NPP decrease across the globe (Zhao and Running, 2010; Pei et al., 2013; Lai et al., 2018). However, other potential factors could be also responsible for NPP variation. For example, heatwaves induce immensely the soils water depletion producing NPP decreases (Bastos et al., 2013). Southern Tunisia withstood drastic heatwaves in 2007, 2008 and 2013 where the maximum daily temperature exceeded 47°C. A clear NPP decrease coincided with these heatwaves. Other natural disturbances from meteorological extremes processes and climate variability such as soil erosion, land degradation and floods could generate different changes level to the growing season start and/or end, or abnormally warmer or colder period resulting in NPP fluctuation. Furthermore, human activities like land management and LULC conversion proved to be among the NPP decline key factors (Li et al., 2020a). Hudak (1999), indicated that Africa contribute to 36% of the world's degraded lands resulting from overgrazing. In southern Tunisia, the increasing anthropogenic pressure generated mainly by overgrazing stimulate the NPP decline (Le Hou  rou, 2009; Gamoun et al., 2015). Also, the withdrawal of croplands and rangelands in favor of bare soils and urban lands, poor irrigation and land management, inappropriate farming combined with the

climatic aridity could profound the NPP loss. Although, Li et al. (2019) found that the human impact on NPP variation is relatively low comparing to droughts impacts in a short-term period. These set of indirect effects should be considered in future studies to ensure more accurate NPP variations assessment in response to droughts.

5.5. Conclusions

Droughts pose a significant environmental challenge in Tunisia and are expected to become more frequent due to climate change. However, research on drought impacts, particularly their effects on NPP, remains lacking and unclear. The Jeffara region, located in southeastern Tunisia, spans various natural landscapes that exhibit clear signs of degradation, exacerbated by harsh climatic and hydrological conditions. This has led to a decline in land productivity, resulting in substantial ecological consequences.

This study aimed to investigate the effects of droughts on NPP in southeastern Tunisia, where existing research has provided limited insights. The methodological approach combined remote sensing and hydrological data analysis. MODIS-derived EVI datasets were utilized as NPP indicators and integrated with SPI to assess NPP responses to drought conditions. This approach enabled the evaluation of drought impacts on NPP and the identification of drought-prone areas within the Jeffara region.

The spatiotemporal analysis of drought occurrences from 2000 to 2016 revealed that the region experienced frequent drought events. These events were more common in the Jeffara plain, located in the eastern part of the study area, which has lower topographic relief compared to the Dahar plateau. Despite the region's aridity, extreme drought events were relatively rare, with a probability of occurrence below 2%. However, moderate and severe droughts were more prevalent, occurring with probabilities of 10% and 7%, respectively. By analyzing annual SPI values alongside NPP anomalies from 2000 to 2016, this study confirmed that droughts play a crucial role in driving NPP fluctuations within Tunisia's arid ecosystems.

The overall trend of NPP remained stable from 2002 to 2016, with the exception of a decline during the extreme droughts of 2000–2001. This finding underscores that NPP responses to droughts are primarily influenced by the intensity and duration of drought events. A key outcome of this study was the identification of a lagging effect of droughts on NPP in Tunisian ecosystems. In general, droughts lead to a decline in NPP on an annual timescale, with their effects becoming more apparent in the following year.

The relationship between SPI and lagged NPP anomalies at a 12-month scale revealed a significant positive correlation, with coefficients ranging from 0.5 to 0.9 (p -value < 0.05). However, when the lag effect was not considered, the correlation weakened across most study sites and was even negative in some cases. These findings emphasize the importance of accounting for the delayed impact of droughts on land productivity in arid ecosystems. Thus, SPI and NPP anomalies prove to be effective indicators for assessing drought effects on land productivity. The results of this research provide a valuable scientific foundation for regional planners and policymakers, aiding in the development of adaptation measures, ecosystem restoration strategies, and comprehensive drought management plans in arid and semi-arid regions.

Chapter 6 Drought Variability and Land Degradation: Integrating VHI and RUSLE to Assess Soil Erosion and Drought Interactions in Southeastern Tunisia

6.1. Introduction

Drought and soil erosion, exacerbated by droughts, are two interconnected environmental challenges that threaten sustainable development in arid and semi-arid regions (Lucatello and Huber-Sannwald, 2020; Wu et al., 2022; Karimi et al., 2024). As one of the most widespread natural hazards, drought poses severe risks to water supplies, agriculture, and ecosystem stability (Lloyd-Hughes, 2013). Various drought indices have been developed to characterize drought severity, frequency, and spatial extent, ranging from precipitation-based measures like SPI to remote sensing-based indicators that provide broader environmental insights (McKee et al., 1993; Zargar et al., 2011; Bayissa et al., 2021). Remote sensing-based indices are becoming more and more significant tools for drought monitoring, as they overcome data limitations often encountered with meteorological and hydrological drought indicators (Marumbwa et al., 2020).

In addition to droughts, degradation of natural resources in dry regions includes not only droughts but also soil erosion, which is acknowledged as a significant environmental issue in arid regions (Du et al., 2016). Soil erosion, induced by both natural and anthropogenic factors, threatens agricultural productivity, water security, and ecosystem resilience (Eekhout and de Vente, 2022) (Patil, 2018; Qiu et al., 2021). Various methods for calculating soil erosion have been developed in the past ten years. Physical and empirical models are among these methods. Due to its stability and simplicity, Renard (1991)'s RUSLE model remains one of the most widely used models. It provides an estimate of soil erosion rates by integrating key environmental parameters such as rainfall erosivity, slope characteristics, soil erodibility, conservation practices, and land cover (Renard et al., 2017). These variables depend on other variables and vary throughout time and space, requiring thus a careful assessment to understand erosion dynamics in drought-prone landscapes (Prasannakumar et al., 2012).

Tunisia, particularly its southern arid regions, faces severe environmental degradation induced by both prolonged drought and increasing soil erosion. Climate change has made these issues more challenging by increasing soil erosion, decreasing biological diversity, and reducing vegetation cover (Sghaier et al., 2012). Nearly 3 million hectares of agricultural soils in Tunisia, which has over 10 million hectares under agricultural use, are at risk of water erosion (Kefi et al., 2012). In southeastern Tunisia, water erosion remains a predominant driver of land degradation, fueled by intense but sporadic rainfall, vulnerable precarious vegetation cover, with unstructured low-depth soils (Batjes, 2010). The selected basin, Negueb watershed, is a key study area that exemplifies these challenges, exhibiting severe soil erosion alongside recurrent droughts that disrupt local agriculture and water resources (Terwayet Bayouli et al., 2023b). The Tunisian Ministry of Equipment and Housing (MEHAT), claimed in 2012 that Medenine was at risk from significant water erosion ($1.06 \times 10^7 \text{ m}^3 \text{ y}^{-1}$) that affected roughly 63% of its land. At the same time, the region was dealing with increasingly severe droughts that had wider negative effects on water supplies and agriculture. At the same time, the region is experiencing increasing severe droughts that had wider negative effects on water supplies and agriculture (Verner et al., 2018a).

Drought and soil erosion are interlinked in a number of ways. One of the main causes of land degradation is soil erosion, which is brought on by previous dry circumstances that had a detrimental impact on ecosystem services (Peri et al., 2021). By increasing sand and runoff activities, droughts worsen soil erosion (Trnka et al., 2016; Middleton and Kang, 2017). Furthermore, droughts reduce soil organic carbon by impeding vegetation growth, particularly in arid areas. When paired with the disturbance of soil structure, this decrease in carbon matter creates a negative feedback loop that intensifies soil erosion (Deng et al., 2021). Droughts indeed contribute to the withdrawal of different land use categories such as croplands and rangelands for bare soils, leading to disturbance of the water cycle, promoting hence the erodibility of soils (Baumhardt and Blanco-Canqui, 2014). Furthermore, because of the changed hydrology that results in lower soil quality, soil erosion may also make people more susceptible to droughts by intensifying the consequences of drought (Reichhuber et al., 2019b). Despite the significance of the problem, the interconnected relationship between soil erosion processes and droughts remains insufficiently explored. Few studies have specifically investigated the connection between soil erosion and droughts on a global scale. For example, Santra, and Santra Mitra (2020) demonstrated that in

west Bengal, India, regions with high drought frequency saw nearly twice as much soil loss as those with low drought frequency. According to other research conducted in western India and southwest China, soil erosion and droughts are positively correlated (Yu et al., 2021; Masroor et al., 2022).

Despite the alarming deterioration of natural resources in the Negueb watershed, which is mostly caused by soil erosion and droughts, there are still very few studies that have been done to evaluate them, and the connection between soil erosion and droughts is still unclear. Therefore, techniques that integrate the main causes of drought and erosive processes at the same time must be developed. Given these pressing challenges, this research is consecrated to: (1) Evaluate droughts in southeastern Tunisia by utilizing remote sensing techniques and precipitation-based drought indices (2000-2016). (2) Estimate soil loss rates through the application of the RUSLE model. (3) Unravel the connection between soil erosion and drought, examining the impact of prolonged drought conditions on soil degradation processes.

By integrating drought and erosion assessments, this chapter provides a comprehensive perspective on land degradation risks in Tunisia's arid landscapes. The findings will contribute to natural resource management strategies and inform policies aimed at mitigating the compounded impacts of drought and soil erosion in fragile ecosystems.

6.2. Datasets

6.2.1. Study Area

As the southern region of Tunisia has been identified as a high drought-prone area (chapter 3 and 4), the study narrows its focus to southeastern Tunisia at a finer spatial scale to Negueb watershed is part of the larger Jeffara Basin, one of Tunisia's most significant basins due to its extensive and dense drainage network (Fig. 6-1). The studied watershed is situated in southeastern Tunisia between the parallels 33°25'60" North and the meridians 10°22'0" East (Terwayet Bayouli et al., 2023b). The upstream is located in the wettest area and has a fairly steep slope, while the foothills and downstream has medium to low slopes. The watershed faces two completely different climates, which are arid and Mediterranean, with an average yearly temperature of approximately 22°C. The study area receives about 200 mm/year, and around 30-day precipitation, which is characterized by spatially and temporally irregular, torrential

precipitation events (Terwayet Bayouli et al., 2023b). The hydrographic network follows a hierarchical structure, with multiple wadis originating from the Matmatas Mountains (Terwayet Bayouli et al., 2023b). The soil's horizons are often shallow, stony and unstructured with sandy to fine sandy texture (Taamallah, 2003). Vegetation is sparse, consisting mainly of *Chamaephyte* steppes, making the watershed highly susceptible to severe soil erosion and recurrent droughts (Gamoun, 2016b). LULC within the watershed primarily consist of rangelands, croplands, and bare soils. Due to ongoing environmental degradation, sustainable land management strategies have been implemented to mitigate land degradation and tackle the impacts of drought (Terwayet Bayouli et al., 2023b).

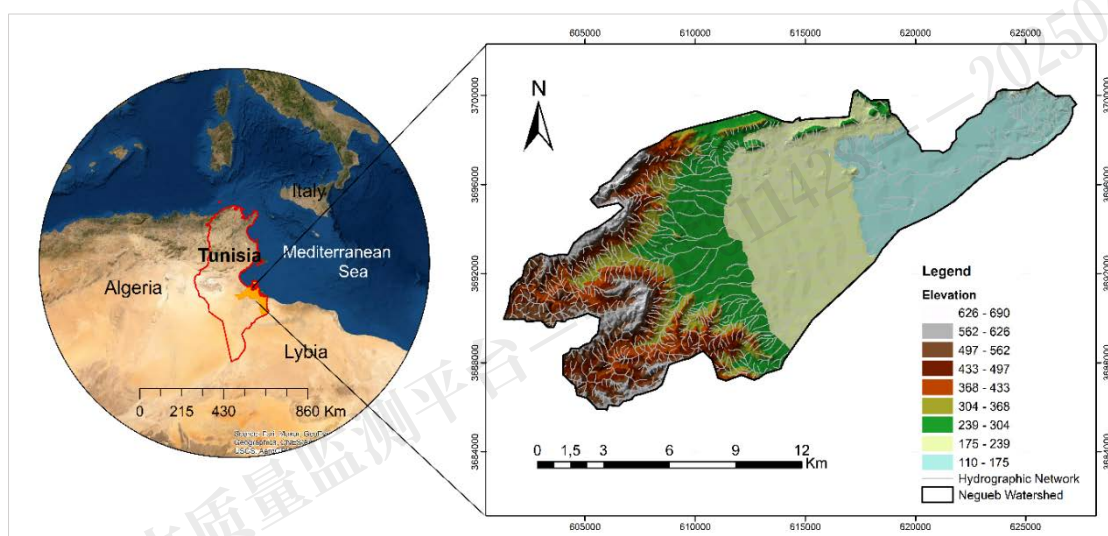


Figure 6-1. Study area localization along with elevation distribution in meters

6.2.2. Datasets

The current study integrates soil erosion and drought, which are considered as the most challenging land degradation issues impacting southeast Tunisia. For drought conditions monitoring, LST and EVI from MOD11A2 and MOD13A2 products, each with a spatial resolution of 1km, for the period 2000–2016 were acquired respectively from the NASA Earthdata portal (<https://www.earthdata.nasa.gov/eosdis/daacs/laads>). VCI was derived from EVI, while TCI was obtained from LST data, and both indices were subsequently combined for VHI computation (Terwayet Bayouli et al., 2023b). For estimating soil erosion processes, the rainfall erosivity factor of RUSLE equation was determined using a mean of historical precipitation datasets over 48 years, from 1968 to 2016, retrieved from the hydrological databases of the Tunisian General

Directorate of Water Resources (Terwayet Bayouli et al., 2023b). The topographic factor was computed using an SRTM-derived digital elevation model (DEM), at a spatial resolution of 30 m. Additionally, the regional agricultural map provided by the Regional Commission for Agricultural Development (CRDA) was used to obtain hydrographic and soil characteristics of the watershed (Terwayet Bayouli et al., 2023b). To generate a land cover classification, Landsat 8 OLI/TIRS imagery (30 m resolution) was utilized. All datasets were integrated within a GIS framework to apply the RUSLE model and determine soil erosion rates within Nagueb watershed.

6.3. Methodology

6.3.1. Drought Characterization

The key attributes used to characterize droughts are frequency, intensity, duration, and spatial extent (Terwayet Bayouli et al., 2023b). In this study, these characteristics were analyzed during 2000-2016 study period (Table 6-1).

Table 6-1. Drought characteristics overview

Drought parameter	Equation	Notation and units
Area	$D = \frac{\sum_{i=1}^n da}{n} \times 100$	D_a = drought area (%) da = number of pixels with VHI values >40 n = total number of pixels
Duration	$D = \frac{\sum_{i=1}^n di}{n}$	D = drought duration (months) di = duration of i th drought event n = total number of drought events
Frequency	$F = \frac{n_m}{N_m} \times 100$	F = drought frequency (%), n_m = number of drought months, N_m = total number of months
Intensity	$I = \left \frac{1}{N} \sum_{i=1}^n VHI_i \right $	I = drought intensity (dimensionless) n = number of drought occurrences in months with VHI > 40, VHI_i = VHI value below the threshold (40).

4.3.1.1. Drought indices

VHI were calculated by combining TCI ad VCI indices for the month of March during the assessment period (2000-2016), following the method described in chapter

4, section 4.2.2.1. Accordingly, drought events were identified, and the spatiotemporal drought consistency was assessed through the determination of the mean VHI and VHI coefficient of variation (CV).

6.3.2. Estimation of soil loss rates

RUSLE model (Equation 6-1) estimates soil erosion as a function of five key input factors represented in raster format.

$$A = R \times K \times LS \times C \times P \quad (6-1)$$

Where LS stands for slope length steepness, C for cover management, P for conservation support practice, R for rainfall erosivity, expressed in MJ mm ha⁻¹ h⁻¹y⁻¹, and K for soil erosivity, expressed in t ha h ha⁻¹ MJ⁻¹ mm⁻¹, while A is the average soil loss per unit area per year, expressed as t ha⁻¹ y⁻¹.

To enhance the applicability of the RUSLE model in the study area, input parameters were meticulously chosen from reliable local databases and validated through field measurements. A structured geodatabase was developed, integrating codified RUSLE factor data. To ensure consistency in spatial analysis, all datasets were standardized to a 20 m cell size within a GIS environment (Terwayet Bayouli et al., 2023b). The overall methodological framework is illustrated in Figure. 6-2.

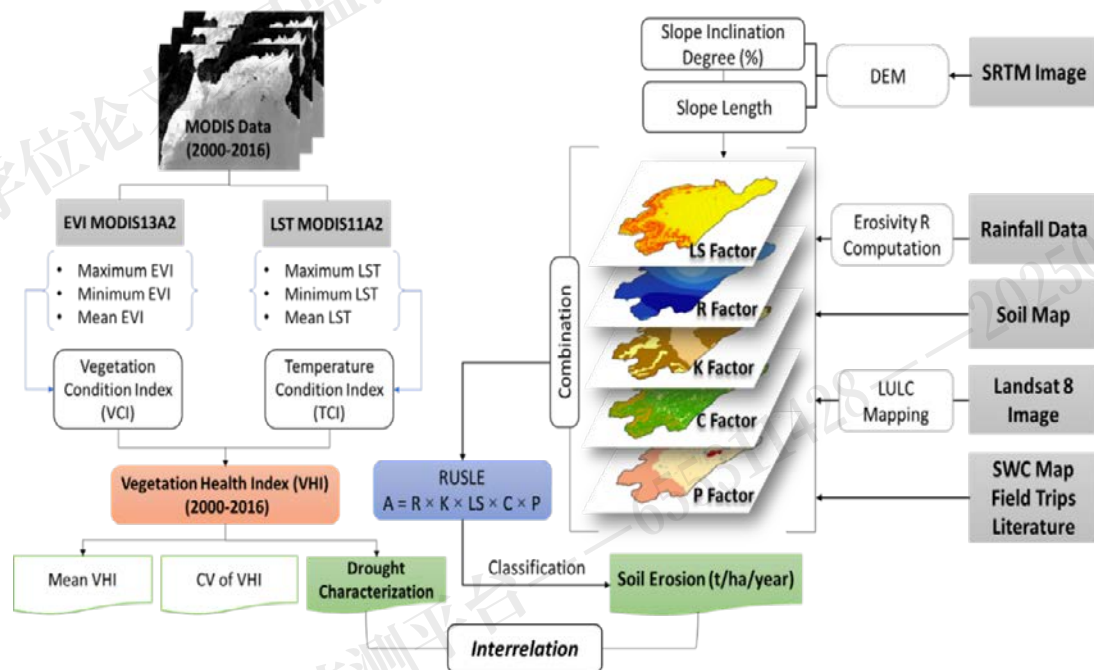


Figure 6-2. Methodology flowchart of this study.

4.3.2.1. R determination

R represents the potential of rainfall to induce soil erosion. It quantifies the impact of precipitation and the resulting runoff on soil degradation (Brychta et al., 2022). Over a specific period, R is computed analytically using the formula proposed by Wischmeier, and Smith (1978) (Equation.6-2).

$$R = Ec \times 100 \quad (6-2)$$

Where R refers to the erosivity expressed in $\text{MJ} \cdot \text{mm} \cdot \text{ha}^{-1} \cdot \text{h}^{-1}$, Ec represents the kinetic energy of rainfall, expressed in $\text{MJ} \cdot \text{ha}^{-1}$ and $I30$ refers to the peak rainfall intensity within a 30-minute period, measured in $\text{mm} \cdot \text{h}^{-1}$.

Whereas lacking rainfall data from the study area could generate inaccuracies in the analytical R calculation, straightforward empirical methods (Table 6-2) that correlated R with easier-to-acquire factors like the Fournier factor (F) or the mean yearly rainfall (P) (Terwayet Bayouli et al., 2023b).

Table 6-2. Adopted equations for calculating the erosivity factor (R)

Model	Author
$R = 0.264 \times \text{MFI}^{1.5}$	Arnoldus (1977)
$R = 587.9 - 1.219 \times P + 0.004105 \times P^2$ for $P > 850 \text{ mm} \cdot \text{y}^{-1}$	Renard Kenneth G and Freimund (1994)
$R = 0.87 - 0.037 \times P + 0.016 \times P^2$	Pepin Y and Hammouda (2012)
$R = 168.42 + 3.27 \times \text{MFI}$	Roose (1977)
$R = 15.485 + 0.602 \times \text{MFI}$	DVWK (1990)
$R = 36.46 + 1.84 \times \text{MFI} - 0.158 \times P$	Ferro et al. (1991)
$R = b_0 + b_1 \times P + b_2 \times P^2$ for $P > 850 \text{ mm/y}$, $b_0 = 0.0483 \text{ MJ} \cdot \text{ha}^{-1} \cdot \text{h}^{-1}$, and $b_1 = 1.610$	Yang et al.(2003)
$R = b_0 \times P^2 \times \sqrt{P} \times (\alpha \times b_1 \times L)$, $\alpha = 2$, $b_0 = 0.117$ and $b_1 = -0.015$	Diodato and Bellocchi (2010)
$R = 1.042 \times \text{MFI}^{1.59}$	Belaid and Habaieb (2015)

*MFI is the Modified Fournier index defined as: $\text{MFI} = \frac{\sum_{i=1}^{12} P_i^2}{P}$ where P_i : average monthly precipitation
P: average annual precipitation. L is the station longitude, $\text{MJ} \cdot \text{ha}^{-1} \cdot \text{h}^{-1}$: megajoule per hectare per hour, mm/y : millimeter per year. These formulas, tailored for Mediterranean climatic conditions, were utilized at the 12 rain gauge stations within the study watershed, based on the availability of rainfall data (Terwayet Bayouli et al., 2023b).

6.3.2.2. K determination

The soil erodibility factor K refers to the cohesiveness of soil particles and their resistivity to erosion. The permeability of soils, their structure, texture, as well as the amount of organic matter, are the main factors that affect it (Terwayet Bayouli et al., 2023b). In the current study, the French soil classification system CPCS (1967) was

used to generate the soil map of the watershed (Terwayet Bayouli et al., 2023b). K values were recognized based on organic matter and soil texture following the table proposed by Renard, (1991) (Table 6-3).

Table 6-3. Soil classes of erodibility factor (K) identified in the study area.

Soil Type	K
Raw mineral soils (Lithosols/ Regosols)	0.036
Composed Soils	0.050
Isohumic soils	0.054
Slightly evolved alluvial soils	0.080

4.3.2.3. LS determination

LS factor accounts for the slope length and slope steepness, which are key in sedimentation and transport of eroded particles within a watershed, accelerating soil loss rates per unit area (Liu et al., 2000). It is calculated as a ratio comparing soil loss on a given slope length (L) and steepness (S) to the soil loss from a reference slope (Eq.6-3), based on a 22.1 m-length and a 9% steepness (Renard, 1997).

$$LS = \left(\frac{\lambda}{22.13} \right)^m \times S \quad (6-3)$$

Where LS is the slope length-steepness factor (dimensionless); λ slope length (m); $S = 16.8 \times \sin \theta + 0.03$, if $\theta < 9^\circ$ or $S = 16.8 \times \sin \theta - 0.5$ if $\theta > 9^\circ$ and θ slope angle (McCool et al., 1987); $m = \beta / (1 + \beta)$, m variable slope-length factor and β variable slope gradient.

Determining LS factor can be complex, as it involves multiple parameters, especially terrain slopes. In this study, LS factor was calculated using DEM from SRTM image, by means of SAGA GIS software. This latter employs DEM to derive the parameters needed for LS determination, including the slope orientation, degree, and cumulative length (Terwayet Bayouli et al., 2023b).

4.3.2.4. C determination

Vegetation cover plays a significant role in controlling soil erosion, where soils with dense vegetation are more resistant to erosion compared to those with sparse vegetation (Masroor et al., 2022). Cover management (C) represents the amount of soil erosion occurring on land with a particular vegetation type relative to the erosion rates observed on bare or tilled soil (Wischmeier and Smith, 1978). The

C-factor map for our study area was created using a LULC map that was generated from a 2014 Landsat 8 image (corresponding to a normal hydrological year) at a resolution of 30 meter (Terwayet Bayouli et al., 2023b). The map was produced through supervised classification using the ‘Winner Takes All’ algorithm, supported by field visits and terrain observations. C-factor values classification was based on the work of Cormary, and Masson (1964), who ascribed these values for various land cover types in Tunisia (Table 6-4).

Table 6-4. Classification of C factor in the study area

LULC	C factor
Agriculture behind <i>Jessours</i>	0.18
Agriculture behind <i>Tabia</i>	0.18
Rangelands	0.55
Bedrock	0.9
Bare Soils	1

4.3.2.5. P determination

P factor referring to the conservation support practice, accounts for the effectiveness of erosion control measures that reduce the impact of surface runoff on soil erosion (Renard, 1991). It varies from 0 (high erosion resistance) to 1 (absence of protective practices) (Fang et al., 2019). For this study, the map displaying the P-factor spatial variability was generated on the previous generated LULC map and field observations. In the Negueb watershed, the primary anti-erosion measures include bench terraces, contour farming, and retention bench terraces, applied at various slope gradients (Terwayet Bayouli et al., 2023b). The P-values used for the study (Table 6-5) were derived from research conducted by the Food and Agriculture Organization (FAO) in Tunisia (FAO, 1977; Terwayet Bayouli et al., 2023b).

Table 6-5. P Factor for the different management practices in the watershed based on slope categories

Management Type	P
Bench terraces with or without plantations (5%-10%)	0.10
Bench terraces with or without plantations (15%-20%)	0.16
Bench terraces with or without plantations (20%-30%)	0.18
Retention bench terraces (0% - 5%)	0.10
Retention bench terraces (5% - 15%)	0.12
Retention bench terraces (15% - 25%)	0.16

6.3.3. RUSLE model outcome's reliability assessment

The validation of RUSLE outputs can be challenging in arid regions owing to the natural setting of these areas like sparse vegetation, low and irregular rainfall, and elevated evapotranspiration, and most importantly due to the lack of sufficient data for direct comparison of model predictions with observed soil erosion (Lazzari et al., 2015). To validate soil loss rates estimates generated by RUSLE model for this study, we conducted extensive field surveys, and we utilized high-resolution remote sensing imagery from Google Earth imagery and TerraIncognita to perform visual interpretation aiming at spatial-scale verification. Then, erosion-prone areas were ground-truthed by accurately locating points in the field using GPS, and these points were compared with the erosion risk map (Terwayet Bayouli et al., 2023b). Erosion categories were assigned attributed on field observations, with local experts categorizing the points into three groups: "low" representative of minimal erosion from sheet erosion, "moderate" indicating both sheet and rill erosion causing some topsoil loss, and "severe" representing significant topsoil removal due to both sheet and rill erosion (Fig. 6-8). Lastly, the validity of the RUSLE model was supported by comparing the results with similar local studies (MEHAT, 2012; Jemai et al., 2021).

6.4. Results

6.4.1. Analysis of drought conditions

VHI was adopted to evaluate the drought conditions in this study during 2000-2016 period, as it reflects the vegetation cover state, which is an important attribute for soil erosion (Fig. 6-3). The period November-April corresponds to the growing season in Tunisia, with March rainfall being particularly key factor in soil moisture, as reported by local agricultural experts. Therefore, VHI were generated based on the combination of VCI and TCI maps for the month of March, to assess drought conditions. VCI and TCI values varied between 0% and 100% (Masroor et al., 2022), with VCI showing more pronounced spatial variations compared to TCI. Similarly, VHI values ranged from 0% to 100%, where 0% represented the driest condition and 100% indicated a drought-free state. The closer the VHI is to 0, the more severe the drought, while higher VHI values reflect less intense drought conditions. The thresholds applied for VHI classification are outlined in Table 4-1 in Chapter 4, section 4.2.2.1 (Kogan, 1995). VHI analysis revealed that the study area experienced recurring droughts of varying

frequency and intensity throughout the 2000-2016 study period (Figure. 6-3). The most severe drought, indicated by the lowest VHI value, occurred in March 2001, impacting almost the whole watershed. On the other hand, 2004 and 2007 saw the highest VHI values recorded (reaching to 100%), indicating drought-free conditions (Terwayet Bayouli et al., 2023b). The CV map (Figure. 6-4a) revealed that drought-prone areas exhibited low CV values, while higher CV values were displayed in regions with mild or no drought conditions (Fig. 6-4a). The CV ranged between 34.8-58.9%, with larger values seen downstream and the lowest values found close to the watershed upstream. This pattern can be attributed primarily to the uneven distribution of rainfall, which led to increased seasonal variability in rainfall across different areas of the studied watershed. Additionally, local installed land management techniques such as *Tabias* and *Jessours* can largely contribute to this spatial variation (Terwayet Bayouli et al., 2023b). The mean VHI map is presented in Figure 6-4b illustrating the average VHI values across all pixels and time periods. The mean VHI was between 42 and 54, with the majority of the map showing high values. The overall average mean VHI accounted for 48. The drought frequency was determined to be 76.5% in the studies watershed. Although this high frequency of droughts in the region, extreme drought events were rare, occurring only twice in 2001 and 2013 (Terwayet Bayouli et al., 2023b). In contrast, severe and moderate droughts were more common, with instances recorded in 2000, 2001, 2002, 2011, 2013, and 2016. Mild droughts were prevalent during the study period, except for the some identified as normal or humid years (i.e. 2004, 2006, 2007, and 2012). The drought frequency map (Figure. 6-4c) represents the overall drought events identified by VHI below 40, during the studied period. Frequency values were between 24% and 36%, with the watershed upstream revealed more frequent droughts compared to the plains. Droughts patterns were predominantly concentrated in the eastern and central regions. The primary factors contributing to these low VHI values in these areas were the topography and low rainfall.

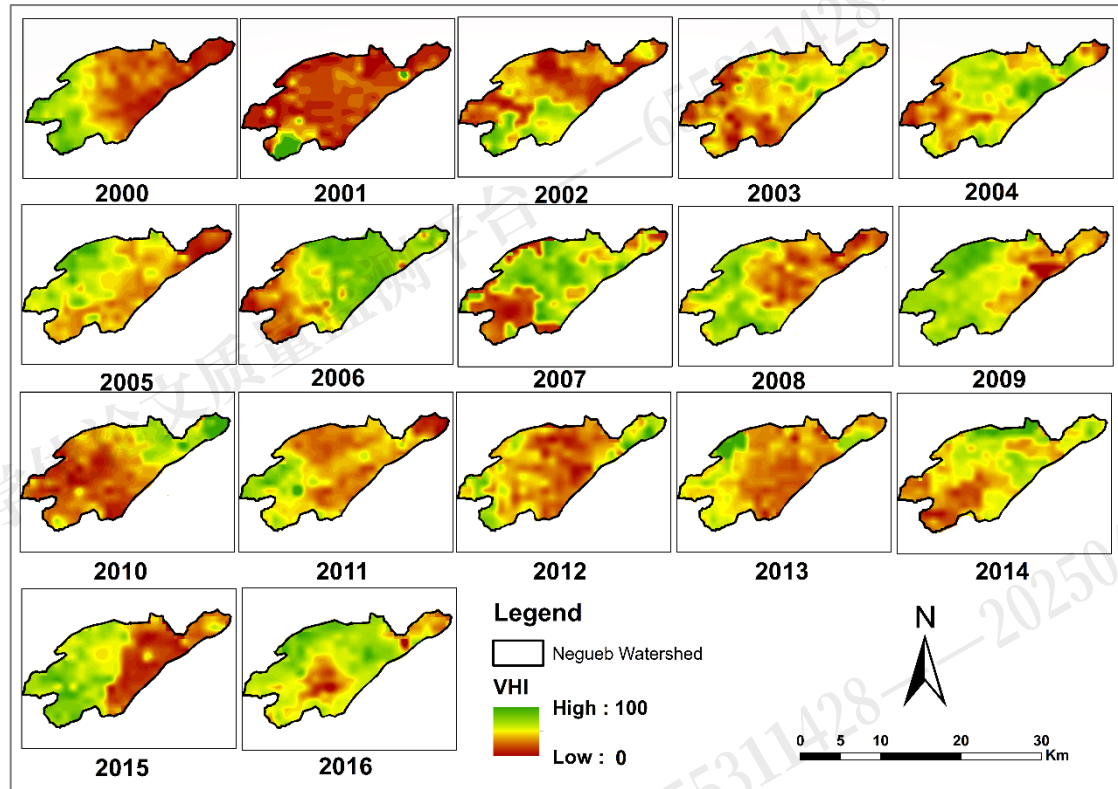


Figure 6-3. VHI-identified drought severity spatiotemporal distribution in the study area.

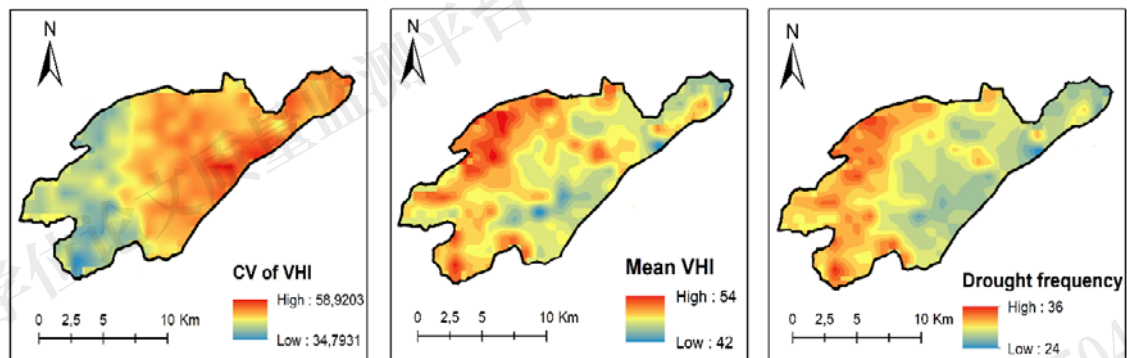


Figure 6-4. (a) Coefficient of variation of VHI, (b) Spatial distribution of VHI mean, and (c) Frequency of VHI-identified droughts.

6.4.2. Determination of erosion rates

In arid southern regions of Tunisia, soil erosion is a significant problem that is worsen by prolonged droughts. In order to guarantee a quantitative and reliable estimation of soil loss in the Negueb watershed, the current study employed the GIS-based RUSLE model.

6.4.2.1. R-factor analysis

In the studied watershed, R-factor varied from 56.36 to 169.03 MJ mm ha⁻¹ h⁻¹ (Fig. 6-5a). It increased across east-west gradient. Particularly, in the northeast and north central regions, it was low downstream of the study area with recorded R values lower than 100. Greater R values were primarily found at higher elevations in foothills and mountainous areas close to the watershed's headwaters (southwest).

6.4.2.2. LS-factor analysis

The watershed's elevation ranges between 110 and 690 m. LS-factor values were between 0.03-13.03 (Fig. 6-5b). While the lower, eastern section of the watershed has long, shallow slopes with LS close to zero, the eastern part has high, steep slopes with LS >10. Most of the watershed was covered by lower LS values (64%), which were concentrated in the central and downstream areas to the east. The second most prevalent LS values in the watershed were those ranging from two to six in the foothills between the plains and mountains (26%). The remainder of LS values (10%) were found in regions classified as mountainous with high elevation and steep high slopes, and they correlated with values greater than 6%. As a result, along east-west gradient, the slope gradient increased. LS seemed to be a main trigger of erosional loss in the watershed and was strongly correlated with soil loss (Terwayet Bayouli et al., 2023b).

4.4.2.3. P-factor analysis

The P factor is influenced by the terrain slope, in addition to the agricultural practices and erosion control measures (Ganasri and Ramesh, 2016). P values were assigned based on slope variations in this study. Low to moderate values (0.1–0.18) were attributed to areas with gentle to moderate slopes, mainly distributed across the plains and foothills in the southeastern part of the watershed (Fig. 6-5c). Greater P values (0.35) (0.35) were allocated to steeply sloping areas, especially in the northern areas and the upstream. Overall, the majority of the study area (56%) exhibited a low P-factor of 0.1.

4.4.2.4. K-factor analysis

The physical and chemical characteristics of the soil were used to determine K factor (Wischmeier et al., 1971). The K-factor ranged from 0.036 to 0.08 across the

study area (Figure. 6-5d). Analysis revealed that 38% of the watershed's surface was made up of raw mineral soils, which had an erodibility index of 0.036. There were also developed alluvial soils (35.5%) with a higher erodibility index (0.08) as a result of water erosion. Due to the combination of raw mineral soils and alluvial soils, the K values in the western upstream region were both extremely low and extremely high. Throughout the plains, regions with isohumic soils ($K = 0.54$) were observed. These shallow soils, which range in texture from sandy to sandy-loamy, were created on glaciers that sloped gently. They are regularly eroded by hydrologic processes, which leaves coarse materials on the surface.

4.4.2.5. C-factor analysis

The C-factor ranged from 1 (bare soils) to 0.18 (arboriculture). The map of C-factor was generated to illustrate the varying susceptibility of different LULC to erosion (Figure. 6-5e). Based on LULC classification, five distinct categories were identified: croplands behind *Tabias*, rangelands, croplands behind *Jessours*, bare soils, and bedrock. The classes of croplands behind *Jessours* and *Tabias*, exhibited the lowest values (0.18), indicating minimal susceptibility to erosion, while a C-factor of 0.55, was assigned to rangelands class. C values reached a peak of 1 for bare soils, which indicated the greatest erosion susceptibility and were distributed over western portion of the watershed upstream.

4.4.2.6. Quantification of soil erosion rates

By integrating thematic RUSLE factor layers, potential soil erosion was estimated in $\text{t ha}^{-1} \text{y}^{-1}$ per spatial unit. According to the soil loss map, the rate of erosion varied between 0 and $72 \text{ t ha}^{-1} \text{y}^{-1}$ (Figure. 6-5) with an average rate of $1.8 \text{ t ha}^{-1} \text{y}^{-1}$. For improved spatial interpretation, erosion was classified into three categories. The first category, representing low erosion rates less than $8 \text{ t ha}^{-1} \text{y}^{-1}$, included 86.6% of total watershed area, primarily affected the downstream regions and low-elevation plains. Rates between 8 and $20 \text{ t ha}^{-1} \text{y}^{-1}$ were categorized to express medium levels of erosion, and made up 10.5% of the catchment and was located mostly in the foothills. Values beyond $>20 \text{ t ha}^{-1} \text{y}^{-1}$ were considered within extreme erosion class, made up 2.9% of the total area of watershed and was mostly found in the mountainous areas upstream. These areas are distinguished by their rough topography, with steep slopes, and delicate soils. Greater LS and K levels were significantly correlated with areas experiencing

substantial soil erosion. Within the watershed, a clear gradient of declining erosion rates was seen from upstream to downstream and from west to east.

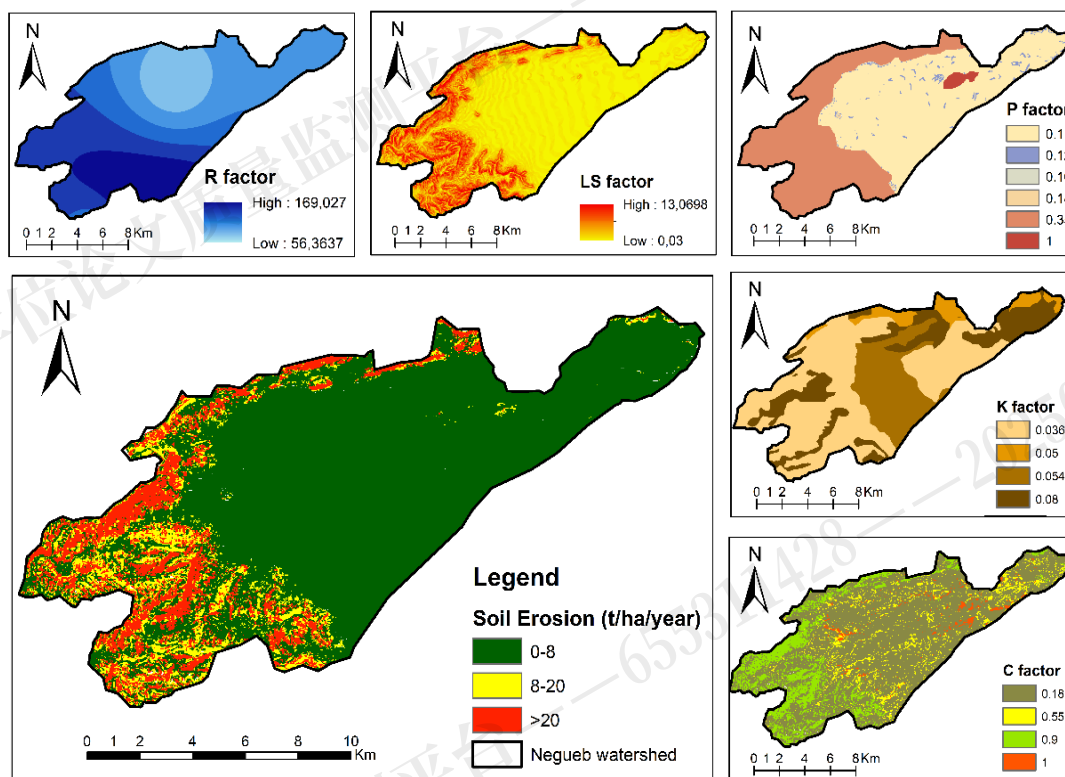


Figure 6-5. Soil erosion map with (a) R-factor, (b) LS-factor, (c) P-factor, (d) K-factor, (e) C-factor

6.4.3. Linkages between soil erosion processes and drought conditions

The relationship between soil loss rates and VHI values was assessed using Pearson correlation, in this study. Correlation analysis between soil erosion and droughts returned a mean r value of 0.3, pointing out towards a weak yet positive correlation in the studied watershed. However, the spatial distribution of correlation coefficients revealed a strong positive association in certain areas of the watershed (Figure. 6-6). The correlation coefficients (r) varied significantly across the study area, ranging between -0.5 and 0.8, with stronger correlations were recorded often in the upstream. This region corresponds to regions experiencing moderate to severe droughts and elevated erosion rates. The study area exhibits diverse geomorphological and ecological characteristics along a west-east gradient, transitioning from mountainous terrain to heterogeneous foothills and ultimately to gently sloping plains. This geographical diversity likely contributes to the overall weak correlation observed (Terwayet Bayouli et al., 2023b). In general, higher elevation areas had worse instances of drought and

greater levels of soil erosion, whereas lower elevation regions, such as the foothills and plains, recorded lower erosion and moderate or even no drought conditions (Figures. 6-3 and 6-5). The significant elevation variation within the watershed caused correlation values to decline. As a result, the geographical disparity in correlation should be taken into account when interpreting the results.

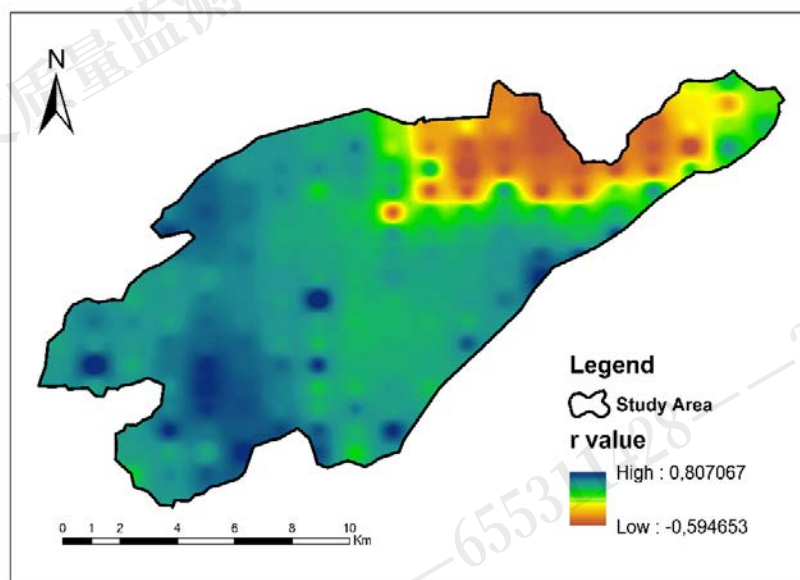


Figure 6-6. Spatial correlation between soil erosion processes and drought conditions

6.5. Discussion

6.5.1. Drought conditions

Climate change-related challenges such as drought and soil erosion pose significant threats to local communities and ecosystems in arid and semi-arid regions, particularly as their frequency and intensity continue to rise (Pachauri et al., 2014; Liu et al., 2023a). As such, it is crucial to monitor these phenomena and understand their interrelationship to mitigate their adverse effects. In this study, VHI was used to assess drought occurrence, highlighting that drought events were recurrently occurring (Figure. 6-3). Additionally, the spatial and temporal frequency assessment showed a higher occurrence of drought in elevated terrains, including mountainous and foothills. These results are consistent with earlier studies suggesting that higher-altitude zones are particularly sensitive to climate variability, especially drought, when compared to lower-lying regions (Rangwala and Miller, 2012; Alamdarloo et al., 2018; Dahal et al., 2021). According to Feng et al. (2020) drought occurrence is closely linked to altitude-related climate patterns. Arid mountain areas face heightened risk due to warming

trends at elevated levels, which can interfere with natural water processes and reduce streamflow. Moreover, the influence of the Mediterranean Sea plays a notable role in shaping annual rainfall patterns across the study region. Episodes of severe dryness were rare, with a frequency of approximately 18%, which corresponds with earlier findings from Tunisia and similar Mediterranean settings (Feki et al., 2012; Bayer Altin and Altin, 2021). The study area endured its most severe drought in terms of intensity, duration, and magnitude between 2000 and 2002. The primary drivers of this extreme dry period were variable intraseasonal rainfall and consistently low average seasonal precipitation (Terwayet Bayouli et al., 2023b). The impacts of this drought were evident in the reduced agricultural productivity at national scale (FAO, 2016; Ghoneim et al., 2017; Verner et al., 2018a). The findings were consistent across the Mediterranean region (Cook et al., 2016; Ballah and Benaabidate, 2021). The VHI-identified droughts closely mirrored the events detected by the SPI throughout the same geographical zone (Terwayet Bayouli et al., 2023a), supporting VHI use as a dependable metric for drought monitoring in Tunisia. Observations of VHI during the vegetation growth period revealed heightened sensitivity to drought in arid zones, particularly when vegetation was at its peak growth stage, echoing conclusions drawn by Bento et al. (2020). Furthermore, regions dominated by sandy soils were subjected to more intense drought stress than those with silty-clay compositions, which are known for superior moisture retention. This highlights the important influence of soil properties in shaping the spatial patterns of drought severity.

6.5.2. Soil erosion processes

The rainfall erosivity (R) findings highlight a noticeable rise through east-west direction along the elevation gradient, which appeared to affect the spatial variation in erosivity. K factor was divided into four principal classes, indicating that the majority of soils had moderate to high levels of permeability. Sandy soils were generally more prone to erosion, whereas clay-rich soils demonstrated greater resistance to erosive forces. This underscores the importance of soil composition and organic carbon levels as critical factors influencing erosion severity (Duiker et al., 2001). Arid weather patterns can significantly impact the soil erodibility factor (K), potentially increasing its sensitivity to erosion, making it essential to account for drought impacts for a more precise assessment (Baskan, 2021). The spatial distribution of the LS-factor closely aligned with the overall topography, reflecting the increased waterflow velocity. As

such, the LS-factor played a significant role in driving soil erosion processes. The impact of LULC on soil erosion became more evident through the assessment of the C-factor (He et al., 2020). Lower C-factor values were observed in flatland areas with dense vegetation cover, whereas elevated values were noted in barren and hilly terrains, suggesting higher susceptibility to erosion. These findings reinforce the utility of remote sensing approaches in evaluating the C-factor (Guo et al., 2023). To address this issue, Soil and Water Conservation (SWC) practices—such as bench terraces and retention ditches—have been adopted in the region as part of a broader national effort to control erosion. Despite the higher P values observed in the upstream and northern regions, these areas experienced greater soil erosion vulnerability. This can be attributed to the decreasing efficiency of conservation systems as the slope surges. Hence, it is advised to adopt more appropriate soil management methods and preserve the soil management structures in the upstream regions, where erosion risk is more pronounced, in order to reduce soil loss. Soil erosion generally increased from east to west in the study area, with the low-lying plains region experiencing a relatively low risk of erosion.

The annual soil loss ranged between 0.01 and 72 t ha⁻¹ y⁻¹, with an average of 1.8 t ha⁻¹ y⁻¹. These findings align with previous studies conducted in Tunisia in the context of arid conditions (Chafai et al., 2020). The high soil erosion risk in the upstream areas can be attributed to both drought susceptibility and dry soil characteristics. These regions, endowed with high relief and steep slopes, are predominantly covered by arboriculture and sparse grasslands. Additionally, the area is made up of raw mineral soils and minimally developed alluvial soils, both of which are highly prone to erosion. Although all five RUSLE factors contribute to the soil erosion rate, their influence varies in magnitude. The LS-factor was found to be the most significant driver of soil erosion in the Negueb watershed, supporting previous studies that suggest erosion increases exponentially with slope steepness (Markose and Jayappa, 2016). R and K factors were also recognized as critical elements that exacerbate soil erosion in the area. On the other hand, the C and P factors had less impact, C and P factors had a relatively lesser inducing impact. Jemai et al. (2021) found that mountainous areas of El-Hamma basin, were at higher risk of erosion. This region is geographically adjacent to our study area with comparable climatic and environmental traits, confirming thus the reliability of our results. Similar trends of vulnerability of mountainous regions, have been reported in north African countries and global studies (Amellah and el Morabiti, 2021; Zhao et al., 2021; Bensekhria and Bouhata, 2022).

We carried out a spatial-scale verification to corroborate soil erosion results through visual interpretation using high-resolution imagery from Google Earth and TerraIncognita and further confirmed the findings by field visits for observing the affected areas. The validation process showed the areas that are most susceptible to erosion were found upstream along the mountains marking the western frontier of the watershed (Figures. 6-7 and 6-8). This was attributed to thin surfaces of soils and scant vegetation cover. The hillsides were predominantly bare, exhibiting important water erosion features, with uncovered weathered limestone. Large chunks of limestone detached, forming scree slopes at the foothills. Water erosion further carved deep valleys into the fine rocks, and the material was deposited downstream, forming alluvial cones or loess layers (Terwayet Bayouli et al., 2023b).

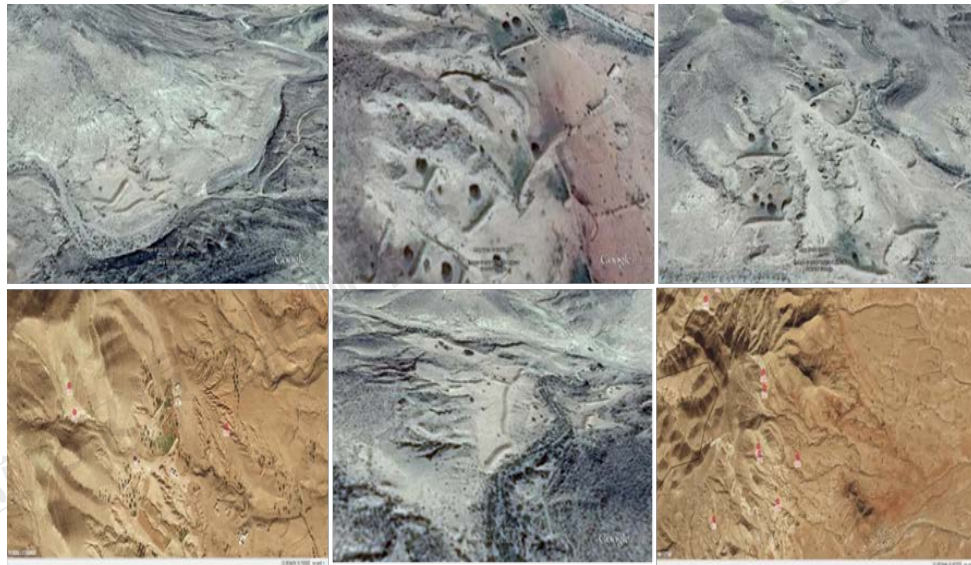


Figure 6-7. Erosion-affected areas identification through Google Earth and Terra Incognita.

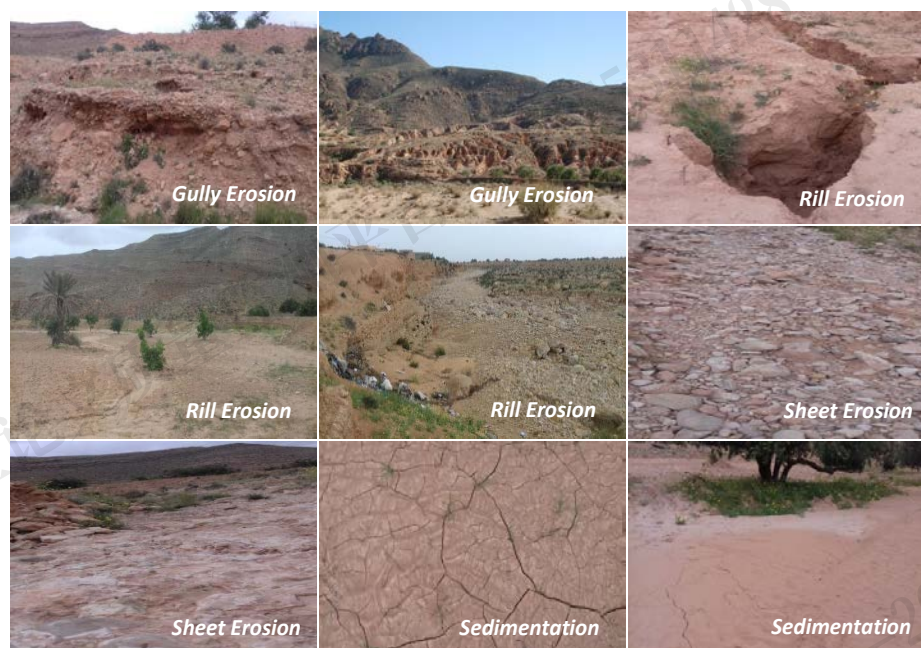


Figure 6-8. In-situ validation of erosion-affected regions in the study area.

The areas most impacted by droughts were also the ones experiencing the greatest soil loss rates, predominantly situated upstream (Fig. 6-4c and 6-5). The severe soil erosion in these mountainous regions can primarily be ascribed to the combined effects of dry soil conditions, high vulnerability to drought, minimal vegetation cover, and insufficient land management practices.

6.5.3. Relationships between drought conditions and soil erosion processes

The investigation into the correlation between the average soil erosion rate and drought in the study area exhibited a positive but weak relationship ($r = 0.3$) between soil erosion and drought intensity. While this general correlation was not very strong, spatial analysis highlighted a more pronounced association was observed in the upstream areas, which were also the most susceptible to drought and soil erosion. Similar findings were reported by Yu et al. (2021), who found a correlation of $r = 0.27$ between drought frequency and soil erosion intensity, indicating a comparable trend. Soil erosion in the study area exacerbated the effects of drought, negatively impacting both local and regional ecosystem services. Intense erosion leads to the reduction of soil depth, which in turn lowers the soil's ability to retain moisture (Magesh and Chandrasekar, 2016). This decrease in soil moisture, especially over extended durations, inhibits vegetation growth and leads to the decline of natural vegetation (Zhu et al., 2022). Consequently, these areas become more vulnerable to

additional drought conditions. Conversely, droughts can exacerbate soil erosion, as dry soils are more susceptible to wind and water-induced mass movement. The loss of vegetation cover reduces water infiltration and increases surface runoff (Le Houérou, 1996; Hewelke et al., 2022). Together, these factors lead to soils that are increasingly prone to rapid erosion (Ahn et al., 2013; Bombino et al., 2019). The irregular and torrential rainfall regime in the study area, following dry periods, trigger soil detachment and the loss of fertile topsoil, which threatens land productivity, especially in arid regions. During field visits for RUSLE model validation, uncontrolled runoff from terraces in mountainous areas was observed to contribute to gully formation (Fig. 6-8). Rill and sheet erosion were more common due to land use changes, improper soil management, and human activities, particularly livestock overgrazing. To address soil loss, implementing effective land management practices is essential for future interventions. Enhancing soil resilience via compost, organic manures, and soil conservation measures on sloping lands is suggested as a "triple win" strategy. This approach can help reverse land degradation, improve climate resilience, and promote sustainability in vulnerable arid regions.

6.6. Conclusion

This study investigated the key drivers contributing to soil erosion processes and drought conditions occurrences across Negueb basin, in southeastern Tunisia. Droughts were assessed through VHI, which is derived from the combination of VCI and TCI for the period from 2000 to 2016. To estimate soil erosion, the RUSLE model was applied, and the relationship between drought and soil erosion was analyzed through Pearson's correlation coefficient.

The results indicated that droughts were a recurring event throughout the 2000-2016 period. Additionally, spatial analysis of drought across the studied basin revealed that the upstream regions, particularly the elevated western areas, were more susceptible. The average yearly soil erosion rate was $1.8 \text{ t ha}^{-1} \text{ y}^{-1}$, with LS identified as the primary contributor to soil erosion. When examining the link between droughts and soil erosion, a slight positive correlation of 0.3 was found. However, the spatial correlation between soil loss and droughts varied between -0.5 and 0.8, showing strong and significant correlations in areas frequently experiencing droughts and higher rates

of soil erosion. This indicates a connection between soil erosion and drought occurrences.

RUSLE model and VHI were effective tools for evaluating soil erosion and droughts, highlighting the upstream areas as highly vulnerable to climatic changes and key targets for soil conservation efforts. However, there were limitations in accurately modeling erosion risks using RUSLE, such as the resolution and validation of input data due to the scarcity of local data. To address this, it is recommended to prioritize high-resolution imagery, especially for vegetation, as it plays a crucial role in both soil erosion and droughts. Future research should focus on assessing the impacts of erosion and drought on land productivity in Tunisia's arid ecosystems.

Chapter 7 The increasing risk of future drought risk in Tunisia: CMIP6 Model-Based projections under Shared Socioeconomic Pathways (SSPs) Scenarios

7.1. Introduction

Climate warming is an unequivocal global challenge, with global temperatures increasing at unprecedented pace since 2012, driving the intensification of hydroclimatic extremes worldwide (Hu and Fedorov, 2017). Among these extremes, drought has emerged as one of the most destructive threats to water security, ecosystems, and socio-economic stability (Lee et al., 2023). The Mediterranean Basin, already recognized as a climate change hotspot, faces disproportionate risks due to accelerated warming and precipitation declines (Sellami et al., 2016; Cos et al., 2022; Urdiales-Flores et al., 2024).

In this context, Tunisia exemplifies a critical case, where observed warming rates of approximately 0.50°C per decade since the 1970s exceed global averages by over 300%, and projections indicate a further rise of 3–7°C by 2100 under high-emission scenarios (Pachauri et al., 2014; Cos et al., 2022). Concurrently, annual rainfall is expected to decline by 10–40%, with pronounced reductions during the wet (October–April) and dry (May–September) seasons (World Bank, 2024). These trends, driven by higher global temperatures and increased rainfall unpredictability, are expected to trigger more frequent and severe drought events, thereby diminishing freshwater availability and exacerbating hydrological and environmental impacts (Candela et al., 2012; Copetti et al., 2013; Van Loon, 2015; Porciello et al., 2020; Tramblay et al., 2020a; Mohamed et al., 2023b). For instance, the Global Drought Observatory claimed that the 2022 Mediterranean drought, exacerbated by persistent precipitation deficits and heatwaves, led to substantial declines in crop yields and river discharge. Tunisia, in particular, has faced recurrent droughts, leading to reduced water storage, freshwater availability, and agricultural productivity (Terwayet Bayouli et al., 2022).

Several studies have focused on predicting climate variability to enhance drought prediction, monitoring, and early warning systems (Foughali et al., 2015; Besbes and Chahed, 2023; Mlayah et al., 2024). Verner et al. (2018a) found that recurrent extreme climate events, particularly multiyear droughts, along with high rainfall variability significantly challenge Tunisian agriculture, especially for key crops such as wheat and

dairy. This growing threat underscores the necessity of thoroughly investigating the expected shifts in drought conditions induced by climate change across Tunisia. Yet, research specifically exploring future drought characteristics, including their spatial extent, frequency, and severity, remains limited.

The Coupled Model Intercomparison Project (CMIP), organized by the World Climate Research Program (WCRP), has provided extensive outputs from GCMs that underpin influential climate change assessment reports (e.g., IPCC AR4 and AR5). These models have historically been used to evaluate potential future changes in climate variables under various emission scenarios, including the Special Report on Emission Scenarios (SRESs) and Representative Concentration Pathways (RCPs) (Taylor et al., 2012; Raju and Kumar, 2020). Previous studies largely relied on CMIP3 and CMIP5 outputs, which indicated an increased frequency and duration of droughts, particularly in mid-latitude regions (Christensen et al., 2007; Amorim Borges et al., 2014), although substantial uncertainties remain (Knutti and Sedláček, 2013; Carvalho et al., 2022). Recent updates provided by CMIP6, which now incorporate shared socioeconomic pathways (SSPs) and offer improved horizontal resolution and representation of synoptic processes (Eyring et al., 2016), have enhanced the accuracy in simulating historical climate variables, extremes, and ocean-atmosphere interactions (Tokarska et al., 2020; Song et al., 2021; Su et al., 2021; Gulakhmadov et al., 2025). Moreover, CMIP6 models indicate a higher occurrence of reliable interactions among ENSO, the Indian Ocean Dipole (IOD), and other climatic phenomena compared with their CMIP predecessors (McKenna et al., 2020). Although several studies have applied CMIP6 models to project global and regional drought trends, with robust drying trends over large parts of the Mediterranean, including Tunisia (Cook et al., 2020; Dong et al., 2022; Seker and Gumus, 2022; Essa et al., 2023; Xu et al., 2023a), no studies have yet specifically addressed drought characteristic projections for Tunisia using CMIP6 models. Previous research has been spatially constrained and conducted at small-scale catchments (Lamon et al., 2014; Moussa et al., 2018; Mlayah et al., 2024), which may overlook localized variations and short-term extreme events. Consequently, a significant gap remains in understanding the full scope of future droughts characteristics in Tunisia, including their spatial extent and severity.

This study addresses this critical gap by providing a comprehensive assessment of future drought characteristics in Tunisia, including duration, extent, frequency, and intensity, across multiple spatiotemporal scales. Using the ensemble mean of three

GCMs from CMIP6 models, this study projects drought conditions for three time periods: 2016–2040 (near-term), 2041–2070 (mid-term), and 2071–2100 (long-term), under the contrasting SSP1-2.6 and SSP5-8.5 scenarios. As the first in-depth investigation of future drought characteristics in Tunisia using CMIP6 models across SSPs scenarios, our study provides important insights into the potential effects of climate change on regional drought risk., thereby providing a robust scientific foundation for policymakers and stakeholders to enhance disaster risk management, implement mitigation measures, and develop proactive adaptation and resilience strategies.

The chapter is structured as follows: Section 2 delves into specifics regarding the data sources, Global Climate Models (GCMs), Shared Socioeconomic Pathways (SSPs), study timeframe, and methodologies for drought analysis. Section 3 showcases the findings of future drought spatiotemporal series and changes in significant drought characteristics, followed by discussions. Finally, Section 4 provides a summary and concluding remarks, and suggests targeted management strategies to prevent and mitigate drought impacts in Tunisia

7.2. Data and Methods

7.2.1. Datasets, Models, SSPs, and Projecting Periods

This study evaluates future climate change impacts on Tunisia using multiple GCMs and SSPs (Fig. 7-1). CMIP6-model outputs, including monthly precipitation, maximum temperature, and minimum temperature, for historical (1981-2015) and future (2016-2100) periods were obtained from <https://esgf-data.dkrz.de/search/cmip6-dkrz/>. CMIP6 datasets were chosen for their bias-correction using historical data and proven reliability, especially for temperature projections critical to drought drought assessments (Mathbout et al., 2023; Varghese et al., 2025).

Three GCMs simulations, namely IPSL-CM6A-LR, MIROC6, and MRI-ESM2-0, were selected based on their demonstrated accuracy in capturing historical climate variability and simulating key variables like precipitation and temperature in Tunisia (Lamon et al., 2014; Besbes and Chahed, 2023; Achour et al., 2024). These GCMs are recognized for their higher resolution, improved representation of land-atmosphere interactions, and global performance, critical for drought-related assessments over the Mediterranean (Drobinski et al., 2020; Seker and Gumus, 2022). Data were downscaled

to 0.25° resolution and the ensemble mean of GCMs was used to assess future drought characteristics changes, improving the reliability of projections by reducing individual model biases. Details of the selected models are provided in Table 7-1.

Five SSPs outline diverse socioeconomic and climate trajectories: SSP1 (sustainability), SSP2 (middle), SSP3 (regional rivalry), SSP4 (inequality), and SSP5 (fossil fuel development) (Huang et al., 2019). This study focuses on SSP1-2.6 and SSP5-8.5, updated versions of RCP2.6 and RCP8.5, respectively, from CMIP5, selected to explore the extremes of potential climate futures: SSP1-2.6 represents a low-emission pathway aligned with global mitigation goals, while SSP5-8.5 reflects a high-emission trajectory with minimal mitigation efforts (Meinshausen et al., 2019). These scenarios were chosen to capture the upper and lower bounds of future drought impacts.

Table 7-1. Climate models description

Climate model	Ensemble member	Resolution
MRI-ESM2-0	r1i1p1f1	1.12° × 1.12° re-gridded to 0.25°
IPSL-CM6A-LR	r2i1p1f1	2.50° × 1.26° re-gridded to 0.25°
MIROC6	r1i1p1f1	1.40° × 1.40° re-gridded to 0.25°

The study spans 1981-2100, with a baseline period (1981-2015) and future projections (2016-2100), for climate analysis. Given GCM simulations began in 2015, historical data for 1981-2015 were sourced from ERA5 (<https://app.climateengine.com/climateEngine>) (Section 3.2.1.3 of Chapter.3, Figure A.9 in Appendix chapter).

To evaluate GCMs accuracy, SPEI values were compared with SPEI derived from in-situ observations from 22 rain gauge stations provided by the General Directorate of Water Resources of Tunisia (DGRE) during 2015–2021 (Figure A.8. in appendix chapter). Validation metrics included the correlation coefficient (CC) (Equation. 7-1), root-mean-square error (RMSE) (Equation. 7-2), mean absolute error (MAE) (Equation. 7-3), and mean bias error (MBE) (Equation. 7-4), defined as:

$$CC = \frac{\sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} \quad (7-1)$$

with:

- x_i and y_i are the modeled and observed values, respectively.

- $\bar{x}$ and $\bar{y}$ are the mean modeled and observed values.
- n is the number of data points.

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (x_i - y_i)^2}{n}} \quad (7-2)$$

RMSE provides an indication of the average magnitude of errors.

$$MAE = \frac{1}{n} \sum_{i=1}^n |x_i - y_i| \quad (7-3)$$

MAE measures the absolute differences between modeled and observed values.

$$MBE = \frac{1}{n} \sum_{i=1}^n (x_i - y_i) \quad (7-4)$$

MBE indicates the systematic bias in the model predictions.

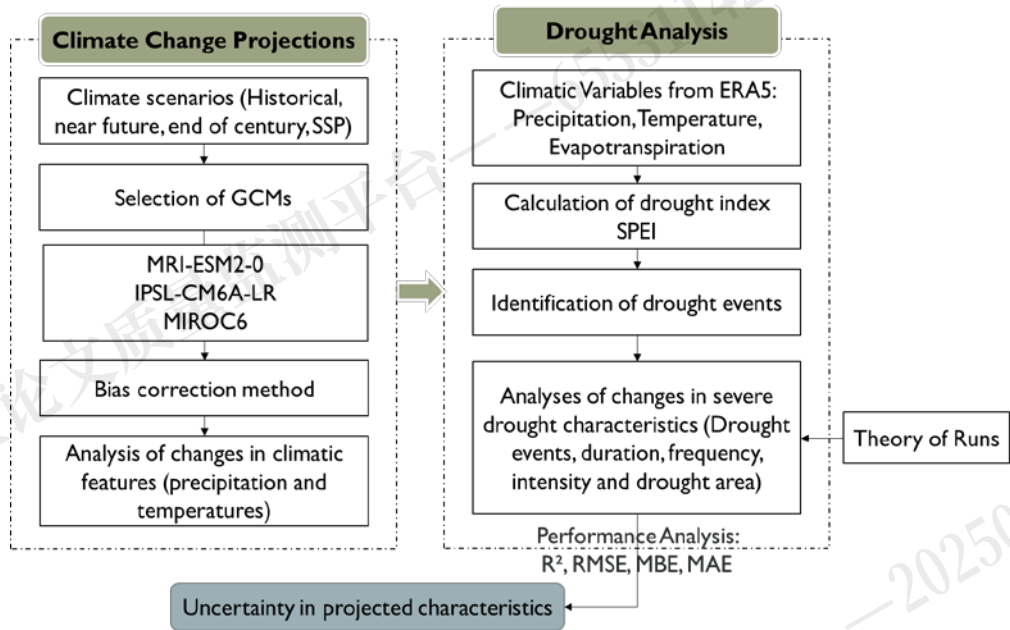


Figure 7-1. Methodological flowchart of future drought projection.

7.2.2. Drought characterization

SPEI was used to quantify future droughts, incorporating precipitation and potential evapotranspiration (PET) to assess water balance (P-PET) and temperature effects (Vicente-Serrano et al., 2010). SPEI-12 was computed using ensemble averages of maximum and minimum temperatures and precipitation from all GCMs., chosen for its effectiveness in arid regions like Tunisia. Drought conditions were identified using

a threshold of $SPEI < -0.5$ (refer to Table.3-2 in chapter.3, section 3.2.2.2), following for computational procedures cited in chapter.4, section 4.2.2.1.

Moreover, drought characteristics, including intensity, events, area, duration, and frequency (refer to Chapter 6, Section 6.3.1), are key indicators of drought severity and climate change impacts (Cook et al., 2014; Liu et al., 2019b). A boxplot was used to compare projected changes in these characteristics from the baseline (1981–2015) to long-term projections (2016–2100) under various SSPs, with percentage deviations highlighting future drought frequency and severity over the 21st century.

7.2.3. Analysis of Projected Drought Uncertainty

The assessment of future climate change projections involves inherent uncertainties originating from various sources. To address these uncertainties, an ensemble-based experiment was conducted, computing SPEI for each GCM. The minimum and maximum SPEI values defined the uncertainty range, while the mean of the three models represented average trends in future drought conditions under SSPs 1-2.6 and 5-8.5 during 1981-2100, providing insight into potential uncertainty ranges associated with GCM projections.

7.3. Results

7.3.1. Evaluation of GCMs Models

The reliability of GCMs was examined using various evaluation metrics (Table 7-2) for SPEI, with values computed for 2015-2021 to evaluate model performance. Among the models, MIROC-6 under SSP5-8.5 exhibited the highest correlation with the observed data ($r = 0.7$), effectively capturing both the trend and magnitude of SPEI variability. It also exhibited the lowest RMSE and MAE, highlighting higher accuracy in replicating observed patterns. However, its MBE (1.03) indicated a systematic overestimation of SPEI, potentially underestimating drought severity in some scenarios, consistent with He et al. (2023). IPSL under SSP1-2.6 revealed the smallest bias, reflecting moderate SPEI overestimation, which could be advantageous in certain predictive contexts. Conversely, IPSL under SSP5-8.5 showed higher RMSE and MAE, indicating reduced accuracy compared to MIROC-6 but better performance than MRI. MRI models generally showed significant overestimation biases, lower correlations, and higher RMSEs, resulting in less reliable predictions. Overall, MIROC-6 under SSP5-8.5 proved the most reliable model, despite its tendency to overestimate SPEI.

Table 7-2. Validation Metrics for SPEI values derived from GCMs under SSP1-2.6 and SSP5-8.5 during 2015–2021.

GCM Model and Scenario	Mean	Standard Deviation	R	RMSE	MAE	MBE
IPSL SSP5-8.5	0.965	0.401	0.283	1.313	1.114	0.969
MIROC SSP5-8.5	1.020	0.766	0.696	1.222	1.049	1.025
MRI SSP5-8.5	1.138	0.442	0.131	1.560	1.304	1.143
IPSL SSP1-2.6	0.578	0.837	0.288	1.516	1.113	0.582
MIROC SSP1-2.6	1.074	0.762	0.118	1.551	1.305	1.078
MRI SSP1-2.6	1.114	0.906	0.275	1.562	1.261	1.118

Figure 7-2 compares SPEI values from different models against in-situ observations across scenarios. MIROC-6 under SSP5-8.5 clustered closely around the line of perfect agreement, confirming its high correlation, though a subset of points above this line underscores its overestimation tendency. In contrast, MRI under SSP5-8.5 displayed a scattered distribution, indicating poorer agreement with observations, while IPSL model under SSP1-2.6 showed a similar scatter but with reduced dispersion, reflecting moderate overestimation. Discrepancies among models likely stem from observational uncertainties, differences in averaging period, and the inherent variability of high-emission pathways, which amplify extreme climatic conditions. Moreover, the incomplete representation of teleconnection effects (e.g., ENSO) may further contribute to deviations from observed values.

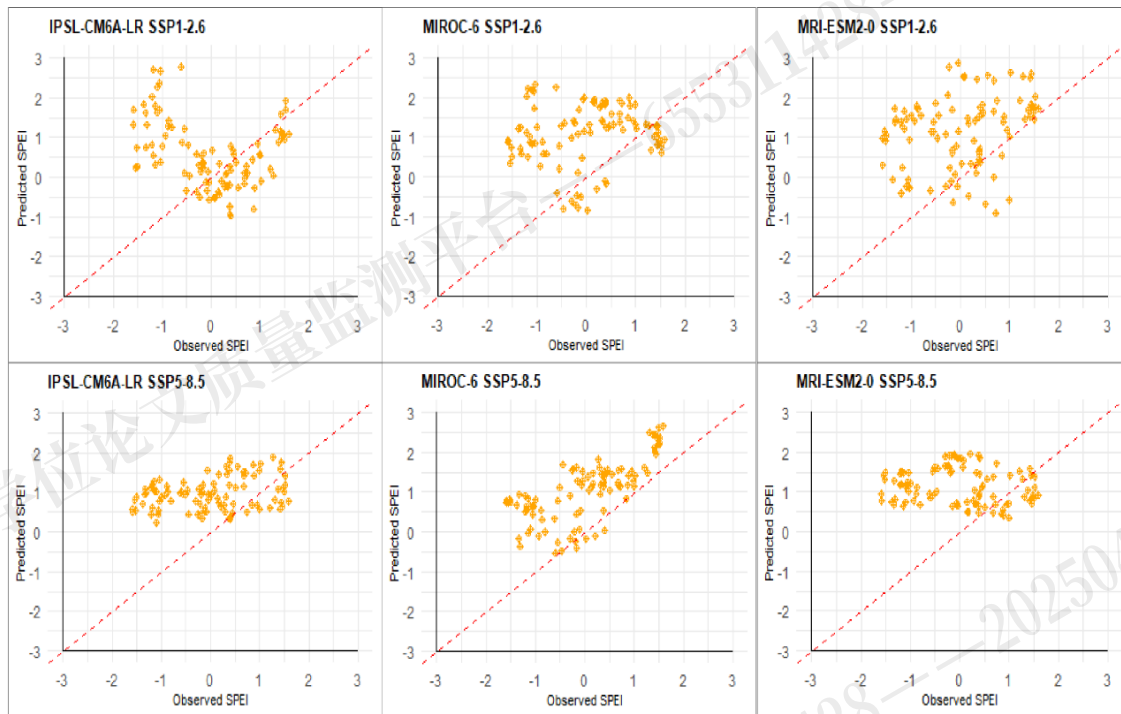


Figure 7-2. Comparative Scatter Plots of Modeled vs. Observed SPEI Across the Different used GCMs (IPSL, MIROC-6, and MRI) under scenarios SSP1-2.6 and SSP5-8.5.

7.3.2. Long-Term Drought Projections Using SPEI

Figure 7-3 presents the ensemble-projected SPEI time series for Tunisia (1981-2100) under both SSPs scenarios. Combining precipitation and PET, SPEI effectively measures moisture deficits and surpluses, crucial for assessing climate change impacts on water resources (Vicente-Serrano et al., 2010; Liu et al., 2021). The ensemble projections from the three CMIP6 models a steady decline in SPEI values over time, signaling an overall intensification of drought severity and frequency.

Historical patterns (e.g., sharp declines in the early 1990s and mid-2000s) persist into the near-term, but mid- and late-century projections under SSP5-8.5 reveal greater variability and more frequent drought events compared to SSP1-2.6. In particular, SSP5-8.5 exhibits extreme fluctuations and prolonged recovery periods, suggesting enhanced climatic instability and drier conditions. These projections align with regional studies reporting continuous negative SPEI trends in Tunisian forests (Touhami et al., 2023) and historical evidence of severe droughts since the 15th century (Touchan et al., 2008), along with documented impacts on soil moisture and agricultural productivity (Terwayet Bayouli et al., 2022).

Overall, the convergence of historical and near-term trends under both scenarios implies that early mitigation efforts may require time to yield significant benefits.

However, the cumulative effects of greenhouse gas emissions under SSP5-8.5 lead to markedly more severe droughts than those projected under SSP1-2.6, underscoring the critical importance of emissions reductions and robust adaptation strategies for safeguarding Tunisia's water resources and ecosystems.

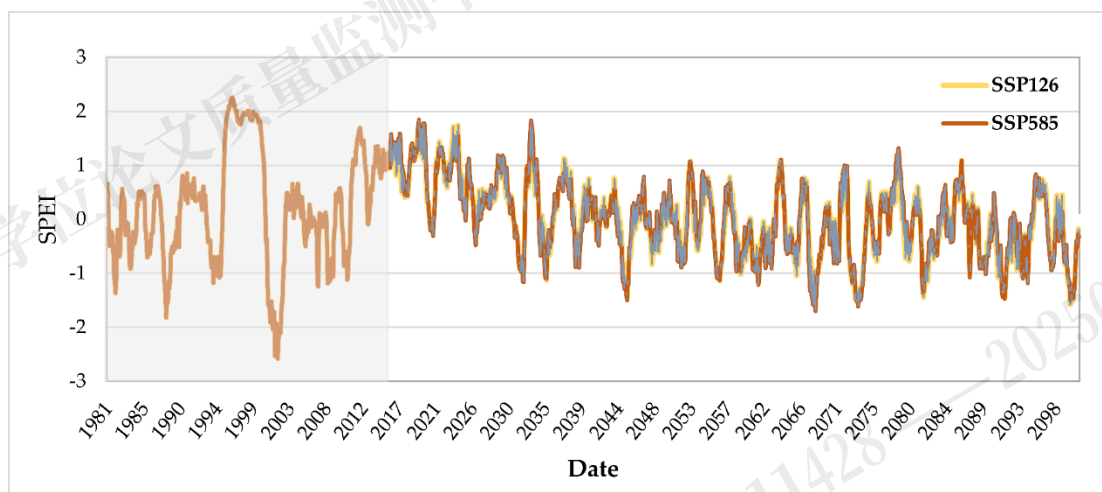


Figure 7-3. Projected SPEI time series under SSP1-2.6 and 5-8.5 scenarios during 1981-2100. The data reflects the GCMs ensemble mean. The grey zone marks the baseline period, while the dotted vertical line separates the study periods: near-term (2016–2040), mid-term (2041–2070), and late-term (2071–2100).

7.3.3. Uncertainty Analysis of SPEI-Based Drought Projections

Understanding SPEI projection uncertainties is crucial for effective climate adaptation and mitigation. Droughts are expected to increase in frequency and severity by the late 21st century, as indicated by mean SPEI values under SSP scenarios. Figure 7-4 illustrates the uncertainty analysis of mean SPEI time series, highlighting an upward uncertainty trend alongside future drought projections, driven mainly by precipitation variability rather than temperature. This aligns with Gao et al. (2020), who identified internal climate variability and GCM datasets as major uncertainty sources for mid- and late-century projections.

Additional contributors include variability in precipitation and PET datasets, as well as large-scale oscillations like ENSO and La Niña (Nguyen et al., 2021). Regional complexities, including Tunisia's terrain, soil heterogeneity, and climate interactions, further amplify uncertainties (Batjes, 2010; Verner et al., 2018a). GCM assumptions and initial conditions also introduce variability in drought projections (Ghosh and Mujumdar, 2007; Duan and Mei, 2014).

Despite uncertainties, the projections consistently indicate worsening drought conditions for Tunisia, particularly towards the century's end, emphasizing the importance of tailored hydroclimatic and geographic considerations. Addressing these uncertainties necessitates rigorous evaluations of precipitation variability, regional climatic interactions, and model assumptions.

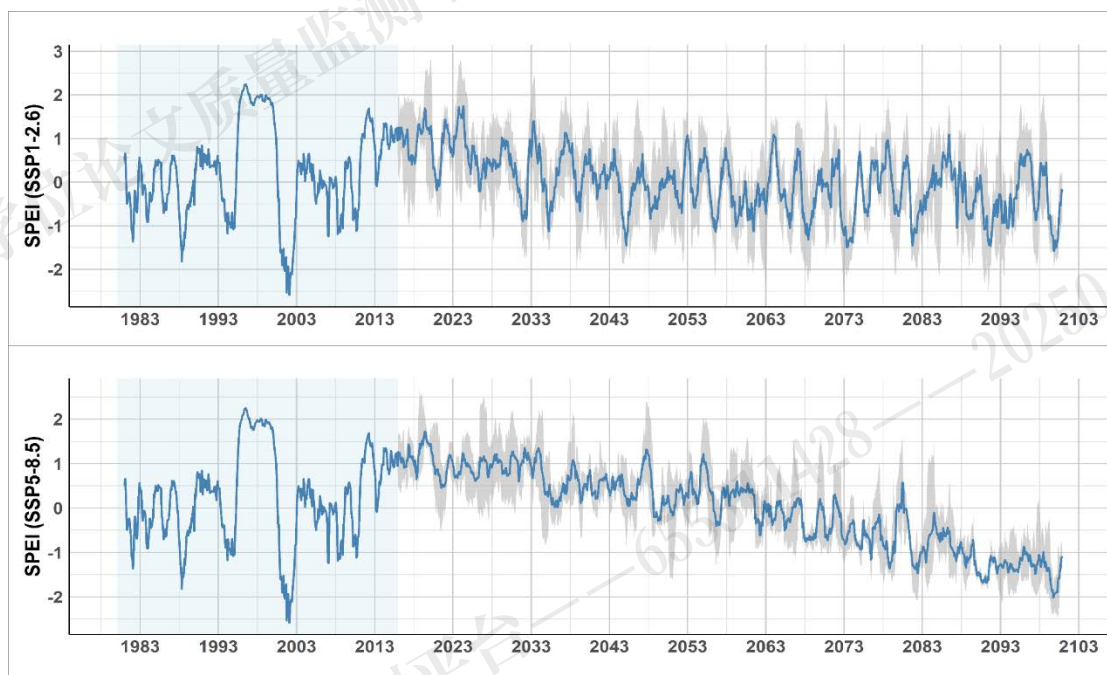


Figure 7-4. Projected SPEI time series (1981–2100). The blue area denotes the baseline period, while the gray shading indicates uncertainty, bounded by the GCMs' minimum and maximum SPEI values. The mean is derived from the ensemble of SPEI values across GCMs.

7.3.4. Future SPEI changes

Changes in precipitation and PET influence dry and/or wet events occurrences (Yang et al., 2024; Fanta et al., 2025). Figure 7-5 compares projected SPEI-based dryness and wetness trends in Tunisia to the baseline period. Significant shifts in dryness align with precipitation and PET variations relative to the reference period.

Under SSP1-2.6, baseline conditions initially showed high variability, with lower outliers indicating extreme dry events (Fig. 7-5a). The boxplot showed a slight shift toward wetter conditions emerged in the mid- and late 21st century, with the median slightly above zero. The Interquartile Range (IQR) indicates reduced variability, suggesting more stable conditions with fewer extreme events. In the late-term period, the median returns closer to zero, and the IQR widens, indicating increased variability and the potential for future dry spells.

For SSP5-5.8 (Fig. 7-5b), slight change towards wetter conditions was observed in the near-term, with significant variability indicated by wide IQR. The presence of extreme and dry events is possible as suggested by lower and upper outliers. Mid-century conditions showed stabilization with fewer extremes, while the late-century period approached baseline, with reduced variability and fewer extremes. This suggests greater extreme monthly variations as emissions increase. Overall, SSP5-8.5 exhibited less variability than SSP1-2.6, highlighting a more stable prediction of future drought characteristics.

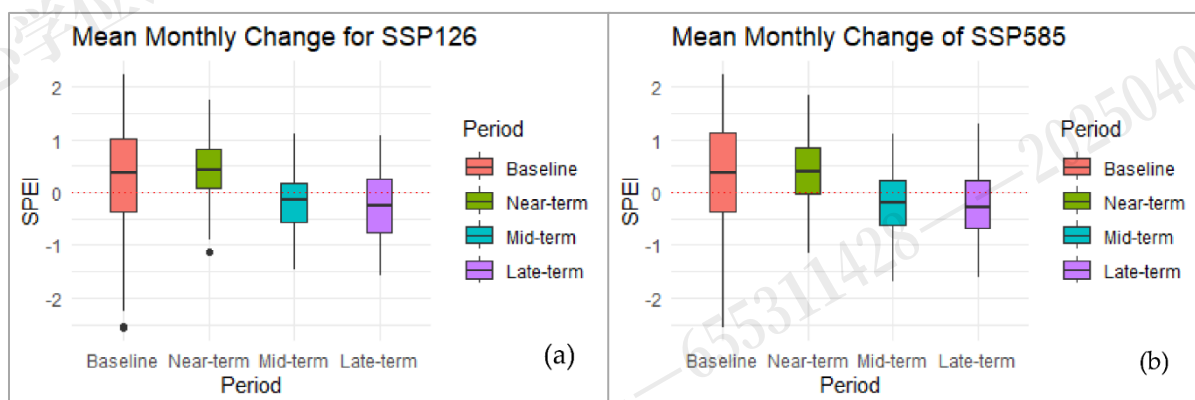


Figure 7-5. SPEI-detected changes in dryness and/or wetness for 2016–2040, 2041–2070 and 2071–2100 periods under SSP1-2.6 (a) and SSP5-8.5 (b) scenarios, compared to the baseline across Tunisia. Boxes show the interquartile range (IQR), horizontal lines indicating the median, whiskers extend to values within 1.5 times IQR, and crosses mark outliers beyond this range.

7.3.5. Long-Term Changes of Projected Drought-affected Area

7.3.5.1. Projections of Changes in Drought Area

Figure 7-6 presents the projected percentage of Tunisia's drought-affected area ($SPEI \leq -0.5$) between 1981 and 2100, under SSP1-2.6 and SSP5-8.5 climate projecting scenarios. The mean changes in areas impacted by droughts were calculated for each period relative to the baseline (1980–2015). Under SSP1-2.6, drought area remains relatively stable, with mean values fluctuating between 35%–40% across all periods, with a narrow uncertainty range, indicating greater confidence in these projections. In the near-term, drought areas decrease by 5.72%, averaging a change of 37.44%, and a peaking at 41.24% in 2039. However, by the mid-term, drought areas begun to increase slightly 3.01%, reaching 48.22% in 2066, and further expand by 6.60% in the long-term, to 46.52% in 2097. In contrast, SSP5-8.5 shows a consistent upward trend across

all periods. In the near-term, areas increase by 3.12%, with mean values ranging from 40% to 45%. By the mid-term, drought areas rise by 6.70%, reaching 44.86% in 2070, and a further 7.67% increase in the long-term, peaking at 46.08% in 2091. The uncertainty range expands significantly in the long-term, from 35% to over 55%, reflecting higher variability and greater uncertainty under high-emission conditions.

In summary, drought-affected areas are expected to increase more rapidly under SSP5-8.5, with emissions playing a key role in this trend. SSP1-2.6 shows more stability, underscoring the importance of emissions reductions in mitigating future drought risks. Rising temperatures, which enhance evaporation, are the primary driver of these changes.

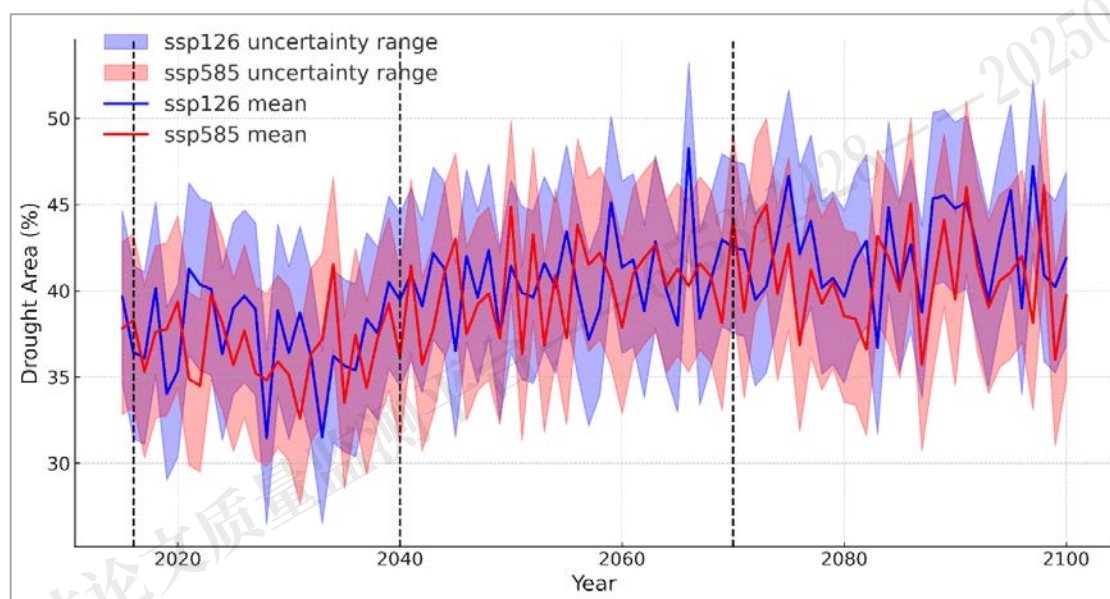


Figure 7-6 Projected drought area time series (%) from 1981 to 2100 under SSP1-2.6 and SSP5-8.5. The shaded area represents uncertainty range, bounded by the minimum and maximum GCM values for each SSP. The black dashed vertical lines mark the projection periods: Near-term (2016–2040), mid-term (2041–2070), and long-term (2071–2100).

7.3.5.2. Projected Variations in Drought Severity Categories

To gain a deeper understanding of future drought conditions in Tunisia, drought severity levels (mild, moderate, severe, extreme) have been analyzed across different periods (Fig. 7-7). In the near-term, mild drought affects about 31% of the area under both scenarios, peaking in 2021 at 75% for SSP1-2.6 and 74.2% for SSP5-8.5, suggesting minimal impact from emissions. By mid-century, differences between scenarios become more pronounced. Mild droughts increase slightly, reaching 31.4% under SSP1-2.6 and 32% under SSP5-8.5. Notably, Moderate and severe droughts show

more variability, with SSP5-8.5 peaking at 81.7% for mild droughts in 2068, compared to 72.6% under SSP1-2.6 in 2070, indicating higher emissions lead to more intense droughts. By the End-century period, SSP5-8.5 consistently shows larger drought-affected areas across all categories. Mild and moderate droughts cover 32.2% and 15.6% of the area on average, peaking at 78.9% in 2085, and portraying increasing trends particularly after mid-century.

Severe and extreme droughts under SSP5-8.5 record the most significant increase, with nearly triple the area affected for severe drought (10.8% mean) and more than quadruple for extreme drought (4.9% mean), intensifying sharply in the latter half of the century. These findings imply that higher emissions (SSP5-8.5), will lead to significantly larger areas affected by severe droughts as the century progresses. While near-term projections show limited divergence, mid- and end-century projections highlight the increasing impact of higher emissions on drought severity.

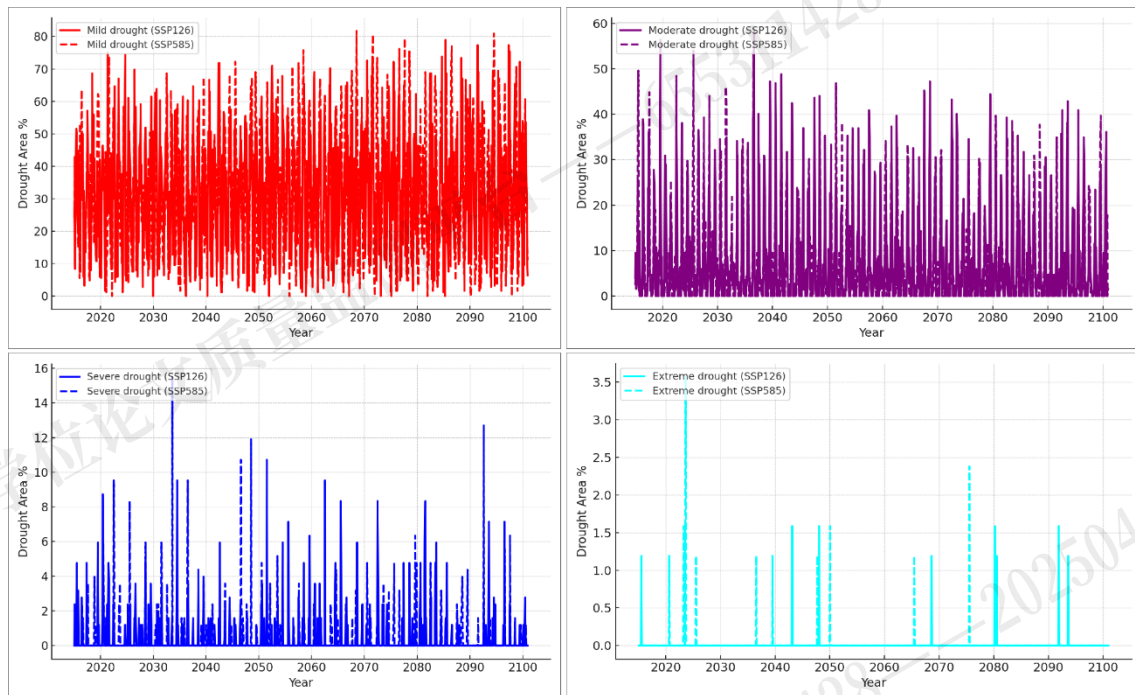


Figure 7-7. Projected drought severity under SSP1-2.6 and SSP5-8.5, categorized as mild (SPEI -0.5 to -1), moderate (SPEI -1 to -1.5), severe (SPEI -1.5 to -2), and extreme (SPEI $\leq$ -2).

7.3.5.3. Monthly Drought Area Change Projections

Figure 7-8 illustrates projections of monthly drought area changes ($\text{SPEI} \leq -0.5$) in Tunisia from 1981 to 2100 under SSP1-2.6 and SSP5-8.5 scenarios. Both scenarios indicate a gradual increase in drought-prone regions from the 2020s to the 2080s, with

a more pronounced expansion under SSP5-8.5. Under SSP1-2.6, July consistently emerges as the month with the most significant drought area changes, peaking at 80–85% across all periods, followed by June and August (70–80%). Under SSP5-8.5, August shows the most substantial drought area changes, reaching 80% in the 2020s, 85% in the 2050s, and 90% in the 2080s, with July also exhibiting substantial changes. April shows a notable increase, from 20% in the 2020s to 50–55% in the 2080s. May, September, October, and November also reveal significant increases as the century progresses.

Overall, mid-year months (May–September) show the highest increases in drought area for all periods and scenarios, though SSP1-2.6 demonstrates a relatively smaller and more stable increase compared to SSP5-8.5. The greater spread and presence of outliers in SSP5-8.5 suggest more extreme and unpredictable drought conditions under higher emissions. These months are critical for drought management due to the largest changes observed.

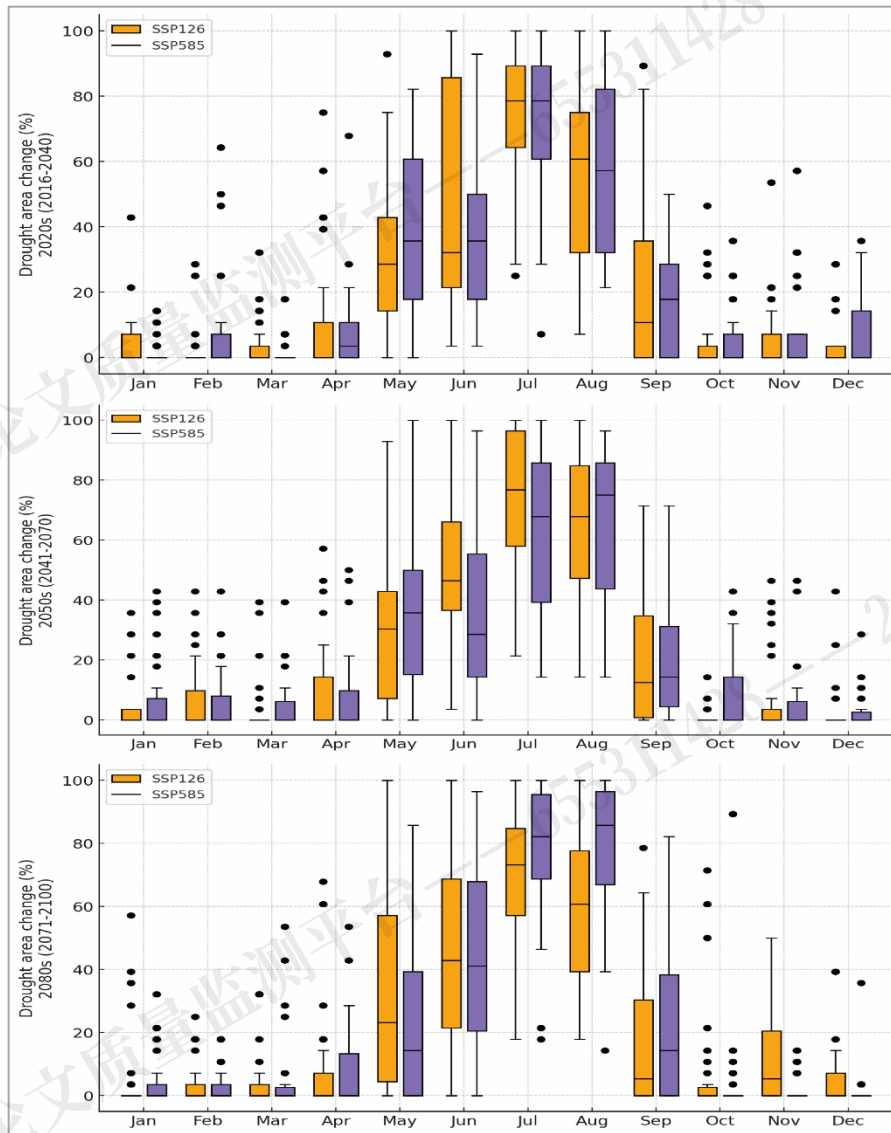


Figure 7-8. Monthly variation in drought area (%) ($\text{SPEI} \leq -0.5$) during near-term (a), mid-term (b), and long-term (c), across Tunisia under SSP1-2.6 and SSP5-8.5.

7.3.6. Spatially Projected Changes of Drought

7.3.6.1. Drought Characteristics during the Baseline Period

Drought analysis for Tunisia during the baseline period (1981–2015) reveals significant spatial variability in drought characteristics (Fig. 7-9). Average drought duration ranged from 1.89 to 5.35 months, with prolonged droughts concentrated in the arid southern and central regions. Drought frequency was highest in the western and central areas, reaching up to 92 events, while the northeastern regions experienced fewer events, with a minimum of 34. The southern and central regions also had the highest frequency (up to 44.53%), indicating persistent droughts. Intensity was most

severe in southern and central Tunisia, with values reaching -1.00, while northern regions saw less severe droughts (-0.51). These findings underscore the heightened vulnerability of southern and central Tunisia to prolonged, frequent, and severe drought conditions.

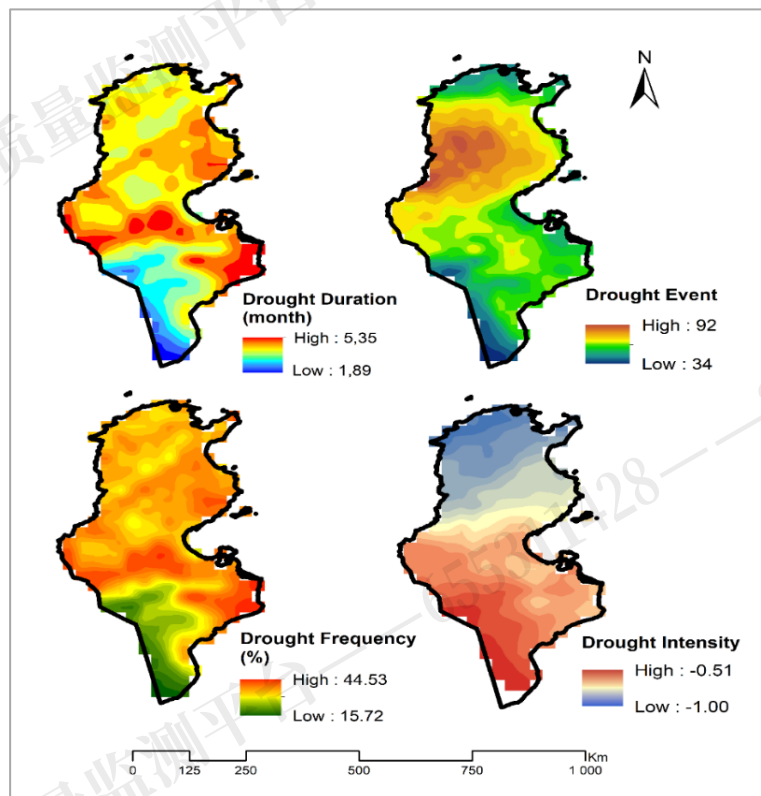


Figure 7-9. Spatial distribution of drought characteristics in Tunisia during the baseline period (1981–2015).

7.3.6.2. Drought Characteristics During the projecting Periods

- **Drought Intensity**

Figure 7-10 illustrates future drought intensity (%) variations across three periods (2016–2040, 2041–2070, and 2071–2100) under SSP1-2.6 and SSP5-8.5, relative to the baseline period (1981–2015). Drought intensity in Tunisia shows distinct spatiotemporal patterns. Under SSP1-2.6, intensity steadily increases, with the southern regions experiencing the most significant changes, reaching 170.34% by the long-term. The north experiences more moderate increases, (24%-85% in the near term, and 70–110% by 2100). SSP5-8.5 results in a more dramatic rise in intensity, especially in the south, peaking at 251.87%. Central Tunisia follows a similar trend. Both scenarios show a north-to-south intensity gradient, with SSP5-8.5 resulting in more severe droughts,

especially in the long-term, highlighting the impact of high emissions on drought severity in Tunisia.

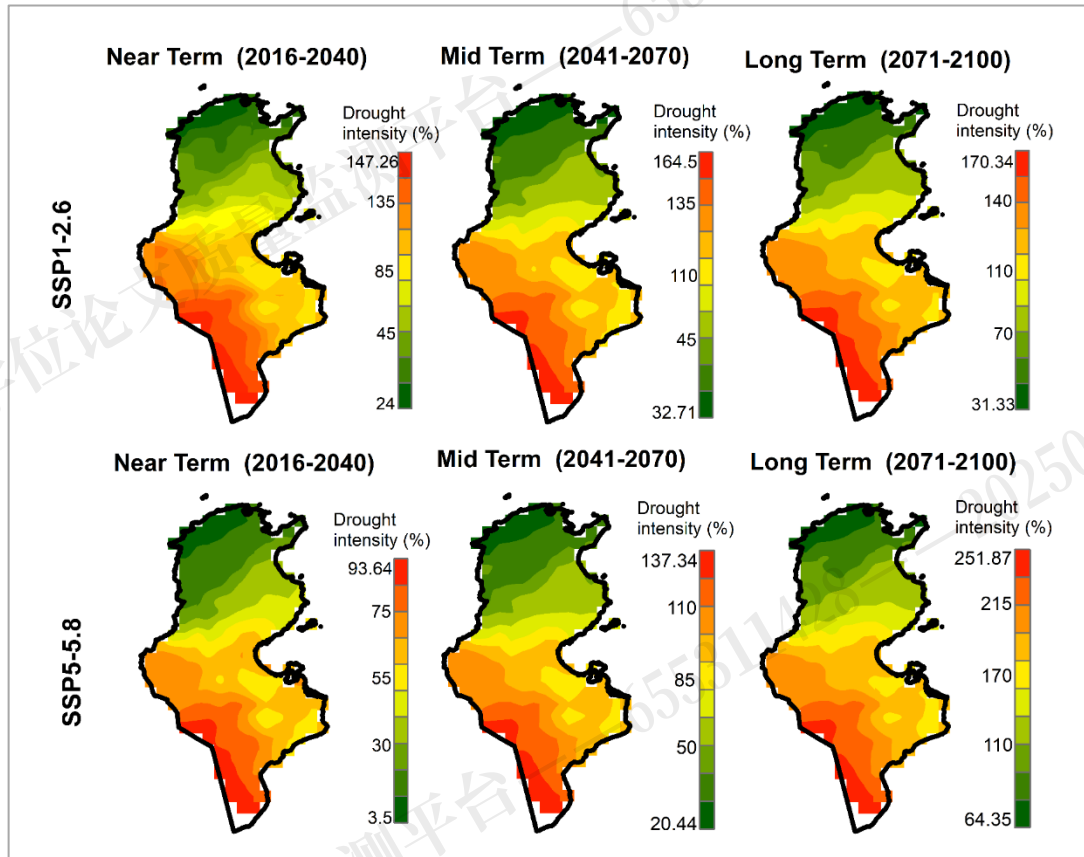


Figure 7-10. Projected changes in drought intensity (%) under SSP1-2.6 and SSP5-8.5 scenarios for the near-term (2015–2040), mid-term (2041–2070), and late-term (2071–2100) periods.

• Drought Duration

Figure 7-11 displays the spatial variations in future drought durations (2016–2100) relative to the baseline period under SSP1-2.6 and SSP5-8.5. Under SSP1-2.6, drought duration initially decreases in the near term in the extreme north and southeast, with reductions from -2.07 to -8.73 months. In the mid-term, reductions decrease, and the central and southern regions return closer to baseline conditions. By the long-term, drought duration increases slightly (0.55 to 1.74 months), particularly in the southwest and central regions, signaling longer droughts. Conversely, SSP5-8.5, shows consistent increases in drought duration. The near-term sees reductions in the extreme north and southeast (-11.57 months), while southwestern and central areas face increases of up to 3.45 months. The disparity grows in the mid-term, with northern and central regions experiencing increases of 0.81 to 2.67 months. By the long-term, droughts become more

prolonged, particularly in the north and central areas (up to 2.72 months). Overall, SSP5-8.5 projects more severe and prolonged droughts, while SSP1-2.6 offers short-term relief but signals longer droughts in the long term, especially in central and southwestern regions, highlighting their vulnerability to extended droughts.

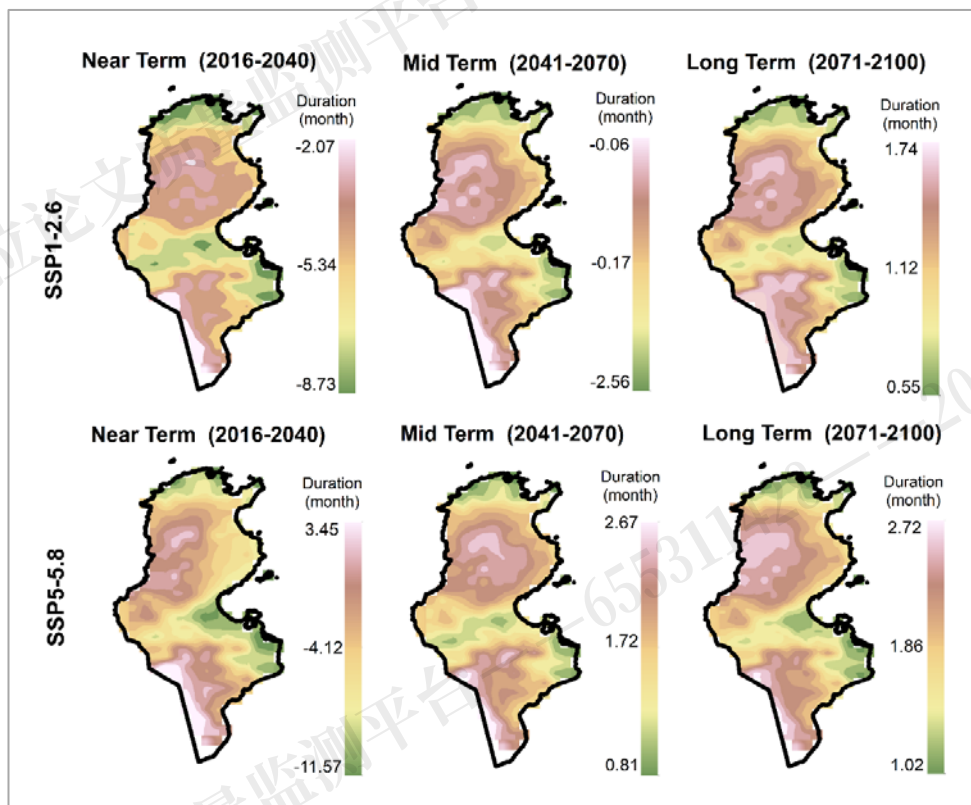
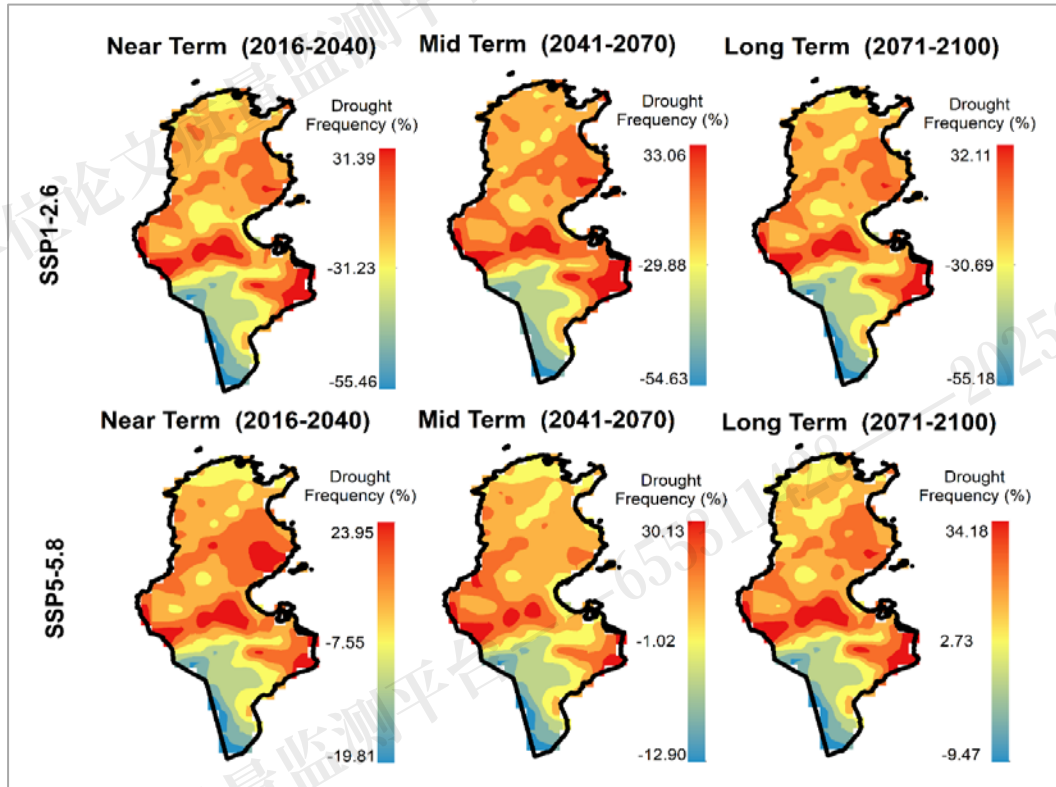


Figure 7-11. Projected changes in drought duration (months). under SSP1-2.6 and SSP5-8.5 scenarios for the near-term (2015–2040), mid-term (2041–2070), and late-term (2071–2100) periods.

• Drought Frequency

Figure 7-12 illustrates projected changes in drought frequency (%) across Tunisia for the periods near-, mid-, and long-term periods, relative to the baseline. Under SSP1-2.6, drought frequency increases from the near- to mid-century, with slight decreases in the long-term. Hotspots of increased frequency (up to 33.06%) are observed in southeastern, northeastern, and central-southern regions during the mid- and long-term. The extreme southwestern regions, corresponding to the Great Eastern Erg of the Sahara Desert, experience significant reductions, particularly in the near-term (-55.46%), which persist over time. Conversely, SSP5-8.5 exhibits a consistent upward trend in drought frequency across all periods. Southeastern and northeastern regions show moderate increases, reaching 23.95% in the near-term, 30.13% in the mid-term,

and 34.18% in the long-term. Central areas also experience increases, albeit to a lesser extent, while the extreme northern regions see mild decreases. Overall, northeastern and southeastern Tunisia are projected to face the most significant increases in drought frequency, with more pronounced impacts under SSP5-8.5. Meanwhile, the extreme



southwestern regions benefit from reduced drought frequency SSP1-2.6. These findings underscore the critical role of emission pathways in shaping future drought conditions.

Figure 7-12. Projected changes in drought frequency (%), under SSP1-2.6 and SSP5-8.5 scenarios, across the near-term (2015–2040), mid-term (2041–2070), and late-term (2071–2100).

7.3.7. Long-Term Temporal Changes in Drought Characteristics

Table 7-3 summarizes the mean projected drought characteristics under SSP1-2.6 and SSP5-8.5, revealing an overall increasing trend across future periods. In the near-term, mean drought occurrences drop to 9 events under both scenarios, indicating an initial reduction potentially due to mitigation measures (Fig. 7-13b). However, a sharp increase occurs in the mid-term, with frequencies reaching 20 for SSP1-2.6 and 17 for SSP5-8.5, suggesting intensifying drought-driving factors over time. During 2071–2100, SSP1-2.6 shows a slight decline in frequency (18 events), while SSP5-8.5 rises to 19 events, reflecting a stabilization yet sustained higher frequency compared to the

near-term. These trends highlight the persistent impact of emissions pathways on drought frequency, with SSP5-8.5 showing more pronounced long-term impacts.

Table 7-3. Summary of drought characteristics for baseline and future periods under SSP1-2.6 and SSP5-8.5.

Period	Scenario	Frequency	Mean Duration (months)	Mean Intensity	Total Severity
Baseline	SSP585	16	11.31	-0.469	-132.54
Near-term	SSP126	9	5.89	-0.218	-21.23
Near-term	SSP585	9	7.56	-0.252	-30.97
Mid-term	SSP126	20	11.15	-0.289	-108.35
Mid-term	SSP585	17	12.94	-0.374	-126.14
Late-term	SSP126	18	12.39	-0.421	-151.78
Late-term	SSP585	19	11.74	-0.471	-141.70

Drought intensity initially increases, in the near-term under both SSP1-2.6 and SSP5-8.5 but declines slightly in the mid-term, indicating less severe conditions in the near future (Fig. 7-13a). However, long-term projections show intensifying droughts, with SSP5-8.5 exhibiting the highest severity, reaching an intensity of -0.47, underscoring the impact of aggressive warming under this scenario.

During the baseline period, droughts had the longest duration, (Fig. 7-13c), averaging 11.31 months. In the near-term, durations drop to 5.89 months (SSP1-2.6) and 7.56 months (SSP5-8.5). However, they increase in the mid-term to 11.15 and 12.94 months, respectively, and approach year-long averages by the late-term, signaling persistent drought challenges.

A notable improvement was recorded in the total severity (Fig. 7-13d), from -132.5 in the baseline period to -21.23 for SSP1-2.6 and -30.97 for SSP5-5.8 in the near-term. For the mid-term period, the severity significantly increases again to reach -108.35 for SSP1-2.6 and -126.14 for SSP5-8.5. The total severity continues to increase during the late-term period, peaking at -151.78 for SSP1-2.6.

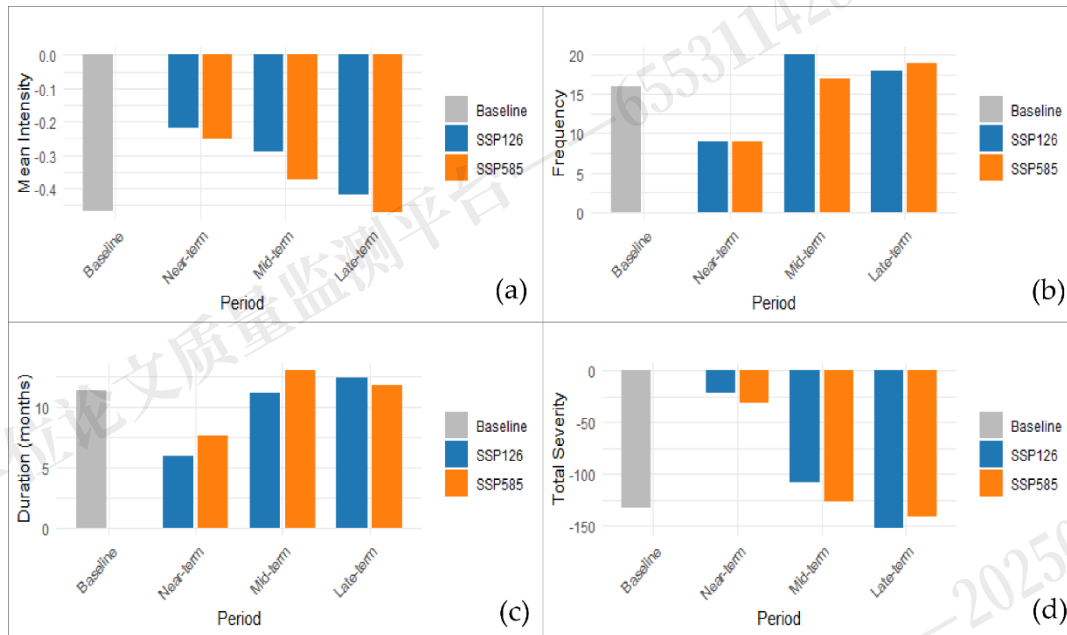


Figure 7-13. SPEI-identified changes in drought characteristics (a) intensity, (b) frequency, (c) duration, and (d) severity for the 2016–2040, 2041–2070, and 2071–2100 periods under the SSP1-2.6 and SSP5-8.5, relative to the reference period.

7.4. Discussion

SPEI-based drought projections clearly indicate a significant intensification of drought in Tunisia throughout the 21st century under both SSP scenarios. Under SSP5-8.5 pathway, our simulations reveal a rapid expansion in drought area and severity, underscoring increasing future risk. Whereas under SSP1-2.6 pathway, the progression is more gradual. Months from April to October, crucial for agriculture and water resources, were identified as critical periods for drought mitigation and targeted adaptation strategies. These trends align with global studies, linking warming to expanding drought-affected areas through increased evaporation, shifting precipitation, and land degradation (Dai, 2013; Li et al., 2024).

Under SSP1-2.6, the gradual increase in drought severity suggests a manageable progression, offering opportunities for mitigation through emissions reduction, improved irrigation, and drought-resistant crops. In contrast, SSP5-8.5 projects a sharp and rapid intensification, with larger drought-affected areas and more frequent extreme events, aligning with global trends linking high emissions to severe droughts in arid regions (Li et al., 2021). This accelerated trajectory limits adaptation time, underscoring the critical need for immediate mitigation measures to prevent severe consequences, including food insecurity, water conflicts, and rural displacement. The contrasting

projections between scenarios highlights the importance of emissions reduction, to slow drought progression and enable effective adaptation. Prompt and targeted policy action, particularly in agriculture-dependent regions, is essential to mitigate the severe socioeconomic impacts of higher-emission pathways.

Temporal projections indicate a marked rise in drought frequency, duration, and intensity, particularly post-2050. While near-term variability may temporarily reduce drought duration, long-term trends point to more prolonged and severe droughts, with SSP5-8.5 exacerbating these impacts. Rising temperatures, altered precipitation, and increased evaporation, particularly in Tunisia's arid and semi-arid regions, are the primary drivers, consistent with global and regional studies. For example, Vicente-Serrano et al. (2022) report increasing drought severity, particularly regions with high evaporative demand, while Trambly et al. (2020b) highlight more frequent and prolonged droughts in the Mediterranean, including Tunisia, due to higher temperatures and reduced precipitation. Similarly, Cook et al. (2018a) emphasize the growing influence of warming on drought severity, with soil moisture as a key indicator.

Spatial analysis reveals southeastern and northeastern Tunisia as hotspots for increased drought frequency and duration, particularly under SSP5-8.5. Similar patterns have been observed across Africa (Nooni et al., 2022), and neighboring countries (Bouabdelli et al., 2020; Gumus et al., 2024). Essa et al. (2023) also projected rising drought frequency across the Mediterranean, with southern countries, including Tunisia, facing the highest risks, supporting our findings under SSP5-8.5. A slight reduction in drought frequency in parts of the southwestern Sahara under SSP1-2.6 may reflect localized atmospheric influences (Pausata et al., 2020; Nasa, 2024).

Nevertheless, drought intensity is projected to rise across all regions, with central and southern Tunisia most affected under SSP5-8.5. Consistent with our findings, the mediterranean basin is projected to experience significant warming and drying trends, with precipitation shortages and increases in temperature and evapotranspiration, during the 21st century over all seasons, particularly under SSP5-8.5 (Cos et al., 2022; Arjdal et al., 2023; Ferreiro-Lera et al., 2024). Li et al. (2021) further confirmed that 93.2% of arid regions under SSP5-8.5 will experience intensified droughts by the late 21st century. Worsening drought conditions pose severe risks to Tunisia's agriculture, water resources, and socio-economic stability. Rain-fed agriculture, already strained by water scarcity, faces increasing vulnerability, with reduced crop yields expected in central and southern regions (Mougou et al., 2011; Radhouane, 2013). The FAO has

warned of declining agricultural productivity in North Africa, further aggravated by intensified droughts (Turrall et al., 2011). These projections emphasize the urgency for comprehensive adaptation strategies to address Tunisia's water crisis, driven by low rainfall, high evaporation, and groundwater overextraction.

The divergent outcomes between SSP1-2.6 and SSP5-8.5 underscore the critical importance of global emissions reductions and robust adaptation measures. Immediate policy actions, such as upgrading irrigation infrastructure, adopting drought-resilient crop varieties, and implementing integrated water management, are essential to mitigate the projected adverse impacts on Tunisia's water resources, agriculture, and overall socioeconomic stability. These findings provide a robust scientific basis for decision-makers, emphasizing that a multi-model ensemble approach using CMIP6 outputs offers a valuable tool for regional water resource planning in the face of intensifying climate change.

7.5. Conclusion

This study provides the first in-depth assessment of future drought characteristics in Tunisia (1981-2100) using SPEI, derived from three CMIP6-GCMs (MIROC6, MPI-ESM1-2-HR, and IPSL-CM6A-LR) under SSP1-2.6 and SSP5-8.5 scenarios. Anchored by a historical baseline (1981–2015) and divided into near-term (2016–2040), mid-term (2041–2070), and late-term (2071–2100) periods, our projections reveal significant findings:

- MIROC6 under SSP5-8.5 showed the highest reliability ($r = 0.7$, lowest RMSE and MAE), effectively capturing drought trends despite a slight overestimation of SPEI values.
- SPEI projections indicate an intensifying drought, particularly under SSP5-8.5, highlighting the importance of incorporating PET data in assessments.
- Drought severity and spatial extent are projected to increase, affecting up to 78.9% (SSP1-2.6) and 81.7% (SSP5-8.5) of Tunisia by 2100, with mid-year months (spring–summer) experiencing the most significant changes.
- Southern Tunisia faces the highest risk, with drought intensity increasing by 251.87% and frequency by 33.06%, while northern areas experience moderate but prolonged droughts.

- SSP5-8.5 predicts greater variability and extremes, particularly in southern and central Tunisia by 2100, whereas SSP1-2.6 offers short-term relief but suggests worsening conditions long-term.
- Uncertainty analysis indicates that projected drought changes will become more pronounced, particularly from the near to late-term future.

These findings underscore the critical role of emission pathways in shaping future drought conditions in Tunisia, with higher emissions leading to more severe and prolonged droughts, implying Tunisia's acute vulnerability to climate change, highlighting the urgent need for enhanced water resource management, climate-resilient agricultural practices, and comprehensive adaptation strategies. Future investigations should integrate simulations from different models with GGDI to forecast future groundwater droughts.

Chapter 8 Conclusions and Future Perspectives

8.1. General Conclusions

This doctoral dissertation presents a significant advancement in the understanding and monitoring of droughts in Tunisia, a country increasingly vulnerable to climate change. It provides a holistic and integrated assessment of drought dynamics in Tunisia, using an interdisciplinary approach that combines advanced remote sensing data, conventional drought indices, geospatial techniques, and climate models. This thesis addresses the multifaceted nature of droughts, including their different types, characteristics, impacts, propagation mechanisms, linkages with climate teleconnections and land degradation, and projections, offering novel insights and substantial contributions to the field.

Initially, we conducted a systematic review (Chapter 1) of various drought types, characteristics, impacts on land productivity and groundwater, and recent advancements in remote sensing for drought monitoring. We identified notable research gaps in drought monitoring in Tunisia, where studies devoted to droughts remain limited, despite its high susceptibility to drought. This chapter highlights the need for long-term assessments of drought variability and its associated driving factors to mitigate its hazardous impacts. Chapter 3 examines hydro-meteorological droughts using GRACE-derived TWS monthly data, GLDAS water balance factors, and ERA5 climate datasets, and explores their interlinkages with climate teleconnections factors using non-stationary statistical approach. Chapter 4 complemented hydrometeorological drought analysis, by targeting agricultural drought (AD), providing further insights into the complex interaction between different drought types. It introduces a novel integrated framework to analyze propagation pathways between MD and AD, using SPEI and MODIS-derived VHI. Results show high drought vulnerability in southern Tunisia, prompting a regional-scale assessment of drought impacts on NPP (Chapter 5), and its relationships with soil erosion through RUSLE model, in these drought-prone regions. The selected basins in the south are key socio-agroecological zones where water scarcity, drought, and land degradation are most pronounced, making them critical for water and land resource management and representative of broader challenges in Tunisian drylands. Chapter 6 evaluated future

changes in drought characteristics under climate projections using SPEI and ensemble mean of GCMs from CMIP6 (1981-2100) over Tunisia. The main research results of this investigation include the following parts:

Firstly, this study assessed hydro-meteorological droughts in Tunisia, using GRACE-derived TWS monthly data, GLDAS water balance factors, and ERA5 climate datasets. It pioneered GGDI application over Tunisia, linking it to surface and subsurface conditions through various hydro-meteorological indices (SPI, SPEI, and SRI), and exploring teleconnections impacts through wavelet coherence. Significant disparities among indices were revealed, with GGDI exhibiting more severe and prolonged droughts. The driest period (2002-2003) has an average SPEI, SPI, and SRI, of -1.9, while GGDI identified six major GD events (>3 months), the longest lasted 12 months (Nov-2014/Oct-2015), with a total severity of -13.60, and an average GGDI of -1.36. Decreasing GGDI trend, reaching -3.64 in 2017, with notable spatial disparities, emphasized critical groundwater depletion driven by anthropogenic activities and climate change. The first-order Markov chain analysis revealed persistent droughts in Tunisia, with extreme events exhibiting high stability and low recovery rates, with potential for significant shifts in severity, emphasizing the need for improved drought management. ENSO exerted the strongest impact on drought patterns, with El Niño phases associated with severe droughts. This study expands on hydrometeorological drought assessment to examine surface-level impacts of droughts on vegetation and productivity.

Chapter 4 introduces a novel framework integrating multiple statistical methods, i.e. Spearman rank correlation (SRC), Dynamic Time Warping (DTW), and transition probability analysis, to quantify drought propagation time and probabilistic linkages between MD and AD at different severity levels over time, through MODIS-derived VHI and SPEI. DTW reveals a nonlinear MD-AD relationship, with a delayed propagation due to environmental factors, evidenced by the strong alignment between SPEI-12 and VHI-1 (DTW distance: 136.36). Vegetation health is more sensitive to medium- and long-term drought, with stronger SRC at longer SPEI scales (SPEI-6: 0.64 and SPEI-12: 0.5). Propagation time is shorter during warmer seasons, while winter and spring, exhibit longer lags (6-9 months), due to soil moisture persistence and evapotranspiration. The occurrence probability of AD increases with MD intensity, with 19.2% of moderate MDs evolving into severe AD, while severe MD almost always leads to severe AD. Key findings include the identification of southern Tunisia as a

drought-prone region, frequently experiencing moderate to severe droughts, with extreme events in 2001, 2005, 2007, 2011, and 2016, mainly in December-March. These findings emphasize the need for long-term drought impact assessments for on land productivity, through SPEI-6 and SPEI-12, which better predict severe AD, particularly in the vulnerable regions of southern Tunisia.

Consequently, Chapter 5 investigates drought impacts on NPP, specifically at a basin-scale (Jeffara) in southern Tunisia, marking the first study in Tunisia to integrate MODIS EVI (used as NPP proxies) with SPI to assess NPP responses to droughts. Spatiotemporal analysis confirmed the recurrence of droughts in the region, with moderate (10%) and severe (7%) events being frequent, while extreme droughts remained rare (2%). Drought emerged as a key trigger of NPP variations in Tunisian arid ecosystems, with NPP responses depending mainly on drought intensity and duration. A significant lagged effect on land productivity was evident, with SPI showing a strong positive correlation (0.5–0.9, p -value < 0.05) with NPP anomalies at a 12-month scale. Findings further highlight southeastern Tunisia's critical vulnerability to worsening droughts, which not only reduce land productivity but also accelerate land degradation. This underscores the urgent need to explore how drought variability influences land degradation through soil erosion dynamics in these fragile ecosystems.

Therefore, Chapter 6 introduces a comprehensive GIS-based approach to assess both soil erosion and droughts at a basin scale (Negueb) in southern Tunisia, and explore their controlling factors. Drought conditions were analyzed using VHI, combining VCI from MOD13A2 EVI and TCI from MOD11A2 LST, while RUSLE model was applied to evaluate soil erosion processes. The findings highlighted the recurring aspect of droughts, with the upstream western higher elevations being particularly vulnerable. The average soil loss rate was found to be 1.8 t ha⁻¹ y⁻¹, with slope length and steepness (LS) being the main factors contributing to erosion. A significant positive correlation ($r = 0.3$) between droughts and soil erosion was revealed, with spatial correlations ranging from -0.5 to 0.8, strongest in drought-prone, high-erosion areas, indicating that soil erosion is interlinked to droughts. This study supports adaptation strategies in these vulnerable regions, and promotes future research on projecting climate-induced drought changes for early warning systems.

Finally, Chapter 7 provides the first in-depth assessment of future drought characteristics in Tunisia (1981–2100) using SPEI from three CMIP6-GCMs under SSP1-2.6 and SSP5-8.5. Results based on historical data (1981–2015) and projections

for near-term (2016–2040), mid-term (2041–2070), and late-term (2071–2100) show significant trends. MIROC6 under SSP5-8.5 displayed the highest reliability ($r = 0.7$, lowest RMSE and MAE) despite slight SPEI overestimation. Projections suggest worsening droughts, particularly under SSP5-8.5, with severity and extent increasing to 78.9% (SSP1-2.6) and 81.7% (SSP5-8.5) by 2100, especially in spring and summer. Southern Tunisia faces the highest risk, with drought intensity increasing by 251.87% and frequency by 33.06%, while northern regions will experience moderate but prolonged droughts. SSP5-8.5 predicts greater variability and extremes, particularly in southern and central Tunisia by 2100, whereas SSP1-2.6 offers temporary relief but suggests long-term worsening. These findings highlight Tunisia's vulnerability to climate change and the urgent need for improved water management, climate-resilient agriculture, and adaptation strategies.

By considering these various facets, this doctoral thesis lays a foundation for improved drought risk monitoring and a deeper understanding of its causes in Tunisia, emphasizing the integration of advanced remote sensing and comprehensive datasets for effective drought assessment and management, in such data-scarce regions. The findings provide a valuable framework for policy decisions and mitigation strategies, supporting global efforts to achieve SDG 13, SDG 6, and SDG 15, and helping to mainstream disaster risk reduction in the country's development efforts.

8.2. Innovation points of the paper

The inter-dependency of land and water management and the multiple benefits of reversing drought impacts and land degradation for sustainable development, water security and resilience to natural hazards, have been acknowledged and greatly underpinned by recent researchers. Several studies are beginning to acknowledge that achieving sustainable development goals (SDGs), particularly SDG 13, SDG 6 and SDG 15, require significant advancements in the policy operational synergies within the land-water nexus. This doctoral thesis developed a holistic drought assessment that underpins the synergies between droughts mitigation, water security and land degradation neutrality. Its key innovations include:

- A pioneering, comprehensive framework that integrates multiple drought types, meteorological, hydrological, and agricultural, while considering influence of climate forcing and teleconnections in Tunisia. This research serves as a reliable national

database, enhancing the understanding of Tunisia's increasing drought risks under climate change.

- A novel exploration that links between drought, climate change, water scarcity, and land degradation, offering an integrated perspective that has not been explored in such depth in Tunisia or globally. By identifying vulnerable areas and risk factors, this multidimensional approach informs measures for reducing drought-related hazards, facilitating targeted mitigation strategies that take into account the full scope of environmental and socioeconomic challenges for water-scarce regions, particularly in the MENA region.
- A first application of GGDI over Tunisia, linking it to surface and subsurface conditions through hydro-meteorological drought indices. It pioneers the use of the advanced GRACE satellite data to deliver an unprecedented assessment of groundwater drought, and integrate key water balance components from GLDAS which are vital for water resource variability. GGDI proved its effectiveness as a robust tool for groundwater drought monitoring in Tunisia. By complementing traditional drought indices, this approach offers a more comprehensive view of hydrometeorological drought, providing valuable insights for policymakers and water managers in addressing groundwater over-extraction.
- An improved understanding of the teleconnection impacts on hydro-meteorological droughts in Tunisia where this relationship is unprecedentedly studied. It is the first investigation of long-term analyses of teleconnection impacts on hydro-meteorological droughts. It explores key teleconnections, including WeMO, ENSO, NAO, and MO, whose effects on droughts remain largely unexplored. Using advanced statistical techniques, particularly wavelet coherence, the study captures non-stationary behavior and non-linear interactions, offering new insights into climate-drought linkages in the region.
- A novel framework integrating statistical methods (Dynamic Time Warping and Spearman rank correlation), and transition probability analysis, to quantify for the first time, drought propagation time and probabilistic linkages between meteorological and agricultural droughts at different severity levels over time in Tunisia. This study provides novel evidence that MD severity acts as a threshold for AD development. The findings advocate for using medium- to long-term indicators (SPEI-6, SPEI-12) as

stronger predictors of severe agricultural droughts, providing valuable early-warning insights to mitigate crop losses and food security risks.

- This study takes a leading role in unraveling drought impacts on NPP in southeastern Tunisia. The study brings new insights in NPP responses to droughts, showing resilience in arid ecosystems and informs the importance of arid ecosystems to resist dry periods.

- This study uniquely integrates drought with soil erosion through a comprehensive remote sensing and GIS-based approach, advancing the understanding of their relationship under arid climate conditions, a largely unexplored area worldwide. Applying the RUSLE model and VHI, findings highlight that drought and soil erosion are closely interconnected in Tunisia's dry regions. This interdisciplinary approach offers practical insights for developing targeted soil conservation and mitigating land degradation mitigation strategies.

- A first study, utilizing advanced CMIP6 model simulations across SSPs scenarios to address the impact of future climate change on drought characteristics in Tunisia. Findings reveal increasing drought severity, frequency, and duration, with SSP5-8.5 showing the most extremes, supporting long-term climate adaptation planning.

In summary, by considering these various facets of drought, this dissertation embraces a holistic lens to capture a nuanced understanding of droughts in Tunisia. It introduces several methodological and analytical advancements in drought assessment, ranging from satellite-based indices to climate model projections, particularly useful in regions where hydrometeorological databases are lacking like Tunisia. By integrating diverse datasets and analyzing climate-induced water resource changes, it enhances drought monitoring and management. The findings offer valuable insights for policymakers and adaptation planners, supporting sustainable resilience strategies in arid regions

8.3. Future perspective

Despite the significance of the current research in addressing the main purposes, there are some future directions should be targeted to provide a more robust and holistic understanding of droughts in Tunisia:

- Along with climatic factors, anthropogenic activities were found to have a significant impact on triggering groundwater drought, creating hence a more complicated relationship between meteorological and groundwater droughts. Consequently, a future research direction encompasses developing quantitative methods to analyze the impact of human activities on groundwater drought onset. Also, it is recommended to further investigate the relationship between hydrometeorological factors using non-linear methods such as three-parameter power function to better characterize the relationship between the different types of droughts in Tunisia.
- Although outcomes from GRACE and GLDAS were basically consistent endorsed by a significant high correlation, proving that the linear interpolation method was acceptable. However, it is still recommended to adopt nonlinear approaches such as machine learning, deep learning, and artificial neural network, as better alternative for handling missing data in GRACE-derived data.
- Application of the methodology adopted in this study to investigate the relationship between droughts and soil erosion in different watersheds located in different bioclimatic zones across the globe, to further investigate and validate the relationship between droughts and land degradation.
- The formulation of a national policy or strategy for drought management, based on scientifically validated hydrometeorological drought onset indicators tailored to Tunisia's conditions, is of utmost priority in light of climate variability. It is crucial to integrate the immediate risks of drought with the necessity to adapt to long-term climate changes within both national and subnational strategies and management plans. This dissertation integrates various meteorological, hydrological, and agricultural drought indicators, providing a basis for coordinating and cooperating among the relevant agencies. To support this, creating an integrated data-sharing platform that connects agricultural, hydrological, and meteorological data from different agencies, incorporating results from this dissertation, could be established.
- Finally, it is essential to strengthen the link between technical and institutional efforts to develop a simple yet robust early warning system, including mechanisms to trigger comprehensive drought management coordination before significant drought impacts occur. Continued technical assistance, capacity building, and regular information exchange with institutions and experts in the MENA region and beyond could greatly enhance these efforts.

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Appendix

Table A.1. Summary of drought indices, sources and selection criteria

Index	Input Data	Selection criteria	Purpose	Reference
GGDI	GRACE-derived monthly TWSA, GLDAS water balance factors	- Directly reflect surplus and deficit in GWS. - Comprehensive normalized indicator integrating TWS changes, crucial for groundwater-dependent regions like Tunisia, with soil moisture, and canopy water storage. - Evaluate applicability in Tunisia, a heavily groundwater-dependent region.	Groundwater drought Assessment.	(Thomas et al., 2017)
SPI	ERA5-monthly Precipitation	- Long-term precipitation data - widely validated, ideal for arid climates. - flexibility in temporal scales - Simplicity and reliability: standard monitoring tool.	Meteorological droughts assessment	(McKee et al., 1993)
SPEI	Precipitation, temperature maximum and minimum from ERA5	- Combines precipitation and PET to assess drought severity under changing temperatures, making it ideal for climate-impacted regions like Tunisia.	Improved meteorological droughts assessment	(Vicente-Serrano et al., 2010)

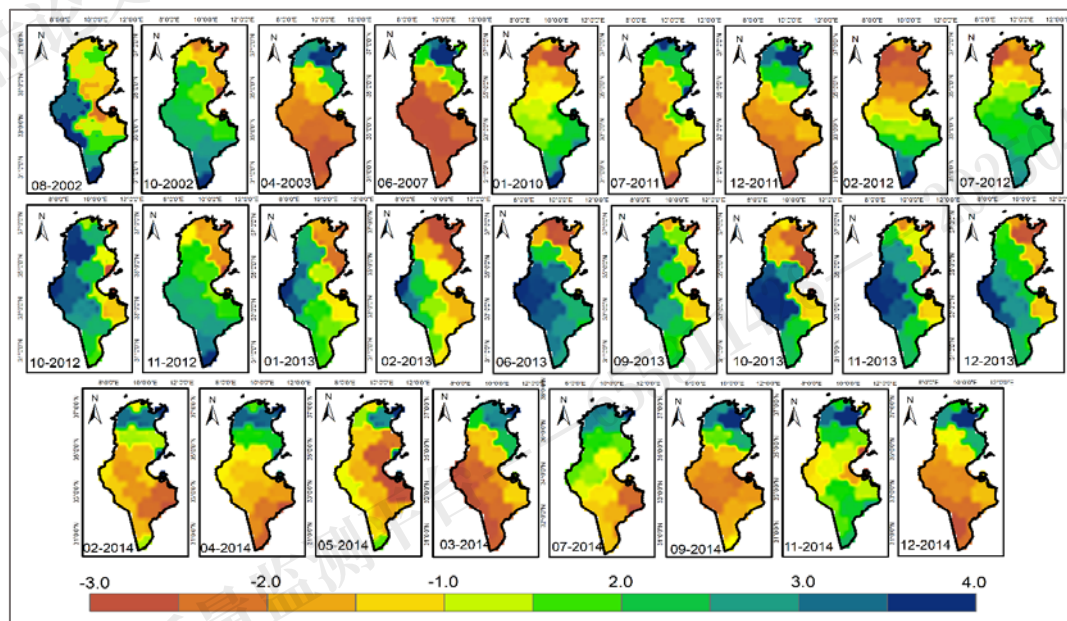


Figure A.1. Spatial distribution of groundwater drought events (in terms of duration and intensity) identified by GGDI in Tunisia during 2002-2017. (Continued)

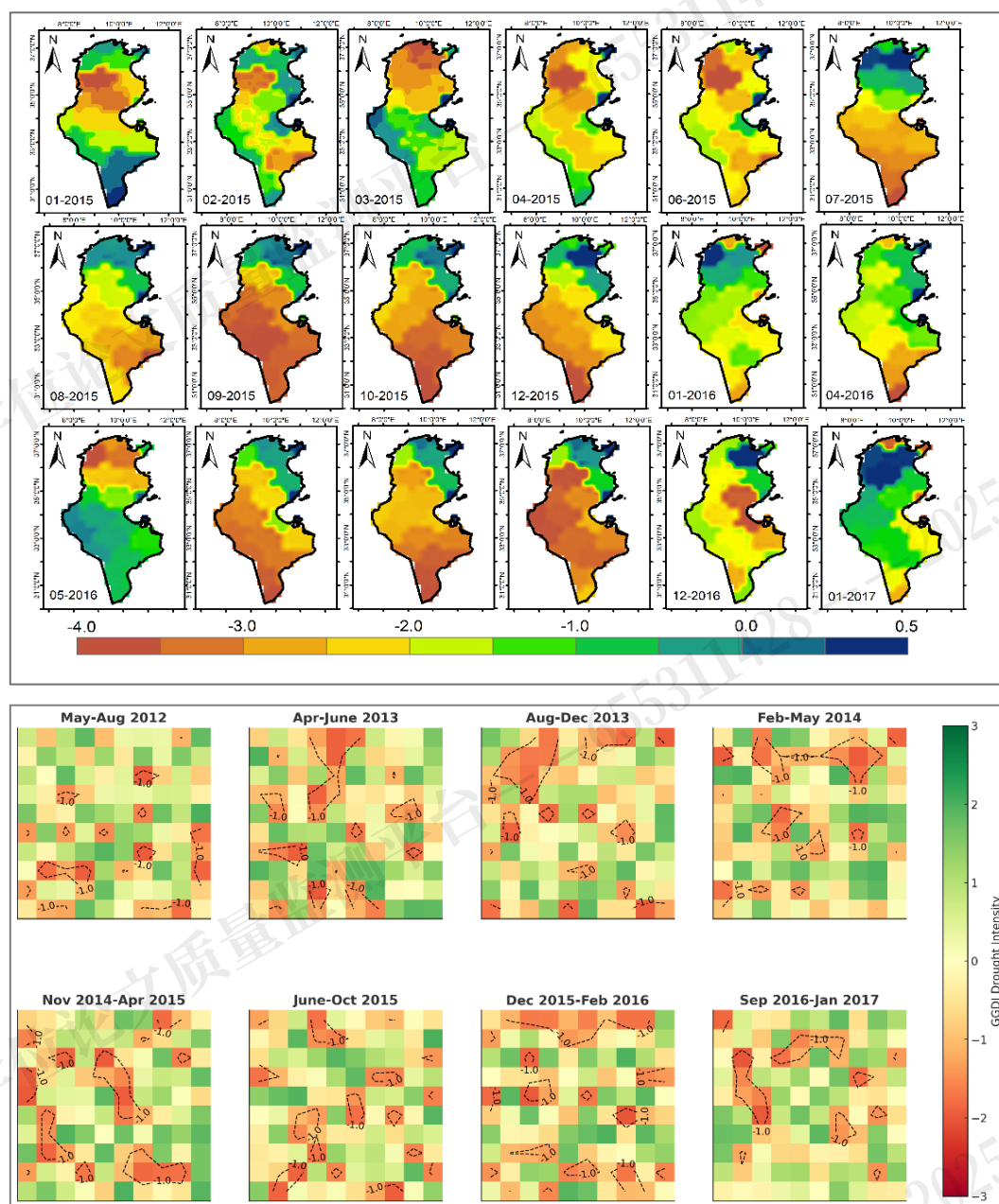


Figure A.1. Spatial distribution of groundwater drought events (in terms of duration and intensity) identified by GGDI in Tunisia during 2002-2017. (Continued)

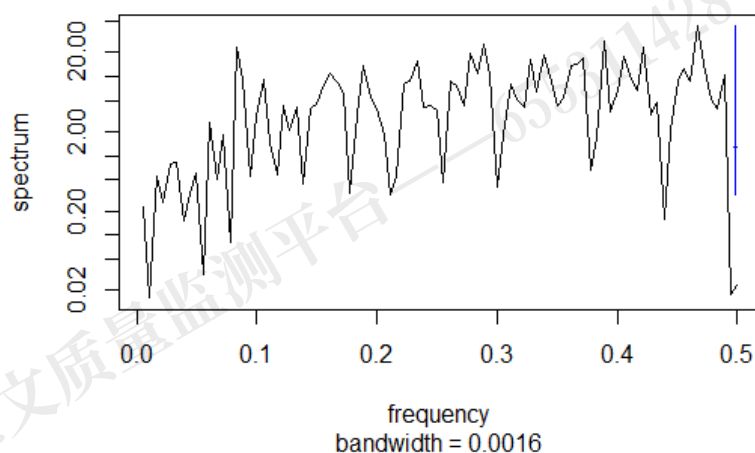


Figure A. 2. Periodicity check for GWSA time series

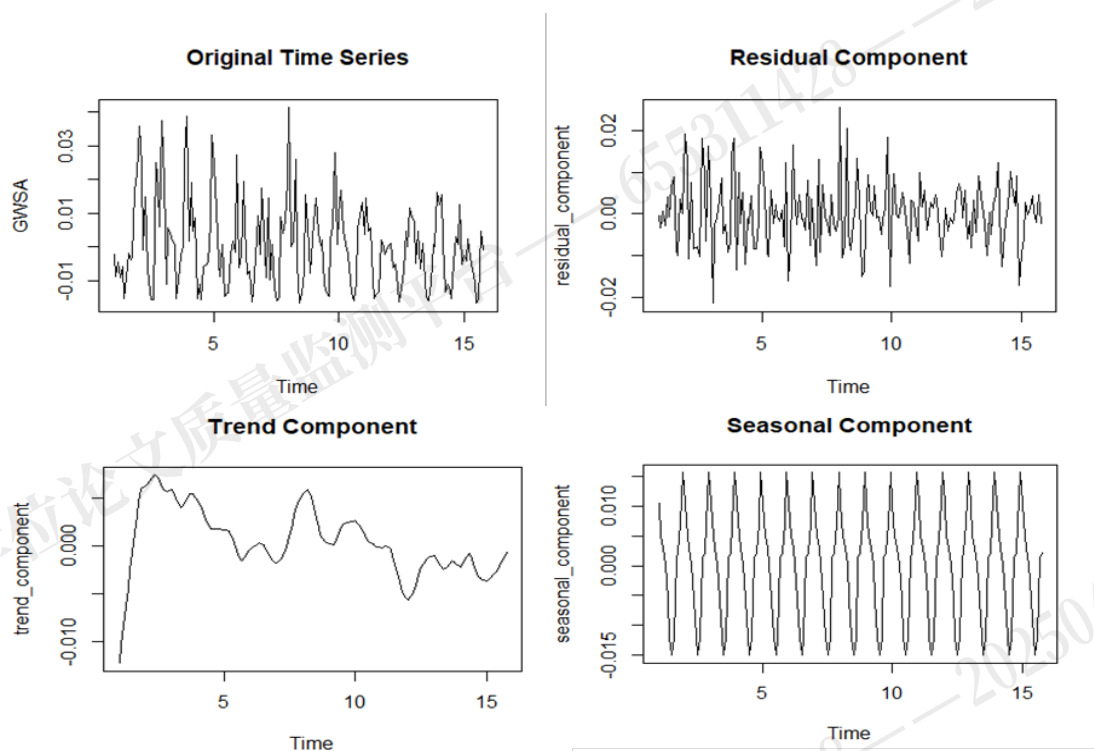


Figure A. 3. Seasonal-Trend decomposition using LOESS (STL) of GWSA time series

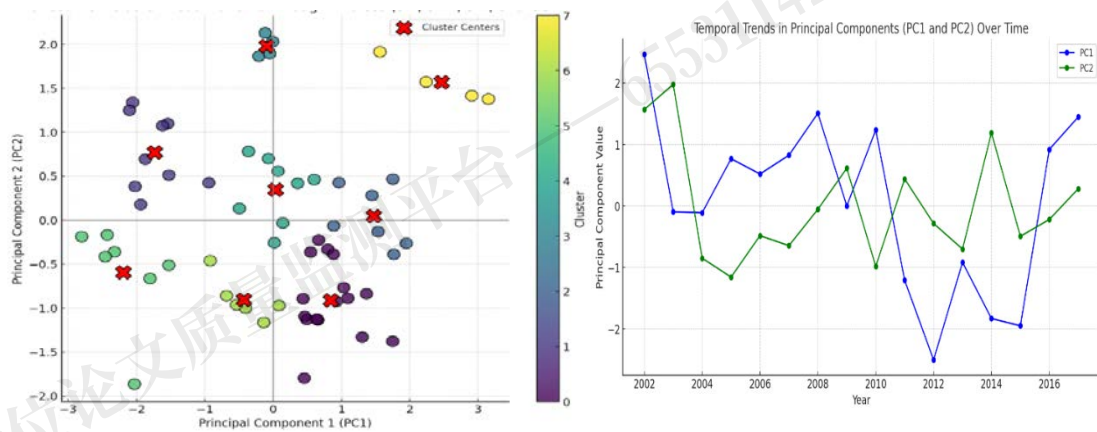


Figure A. 4. PCA cluster visualization of different hydrometeorological drought indices (SPI, SPEI, SRI, and GDI)

A4. SPEI-based spatiotemporal drought analysis

The analysis of SPEI trends from 2001 to 2020 over Tunisia revealed significant variability in drought and wet conditions over time. The temporal trends indicate an increasing intensity of dry conditions, particularly in recent years, with a noticeable increase in extreme drought events in 2001, 2007, 2016, 2019, and 2020. These years exhibited widespread and intense drought conditions, with 2016 standing out as a particularly severe period, along with the extreme drought patterns observed in 2001, 2019 and 2020.

SPEI patterns showed cyclical patterns, with alternating periods of wet and dry conditions. For example, extreme wet conditions were observed in 2001-07 (SPEI: 6.95), 2009-07 (SPEI: 6.55), and 2014-07 (SPEI: 6.51), while severe dry conditions occurred in 2005-05 (SPEI: -2.55), 2012-12 (SPEI: -2.69), and 2019-12 (SPEI: -1.85). These fluctuations suggest that drought and wet conditions are recurring phenomena rather than a unidirectional trend toward increasing drought intensity.

Seasonal variability is evident, with wet conditions more frequently occurring during the summer months, such as July 2001, July 2009, and July 2014, while dry conditions are more common in the winter months, such as December 2005, December 2012, and December 2019. This aligns with typical climatic patterns, where higher temperatures and increased evapotranspiration during summer can exacerbate drought conditions, while reduced precipitation in winter may lead to drier periods.

Spatial distribution analysis highlights that the northern region tends to experience milder wet and dry fluctuations, while the central and southern regions are more prone to prolonged drought conditions. Over the long term, drought frequency and intensity tended to increase, indicating worsening climatic conditions. Southern region is the most affected by droughts, followed by the central region, while the northern part remains relatively less impacted.

Overall, SPEI trends indicated cyclical patterns of wet and dry conditions, with severe drought events recurring every few years, such as in 2001, 2005, 2012, and 2019. These findings suggest an urgent need for climate adaptation strategies to mitigate increasing drought risks, particularly in the southern regions.

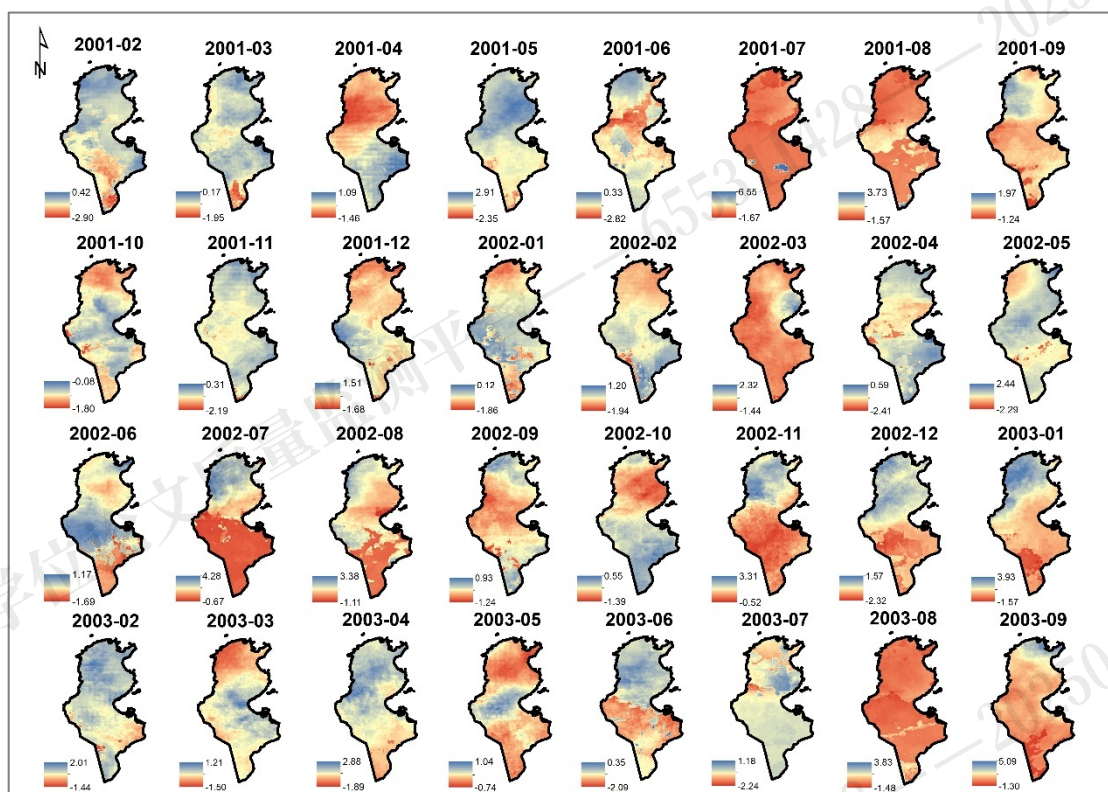


Figure A.4. Spatial distribution of meteorological drought identified by SPEI in Tunisia during 2001-2020. (Continued)

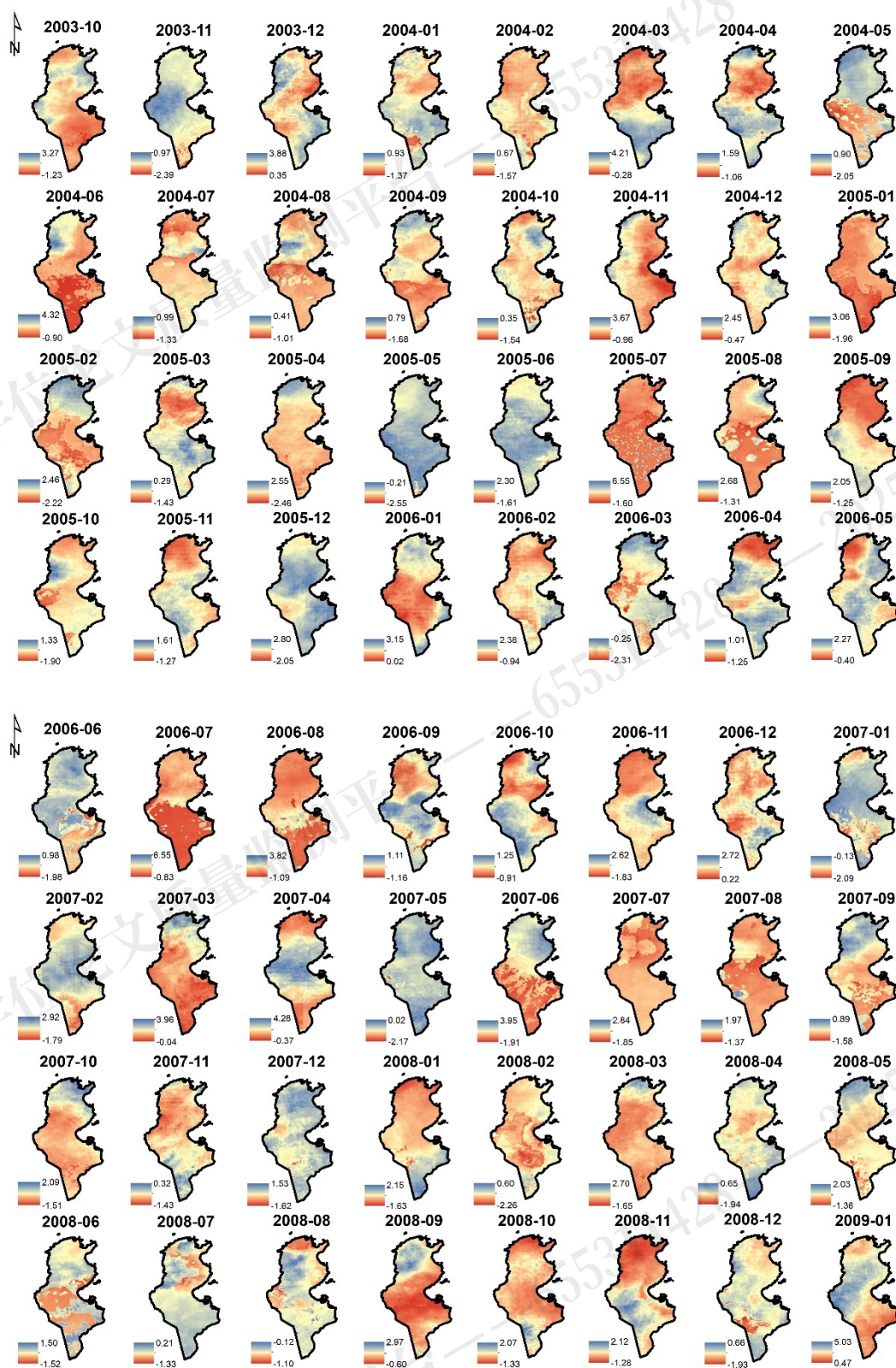


Figure A.4. (Continued).

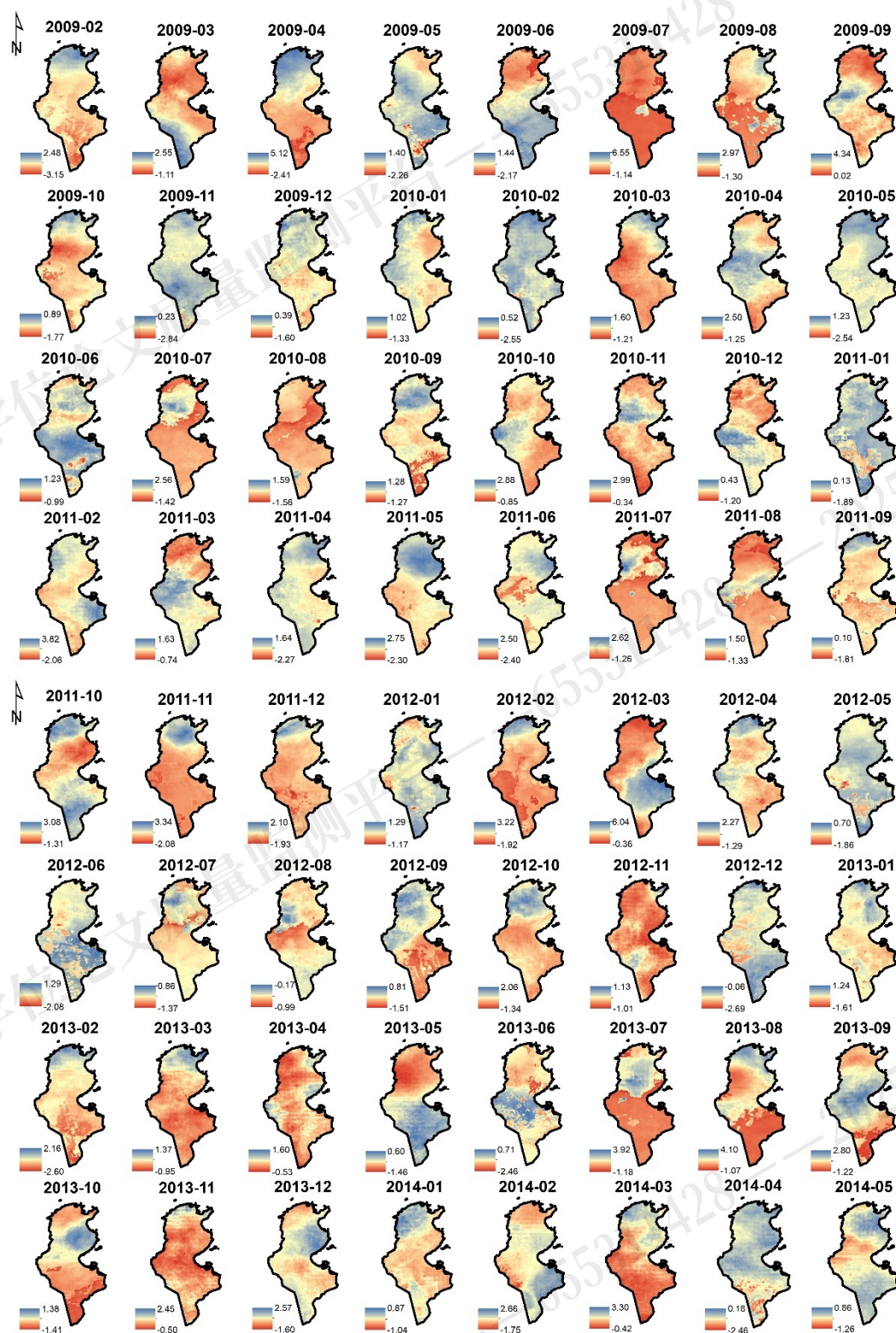


Figure A.4. (Continued).

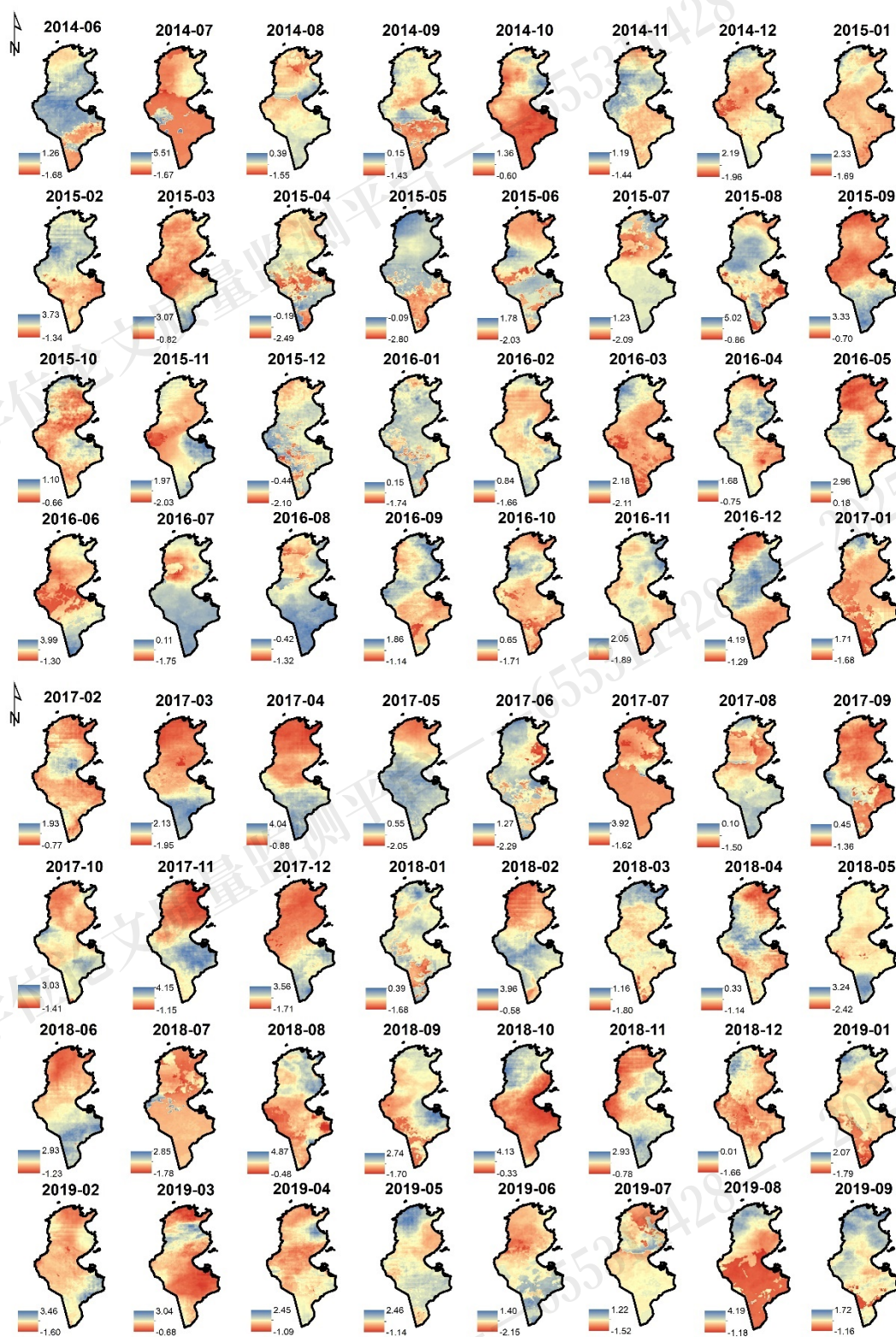


Figure A.4. (Continued).

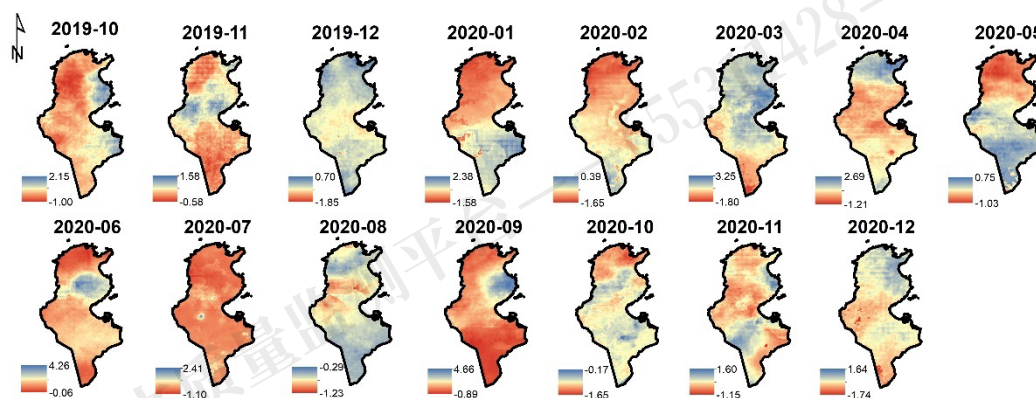


Figure A.4. (Continued).

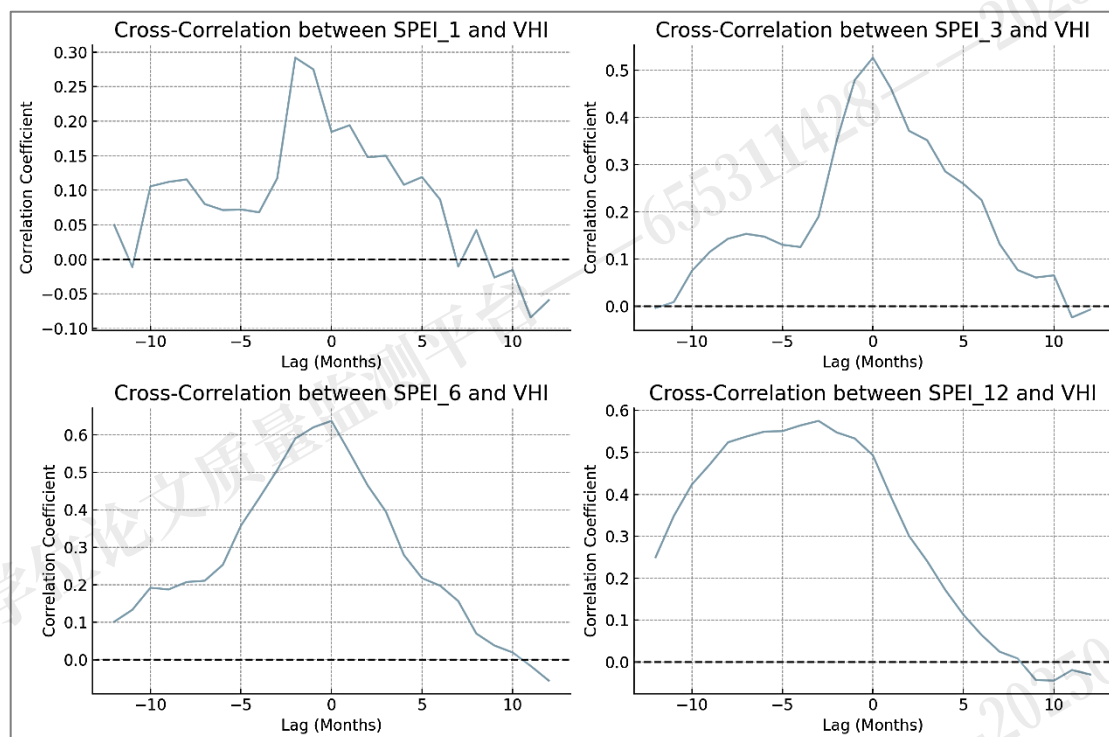


Figure A.5. Cross-Correlation analysis between VHI-1 and SPEI time series at different time-scales over Tunisia from 2001 to 2020.

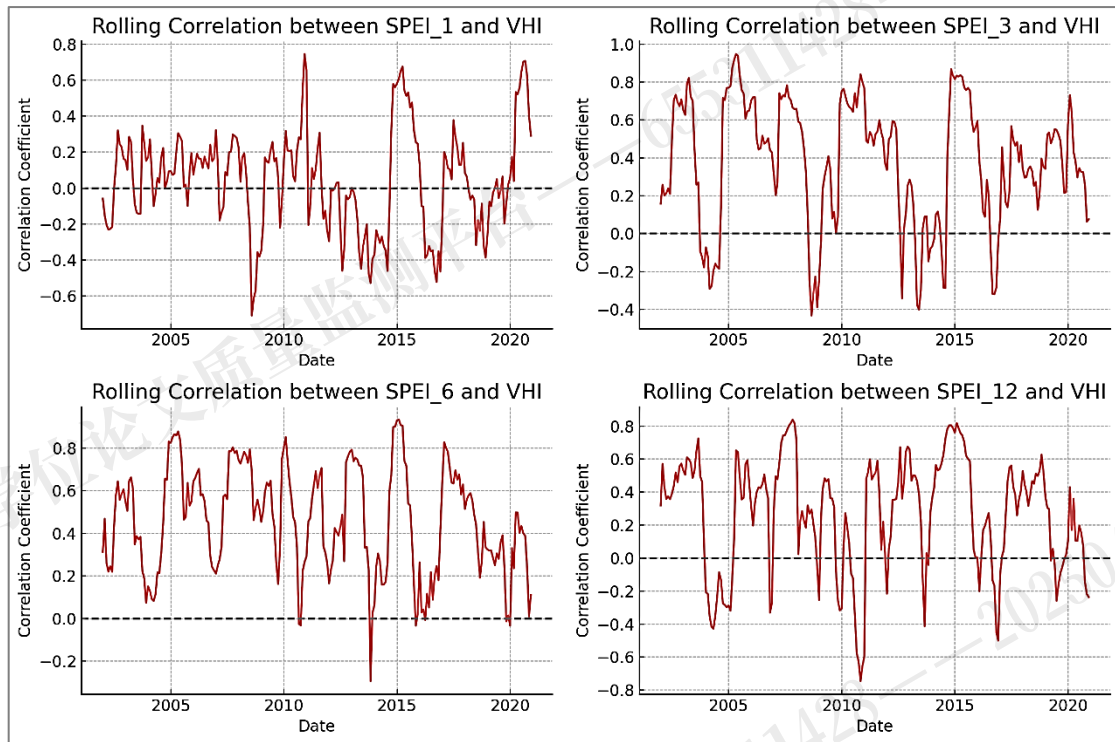


Figure A.6. Rolling Correlation analysis between VHI-1 and SPEI time series at different time-scales over Tunisia from 2001 to 2020.

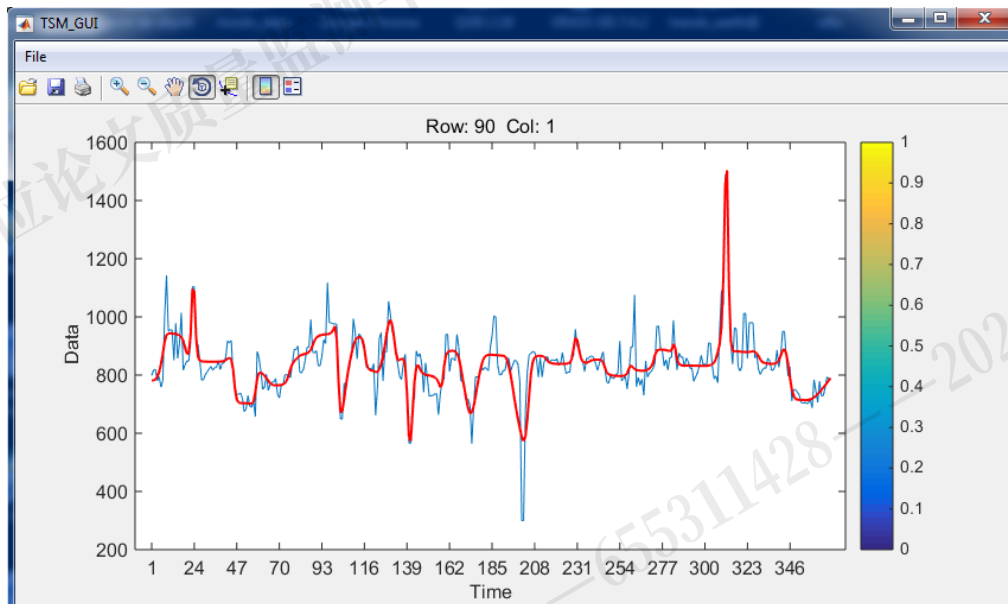


Figure A.7. Extraction of seasonal dynamics metrics from MOD13Q1 scaled EVI data time series through TIMESAT (TSM_GUI command).

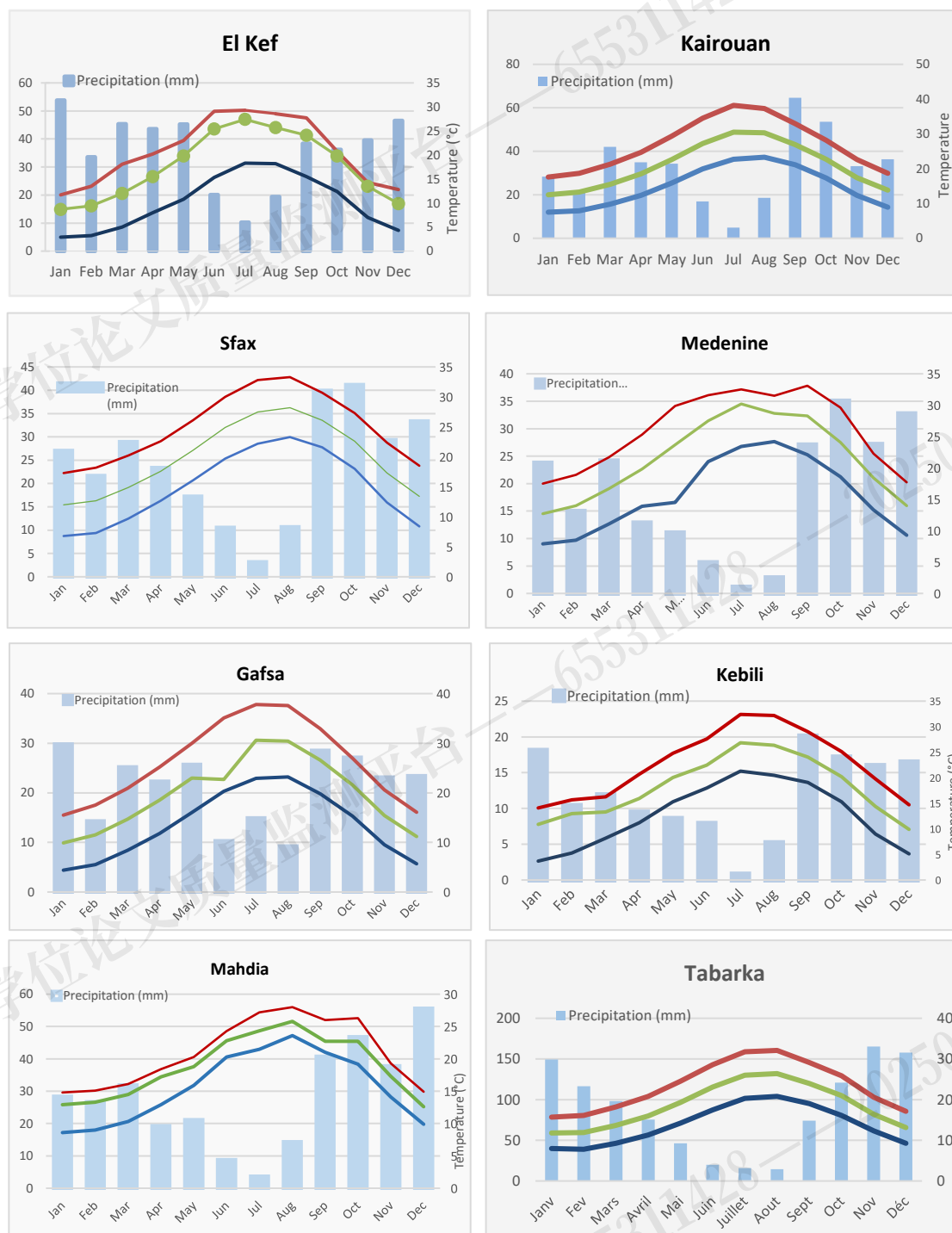


Figure A.8. Climate normal (1981-2021) of precipitation and maximum, mean, and minimum temperatures for some of the selected rain gauge stations in Tunisia.

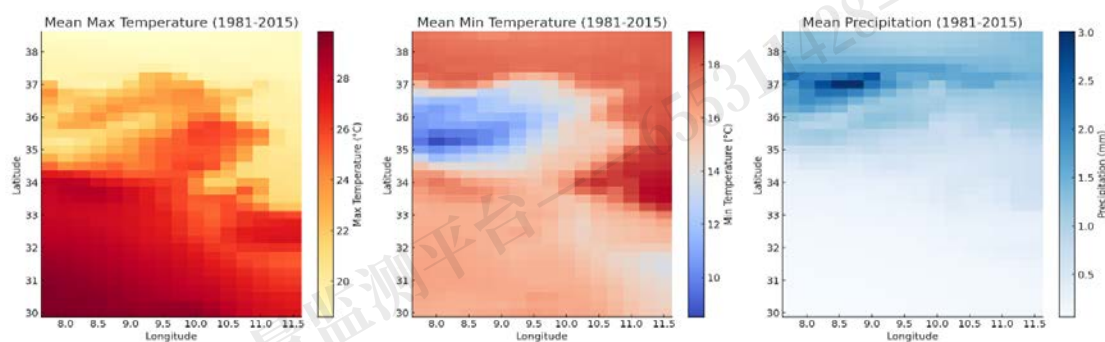


Figure A.9. Climatological Trends in Tunisia: Mean, maximum temperature, minimum temperature, and precipitation for the baseline period (1981–2015)

